

gilmore-lie-groups: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00001	10	1.000	$U(2) \rightarrow H_4$	$U(2) \rightarrow H_4$	13.5 Conclusion
gilmore-lie-groups_F00002	10	1.000	$S_0(3)$	$SO(3)$	14.7 Relation with dynamics i
gilmore-lie-groups_F00003	10	1.000	$S_0(4)$	$SO(4)$	14.8 DeSitter symmetry $SO(4,$
gilmore-lie-groups_F00004	10	0.997	$S_0(4,1)$	$SO(4,1)$	14.10 Spin angular momentum 14.11 Spectrum generating group
gilmore-lie-groups_F00005	10	1.000	$S_0(4,2)$	$SO(4,2)$	14.11 Spectrum generating group 14.12 Conclusion
gilmore-lie-groups_F00006	15	0.764	$(+,-,\times,\div)$	$(+,-,\times,\div)$	to decide on the solvability of finite degree polynomial equations; then “finite groups” (i.e. groups depending continuously on one or more real or complex vari-
gilmore-lie-groups_F00007	15	0.764	n	n	to decide on the solvability of finite degree polynomial equations; then “finite groups” (i.e. groups depending continuously on one or more real or complex vari-
gilmore-lie-groups_F00008	17	1.000	G	G	(i) G_{i+1} is an invariant subgroup of G_i ;
gilmore-lie-groups_F00009	17	1.000	$G=G_{\{0\}} \supset G_{\{1\}} \supset$	$G = G_0 \supset G_1 \supset$	(i) G_{i+1} is an invariant subgroup of G_i ;
gilmore-lie-groups_F00010	17	1.000	$\cdots \supset G_{\{\omega\}}=I$	$\cdots \supset G_\omega = I$	(ii) each factor group G_i/G_{i+1} is comm
gilmore-lie-groups_F00011	17	1.000	$G_{\{i+1\}}$	G_{i+1}	In the statement of this theorem the fi
gilmore-lie-groups_F00012	17	1.000	$G_{\{i\}}$	G_i	In the statement of this theorem the fi
gilmore-lie-groups_F00013	17	1.000	$G_{\{i\}} / G_{\{i+1\}}$	G_i/G_{i+1}	“solvable by radicals.” This means that in addi
gilmore-lie-groups_F00014	17	0.998	$G=\left\{g_{\{1\}}, g_{\{2\}}, \ldots\right\}$	$G = \{g_1,g_2,\ldots\}$	
gilmore-lie-groups_F00015	18	1.000	$g_{\{i\}} \in G, g_{\{j\}} \in G$	$g_i \in G, g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \cdot g_j \in$
gilmore-lie-groups_F00016	18	1.000	$g_{\{i\}} \cdot g_{\{j\}} \in G$	$g_i \cdot g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \cdot g_j \in$
gilmore-lie-groups_F00017	18	1.000	$g_{\{i\}} \in G, g_{\{j\}} \in G, g_{\{k\}} \in G$	$g_i \in G, g_j \in G, g_k \in G$	(ii) Associativity: for all $g_i \in G, g_j \in G, g_k \in$
gilmore-lie-groups_F00018	18	0.835	I	I	(iii) Identity: there is a group operation, $\cdot$ (identity operator), with the property that
gilmore-lie-groups_F00019	18	1.000	$g_{\{i\}}$	g_i	$\left\{g_i, g_j, g_k, \ldots\right\}$
gilmore-lie-groups_F00020	18	1.000	$g_{\{i\}}^{-1}$	g_i^{-1}	$\left\{g_i, g_j, g_k, \ldots\right\}$
gilmore-lie-groups_F00021	18	1.000	$z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}$	$z_1, z_2, \ldots, z_n$	$\left\{z_1, z_2, \ldots, z_n\right\}$
gilmore-lie-groups_F00022	18	0.658	$P_{\{n\}}$	P_n	The permutation group S_n has a particularly convenient representation in terms

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gilmore-lie-groups-F00023	18	0.658	$S_{\{n\}}$	S_n	The permutation group S_n has a particularly convenient representation in terms
gilmore-lie-groups-F00024	18	0.658	$n!$	$n!$	The permutation group S_n has a particularly convenient representation in terms
gilmore-lie-groups-F00025	18	0.996	$n \times n$	$n \times n$	$\begin{matrix} \Gamma & 1 & 0 & 0 \\ \Gamma & 0 & 1 & 0 \\ \Gamma & 0 & 0 & 1 \end{matrix}$
gilmore-lie-groups-F00026	18	1.000	$6=3!3 \times 3$	$6 = 3!3 \times 3$	$I \rightarrow \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} \quad (123) \rightarrow \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} \quad (321) \rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$
gilmore-lie-groups-F00027	18	1.000	$S_{\{3\}}$	S_3	$\begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$
gilmore-lie-groups-F00028	19	0.998	$z_{\{1\}}$	z_1	The symbol (123) means that the first root, z_1 , is replaced by z_2 , z_2 is replaced by
gilmore-lie-groups-F00029	19	0.998	$z_{\{2\}}, z_{\{2\}}$	z_2, z_2	The symbol (123) means that the first root, z_1 , is replaced by z_2 , z_2 is replaced by
gilmore-lie-groups-F00030	19	0.999	$z_{\{3\}}$	z_3	z_3 , and z_3 is replaced by z_1
gilmore-lie-groups-F00031	19	0.768	$\times$	$\times$	tion of a group, $G \rightarrow \Gamma(G)$, is that the product of matrices representing two group operations $\Gamma(g_i) \times \Gamma(g_j) \in \Gamma(G \times G)$ is equal to the matrix representing the product of these
gilmore-lie-groups-F00032	19	1.000	$G \rightarrow \Gamma(G)$	$G \rightarrow \Gamma(G)$	operations in the group $(g_i \times g_j) \in G \times G$ to group operations $g_i, g_j \in G$. This permutation representation of S_3 is 1:1 or a faithful representation of S_3 .
gilmore-lie-groups-F00033	19	0.988	$\left(\Gamma(g_i) \times \Gamma(g_j)\right)$	$(\Gamma(g_i) \times \Gamma(g_j))$	since knowledge of the 3×3 matrix uniquely identifies the original group operation
gilmore-lie-groups-F00034	19	0.999	$\left(\Gamma(g_i \cdot g_j)\right)$	$(\Gamma(g_i \cdot g_j))$	in S_3 .
gilmore-lie-groups-F00035	19	0.999	$g_i, g_j \in G$	$g_i, g_j \in G$	in S_3 .
gilmore-lie-groups-F00036	19	0.874	3×3	3×3	under the group multiplication in G .
gilmore-lie-groups-F00037	19	1.000	H	H	particular suberoud is denoted A_3 (alternating group). It consists of those oper-
gilmore-lie-groups-F00038	19	1.000	$A_{\{3\}}$	A_3	$S_2(12) = \{I, (12)\}$
gilmore-lie-groups-F00039	19	1.000	$A_{\{3\}} \subset S_{\{3\}}$	$A_3 \subset S_3$	
gilmore-lie-groups-F00040	19	1.000	$S_{\{2\}}(i \ j)$	$S_2(ij)$	
gilmore-lie-groups-F00041	20	0.706	$(123) \cdot (123)=(321)$	$(123) \cdot (123) = (321)$	$(12) = (23)$, etc. In fact, the two operations (123) , (12) generate S_3 by taking products of various lengths in various order
gilmore-lie-groups-F00042	20	1.000	$G, H_{\{1\}} \subset G$	$G, H_1 \subset G$	$g^{1118} = 112 \quad (1.2)$
gilmore-lie-groups-F00043	20	1.000	$H_{\{2\}} \subset G$	$H_2 \subset G$	$g^{1118} = 112 \quad (1.2)$
gilmore-lie-groups-F00044	20	1.000	$g \in G$	$g \in G$	Example The subgroups S_2
gilmore-lie-groups-F00045	20	1.000	$S_{\{2\}}(12)$	$S_2(12)$	On the other hand, the alternating group $A_3 \subset S_3$ is self-conjugate , operation in $G = S_3$ carries merely to permute the group operations in
gilmore-lie-groups-F00046	20	1.000	$S_{\{2\}}(13)$	$S_2(13)$	On the other hand, the alternating group $A_3 \subset S_3$ is self-conjugate , operation in $G = S_3$ carries merely to permute the group operations in
gilmore-lie-groups-F00047	20	1.000	$G=S_{\{3\}}$	$G = S_3$	A subgroup $H \subset G$ which is self-conjugate under all operations in G is called
gilmore-lie-groups-F00048	20	1.000	$H \subset G$	$H \subset G$	

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gilmore-lie-groups_F00049	21	■ 1.000	f	f	$\times$ is called a homomorphism if it preserves group multiplication:
gilmore-lie-groups_F00050	21	■ 1.000	$g_{\{1\}}, g_{\{2\}}, \ldots$	$g_1, g_2, \dots$	$\times$ is called a homomorphism if it preserves group multiplication:
gilmore-lie-groups_F00051	21	■ 0.999	$h_{\{1\}}, h_{\{2\}}, \ldots$	$h_1, h_2, \dots$	
gilmore-lie-groups_F00052	21	■ 1.000	$h_{\{i\}} \in H$	$h_i \in H$	unique inverse image $g \in G$ ($h_1 = f(g_1)$ and $h_2 = f(g_2)$) $h_1 = h_2 \Rightarrow g_1 = g_2$ and mapping f is an isomorphism .
gilmore-lie-groups_F00053	21	■ 1.000	$g_{\{j\}} \in G$	$g_j \in G$	
gilmore-lie-groups_F00054	21	■ 1.000	$h \in H$	$h \in H$	
gilmore-lie-groups_F00055	21	■ 0.933	$g \in G \backslash \left(h_{\{1\}} = f \backslash \left(g_{\{1\}} \right) \right) \backslash \text{right}.$	$g \in G \left(h_1 = f \left(g_1 \right) \right)$	Example The S.T mapping f of S_3 onto S_2 given by
gilmore-lie-groups_F00056	21	■ 0.933	$\backslash \text{left} . h_{\{2\}} = f \backslash \left(g_{\{2\}} \right) \backslash \text{right} , h_{\{1\}} = h_{\{2\}} \Rightarrow g_{\{1\}} = g_{\{2\}} \Rightarrow g_{\{1\}} = g_{\{2\}} \backslash \text{right})$	$h_2 = f \left(g_2 \right) , h_1 = h_2 \Rightarrow g_1 = g_2$	Example The S.T mapping f of S_3 onto S_2 given by
gilmore-lie-groups_F00057	21	■ 1.000	$S_{\{2\}}$	S_2	$\begin{array}{c} I, (123), (321) \longrightarrow I \\ (12), (23), (31) \longrightarrow (12) \end{array}$
gilmore-lie-groups_F00058	22	■ 0.987	h	h	product of an element h in the subgroup H with a group element in a quotient, or coset (denoted G/H). A coset is a subset of G . If the order of G is $ G $ (S_3 has $3! = 6$ group elements, so the order of S_2 is 6), then the order of G/H is
gilmore-lie-groups_F00059	22	■ 1.000	G / H	G/H	or coset (denoted G/H), a coset is a subset of G . If the order of G is $ G $ (S_3 has $3! = 6$ group elements, so the order of S_2 is 6), then the order of G/H is
gilmore-lie-groups_F00060	22	■ 1.000	$ G \backslash \left(S_{\{3\}} \right) \backslash \text{right}.$	$ G \left(S_3 \right)$	or coset (denoted G/H), a coset is a subset of G . If the order of G is $ G $ (S_3 has $3! = 6$ group elements, so the order of S_2 is 6), then the order of G/H is
gilmore-lie-groups_F00061	22	■ 0.637	$3!=6$	$3! = 6$	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups_F00062	22	■ 0.866	$ G / H = G / H $	$ G/H = G / H $	$H = S_2(23) = \{I, (23)\}$ we have
gilmore-lie-groups_F00063	22	■ 0.866	$H=A_{\{3\}}=\{I$	$H = A_3 = \{I$	$H = S_2(23) = \{I, (23)\}$ we have
gilmore-lie-groups_F00064	22	■ 0.866	$\backslash \}$	$\}$	$H = S_2(23) = \{I, (23)\}$ we have
gilmore-lie-groups_F00065	22	■ 0.992	$H=S_{\{2\}}(23)=\{I, (23)\}$	$H = S_2(23) = \{I, (23)\}$	$\begin{array}{c} G/H \\ I \end{array}$
gilmore-lie-groups_F00066	22	■ 1.000	$ G / H $	$ G / H $	the subgroup A_3 we could equally well have chosen $G/H = S_3/A_3 = \{I, (123), (321)\}$ or $H = (23)$, for $S_3(23)$ we could equally well have chosen $G/H = S_3/S_3(23) = \{I, (123), (321)\}$.
gilmore-lie-groups_F00067	22	■ 0.692	$G / H=S_{\{3\}} / A_{\{3\}}=\{I$	$G/H = S_3/A_3 = \{I$	$\{I, (123), (321)\}$.
gilmore-lie-groups_F00068	22	■ 0.997	$\backslash \{I, (23)\} \backslash \}$	$\{I, (23)\}$	In general, it is not possible to choose the group elements in G/H so that they
gilmore-lie-groups_F00069	22	■ 0.997	$S_{\{2\}}(23)$	$S_2(23)$	In general, it is not possible to choose the group elements in G/H so that they
gilmore-lie-groups_F00070	22	■ 0.997	$G / H=S_{\{3\}} / S_{\{2\}}(23)=$	$G/H = S_3/S_2(23) =$	In general, it is not possible to choose the group elements in G/H so that they
gilmore-lie-groups_F00071	22	■ 0.556	$\backslash \{I, (123), (321)\} \backslash \}$	$\{I, (123), (321)\}$	form a sub;
gilmore-lie-groups_F00072	22	■ 1.000	$S_{\{3\}} / A_{\{3\}}$	S_3/A_3	then the group G is the direct product of the invariant subgroup H with the factor group G/H : $G = G/H \times H$
gilmore-lie-groups_F00073	22	■ 0.630	$\backslash \operatorname{operatorname{group}} G / H: G=G / H \times H$	group $G/H : G = G/H \times H$	The general n th degree pol
gilmore-lie-groups_F00074	22	■ 1.000	k	k	$I_1 = \sum z_i = z_1 + z_2 + \dots + z_n$

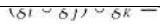
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups-F00075	22	■ 1.000	$I_{\{k\}}$	I_k	$I_1 = \sum z_i = z_1 + z_2 + \cdots + z_n$
gilmore-lie-groups-F00076	22	■ 1.000	$z_{\{i\}}$	z_i	$I_1 = \sum z_i = z_1 + z_2 + \cdots + z_n$
gilmore-lie-groups-F00077	23	■ 0.992	$I_{\{k\}}(k=1,2, \ldots, n)$	$I_k(k = 1, 2, \ldots, n)$	The n functions I_k ($k = 1, 2, \ldots, n$) of the n roots $(z_1, z_2, \ldots, z_n)$ are symmetric:
gilmore-lie-groups-F00078	23	■ 0.992	$\left(z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}\right)$	$(z_1, z_2, \ldots, z_n)$	The n functions I_k ($k = 1, 2, \ldots, n$) of the n roots $(z_1, z_2, \ldots, z_n)$ are symmetric:
gilmore-lie-groups-F00079	23	■ 1.000	$f\left(z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}\right)$	$f(z_1, z_2, \ldots, z_n)$	any function $f(z_1, z_2, \ldots, z_n)$ that is invariant under S_n can be written as a function
gilmore-lie-groups-F00080	23	■ 1.000	$I_{\{1\}}, I_{\{2\}}, \ldots, I_{\{n\}}$	$I_1, I_2, \ldots, I_n$	of the invariants $I_1, I_2, \ldots, I_n$. The invariants are easily expressed in terms of the roots (see Box 11.15). The inverse map, that of expressing the roots in terms
gilmore-lie-groups-F00081	23	■ 1.000	$G_{\{\omega-1\}} \supset G_{\{\omega\}}=I$	$G_{\omega-1} \supset G_{\omega} = I$	the $ G_{\omega-1} / G_{\omega} $ inequivalent one-dimensional representations of $G_{\omega-1}$. Linear
gilmore-lie-groups-F00082	23	■ 1.000	$G_{\{\omega-1\}} / G_{\{\omega\}}=G_{\{\omega-1\}}$	$G_{\omega-1}/G_{\omega} = G_{\omega-1}$	combinations of the roots z_i are identified that transform under (i.e., are basis
gilmore-lie-groups-F00083	23	■ 1.000	$\left G_{\{\omega-1\}}\right /\left G_{\{\omega\}}\right $	$ G_{\omega-1} / G_{\omega} $	functions for) the one-dimensional irreducible representations of $G_{\omega-1}$. These functions are
gilmore-lie-groups-F00084	23	■ 1.000	$G_{\{\omega-1\}}$	$G_{\omega-1}$	functions for) the one-dimensional irreducible representations of $G_{\omega-1}$. These functions are
gilmore-lie-groups-F00085	23	■ 1.000	$G_{\{\omega\}}=I$	$G_{\omega} = I$	Next, the next pair of group
gilmore-lie-groups-F00086	23	■ 1.000	$G_{\{\omega-2\}} \supset G_{\{\omega-1\}}$	$G_{\omega-2} \supset G_{\omega-1}$	i) symmetric under $G_{\omega-1}$, ii) not all symmetric under $G_{\omega-2}$
gilmore-lie-groups-F00087	23	■ 1.000	$G_{\{\omega-2\}}$	$G_{\omega-2}$	These solutions are then fed dov
gilmore-lie-groups-F00088	23	■ 1.000	$G=G_{\{0\}} \supset$	$G = G_0 \supset$	he roots $z_1, z_2, \ldots, z_n$ and functions of the invariants $I_1, I_2, \ldots, I_n$.
gilmore-lie-groups-F00089	23	■ 1.000	$G_{\{1\}}$	G_1	This brief description will now be illustrated by using Galois theory to solve
gilmore-lie-groups-F00090	24	■ 1.000	$\Gamma^{\{1\}}, \Gamma^{\{2\}}$	Γ^1, Γ^2	$\left[u_2\right]^{\overline{-}}$
gilmore-lie-groups-F00091	24	■ 1.000	$r_{\{1\}}-r_{\{2\}}$	r_1-r_2	$(r_1-r_2)\left\{\begin{array}{l} \longrightarrow+(r_1-r_2) \\ \longrightarrow-(r_1-r_2) \end{array}\right.$ (1.20)
gilmore-lie-groups-F00092	24	■ 0.899	$\left(r_{\{1\}}-r_{\{2\}}\right)$	(r_1-r_2)	$\begin{array}{l} \longrightarrow+(r_1-r_2) \\ \longrightarrow-(r_1-r_2) \end{array}$
gilmore-lie-groups-F00093	24	■ 1.000	$\Gamma^{\{2\}}$	Γ^2	$(r_1-r_2)^{\overline{-}}=r_1^{\overline{-}}-2 r_1 r_2+r_2^{\overline{-}}$
gilmore-lie-groups-F00094	24	■ 1.000	$\Gamma^{\{1\}}$	Γ^1	$(r_1-r_2)^{\overline{-}}=r_1^{\overline{-}}-2 r_1 r_2+r_2^{\overline{-}}$
gilmore-lie-groups-F00095	24	■ 1.000	$\left(r_{\{1\}}-r_{\{2\}}\right)^{\{2\}}$	$(r_1-r_2)^2$	$=r_1^2+2 r_1 r_2+r_2^2-4 r_1 r_2=I_1^2-4 I_2=D$ (1.21)
gilmore-lie-groups-F00096	24	■ 0.999	$I_{\{1\}}, I_{\{2\}}$	I_1, I_2	
gilmore-lie-groups-F00097	25	■ 1.000	D	D	where D is the discriminant of the quadratic equation. Since $(r_1-r_2)=\pm \sqrt{D}$,
gilmore-lie-groups-F00098	25	■ 1.000	$\left(r_{\{1\}}-r_{\{2\}}\right)=\pm \sqrt{D}$	$(r_1-r_2)=\pm \sqrt{D}$	where D is the discriminant of the quadratic equation. Since $(r_1-r_2)=\pm \sqrt{D}$,
gilmore-lie-groups-F00099	25	■ 1.000	x	x	$\frac{1}{5} I_1$
gilmore-lie-groups-F00100	25	■ 1.000	z	z	$z=x+\frac{1}{5} I_1$

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gilmore-lie-groups_ F00101	26	■ 1.000	$S_{\{3\}} / A_{\{3\}} = S_{\{2\}}$	$S_3/A_3 = S_2$	equation can be solved.
gilmore-lie-groups_ F00102	26	■ 1.000	$A_{\{3\}} / I = A_{\{3\}}$	$A_3/I = A_3$	equation can be solved.
gilmore-lie-groups_ F00103	26	■ 1.000	$A_{\{3\}} \setminus \supseteq I$	$A_3 \supset I$	$I = (123) \cup (321)$ Basic functions
gilmore-lie-groups_ F00104	26	■ 0.877	$v_{\{1\}} = s_{\{1\}} + s_{\{2\}} + s_{\{3\}}$	$v_1 = s_1 + s_2 + s_3$	$\begin{matrix} \Gamma^3 \\ \Gamma^3 \end{matrix} \begin{matrix} 1 & \omega & \omega^* \\ 1 & \omega^2 & \omega \end{matrix} \begin{matrix} v_2 = s_1 + \omega s_2 + \omega^* s_3 \\ v_3 = s_1 + \omega^2 s_2 + \omega s_3 \end{matrix}$ $\omega^3 = \pm 1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$
gilmore-lie-groups_ F00105	26	■ 0.877	ω	ω	$\begin{matrix} \Gamma^3 \\ \Gamma^3 \end{matrix} \begin{matrix} 1 & \omega & \omega^* \\ 1 & \omega^2 & \omega \end{matrix} \begin{matrix} v_2 = s_1 + \omega s_2 + \omega^* s_3 \\ v_3 = s_1 + \omega^2 s_2 + \omega s_3 \end{matrix}$ $\omega^3 = \pm 1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$
gilmore-lie-groups_ F00106	26	■ 0.877	$\omega^{\{2\}}$	ω^2	$\begin{matrix} \Gamma^3 \\ \Gamma^3 \end{matrix} \begin{matrix} 1 & \omega & \omega^* \\ 1 & \omega^2 & \omega \end{matrix} \begin{matrix} v_2 = s_1 + \omega s_2 + \omega^* s_3 \\ v_3 = s_1 + \omega^2 s_2 + \omega s_3 \end{matrix}$ $\omega^3 = \pm 1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$
gilmore-lie-groups_ F00107	26	■ 0.877	$v_{\{2\}} = s_{\{1\}} + \omega s_{\{2\}} + \omega^{\{2\}} s_{\{3\}}$	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	$\begin{matrix} \Gamma^3 \\ \Gamma^3 \end{matrix} \begin{matrix} 1 & \omega & \omega^* \\ 1 & \omega^2 & \omega \end{matrix} \begin{matrix} v_2 = s_1 + \omega s_2 + \omega^* s_3 \\ v_3 = s_1 + \omega^2 s_2 + \omega s_3 \end{matrix}$ $\omega^3 = \pm 1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$
gilmore-lie-groups_ F00108	26	■ 0.877	$\Gamma^{\{3\}}$	Γ^3	$\begin{matrix} \Gamma^3 \\ \Gamma^3 \end{matrix} \begin{matrix} 1 & \omega & \omega^* \\ 1 & \omega^2 & \omega \end{matrix} \begin{matrix} v_2 = s_1 + \omega s_2 + \omega^* s_3 \\ v_3 = s_1 + \omega^2 s_2 + \omega s_3 \end{matrix}$ $\omega^3 = \pm 1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$
gilmore-lie-groups_ F00109	26	■ 0.877	$v_{\{3\}} = s_{\{1\}} + \omega^{\{2\}} s_{\{2\}} + \omega s_{\{3\}}$	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$	$\begin{matrix} \Gamma^3 \\ \Gamma^3 \end{matrix} \begin{matrix} 1 & \omega & \omega^* \\ 1 & \omega^2 & \omega \end{matrix} \begin{matrix} v_2 = s_1 + \omega s_2 + \omega^* s_3 \\ v_3 = s_1 + \omega^2 s_2 + \omega s_3 \end{matrix}$ $\omega^3 = \pm 1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$
gilmore-lie-groups_ F00110	26	■ 1.000	$(123)^{\{-1\}}$	$(123)^{-1}$	
gilmore-lie-groups_ F00111	26	■ 1.000	$v_{\{2\}}$	v_2	
gilmore-lie-groups_ F00112	27	■ 1.000	$v_{\{1\}}$	v_1	Since v_1 is symmetric under both A_3 and S_3 , it can be expressed in terms of the
gilmore-lie-groups_ F00113	27	■ 1.000	$v_{\{3\}}$	v_3	the remaining functions, v_2 and v_3 , are symmetric under I but not under A_3 . We now proceed to the next group-subgroup pair: $S_3 \supset A_3$. To construct func-
gilmore-lie-groups_ F00114	27	■ 1.000	$S_{\{3\}} \setminus \supseteq A_{\{3\}}$	$S_3 \supset A_3$	the next proceeds to the next group-subgroup pair: $S_3 \supset A_3$. To construct func-
gilmore-lie-groups_ F00115	27	■ 1.000	$S_{\{2\}} = S_{\{3\}} / A_{\{3\}}$	$S_2 = S_3/A_3$	form
gilmore-lie-groups_ F00116	27	■ 1.000	$v_{\{2\}}^{\{3\}}$	v_2^3	form

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gilmore-lie-groups_ F00117	27	■ 1.000	$v_{\{3\}}^{\wedge\{3\}}$	v_3^3	form																				
gilmore-lie-groups_ F00118	27	■ 1.000	$J_{\{1\}}, J_{\{2\}}$	J_1, J_2	(of orders 1, 2, 3) of the original cubic equation (1.29) as follows:																				
gilmore-lie-groups_ F00119	27	■ 0.995	$I_{\{1\}}, I_{\{2\}}, I_{\{3\}}$	I_1, I_2, I_3																					
gilmore-lie-groups_ F00120	27	■ 0.995	$J_{\{1\}}$	J_1																					
gilmore-lie-groups_ F00121	27	■ 0.995	$J_{\{2\}}$	J_2																					
gilmore-lie-groups_ F00122	27	■ 1.000	$I_{\{1\}}=s_{\{1\}}+s_{\{2\}}+$	$I_1 = s_1 + s_2 +$	$z = y + \frac{1}{2}I_1$ (1.38)																				
gilmore-lie-groups_ F00123	27	■ 1.000	$s_{\{3\}}=0$	$s_3 = 0$																					
gilmore-lie-groups_ F00124	28	■ 1.000	$J_{\{1\}}=v_{\{2\}}^{\wedge\{3\}}+v_{\{3\}}^{\wedge\{3\}}$	$J_1 = v_2^3 + v_3^3$	as follows																				
gilmore-lie-groups_ F00125	28	■ 1.000	$J_{\{2\}}=v_{\{2\}}^{\wedge\{3\}}\ v_{\{3\}}^{\wedge\{3\}}$	$J_2 = v_2^3 v_3^3$	as follows																				
gilmore-lie-groups_ F00126	28	■ 1.000	$I_{\{2\}}^{\wedge\{\backslash\mathrm{prime}\}}, I_{\{3\}}^{\wedge\{\backslash\mathrm{prime}\}}$	I_2', I_3'	as follows																				
gilmore-lie-groups_ F00127	28	■ 1.000	$v_{\{2\}}^{\wedge\{3\}}, v_{\{3\}}^{\wedge\{3\}}$	v_2^3, v_3^3	$x^2 - (-27I_3')x + (-27I_2'^3) = 0$																				
gilmore-lie-groups_ F00128	28	■ 1.000	$s_{\{1\}}, s_{\{2\}}, s_{\{3\}}$	s_1, s_2, s_3	$\left \begin{array}{ccc c}1&\omega&\omega^2&\\1&\omega^2&\omega&\end{array}\right \left \begin{array}{c}s_2\\s_3\end{array}\right =\left \begin{array}{c}v_2\\v_3\end{array}\right $																				
gilmore-lie-groups_ F00129	28	■ 1.000	$v_{\{1\}}, v_{\{2\}}, v_{\{3\}}$	v_1, v_2, v_3	$\left \begin{array}{ccc c}1&\omega&\omega^2&\\1&\omega^2&\omega&\end{array}\right \left \begin{array}{c}s_2\\s_3\end{array}\right =\left \begin{array}{c}v_2\\v_3\end{array}\right $																				
gilmore-lie-groups_ F00130	29	■ 1.000	$z=z^{\wedge\{\backslash\mathrm{prime}\}}+\frac{1}{4}\ I_{\{1\}}$	$z = z' + \frac{1}{4}I_1$	$(z' - t_1)(z' - t_2)(z' - t_3)(z' - t_4) = z'^4 - I_1' z'^3 + I_2' z'^2 - I_3' z' + I_4' =$																				
gilmore-lie-groups_ F00131	29	■ 1.000	$S_{\{4\}}$	S_4	(1) In the permutation matrix representation. The four-group (vierergruppe, Klein group, Klein four group) V_4 is $\{I, (12)(34), (13)(24), (14)(23)\}$. The chain																				
gilmore-lie-groups_ F00132	29	■ 1.000	$A_{\{4\}}$	A_4	$\{I, (12)(34), (13)(24), (14)(23), (1234), (1243), (1342), (1432)\}$.																				
gilmore-lie-groups_ F00133	29	■ 0.957	$V_{\{4\}}$	V_4																					
gilmore-lie-groups_ F00134	29	■ 0.957	$\backslash\{I, (12)(34), (13)(24), (14)(23)\}$	$\{I, (12)(34), (13)(24), (14)(23)\}$																					
gilmore-lie-groups_ F00135	30	■ 1.000	$S_{\{4\}}\ /\ A_{\{4\}}=S_{\{2\}}$	$S_4/A_4 = S_2$	(i) A_4 is invariant in S_4 and $S_4/A_4 =$																				
gilmore-lie-groups_ F00136	30	■ 0.987	$A_{\{4\}}\ /\ V_{\{4\}}=C_{\{3\}}=\backslash\{I$	$A_4/V_4 = C_3 = \{I$	(ii) V_4 is invariant in A_4 and $A_4/V_4 = C_3 = \{I, (254), (452)\}$ (iii) I is invariant in V_4 and $V_4/I = V_4 = \{I, (12)(34), (13)(24), (14)(23)\}$.																				
gilmore-lie-groups_ F00137	30	■ 0.987	$V_{\{4\}}\ /\ I=V_{\{4\}}=\backslash\{I$	$V_4/I = V_4 = \{I$																					
gilmore-lie-groups_ F00138	30	■ 0.669	$w_{\{1\}}=t_{\{1\}}+t_{\{2\}}+t_{\{3\}}+t_{\{4\}}$	$w_1 = t_1 + t_2 + t_3 + t_4$	$\Gamma^1$$\Gamma^2$$\Gamma^3$$\Gamma^4$ <table><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>$w_1 = t_1 + t_2 + t_3 + t_4$</td></tr><tr><td>1</td><td>1</td><td>-1</td><td>-1</td><td>$w_2 = t_1 + t_2 - t_3 - t_4$</td></tr><tr><td>1</td><td>-1</td><td>1</td><td>-1</td><td>$w_3 = t_1 - t_2 + t_3 - t_4$</td></tr><tr><td>1</td><td>-1</td><td>-1</td><td>1</td><td>$w_4 = t_1 - t_2 - t_3 + t_4$</td></tr></table> The linear combinations of these roots that transform under each of the	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$
1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$																					
1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$																					
1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$																					
1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$																					

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gilmore-lie-groups_ F00139	30	0.669	$w_{\{2\}}=t_{\{1\}}+t_{\{2\}}-t_{\{3\}}-t_{\{4\}}$	$w_2 = t_1 + t_2 - t_3 - t_4$	$\begin{array}{c cccc} \Gamma^1 & 1 & 1 & 1 & 1 \\ \Gamma^2 & 1 & 1 & -1 & -1 \\ \Gamma^3 & 1 & -1 & 1 & -1 \\ \Gamma^4 & 1 & -1 & -1 & 1 \end{array} \quad \begin{array}{l} w_1 = t_1 + t_2 + t_3 + t_4 \\ w_2 = t_1 + t_2 - t_3 - t_4 \\ w_3 = t_1 - t_2 + t_3 - t_4 \\ w_4 = t_1 - t_2 - t_3 + t_4 \end{array}$ The linear combinations of these roots that transform under each of the
gilmore-lie-groups_ F00140	30	0.669	$w_{\{3\}}=t_{\{1\}}-t_{\{2\}}+t_{\{3\}}-t_{\{4\}}$	$w_3 = t_1 - t_2 + t_3 - t_4$	$\begin{array}{c cccc} \Gamma^1 & 1 & 1 & 1 & 1 \\ \Gamma^2 & 1 & 1 & -1 & -1 \\ \Gamma^3 & 1 & -1 & 1 & -1 \\ \Gamma^4 & 1 & -1 & -1 & 1 \end{array} \quad \begin{array}{l} w_1 = t_1 + t_2 + t_3 + t_4 \\ w_2 = t_1 + t_2 - t_3 - t_4 \\ w_3 = t_1 - t_2 + t_3 - t_4 \\ w_4 = t_1 - t_2 - t_3 + t_4 \end{array}$ The linear combinations of these roots that transform under each of the
gilmore-lie-groups_ F00141	30	0.669	Γ^4	Γ^4	$\begin{array}{c cccc} \Gamma^1 & 1 & 1 & 1 & 1 \\ \Gamma^2 & 1 & 1 & -1 & -1 \\ \Gamma^3 & 1 & -1 & 1 & -1 \\ \Gamma^4 & 1 & -1 & -1 & 1 \end{array} \quad \begin{array}{l} w_1 = t_1 + t_2 + t_3 + t_4 \\ w_2 = t_1 + t_2 - t_3 - t_4 \\ w_3 = t_1 - t_2 + t_3 - t_4 \\ w_4 = t_1 - t_2 - t_3 + t_4 \end{array}$ The linear combinations of these roots that transform under each of the
gilmore-lie-groups_ F00142	30	0.669	$w_{\{4\}}=t_{\{1\}}-t_{\{2\}}-t_{\{3\}}+t_{\{4\}}$	$w_4 = t_1 - t_2 - t_3 + t_4$	$\begin{array}{c cccc} \Gamma^1 & 1 & 1 & 1 & 1 \\ \Gamma^2 & 1 & 1 & -1 & -1 \\ \Gamma^3 & 1 & -1 & 1 & -1 \\ \Gamma^4 & 1 & -1 & -1 & 1 \end{array} \quad \begin{array}{l} w_1 = t_1 + t_2 + t_3 + t_4 \\ w_2 = t_1 + t_2 - t_3 - t_4 \\ w_3 = t_1 - t_2 + t_3 - t_4 \\ w_4 = t_1 - t_2 - t_3 + t_4 \end{array}$ The linear combinations of these roots that transform under each of the
gilmore-lie-groups_ F00143	30	0.971	$w_{\{2\}}, w_{\{3\}}, w_{\{4\}}$	w_2, w_3, w_4	to construct from these linear combinations functions that are
gilmore-lie-groups_ F00144	30	1.000	$A_{\{4\}} \supset V_{\{4\}}$	$A_4 \supset V_4$	(i) symmetric under V_4 ,
gilmore-lie-groups_ F00145	30	1.000	$A_{\{4\}} / V_{\{4\}}$	A_4/V_4	sum of the roots is zero, the three functions w_2, w_3, w_4
gilmore-lie-groups_ F00146	30	0.999	$w_{\{1\}}=I_{\{1\}}$	$w_1 = I_1$	$w_2 = (t_1 + t_2 - t_3 - t_4) = 4(t_1 + t_2) - (t_1 + t_2 + t_3 + t_4) = 4w_1 - w_1$
gilmore-lie-groups_ F00147	30	0.999	$w_{\{2\}}^2, w_{\{3\}}^2, w_{\{4\}}^2$	w_2^2, w_3^2, w_4^2	$w_2 = (t_1 + t_2 - t_3 - t_4) = 4(t_1 + t_2) - (t_1 + t_2 + t_3 + t_4) = 4w_1 - w_1$
gilmore-lie-groups_ F00148	30	1.000	$w_{\{j\}}^2 (j=2,3,4)$	$w_j^2 (j = 2, 3, 4)$	
gilmore-lie-groups_ F00149	30	1.000	$C_{\{3\}}=A_{\{4\}} / V_{\{4\}}$	$C_3 = A_4/V_4$	
gilmore-lie-groups_ F00150	31	1.000	$J_{\{k\}}$	J_k	symmetric functions (coefficients) of the original quartic equation (1.45) or (1.46).
gilmore-lie-groups_ F00151	31	1.000	$C_{\{3\}}$	C_3	symmetric functions (coefficients) of the original quartic equation (1.45) or (1.46).
gilmore-lie-groups_ F00152	31	1.000	$S_{\{4\}} \supset A_{\{4\}}$	$S_4 \supset A_4$	If the three solutions of the resolvent cubic equation are called y_2, y_3, y_4 , then
gilmore-lie-groups_ F00153	31	1.000	$y_{\{2\}}, y_{\{3\}}, y_{\{4\}}$	y_2, y_3, y_4	$m_3 = +\sqrt{y_3}$
gilmore-lie-groups_ F00154	31	1.000	$w_{\{2\}} w_{\{3\}} w_{\{4\}}=8 I_{\{3\}}^{\prime}$	$w_2 w_3 w_4 = 8I_3'$	$\Gamma^1 \Gamma^2 \Gamma^3 \Gamma^4 = \Gamma^1 \Gamma^2 \Gamma^3 \Gamma^4 = \Gamma^1 \Gamma^2 \Gamma^3 \Gamma^4 = \Gamma^1 \Gamma^2 \Gamma^3 \Gamma^4$
gilmore-lie-groups_ F00155	31	1.000	$\pm \sqrt{y_{\{j\}}}$	$\pm \sqrt{y_j}$	$\Gamma^1 \Gamma^2 \Gamma^3 \Gamma^4 = \Gamma^1 \Gamma^2 \Gamma^3 \Gamma^4 = \Gamma^1 \Gamma^2 \Gamma^3 \Gamma^4 = \Gamma^1 \Gamma^2 \Gamma^3 \Gamma^4$

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gilmore-lie-groups_F00181	36	■ 0.996	$v_{\{2\}}, v_{\{3\}}=\frac{1}{\{2\}}(3 \pm$	$v_2, v_3 = \frac{1}{2}(3 \pm$	
gilmore-lie-groups_F00182	36	■ 1.000	$i \sqrt{3}$	$i5\sqrt{3}$	
gilmore-lie-groups_F00183	36	■ 0.543	$x+y, x-y, x \ y$	$x+y, x-y, xy$	numbers that can be constructed from integers by addition, subtraction, multiplication,
gilmore-lie-groups_F00184	36	■ 0.543	$x \ / \ y$	x/y	numbers that can be constructed from integers by addition, subtraction, multiplication,
gilmore-lie-groups_F00185	36	■ 1.000	$\sqrt{x}$	$\sqrt{x}$	division, and extraction of square roots is called the set of constructable numbers. This forms a subset of the numbers $x + iy = (x, y)$ in the complex plane. If a number is
gilmore-lie-groups_F00186	36	■ 1.000	$x+iy=(x, \ y)$	$x+iy=(x, y)$	number satisfies an algebraic equation of degree K with integer coefficients, where
gilmore-lie-groups_F00187	36	■ 1.000	K	K	a. Square a circle? For the circle of radius 1 the area is π . Squaring a circle means
gilmore-lie-groups_F00188	36	■ 1.000	$K=2^{\{n\}}$	$K=2^n$	finding an interval of length x ,
gilmore-lie-groups_F00189	36	■ 1.000	π	π	ruled and compass alone.
gilmore-lie-groups_F00190	36	■ 1.000	$x^{\{2\}}-\pi=0$	$x^2-\pi=0$	b. Double the cube? A cube with edge length 1 has volume $1^3=1$. A cube with
gilmore-lie-groups_F00191	36	■ 1.000	$1^{\{3\}}=1$	$1^3=1$	twice the volume has edge length v , where v satisfies $v^3=2=0$. Although the
gilmore-lie-groups_F00192	36	■ 1.000	$x^{\{3\}}-2=0$	$x^3-2=0$	c. Trisect an angle? If 3θ is some angle, the trigonometric functions of 3θ and
gilmore-lie-groups_F00193	36	■ 1.000	$3 \neq 2^{\{n\}}$	$3 \neq 2^n$	$\frac{1}{3}(3\theta)=\theta$ are related by
gilmore-lie-groups_F00194	36	■ 1.000	$3 \ \theta$	3θ	
gilmore-lie-groups_F00195	36	■ 1.000	$\frac{1}{\{3\}}(3 \ \theta)=\theta$	$\frac{1}{3}(3\theta)=\theta$	$\cos(3\theta)+i\sin(3\theta)=(\cos(\theta)+i\sin(\theta))^3$
gilmore-lie-groups_F00196	37	■ 1.000	$\cos \ (3 \ \theta)$	$\cos(3\theta)$	whether $\cos(3\theta)$ is rational or irrational, the equation for $\cos(\theta)$:
gilmore-lie-groups_F00197	37	■ 1.000	$\cos \ (\theta)$	$\cos(\theta)$	whether $\cos(3\theta)$ is rational or irrational, the equation for $\cos(\theta)$:
gilmore-lie-groups_F00198	37	■ 1.000	$\left(x^{\{2\}}+a \ x+b\right)(x+c)=0$	$(x^2+ax+b)(x+c)=0$	For example, if $\cos(3\theta)=0, c=0$ so that $a=0$ and $b=-3/4$. Then $\cos(\theta)=0$
gilmore-lie-groups_F00199	37	■ 1.000	$a, \ b, \ c$	a, b, c	For example, if $\cos(3\theta)=0, c=0$ so that $a=0$ and $b=-3/4$. Then $\cos(\theta)=0$
gilmore-lie-groups_F00200	37	■ 1.000	$\cos \ (3 \ \theta)=0, \ c=0$	$\cos(3\theta)=0, c=0$	or $\pm\sqrt{3}/2$ for $3\theta=\pi/2(+), 3\pi/2(0)$, or $5\pi/2(-)$.
gilmore-lie-groups_F00201	37	■ 1.000	$a=0$	$a=0$	or $\pm\sqrt{3}/2$ for $3\theta=\pi/2(+), 3\pi/2(0)$, or $5\pi/2(-)$.
gilmore-lie-groups_F00202	37	■ 1.000	$b=-3 \ / \ 4$	$b=-3/4$	or $\pm\sqrt{3}/2$ for $3\theta=\pi/2(+), 3\pi/2(0)$, or $5\pi/2(-)$.
gilmore-lie-groups_F00203	37	■ 1.000	$\cos \ (\theta)=0$	$\cos(\theta)=0$	or $\pm\sqrt{3}/2$ for $3\theta=\pi/2(+), 3\pi/2(0)$, or $5\pi/2(-)$.
gilmore-lie-groups_F00204	37	■ 0.986	$\pm \sqrt{3} \ / \ 2$	$\pm\sqrt{3}/2$	
gilmore-lie-groups_F00205	37	■ 0.986	$3 \ \theta=\pi \ / \ 2(+), \ 3 \ \pi \ / \ 2(0)$	$3\theta=\pi/2(+), 3\pi/2(0)$	
gilmore-lie-groups_F00206	37	■ 0.986	$5 \ \pi \ / \ 2(-)$	$5\pi/2(-)$	
gilmore-lie-groups_F00207	38	■ 1.000	$g_{\{i\}}, \ g_{\{j\}}, \ g_{\{k\}}, \ \ldots$	$g_i, g_j, g_k, \dots$	

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gilmore-lie-groups-F00208	38	1.000	$g_{\{i\}} \circ g_{\{j\}} \in G$	$g_i \circ g_j \in G$	
gilmore-lie-groups-F00209	38	1.000	e	e	
gilmore-lie-groups-F00210	38	1.000	$g_{\{i\}} \in G$	$g_i \in G$	(iv) Inverse: every group op
gilmore-lie-groups-F00211	39	1.000	2×2	2×2	Example We consider the set of real 2×2 matrices $SL(2; \mathbb{R})$:
gilmore-lie-groups-F00212	39	1.000	$SL(2; \mathbb{R})$	$SL(2; \mathbb{R})$	Example We consider the set of real 2×2 matrices $SL(2; \mathbb{R})$:
gilmore-lie-groups-F00213	39	1.000	$\alpha, \beta, \gamma, \delta$	$\alpha, \beta, \gamma, \delta$	where $\alpha, \beta, \gamma, \delta$ are real numbers. This set forms a group under matrix multiplication. This is verified by checking that the group axioms are satisfied.
gilmore-lie-groups-F00214	39	0.797	A	A	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups-F00215	39	0.797	B	B	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups-F00216	39	0.797	$A \circ B = C$	$A \circ B = C$	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups-F00217	39	0.797	$\circ$	$\circ$	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups-F00218	39	0.922	C	C	then $\det(C) = \det(A) \det(B) = +1$.
gilmore-lie-groups-F00219	39	0.922	$\operatorname{det}(A) = +1$	$\det(A) = +1$	then $\det(C) = \det(A) \det(B) = +1$.
gilmore-lie-groups-F00220	39	0.922	$\operatorname{det}(B) = +1$	$\det(B) = +1$	then $\det(C) = \det(A) \det(B) = +1$.
gilmore-lie-groups-F00221	39	1.000	$\operatorname{det}(C) = \operatorname{det}(A) \operatorname{det}(B)$	$\det(C) = \det(A) \det(B) = +1$	) Associativity: $(A \circ B) \circ C$ and $A \circ (B$
gilmore-lie-groups-F00222	39	0.998	$(A \circ B) \circ C$	$(A \circ B) \circ C$	
gilmore-lie-groups-F00223	39	0.998	$A \circ (B \circ C)$	$A \circ (B \circ C)$	
gilmore-lie-groups-F00224	39	1.000	$g_{\{i\}} \rightarrow g(x)$	$g_i \rightarrow g(x)$	fold. A manifold is a space that looks Euclidean on a small scale everywhere. For
gilmore-lie-groups-F00225	39	1.000	i	i	example, every point on the surface of a unit sphere $S^2 \subset \mathbb{R}^3$:
gilmore-lie-groups-F00226	39	1.000	$S^2 \subset \mathbb{R}^3 : x^2 + y^2 + z^2 = 1$	$S^2 \subset \mathbb{R}^3 : x^2 + y^2 + z^2 = 1$	has a neighborhood that looks, over small distances, like a pla-
gilmore-lie-groups-F00227	39	1.000	$\mathbb{R}^2$	$\mathbb{R}^2$	Definition An n -dimensional differentiable manifold M^n consists of the follow-
gilmore-lie-groups-F00228	39	1.000	S^2	S^2	ing.
gilmore-lie-groups-F00229	39	1.000	M^n	M^n	
gilmore-lie-groups-F00230	39	1.000	T	T	
gilmore-lie-groups-F00231	39	1.000	U_{α}	U_{α}	
gilmore-lie-groups-F00232	39	0.896	$T : \cup_{\alpha} U_{\alpha} = T$	$T : \cup_{\alpha} U_{\alpha} = T$	
gilmore-lie-groups-F00233	40	1.000	p	p	that is indistinguishable from an open neighborhood of any point in the plane $\mathbb{R}^2$.
gilmore-lie-groups-F00234	40	1.000	ϕ_{α}	ϕ_{α}	$\cup_{\alpha} U_{\alpha} = T$. GIV Smoothness conditions. The homeomorphisms $\phi_{\alpha} : U_{\alpha} \rightarrow \mathbb{R}^n$ satisfy $\phi_{\alpha}^{-1}(\phi_{\alpha}(U_{\alpha} \cap U_{\beta})) = U_{\alpha} \cap U_{\beta}$.

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gilmore-lie-groups-F00235	40	1.000	$\phi_{\alpha}(U_{\alpha}) \cap V_{\alpha} = V_{\alpha} \cap U_{\alpha}$	$\phi_{\alpha}(U_{\alpha}) \cap V_{\alpha} = V_{\alpha} \cap U_{\alpha}$	
gilmore-lie-groups-F00236	40	1.000	V_{α}	V_{α}	
gilmore-lie-groups-F00237	40	1.000	$\phi_{\alpha} \circ \phi_{\beta}^{-1} : \phi_{\beta}(U_{\alpha} \cap U_{\beta}) \rightarrow \phi_{\alpha}(U_{\alpha} \cap U_{\beta})$	$\phi_{\alpha} \circ \phi_{\beta}^{-1} : \phi_{\beta}(U_{\alpha} \cap U_{\beta}) \rightarrow \phi_{\alpha}(U_{\alpha} \cap U_{\beta})$	
gilmore-lie-groups-F00238	40	0.988	U_{β}	U_{β}	Remarks The charts ϕ_{α} allow construction of coordinate systems on the manifold.
gilmore-lie-groups-F00239	40	0.988	R^n	R^n	Remarks The charts ϕ_{α} allow construction of coordinate systems on the manifold.
gilmore-lie-groups-F00240	40	1.000	$\phi_{\alpha} \circ \phi_{\beta}^{-1}$	$\phi_{\alpha} \circ \phi_{\beta}^{-1}$	replaced by other adjectives (C^k , smooth, analytic, ...) in the definition above.
gilmore-lie-groups-F00241	40	1.000	$R^n \rightarrow R^n$	$R^n \rightarrow R^n$	replaced by other adjectives (C^k , smooth, analytic, ...) in the definition above.
gilmore-lie-groups-F00242	40	0.697	C^k	C^k	condition $\det(A) = +1$ places one constraint on these four real variables. Therefore
gilmore-lie-groups-F00243	40	1.000	$x_1 \rightarrow 0$	$x_1 \rightarrow 0$	fold K^+ (plane) $\times S^1$ (circle) (see Fig. 2.2). This is not at all obvious, but will become clear when we discuss the infinitesimal representation of the group in Chapter 4.
gilmore-lie-groups-F00244	40	1.000	$\times S^1$	$\times S^1$	each operation in the group uniquely.
gilmore-lie-groups-F00245	41	0.999	$z^2 - x^2 - y^2 = 1$	$z^2 - x^2 - y^2 = 1$	
gilmore-lie-groups-F00246	41	1.000	$H^2 \times S^1$	$H^2 \times S^1$	
gilmore-lie-groups-F00247	41	0.698	U_{α}	U_{α}	$ x_n - x_n' $), this definition is equivalent to: a space is compact if every infinite sequence
gilmore-lie-groups-F00248	41	0.698	$\bigcup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$	$\bigcup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$	$ x_n - x_n' $), this definition is equivalent to: a space is compact if every infinite sequence
gilmore-lie-groups-F00249	41	1.000	$\left(x - x' ^2 = x_1 - x_1' ^2 + \dots + x_n - x_n' ^2 \right)$	$\left(x - x' ^2 = x_1 - x_1' ^2 + \dots + x_n - x_n' ^2 \right)$	A space is compact if every infinite sequence of points has a subsequence that converges to a point in the space.
gilmore-lie-groups-F00250	41	1.000	$ x_n - x_n' ^2$	$ x_n - x_n' ^2$	converges to a point in the space.
gilmore-lie-groups-F00251	42	1.000	$(g(x), x \in M^n)$	$(g(x), x \in M^n)$	operations $(g(x), x \in M^n)$ and a combinatorial operation defined by $g(x) \circ g(y) =$
gilmore-lie-groups-F00252	42	1.000	$g(x) \circ g(y) =$	$g(x) \circ g(y) =$	operations $(g(x), x \in M^n)$ and a combinatorial operation defined by $g(x) \circ g(y) =$
gilmore-lie-groups-F00253	42	1.000	$g(z)$	$g(z)$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups-F00254	42	1.000	$z \in M^n$	$z \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups-F00255	42	1.000	$x \in M^n$	$x \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups-F00256	42	1.000	$y \in M^n$	$y \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups-F00257	42	1.000	$z = \phi(x, y)$	$z = \phi(x, y)$	through a function $z = \phi(x, y)$
gilmore-lie-groups-F00258	42	1.000	$g(x) \circ g(y) = g(z)$	$g(x) \circ g(y) = g(z)$	(ii) Smoothness of the group inversion map The
gilmore-lie-groups-F00259	42	1.000	$y = \psi(x)$	$y = \psi(x)$	by $g(x)^{-1} = g(y)$, is differentiable.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00260	42	■ 1.000	$g(x)^{-1}=g(y)$	$g(x)^{-1} = g(y)$	It is possible to combine the
gilmore-lie-groups_F00261	42	■ 1.000	$g(\phi(x, y))$	$g(\phi(x, y))$	
gilmore-lie-groups_F00262	42	■ 1.000	ϕ	ϕ	
gilmore-lie-groups_F00263	42	■ 1.000	$x_{\{1\}}$	x_1	
gilmore-lie-groups_F00264	42	■ 1.000	$y_{\{1\}}$	y_1	
gilmore-lie-groups_F00265	42	■ 1.000	$x_{\{2\}}-x_{\{3\}}$	$x_2 - x_3$	
gilmore-lie-groups_F00266	42	■ 1.000	$x_{\{1\}}=0$	$x_1 = 0$	
gilmore-lie-groups_F00267	42	■ 1.000	$y_{\{2\}}-y_{\{3\}}$	$y_2 - y_3$	
gilmore-lie-groups_F00268	42	■ 1.000	$y_{\{1\}}=0$	$y_1 = 0$	
gilmore-lie-groups_F00269	43	■ 0.983	$\left(y_{\{1\}}, y_{\{2\}}, y_{\{3\}}\right)$	(y_1, y_2, y_3)	inverse. We can determine the coordinates (y_1, y_2, y_3) of $[g(x_1, x_2, x_3)]^{-1}$ by setting
gilmore-lie-groups_F00270	43	■ 0.983	$\left[g\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)\right]^{-1}$	$[g(x_1, x_2, x_3)]^{-1}$	inverse. We can determine the coordinates (y_1, y_2, y_3) of $[g(x_1, x_2, x_3)]^{-1}$ by setting
gilmore-lie-groups_F00271	43	■ 0.998	$\left(z_{\{1\}}, z_{\{2\}}, z_{\{3\}}\right)=(1,0,0)$	$(z_1, z_2, z_3) = (1, 0, 0)$	$(z_1, z_2, z_3) = (1, 0, 0)$ and solving for (y_1, y_2, y_3) in terms of (x_1, x_2, x_3) . Or more
gilmore-lie-groups_F00272	43	■ 0.998	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)$	(x_1, x_2, x_3)	$(z_1, z_2, z_3) = (1, 0, 0)$ and solving for (y_1, y_2, y_3) in terms of (x_1, x_2, x_3) . Or more
gilmore-lie-groups_F00273	43	■ 1.000	$[g(x)]^{-1}=g(y)=g(\psi(x))$	$[g(x)]^{-1} = g(y) = g(\psi(x))$	
gilmore-lie-groups_F00274	43	■ 0.964	(x, y, z , θ)	(x, y, z , θ)	parameterizes $SL(2; \mathbb{R})$ is three dimensional. It is $H^{2+} \times S^1$, where H^{2+} is the
gilmore-lie-groups_F00275	43	■ 0.882	$(-x, -y, - z , \theta+\pi)$	$(-x, -y, - z , \theta + \pi)$	upper sheet of the two-sheeted hyperboloid.
gilmore-lie-groups_F00276	43	■ 1.000	$H^{2+} \times S^1$	$H^{2+} \times S^1$	
gilmore-lie-groups_F00277	43	■ 1.000	H^{2+}	H^{2+}	
gilmore-lie-groups_F00278	43	■ 1.000	$\phi(x, y)$	$\phi(x, y)$	
gilmore-lie-groups_F00279	43	■ 1.000	$g(x) \circ g(y)=g(z)=g(\phi(x, y))$	$g(x) \circ g(y) = g(z) = g(\phi(x, y))$	
gilmore-lie-groups_F00280	44	■ 1.000	$[g(x)]^{-1}=$	$[g(x)]^{-1} =$	In addition, the mapping $y = \psi(x)$ of a group operation to its inverse $[g(x)]^{-1} =$
gilmore-lie-groups_F00281	44	■ 1.000	$g(y)=g(\psi(x))$	$g(y) = g(\psi(x))$	$g(y) = g(\psi(x))$ is also differential
gilmore-lie-groups_F00282	44	■ 0.863	$\operatorname{SL}(2 ; \mathbb{R})$	$\operatorname{SL}(2; \mathbb{R})$	trated in Fig. 2.2.
gilmore-lie-groups_F00283	44	■ 1.000	M	M	U in $SU(n)$: $M = SU$. Devise an algorithm for constructing these matrices. Show
gilmore-lie-groups_F00284	44	■ 1.000	$S L(n ; \mathbb{R})$	$SL(n; \mathbb{R})$	U in $SU(n)$: $M = SU$. Devise an algorithm for constructing these matrices. Show
gilmore-lie-groups_F00285	44	■ 1.000	S	S	$S = (MM^T)^{1/2}$ and $U = S^{-1}M$. How do you compute the square root of a matrix? Show that Ω is compact while S and M are not compact.

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gilmore-lie-groups_F00286	44	0.755	0	O	show that O is compact while S and M are not compact.
gilmore-lie-groups_F00287	44	0.755	$S O(n): M=S O$	$SO(n): M=SO$	show that O is compact while S and M are not compact.
gilmore-lie-groups_F00288	44	1.000	$S=\left(M M^t\right)^{1 / 2}$	$S=\left(M M^t\right)^{1 / 2}$	4. Construct the most general linear transformation $(x, y, z) \rightarrow\left(x^{\prime}, y^{\prime}, z^{\prime}\right)$ that leaves
gilmore-lie-groups_F00289	44	1.000	$O=S^{-1} M$	$O=S^{-1} M$	4. Construct the most general linear transformation $(x, y, z) \rightarrow\left(x^{\prime}, y^{\prime}, z^{\prime}\right)$ that leaves
gilmore-lie-groups_F00290	44	1.000	$(x, y, z) \rightarrow\left(x^{\prime}, y^{\prime}, z^{\prime}\right)$	$(x, y, z) \rightarrow\left(x^{\prime}, y^{\prime}, z^{\prime}\right)$	
gilmore-lie-groups_F00291	44	1.000	$M_{\{1\}}$	M_1	
gilmore-lie-groups_F00292	44	1.000	$M_{\{2\}}$	M_2	
gilmore-lie-groups_F00293	44	0.995	$[S O(1,1)]$	$[SO(1,1)]$	
gilmore-lie-groups_F00294	44	0.972	$(c t)^2-x^2$	$(c t)^2-x^2$	
gilmore-lie-groups_F00295	44	0.972	$S O(2)$	$SO(2)$	
gilmore-lie-groups_F00296	44	0.999	x^2+y^2	x^2+y^2	
gilmore-lie-groups_F00297	45	1.000	$(c t)^2-x^2-y^2-z^2$	$(c t)^2-x^2-y^2-z^2$	$(c t)^2-x^2-y^2-z^2$. This is the Lorentz group $O(3,1)$. Four disconnected mani-
gilmore-lie-groups_F00298	45	1.000	$O(3,1)$	$O(3,1)$	$(c t)^2-x^2-y^2-z^2$. This is the Lorentz group $O(3,1)$. Four disconnected mani-
gilmore-lie-groups_F00299	45	0.535	$S L(2 ; \mathbb{C})$	$SL(2; \mathbb{C})$	trices in this group have the structure $\left[\begin{smallmatrix} \alpha & \rho \\ \gamma & \delta \end{smallmatrix}\right]$, where $\alpha, \beta, \gamma, \delta$ are complex numbers
gilmore-lie-groups_F00300	45	0.788	$\left[\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right]$	and $\alpha \delta-\beta \gamma=1$. Define the matrix X by
gilmore-lie-groups_F00301	45	0.987	$\alpha \delta-\beta \gamma=1$	$\alpha \delta-\beta \gamma=1$	$\Gamma_{c t+z}$
gilmore-lie-groups_F00302	45	0.987	X	X	$\Gamma_{c t+z}$
gilmore-lie-groups_F00303	45	1.000	$\mathbf{x}$	$\mathbf{x}$	a. Show that X is hermitian: $X^{\dagger} \equiv\left(X^t\right)^*=X$.
gilmore-lie-groups_F00304	45	1.000	$\mathbf{x}=(x, y, z)$	$\mathbf{x}=(x, y, z)$	a. Show that X is hermitian: $X^{\dagger} \equiv\left(X^t\right)^*=X$.
gilmore-lie-groups_F00305	45	1.000	$\sigma=\left(\sigma_1, \sigma_2, \sigma_3\right)=\left(\sigma_x, \sigma_y, \sigma_z\right)$	$\sigma=\left(\sigma_1, \sigma_2, \sigma_3\right)=\left(\sigma_x, \sigma_y, \sigma_z\right)$	a. Show that X is hermitian: $X^{\dagger} \equiv\left(X^t\right)^*=X$.
gilmore-lie-groups_F00306	45	1.000	$X^{\dagger} \equiv\left(X^t\right)^*=X$	$X^{\dagger} \equiv\left(X^t\right)^*=X$	to construct X .
gilmore-lie-groups_F00307	45	1.000	$g \in S L(2 ; \mathbb{C})$	$g \in SL(2; \mathbb{C})$	coordinates $(x, y, z, c t)$. (They are linearly related by coefficients in the matrix elements $\alpha, \beta, \gamma, \delta$ of σ and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adi
gilmore-lie-groups_F00308	45	1.000	$g^{\dagger} X g=X^{\prime}=H\left(x^{\prime}, y^{\prime}, z^{\prime}, c t^{\prime}\right)$	$g^{\dagger} X g=X^{\prime}=H\left(x^{\prime}, y^{\prime}, z^{\prime}, c t^{\prime}\right)$	coordinates $(x, y, z, c t)$. (They are linearly related by coefficients in the matrix elements $\alpha, \beta, \gamma, \delta$ of σ and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adi
gilmore-lie-groups_F00309	45	0.964	$\left(x^{\prime}, y^{\prime}, z^{\prime}, c t^{\prime}\right)$	$\left(x^{\prime}, y^{\prime}, z^{\prime}, c t^{\prime}\right)$	e. Find the subgroup of $SL(2 ; \mathbb{C})$ that leaves $t^{\prime}=t$. (It is $SU(2) \subset SL(2 ; \mathbb{C})$).
gilmore-lie-groups_F00310	45	0.918	$(x, y, z, c t)$	$(x, y, z, c t)$	. For any $g \in SL(2 ; \mathbb{C})$ write $g=k h$, where $h \in SU(2), h^{\dagger}=h^{-1}, h$ has the
gilmore-lie-groups_F00311	45	1.000	g	g	form $h=\operatorname{EXP}\left(\frac{i}{2} \sigma \cdot \theta\right)$ and $k \in SL(2 ; \mathbb{C}) / SU(2), k^{\dagger}=k^{+1}, k$ has the form
gilmore-lie-groups_F00312	45	1.000	$\alpha^*, \beta^*, \gamma^*, \delta^*$	$\alpha^*, \beta^*, \gamma^*, \delta^*$	form $h=\operatorname{EXP}\left(\frac{i}{2} \sigma \cdot \theta\right)$ and $k \in SL(2 ; \mathbb{C}) / SU(2), k^{\dagger}=k^{+1}, k$ has the form

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gilmore-lie-groups_ F08313	45	■ 1.000	$g^{\dagger}$	$g^{\dagger}$	form $h = \text{EXP}(\frac{i}{2}\sigma \cdot \theta)$ and $k \in SL(2; \mathbb{C})/SU(2)$, $k^{\dagger} = k^{+1}$, k has the form
gilmore-lie-groups_ F08314	45	■ 0.990	$t^{\prime} = t$	$t' = t$	$k = \text{EXP}(\frac{1}{2}\sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F08315	45	■ 0.990	$SU(2) \subset SL(2; \mathbb{C})$	$SU(2) \subset SL(2; \mathbb{C})$	$k = \text{EXP}(\frac{1}{2}\sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F08316	45	■ 0.952	$g = k h$	$g = kh$	and $\mathbf{b}$ are real. Construct $k^{\dagger}H(x, y, z, ct)k = H(x', y', z', ct')$. If this is too diffi-
gilmore-lie-groups_ F08317	45	■ 0.952	$h \in SU(2), h^{\dagger} = h^{-1}, h$	$h \in SU(2), h^{\dagger} = h^{-1}, h$	and $\mathbf{b}$ are real. Construct $k^{\dagger}H(x, y, z, ct)k = H(x', y', z', ct')$. If this is too diffi-
gilmore-lie-groups_ F08318	45	■ 0.461	$h = \operatorname{EXP}\left(\frac{i}{2}\sigma \cdot \theta\right)$	$h = \text{EXP}\left(\frac{i}{2}\sigma \cdot \theta\right)$	cult, choose $\mathbf{b}$ along the z -axis, $\mathbf{b} = (0, 0, b)$. Show that the usual Lorentz transformation law results.
gilmore-lie-groups_ F08319	45	■ 0.461	$k \in SL(2; \mathbb{C}) / \operatorname{EXP}\left(\frac{i}{2}\sigma \cdot \theta\right), k^{\dagger} = k^{+1}, k$	$k \in SL(2; \mathbb{C})/SU(2), k^{\dagger} = k^{+1}, k$	cult, choose $\mathbf{b}$ along the z -axis, $\mathbf{b} = (0, 0, b)$. Show that the usual Lorentz transformation law results.
gilmore-lie-groups_ F08320	45	■ 1.000	$k = \operatorname{EXP}\left(\frac{1}{2}\sigma \cdot \mathbf{b}\right)$	$k = \text{EXP}\left(\frac{1}{2}\sigma \cdot \mathbf{b}\right)$	• Show that the usual Lorentz transformation law results. Applying $L(h')$ after applying $L(h)$ results in (a) $L(h' \perp h)$ (b) two successive
gilmore-lie-groups_ F08321	45	■ 1.000	$\mathbf{b}$	$\mathbf{b}$	• Show that the usual Lorentz transformation law results. Applying $L(h')$ after applying $L(h)$ results in (a) $L(h' \perp h)$ (b) two successive
gilmore-lie-groups_ F08322	45	■ 1.000	θ	θ	• Show that the usual Lorentz transformation law results. Applying $L(h')$ after applying $L(h)$ results in (a) $L(h' \perp h)$ (b) two successive
gilmore-lie-groups_ F08323	45	■ 0.899	$k^{\dagger}H(x, y, z, ct)k = H(x', y', z', ct')$	$k^{\dagger}H(x, y, z, ct)k = H(x', y', z', ct')$	• Applying $\kappa(v)$ after applying $\kappa(v)$ results in (a) $\kappa(v \perp v)$, (b) two successive Lorentz transformations. Show that the velocity addition law for collinear boosts
gilmore-lie-groups_ F08324	45	■ 0.893	$\mathbf{b} = (0, 0, b)$	$\mathbf{b} = (0, 0, b)$	results.
gilmore-lie-groups_ F08325	45	■ 0.458	$k(b')$	$k(b')$	related to the Thomas precession (Gilmore 1974b).
gilmore-lie-groups_ F08326	45	■ 0.458	$k(b)$	$k(b)$	related to the Thomas precession (Gilmore 1974b).
gilmore-lie-groups_ F08327	45	■ 0.458	$(a)k(b' + b)$	$(a)k(b' + b)$	related to the Thomas precession (Gilmore 1974b).
gilmore-lie-groups_ F08328	45	■ 0.981	$\mathbf{b}'$	$\mathbf{b}'$	$f(\theta + 2\pi) = f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_ F08329	45	■ 0.981	$k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$	$k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$	$f(\theta + 2\pi) = f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_ F08330	45	■ 0.981	$\mathbf{b}'', \theta$	$\mathbf{b}'', \theta$	$f(\theta + 2\pi) = f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_ F08331	45	■ 1.000	$\theta \rightarrow \theta' = \theta + k + f(\theta)$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$	
gilmore-lie-groups_ F08332	45	■ 1.000	$0 \leq k < 2\pi$	$0 \leq k < 2\pi$	
gilmore-lie-groups_ F08333	45	■ 1.000	$f(\theta)$	$f(\theta)$	
gilmore-lie-groups_ F08334	45	■ 0.991	$f(\theta + 2\pi) = f(\theta)$	$f(\theta + 2\pi) = f(\theta)$	
gilmore-lie-groups_ F08335	45	■ 0.991	$1 : 1$	$1 : 1$	
gilmore-lie-groups_ F08336	45	■ 1.000	$f(\theta) : df(\theta)/d\theta > -1$	$f(\theta) : df(\theta)/d\theta > -1$	
gilmore-lie-groups_ F08337	46	■ 0.769	(a, b, c, d)	(a, b, c, d)	9. Rational fractional transformations (a, b, c, d) map points on the real line (real pro-
gilmore-lie-groups_ F08338	46	■ 1.000	RP^1	RP^1	jective line RP^1) to the real line as follow

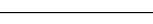
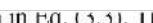
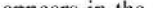
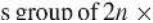
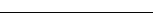
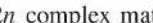
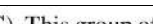
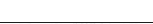
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups-F00339	46	1.000	$(\lambda a, \lambda b, \lambda c, \lambda d) = \lambda(a, b, c, d) (\lambda \neq 0)$	$(\lambda a, \lambda b, \lambda c, \lambda d) = \lambda(a, b, c, d) (\lambda \neq 0)$	The transformations (a, b, c, d) and $(\lambda a, \lambda b, \lambda c, \lambda d) = \lambda(a, b, c, d) (\lambda \neq 0)$ generate identical mappings
gilmore-lie-groups-F00340	46	1.000	A, B, C, D	A, B, C, D	d. Show that the tra
gilmore-lie-groups-F00341	46	0.668	$(\lambda, 0, 0, \lambda)$	$(\lambda, 0, 0, \lambda)$	c. Construct the inverse transformation $x \rightarrow x'$, and show
gilmore-lie-groups-F00342	46	0.999	$x^{\prime} \rightarrow x$	$x' \rightarrow x$	provided $\lambda \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. d. Show that the transformation degeneracy $x' = (a - b - c - d)x = \lambda(a - b - c - d)x$ can
gilmore-lie-groups-F00343	46	0.999	$\lambda(d, -b, -c, a)$	$\lambda(d, -b, -c, a)$	provided $\lambda \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. d. Show that the transformation degeneracy $x' = (a - b - c - d)x = \lambda(a - b - c - d)x$ can
gilmore-lie-groups-F00344	46	1.000	$\lambda \neq 0$	$\lambda \neq 0$	be lifted by requiring that the four parameters a, b, c, d describin
gilmore-lie-groups-F00345	46	1.000	$D = a d - b c \neq 0$	$D = ad - bc \neq 0$	be lifted by requiring that the four parameters a, b, c, d describin
gilmore-lie-groups-F00346	46	1.000	$x^{\prime} = (a, b, c, d) x = \lambda(a, b, c, d) x$	$x' = (a, b, c, d)x = \lambda(a, b, c, d)x$	mations satisfy the constraint $D = ad - bc = 1$.
gilmore-lie-groups-F00347	46	1.000	a, b, c, d	a, b, c, d	e. It is useful to introduce homogeneous coordinates (y, z) and define the real projec-
gilmore-lie-groups-F00348	46	1.000	$D = a d - b c = 1$	$D = ad - bc = 1$	tive coordinate x as the ratio of these homoger
gilmore-lie-groups-F00349	46	0.980	(y, z)	(y, z)	homogeneous coordinates transform linearly under $SL(2; \mathbb{R})$ then the real projec-
gilmore-lie-groups-F00350	46	1.000	$x = y / z$	$x = y/z$	tive coordinates x transform under rational fractional transformations:
gilmore-lie-groups-F00351	46	0.477	x_1, x_2, x_3	x_1, x_2, x_3	$x \rightarrow x' = \frac{x_2 - x_1}{(x - x_1)(x_2 - x_1)}$
gilmore-lie-groups-F00352	46	0.477	∞	∞	$x \rightarrow x' = \frac{x_2 - x_1}{(x - x_1)(x_2 - x_1)}$
gilmore-lie-groups-F00353	46	0.996	$D = 1$	$D = 1$	0. The real projective space RP^n is the space of all straight lines through the origin
gilmore-lie-groups-F00354	46	1.000	$(x_1^{\prime}, x_2^{\prime}, x_3^{\prime})$	(x'_1, x'_2, x'_3)	with $x \neq 0 \leftrightarrow x' \neq 0$ and $x = 0 \leftrightarrow x' = 0$. A straight line through the
gilmore-lie-groups-F00355	46	1.000	RP^n	RP^n	$r \neq 0$ and $v \neq 0$ if (and only if) $v = \lambda r$ for some real scale factor λ
gilmore-lie-groups-F00356	46	0.987	R^{n+1}	R^{n+1}	factor can always be chosen so that y is in the unit sphere in R^{n+1} ; $y \in S^n \subset R^{n+1}$.
gilmore-lie-groups-F00357	46	0.987	$SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$	$SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$	In fact, two values of λ can be chosen: $\lambda = \pm 1 / (\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups-F00358	46	0.987	$x^{\prime} \in R^{n+1}$	$x' \in R^{n+1}$	In fact, two values of λ can be chosen: $\lambda = \pm 1 / (\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups-F00359	46	1.000	$x^{\prime} \neq 0 \rightarrow x \neq 0$	$x' \neq 0 \leftrightarrow x \neq 0$	line containing (x, y, z) can be represented by homogeneous coordinates $(X, Y) =$
gilmore-lie-groups-F00360	46	1.000	$x^{\prime} = 0 \rightarrow x = 0$	$x' = 0 \leftrightarrow x = 0$	$(x, y, z) \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. Such transformations exist if $D = ad - bc \neq 0$.
gilmore-lie-groups-F00361	46	1.000	$x \neq 0$	$x \neq 0$	$(x/z, y/z) \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. Such transformations exist if $D = ad - bc \neq 0$.
gilmore-lie-groups-F00362	46	1.000	$y \neq 0$	$y \neq 0$	$(x/z, y/z) \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. Such transformations exist if $D = ad - bc \neq 0$.
gilmore-lie-groups-F00363	46	1.000	$y = \lambda x$	$y = \lambda x$	$(x/z, y/z) \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. Such transformations exist if $D = ad - bc \neq 0$. Such transformations exist if $D = ad - bc \neq 0$.
gilmore-lie-groups-F00364	46	1.000	y	y	in R^n by $x \mapsto x' = ax$, $ax \in \mathbb{R}^n(x, ax)$. Show that the homogeneous coordinates
gilmore-lie-groups-F00365	46	1.000	$R^{n+1}: y \in S^n \subset R^{n+1}$	$R^{n+1}: y \in S^n \subset R^{n+1}$	in R^n by $x \mapsto x' = ax$, $ax \in \mathbb{R}^n(x, ax)$. Show that the homogeneous coordinates

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gilmore-lie-groups_ F00366	46	■ 1.000	$\backslash\lambda$	λ	
gilmore-lie-groups_ F00367	46	■ 1.000	$\backslash\lambda=\pm 1/\left(\sum_{i=1}^{n+1}x_i^2\right)^{1/2}$	$\lambda=\pm 1/\left(\sum_{i=1}^{n+1}x_i^2\right)^{1/2}$	
gilmore-lie-groups_ F00368	46	■ 1.000	$R^{\{3\}}$	R^3	
gilmore-lie-groups_ F00369	46	■ 0.983	(x, y, z)	(x,y,z)	
gilmore-lie-groups_ F00370	46	■ 0.983	$(X, Y)=$	$(X,Y) =$	
gilmore-lie-groups_ F00371	46	■ 1.000	$(x / z, y / z)$	$(x/z,y/z)$	
gilmore-lie-groups_ F00372	46	■ 1.000	$z \backslash\neq 0$	$z\neq 0$	
gilmore-lie-groups_ F00373	46	■ 1.000	$x \backslashrightarrow x^{\{\prime\}}=M x, M \backslash\in S L(3 ; \backslash\mathbb{R})$	$x\rightarrow x'=Mx, M\in SL(3;\mathbb{R})$	
gilmore-lie-groups_ F00374	47	■ 1.000	$x^{\{\prime\}}$	x'	representing the two lines containing x and x' are related by the linear fractional
gilmore-lie-groups_ F00375	47	■ 1.000	$R P^{\{n\}} \backslashrightarrow R P^{\{n\}}$	$RP^n\rightarrow RP^n$	
gilmore-lie-groups_ F00376	47	■ 0.989	$S L(2 ; \backslash\mathbb{R}) / S O(2) \backslash\mathrm{simeq}\left[\begin{array}{cc} z+x & y \\ y & z-x \end{array}\right]$	$SL(2;\mathbb{R})/SO(2)\simeq\left[\begin{array}{cc} z+x & y \\ y & z-x \end{array}\right]$	manifold in the Minkowski 2 + 1 dimensional space-time with metric (+1, −1, −1)
gilmore-lie-groups_ F00377	47	■ 1.000	$x=r \backslash\cos \backslash\phi, y=r \backslash\sin \backslash\phi$	$x=r\cos\phi,y=r\sin\phi$	$1+r^2$
gilmore-lie-groups_ F00378	47	■ 0.633	$S O(3) / S O(2)$	$SO(3)/SO(2)$	measure on S^2 are obtained from the results above for H^2 by the substitutions $1\pm r^2\mapsto 1\mp r^2$. Show that the disk $0<r<1, 0<\phi<2\pi$ maps onto the un-
gilmore-lie-groups_ F00379	47	■ 0.633	$S^{\{2\}} \backslash\subset R^{\{3\}}$	$S^2\subset R^3$	measure on S^2 are obtained from the results above for H^2 by the substitutions $1\pm r^2\mapsto 1\mp r^2$. Show that the disk $0<r<1, 0<\phi<2\pi$ maps onto the un-
gilmore-lie-groups_ F00380	47	■ 1.000	$z^{\{2\}}+\backslash\left(x^{\{2\}}+y^{\{2\}}\backslash\right)=1$	$z^2+(x^2+y^2)=1$	$1+r^2\mapsto 1-r^2$. Show that the disk $0\leq r\leq 1, 0\leq\phi\leq 2\pi$ maps onto the up- ner hemisphere of the sphere, with $r=0$ mapping to the north pole and $r=1$
gilmore-lie-groups_ F00381	47	■ 1.000	$d s^{\{2\}}=d z^{\{2\}}+\backslash\left(d x^{\{2\}}+d y^{\{2\}}\backslash\right)$	$ds^2=dz^2+(dx^2+dy^2)$	mapping to the equator. Show that the geodesic length from the north pole to the
gilmore-lie-groups_ F00382	47	■ 1.000	$H^{\{2\}}$	H^2	equator along the longitude $\phi=0$ is $s=\int_0^1 d r / \sqrt{1-r^2}=\pi / 2$ and the volume of
gilmore-lie-groups_ F00383	47	■ 1.000	$1+r^{\{2\}} \backslashrightarrow 1-r^{\{2\}}$	$1+r^2\rightarrow 1-r^2$	the hemisphere surface is $V=\int_{r=0}^{r=1} \int_{\phi=0}^{\phi=2 \pi} d V(r, \phi)=\int_0^1 r d r / \sqrt{1-r^2} \int_0^{2 \pi} d \phi=$
gilmore-lie-groups_ F00384	47	■ 1.000	$0 \backslash\leq r \backslash\leq 1,0 \backslash\leq \backslash\phi \backslash\leq 2 \backslash\pi$	$0\leq r\leq 1,0\leq\phi\leq 2\pi$	the hemisphere surface is $V=\int_{r=0}^{r=1} \int_{\phi=0}^{\phi=2 \pi} d V(r, \phi)=\int_0^1 r d r / \sqrt{1-r^2} \int_0^{2 \pi} d \phi=$
gilmore-lie-groups_ F00385	47	■ 1.000	$r=0$	$r=0$	2π .
gilmore-lie-groups_ F00386	47	■ 1.000	$r=1$	$r=1$	2π .
gilmore-lie-groups_ F00387	47	■ 1.000	$\backslash\phi=0$	$\phi=0$	
gilmore-lie-groups_ F00388	47	■ 1.000	$s=\backslash\int_{\{0\}}^{\{1\}} d r / \backslash\sqrt{1-r^{\{2\}}}=\backslash\pi / 2$	$s=\int_0^1 d r / \sqrt{1-r^2}=\pi/2$	
gilmore-lie-groups_ F00389	47	■ 1.000	$V=\backslash\int_{r=0}^{\{r=1\}} \backslash\int_{\phi=0}^{\{\phi=2 \pi\}} d V(r, \backslash\phi)=\backslash\int_{\{0\}}^{\{1\}} r d r / \backslash\sqrt{1-r^{\{2\}}} \backslash\int_{\{0\}}^{\{2 \pi\}} d \backslash\phi=$	$V=\int_{r=0}^{r=1} \int_{\phi=0}^{\phi=2 \pi} d V(r, \phi)=\int_0^1 r d r / \sqrt{1-r^2} \int_0^{2 \pi} d \phi=$	
gilmore-lie-groups_ F00390	47	■ 1.000	$2 \backslash\pi$	2π	

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gilmore-lie-groups_F00391	48	■ 0.969	$G L(n ; \mathbb{F})$	$GL(n; \mathbb{F})$	numbers, and quaternions, and various of their subgroups obtained by
gilmore-lie-groups_F00392	48	■ 1.000	$\mathbb{F}$	$\mathbb{F}$	imposing linear, bilinear and quadratic, and n -linear constraints on these
gilmore-lie-groups_F00393	48	■ 1.000	$H_{\{1\}} \subset G$	$H_1 \subset G$	Examnle If H_1 is the two-dimensional subøroun of $SL(2; \mathbb{R})$ described in (3.1)
gilmore-lie-groups_F00394	48	■ 1.000	$H_{\{12\}}=H_{\{1\}} \cap H_{\{2\}}$	$H_{12} = H_1 \cap H_2$	above and H_2 is the one-dimens
gilmore-lie-groups_F00395	48	■ 1.000	$H_{\{1\}} \cap H_{\{2\}}$	$H_1 \cap H_2$	$H_1 = \{ \begin{pmatrix} \cos \theta & \sin \theta \\ 0 & 1 \end{pmatrix} : \theta \in \mathbb{R} \}$
gilmore-lie-groups_F00396	48	■ 0.950	$H_{\{1\}}$	H_1	
gilmore-lie-groups_F00397	48	■ 0.817	$H_{\{2\}}$	H_2	34
gilmore-lie-groups_F00398	49	■ 1.000	$\pm I_{\{2\}}$	$\pm I_2$	discrete group operations $\pm I_2$.
gilmore-lie-groups_F00399	49	■ 0.950	$(\mathbb{F}=\mathbb{R})$	$(\mathbb{F} = \mathbb{R})$	$(\mathbb{F} = \mathbb{K})$, complex numbers ($\mathbb{F} = \mathbb{C}$), and quaternions ($\mathbb{F} = \mathbb{Q}$). The complex num-
gilmore-lie-groups_F00400	49	■ 0.950	$\mathbb{F}=\mathbb{C}$	$\mathbb{F} = \mathbb{C}$	$(\mathbb{F} = \mathbb{K})$, complex numbers ($\mathbb{F} = \mathbb{C}$), and quaternions ($\mathbb{F} = \mathbb{Q}$). The complex num-
gilmore-lie-groups_F00401	49	■ 0.950	$\mathbb{F}=\mathbb{Q}$	$\mathbb{F} = \mathbb{Q}$	$(\mathbb{F} = \mathbb{K})$, complex numbers ($\mathbb{F} = \mathbb{C}$), and quaternions ($\mathbb{F} = \mathbb{Q}$). The complex num-
gilmore-lie-groups_F00402	49	■ 1.000	$1, \mathcal{I}, J, K$	$1, \mathcal{I}, J, K$	we list, in order, matrix groups on which no constraints are imposed (1), on
gilmore-lie-groups_F00403	49	■ 0.900	$[\operatorname{det}(\mathrm{M})=+1]$	$[\det(\mathbf{M}) = +1]$	1. $GL(n; \mathbb{F})$. General linear group
gilmore-lie-groups_F00404	49	■ 0.999	$G L(1 ; \mathbb{Q})$	$GL(1; \mathbb{Q})$	
gilmore-lie-groups_F00405	49	■ 0.999	1×1	1×1	
gilmore-lie-groups_F00406	50	■ 1.000	$A_{\{i\}^{\{j\}}}$	A_i^j	The determinant of an $n \times n$ matrix A with matrix elements A_i^j is defined by
gilmore-lie-groups_F00407	50	■ 1.000	$\epsilon^{\{i_1 i_2 \cdots i_n\}}$	$\epsilon^{i_1 i_2 \cdots i_n}$	permutation of the integers $1, 2, \dots, n$; -1 for an odd permutation; and 0 if two
gilmore-lie-groups_F00408	50	■ 0.719	$1, 2, \ldots, n ; -1$	$1, 2, \dots, n; -1$	or more values of the indices i_* are equal. With this definition there is no difficulty
gilmore-lie-groups_F00409	50	■ 1.000	$i_{\{*\}}$	i_*	computing the determinant of a matrix containing matrix elements that do not
gilmore-lie-groups_F00410	50	■ 1.000	$U T(p, q)$	$UT(p, q)$	$m = n \qquad p+1 \leq i \leq p+q \qquad (3.6)$
gilmore-lie-groups_F00411	50	■ 1.000	$n \times n (n=p+q)$	$n \times n (n = p + q)$	$m = n \qquad p+1 \leq i \leq p+q \qquad (3.6)$
gilmore-lie-groups_F00412	50	■ 1.000	$U T(1, 1)$	$UT(1, 1)$	$\begin{bmatrix} y' \\ \end{bmatrix} = \begin{bmatrix} 0 & d \\ \end{bmatrix} \begin{bmatrix} y \\ \end{bmatrix} = \begin{bmatrix} dy \\ \end{bmatrix} \qquad (3.7)$
gilmore-lie-groups_F00413	50	■ 0.991	$y=0$	$y = 0$	p and q .
gilmore-lie-groups_F00414	50	■ 0.991	$y=0 \rightarrow y^{\prime}=0$	$y = 0 \rightarrow y' = 0$	p and q .
gilmore-lie-groups_F00415	50	■ 1.000	$x=0$	$x = 0$	
gilmore-lie-groups_F00416	50	■ 1.000	$V_{\{p\}} \oplus V_{\{q\}}$	$V_p \oplus V_q$	

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gilmore-lie-groups_F00417	50	1.000	$V_{\{q\}}$	V_q	
gilmore-lie-groups_F00418	50	1.000	$V_{\{p\}}$	V_p	
gilmore-lie-groups_F00419	50	1.000	q	q	
gilmore-lie-groups_F00420	51	1.000	$H\,T(p,\,q)$	$HT(p,q)$	$m_{ij}-\delta_{ij}=0\qquad\qquad\qquad\begin{matrix}P^{-1}+1\leq i\leq P^{-1}+q\\n+1\leq j\leq n+q\end{matrix}\qquad\qquad(3.8)$
gilmore-lie-groups_F00421	51	1.000	$H\,T(1,1)\backslash\left(m_{22}=1\right)$	$HT(1,1)\,(m_{22}=1)$	$\begin{bmatrix}1&\\&0&1\end{bmatrix}\begin{bmatrix}1&\\&1\end{bmatrix}\begin{bmatrix}1&\\&1\end{bmatrix}$
gilmore-lie-groups_F00422	51	1.000	$x\rightarrow$	$x\rightarrow$	$\begin{bmatrix}1&\\&0&1\end{bmatrix}\begin{bmatrix}1&\\&1\end{bmatrix}\begin{bmatrix}1&\\&1\end{bmatrix}$
gilmore-lie-groups_F00423	51	0.997	$x^{\{\prime\}}=a\,x+b$	$x'=ax+b$	4 $HT(n,\,a)$
gilmore-lie-groups_F00424	51	0.999	$U\,T(p,\,q,\,r)$	$UT(p,q,r)$	
gilmore-lie-groups_F00425	51	1.000	$U\,T(p,\,q+r)\,\cap\,U\,T(p+q,\,r)$	$UT(p,q+r)\cap UT(p+q,r)$	
gilmore-lie-groups_F00426	52	0.460	$S\,U(1,1)$	$SU(1,1)$	structure (3.10) are encountered in treatments of squeezed states of the electromag-
gilmore-lie-groups_F00427	52	1.000	$\operatorname{Sol}(n)=U\,T(1,1,1,\,\ldots,\,1)$	$\operatorname{Sol}(n)=UT(1,1,1,\ldots,1)$	Example We consider the subgroup of 3×3 matrices in $UT(1,1,1)$ of t
gilmore-lie-groups_F00428	52	0.971	$U\,T(1,1,1)$	$UT(1,1,1)$	form
gilmore-lie-groups_F00429	52	1.000	$\hat{n}=a^{\dagger}a$	$\hat{n}=a^{\dagger}a$	tification between operator and matrix groups to develop some powerful operator
gilmore-lie-groups_F00430	52	1.000	$a^{\dagger}$	$a^{\dagger}$	tification between operator and matrix groups to develop some powerful operator
gilmore-lie-groups_F00431	52	1.000	a	a	tification between operator and matrix groups to develop some powerful operator
gilmore-lie-groups_F00432	52	0.994	$I=a\,a^{\dagger}-a^{\dagger}a=\left[a,\,a^{\dagger}\right]$	$I=aa^{\dagger}-a^{\dagger}a=\left[a,a^{\dagger}\right]$	disentangling ineorems.
gilmore-lie-groups_F00433	52	1.000	$\operatorname{Nil}(n)$	$\operatorname{Nil}(n)$	Example Matrices in $Nil(3)$ of the form
gilmore-lie-groups_F00434	52	1.000	$\operatorname{Sol}(n)$	$\operatorname{Sol}(n)$	Example Matrices in $Nil(3)$ of the form
gilmore-lie-groups_F00435	52	0.998	$\operatorname{Nil}(3)$	$\operatorname{Nil}(3)$	$\begin{bmatrix}x&0&0\\0&y&0\\0&0&z\end{bmatrix}$
gilmore-lie-groups_F00436	52	0.976	$a^{\dagger},\,a,\,I$	$a^{\dagger},a,I$	The set of change of basis transformations $\langle p q\rangle=\frac{1}{\sqrt{\pi}}e^{2\pi i pq/h}$ encountered in
gilmore-lie-groups_F00437	52	0.716	$(p$	$(p$	7 $A(n,\,q)$. This group consists of matrices that are the sum of an identity matrix
gilmore-lie-groups_F00438	52	0.716	$q)$	$q)$	7 $A(n,\,q)$. This group consists of matrices that are the sum of an identity matrix
gilmore-lie-groups_F00439	52	0.716	$[p,\,q]=\hbar/i$	$[p,q]=\hbar/i$	7 $A(n,\,q)$. This group consists of matrices that are the sum of an identity matrix
gilmore-lie-groups_F00440	52	1.000	$\langle p\mid q\rangle=\frac{1}{\sqrt{2}}\,e^{2\pi i\,p\,q/h}$	$\langle p\mid q\rangle=\frac{1}{\sqrt{2}}e^{2\pi i pq/h}$	and the upper right hand on diagonal block of a (p,q) -block matrix. Its matrix
gilmore-lie-groups_F00441	52	0.995	$A(p,\,q)$	$A(p,q)$	
gilmore-lie-groups_F00442	52	0.998	$(p,\,q)$	(p,q)	

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gilmore-lie-groups_F00443	53	■ 1.000	$A=B=A$	$AB=BA$	group.
gilmore-lie-groups_F00444	53	■ 1.000	$A(1,1)$	$A(1,1)$	transformations of the x -axis (3.9): $x \rightarrow x' = x + a$. Successive transformations
gilmore-lie-groups_F00445	53	■ 0.798	$x \rightarrow x' = x + a$	$x \rightarrow x' = x + a$	of this type commute
gilmore-lie-groups_F00446	53	■ 1.000	$M^\dagger G M = G$	$M^\dagger G M = G$	but indehinite we can set $G = I_{p,q}$ (9). If G is nonsingular and antisymmetric, we
gilmore-lie-groups_F00447	53	■ 0.679	$G=I_n$	$G=I_n$	can take (10)
gilmore-lie-groups_F00448	53	■ 0.573	$G=I_{p,q}$	$G=I_{p,q}$	$G = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$
gilmore-lie-groups_F00449	53	■ 1.000	I_n	I_n	
gilmore-lie-groups_F00450	54	■ 0.936	$GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	$GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	terize the matrix groups $GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$
gilmore-lie-groups_F00451	54	■ 1.000	C^2	C^2	two-dimensional linear vector space $\mathbb{C}^2$ (or spin states, for example), and quaternion-valued 1×1 matrices that preserve length in a one-dimensional linear vector space
gilmore-lie-groups_F00452	54	■ 1.000	$\mathbb{Q}$	$\mathbb{Q}$	
gilmore-lie-groups_F00453	54	■ 0.808	$SU(1; \mathbb{Q})$	$SU(1; \mathbb{Q})$	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$ (3.16)
gilmore-lie-groups_F00454	54	■ 1.000	R^4	R^4	
gilmore-lie-groups_F00455	54	■ 0.992	$G=I_{p,q}, p+q=n$	$G=I_{p,q}, p+q=n$	condition $M^T G M = G$, where
gilmore-lie-groups_F00456	54	■ 1.000	$p \neq 0, q \neq 0$	$p \neq 0, q \neq 0$	Example The Lorentz group preserves the invariant $x^2 + y^2 + z^2 - (ct)^2$ and is
gilmore-lie-groups_F00457	54	■ 1.000	$x^2 + y^2 + z^2 - (ct)^2$	$x^2 + y^2 + z^2 - (ct)^2$	$\begin{bmatrix} A' & C' \\ -C' & A' \end{bmatrix} \begin{bmatrix} I_3 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} A & B \\ B^T & A \end{bmatrix}$
gilmore-lie-groups_F00458	55	■ 0.999	$N \times N$	$N \times N$	10. Antisymmetric metric-preserving groups The metric G is an $N \times N$ non-
gilmore-lie-groups_F00459	55	■ 1.000	$\det(G) = \det(G^t) = \det(-G) = (-1)^N \det(G), N$	$\det(G) = \det(G^t) = \det(-G) = (-1)^N \det(G), N$	metric matrix can be chosen to have the canonical forms
gilmore-lie-groups_F00460	55	■ 1.000	$N=2n$	$N=2n$	metric matrix can be chosen to have the canonical forms
gilmore-lie-groups_F00461	55	■ 1.000	$i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	equations of motion.
gilmore-lie-groups_F00462	55	■ 0.990	$Sp(2n; \mathbb{R})$	$Sp(2n; \mathbb{R})$	Examlpe The symplectic group $Sp(2; \mathbb{R}) \subset GL(2; \mathbb{R})$ satisfies the constraint
gilmore-lie-groups_F00463	55	■ 0.935	$Sp(2; \mathbb{R}) \subset GL(2; \mathbb{R})$	$Sp(2; \mathbb{R}) \subset GL(2; \mathbb{R})$	$\begin{vmatrix} b & d \\ -d & b \end{vmatrix} = \begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} \begin{vmatrix} c & d \\ d & c \end{vmatrix} = \begin{vmatrix} bc - ad & 0 \\ 0 & bc - ad \end{vmatrix} = \begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ (5.21)
gilmore-lie-groups_F00464	55	■ 0.510	$ad - bc = +1$	$ad - bc = +1$	a singular metric G .
gilmore-lie-groups_F00465	55	■ 0.510	$Sp(2; \mathbb{R}) = SL(2; \mathbb{R})$	$Sp(2; \mathbb{R}) = SL(2; \mathbb{R})$	a singular metric G .
gilmore-lie-groups_F00466	55	■ 1.000	$MGM^t = G$	$MGM^t = G$	
gilmore-lie-groups_F00467	56	■ 0.999	$E(3)$	$E(3)$	R^3 is
gilmore-lie-groups_F00468	56	■ 0.607	$\mathbf{t}$	$\mathbf{t}$	closely similar argument to Lorentz group, consisting of Lorentz transformations ($A \in SO(3, 1)$, $At_3 + A^t = t_3 + (3.17)$) and space-time displacements is isomorphic

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gilmore-lie-groups_ F00469	56	0.792	$A \in SO(3,1), A I_{3,1} A^t = I_{3,1}$	$A \in SO(3,1), A I_{3,1} A^t = I_{3,1}$	
gilmore-lie-groups_ F00470	56	0.608	$\mathbf{v}$	$\mathbf{v}$	
gilmore-lie-groups_ F00471	56	0.608	$(\mathbf{t})$	$(\mathbf{t})$	
gilmore-lie-groups_ F00472	56	0.608	$\left(t_4\right)$	(t_4)	
gilmore-lie-groups_ F00473	57	0.991	$\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] \in S L(2 ; \mathbb{R}) \subset G L(2 ; \mathbb{R})$	$\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] \in S L(2 ; \mathbb{R}) \subset G L(2 ; \mathbb{R})$	
gilmore-lie-groups_ F00474	57	0.998	$U(1,1)$	$U(1,1)$	
gilmore-lie-groups_ F00475	57	1.000	$a^* a-b^* b=+1$	$a^* a-b^* b=+1$	
gilmore-lie-groups_ F00476	57	1.000	$U(n)$	$U(n)$	
gilmore-lie-groups_ F00477	57	1.000	$U^{\dagger} U=I_{n}$	$U^{\dagger} U=I_n$	
gilmore-lie-groups_ F00478	57	1.000	$2 n \times 2 n$	$2 n \times 2 n$	
gilmore-lie-groups_ F00479	57	1.000	$M^t M=I_{2 n}$	$M^t M=I_{2 n}$	
gilmore-lie-groups_ F00480	57	1.000	$\{ \}^{\dagger}$	$\dagger$	
gilmore-lie-groups_ F00481	57	1.000	$\{ \}^t$	t	
gilmore-lie-groups_ F00482	57	1.000	$I_{2 n}$	$I_{2 n}$	
gilmore-lie-groups_ F00483	57	0.712	$S O(2 n)$	$S O(2 n)$	
gilmore-lie-groups_ F00484	57	1.000	$O U(2 n)$	$O U(2 n)$	
gilmore-lie-groups_ F00485	58	0.999	$U(n ; \mathbb{Q})=S p(n)$	$U(n ; \mathbb{Q})=S p(n)$	
gilmore-lie-groups_ F00486	58	1.000	$M^{\dagger} M=I_{2 n}$	$M^{\dagger} M=I_{2 n}$	
gilmore-lie-groups_ F00487	58	1.000	$\mathbb{C}$	$\mathbb{C}$	
gilmore-lie-groups_ F00488	58	0.945	$\operatorname{SU}(2 n)$	$S U(2 n)$	
gilmore-lie-groups_ F00489	58	1.000	$U \operatorname{Sp}(2 n)$	$U \operatorname{Sp}(2 n)$	
gilmore-lie-groups_ F00490	58	1.000	$\mathbb{R}$	$\mathbb{R}$	
gilmore-lie-groups_ F00491	58	1.000	$G L(n ; \mathbb{Z})$	$G L(n ; \mathbb{Z})$	
gilmore-lie-groups_ F00492	58	1.000	$m \in G L(n ; \mathbb{Z}), \operatorname{det}(m)=\pm 1$	$m \in G L(n ; \mathbb{Z}), \operatorname{det}(m)=\pm 1$	
gilmore-lie-groups_ F00493	58	1.000	$S L(n ; \mathbb{Z})$	$S L(n ; \mathbb{Z})$	
gilmore-lie-groups_ F00494	58	1.000	$m \in S L(n ; \mathbb{Z}), \operatorname{det}(m)=+1$	$m \in S L(n ; \mathbb{Z}), \operatorname{det}(m)=+1$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00495	58	1.000	$PSL(n; \mathbb{Z}), n$	$PSL(n; \mathbb{Z}), n$	any matrix composed of integers must be an integer. t
gilmore-lie-groups_F00496	58	1.000	$PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$	$PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$	any matrix composed of integers must be an integer. t
gilmore-lie-groups_F00497	58	0.999	$n=2$	$n = 2$	Let Γ be the subset of $GL(2; \mathbb{Z})$ with determinant ± 1 forming the subgroup $GL(2; \mathbb{Z}) \subset GL(2; \mathbb{Z})$. The modular group $PSL(2; \mathbb{Z})$ is obtained by identifying each pair of
gilmore-lie-groups_F00498	58	0.999	$\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	Let Γ be the subset of $GL(2; \mathbb{Z})$ with determinant ± 1 forming the subgroup $GL(2; \mathbb{Z}) \subset GL(2; \mathbb{Z})$. The modular group $PSL(2; \mathbb{Z})$ is obtained by identifying each pair of
gilmore-lie-groups_F00499	58	1.000	$\operatorname{det}(m)=n$	$\det(m) = n$	matrices in $SL(2; \mathbb{Z})$ of the form $\begin{bmatrix} -a & -b \\ c & d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
gilmore-lie-groups_F00500	58	1.000	$\operatorname{det}(m^{-1})=1/n$	$\det(m^{-1}) = 1/n$	matrices in $SL(2; \mathbb{Z})$ of the form $\begin{bmatrix} -a & -b \\ c & d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
gilmore-lie-groups_F00501	58	1.000	$\operatorname{det}(m)=$	$\det(m) =$	As a hint of the useful properties of these groups, we consider the matrix
gilmore-lie-groups_F00502	58	0.714	± 1	± 1	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in GL(2; \mathbb{Z})$
gilmore-lie-groups_F00503	58	0.714	$GL(2; \mathbb{Z})$	$GL(2; \mathbb{Z})$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in GL(2; \mathbb{Z})$
gilmore-lie-groups_F00504	58	0.714	$SL(2; \mathbb{Z}) \subset$	$SL(2; \mathbb{Z}) \subset$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in GL(2; \mathbb{Z})$
gilmore-lie-groups_F00505	58	1.000	$PSL(2; \mathbb{Z})$	$PSL(2; \mathbb{Z})$	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \in GL(n; \mathbb{Z}) \quad (3.33)$
gilmore-lie-groups_F00506	58	0.906	$SL(2; \mathbb{Z})$	$SL(2; \mathbb{Z})$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
gilmore-lie-groups_F00507	58	0.906	$\left[\begin{array}{ll} -a & -b \\ -c & -d \end{array} \right] \simeq \left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\left[\begin{array}{ll} -a & -b \\ -c & -d \end{array} \right] \simeq \left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
gilmore-lie-groups_F00508	59	1.000	$F(n)$	$F(n)$	where $F(n)$ is the nth Fibonacci number, defined recursively by
gilmore-lie-groups_F00509	59	1.000	$n=1(F(0)=0, F(1)=1)$	$n = 1(F(0) = 0, F(1) = 1)$	in the groups $GL(2; \mathbb{Z}), GL(3; \mathbb{Z}),$ etc. The group $GL(n; \mathbb{Z})$ has important subgroups defined by imposing linear
gilmore-lie-groups_F00510	59	0.998	$GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$	$GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$	with $GL(n; \mathbb{R})$.
gilmore-lie-groups_F00511	59	0.998	$GL(n; \mathbb{R})$	$GL(n; \mathbb{R})$	of Example (5)
gilmore-lie-groups_F00512	59	1.000	$M^t I_n M = I_n$	$M^t I_n M = I_n$	groups $O(n; \mathbb{R}), O(n; \mathbb{Z})$ consisting of $n \times n$ matrices M which satisfy the matrix equation $M^t I_n M = I_n$ in every row or column is zero and the nonzero matrix element is ± 1 . An im-
gilmore-lie-groups_F00513	59	1.000	$O(n; \mathbb{Z})$	$O(n; \mathbb{Z})$	group S_n .
gilmore-lie-groups_F00514	59	1.000	$\operatorname{det}(m)=+1$	$\det(m) = +1$	
gilmore-lie-groups_F00515	60	1.000	A_n	A_n	ple, the alternating group A_n consists of unimodular matrices in P_n :
gilmore-lie-groups_F00516	60	0.932	$O(2; \mathbb{Z})$	$O(2; \mathbb{Z})$	Example The group $O(2; \mathbb{Z})$ consists of the $0 = 2 \times 2$ matrices
gilmore-lie-groups_F00517	60	0.932	$8=2^2 \times 2!$	$8 = 2^2 \times 2!$	Example The group $O(2; \mathbb{Z})$ consists of the $0 = 2 \times 2$ matrices
gilmore-lie-groups_F00518	60	0.974	$O(3; \mathbb{Z})$	$O(3; \mathbb{Z})$	of the six matrices
gilmore-lie-groups_F00519	60	0.974	$2^3 \times 3! = 48$	$2^3 \times 3! = 48$	of the six matrices

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gilmore-lie-groups_F00520	60	■ 0.974	$6=3!$	$6 = 3!$	of the six matrices
gilmore-lie-groups_F00521	60	■ 1.000	$A_3 \subset S_3 \subset O(3; \mathbb{Z})$	$A_3 \subset S_3 \subset O(3; \mathbb{Z})$	
gilmore-lie-groups_F00522	60	■ 0.997	$G_{\{2\}}, F_{\{4\}}, E_{\{6\}}, E_{\{7\}}, E_{\{8\}}$	G_2, F_4, E_6, E_7, E_8	also of great interest, but fall outside the scope of our discussions.
gilmore-lie-groups_F00523	60	■ 1.000	$GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$	$GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$	
gilmore-lie-groups_F00524	60	■ 1.000	$PSL(n; \mathbb{Z})$	$PSL(n; \mathbb{Z})$	
gilmore-lie-groups_F00525	61	■ 1.000	$\mathcal{IJ} = -\mathcal{K}$	$\mathcal{IJ} = -\mathcal{K}$	where $\mathcal{IJ} = -\mathcal{K}$.
gilmore-lie-groups_F00526	61	■ 0.997	$\sigma_x, \sigma_y, \sigma_z$	$\sigma_x, \sigma_y, \sigma_z$	that any pair of the unit quaternions anticommute: i.e., $\{\mathcal{I}, \mathcal{J}\} = \mathcal{IJ} + \mathcal{JI} = 0$.
gilmore-lie-groups_F00527	61	■ 0.924	$\{\mathcal{I}, \mathcal{J}\} = \mathcal{IJ} + \mathcal{JI} = 0$	$\{\mathcal{I}, \mathcal{J}\} = \mathcal{IJ} + \mathcal{JI} = 0$	Show that the unit quaternions $\mathcal{I}, \mathcal{J}, \mathcal{K}$ generate a group of order 8 under multiplication. Show that this group is isomorphic to $U(2; \mathbb{Z})$. Exhibit this isomorphism explicitly.
gilmore-lie-groups_F00528	61	■ 1.000	$\mathcal{I}, J, K$	$\mathcal{I}, J, K$	
gilmore-lie-groups_F00529	61	■ 0.464	$SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$	$SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$	4. Show that the dimensionalities (over $\mathbb{C}$) of the representations ρ_1, ρ_2, ρ_3 are 1, 1, 1.
gilmore-lie-groups_F00530	61	■ 1.000	$p+q=n$	$p + q = n$	Exercise 1.1.1. Let p, q, n be positive integers such that $p+q=n$. Show that the number of partitions of n into p parts is equal to the number of partitions of n into q parts.
gilmore-lie-groups_F00531	61	■ 0.996	$SL_i(n; \mathbb{C})$	$SL_i(n; \mathbb{C})$	$SL_2(n; \mathbb{C}) \det(M) = e^\lambda \quad 2n^2 - 1$
gilmore-lie-groups_F00532	61	■ 0.996	$GL(n; \mathbb{C})$	$GL(n; \mathbb{C})$	$SL_2(n; \mathbb{C}) \det(M) = e^\lambda \quad 2n^2 - 1$
gilmore-lie-groups_F00533	61	■ 1.000	ϕ, λ, r	ϕ, λ, r	restriction $\mathbb{C} \rightarrow \mathbb{R}$? and the field extension $\mathbb{C} \rightarrow \mathbb{Q}$?
gilmore-lie-groups_F00534	61	■ 1.000	$r \neq 0$	$r \neq 0$	restriction $\mathbb{C} \rightarrow \mathbb{R}$? and the field extension $\mathbb{C} \rightarrow \mathbb{Q}$?
gilmore-lie-groups_F00535	61	■ 1.000	$2n^2-1$	$2n^2 - 1$	A subgroup of $UT(1, 1)$ includes matrices of the form $\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}, a \in \mathbb{R}$. Show that the
gilmore-lie-groups_F00536	61	■ 1.000	$SL_3(n; \mathbb{C})$	$SL_3(n; \mathbb{C})$	A subgroup of $UT(1, 1)$ includes matrices of the form $\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}, a \in \mathbb{R}$. Show that the
gilmore-lie-groups_F00537	61	■ 1.000	$SL(n; \mathbb{C}) = SL_1(n; \mathbb{C}) \cap SL_2(n; \mathbb{C})$	$SL(n; \mathbb{C}) = SL_1(n; \mathbb{C}) \cap SL_2(n; \mathbb{C})$	the form $\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ are also included, then the parameterizing manifold consists of how
gilmore-lie-groups_F00538	61	■ 1.000	$\mathbb{C} \rightarrow \mathbb{R}$	$\mathbb{C} \rightarrow \mathbb{R}$	many copies of $\mathbb{R}^1$?
gilmore-lie-groups_F00539	61	■ 1.000	$\mathbb{C} \rightarrow \mathbb{Q}$	$\mathbb{C} \rightarrow \mathbb{Q}$	many copies of $\mathbb{R}^1$?
gilmore-lie-groups_F00540	61	■ 0.936	$\left[\begin{array}{cc} -1 & a \\ 0 & 1 \end{array} \right], a \in \mathbb{R}$	$\left[\begin{array}{cc} -1 & a \\ 0 & 1 \end{array} \right], a \in \mathbb{R}$	
gilmore-lie-groups_F00541	61	■ 1.000	$\mathbb{R}^1$	$\mathbb{R}^1$	
gilmore-lie-groups_F00542	61	■ 1.000	$\left[\begin{array}{cc} 1 & a \\ 0 & -1 \end{array} \right]$	$\left[\begin{array}{cc} 1 & a \\ 0 & -1 \end{array} \right]$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image														
gilmore-lie-groups_ F08543	62	0.837	$p^2+q^2+p\ q$	$p^2 + q^2 + pq$	<table><thead><tr><th>Group</th><th>Dimension</th></tr></thead><tbody><tr><td>$UT(p,q)$</td><td>$p^2 + q^2 + pq$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2 + q^2 + r^2 + pq + pr + qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$pq$</td></tr></tbody></table> ens to these dimensions if the matrix groups are over the	Group	Dimension	$UT(p,q)$	$p^2 + q^2 + pq$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2 + q^2 + r^2 + pq + pr + qr$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	pq
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$A(p,q)$	pq																		
gilmore-lie-groups_ F08544	62	0.837	$p(p+q)$	$p(p+q)$	<table><thead><tr><th>Group</th><th>Dimension</th></tr></thead><tbody><tr><td>$UT(p,q)$</td><td>$p^2 + q^2 + pq$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2 + q^2 + r^2 + pq + pr + qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$pq$</td></tr></tbody></table> ens to these dimensions if the matrix groups are over the	Group	Dimension	$UT(p,q)$	$p^2 + q^2 + pq$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2 + q^2 + r^2 + pq + pr + qr$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	pq
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$Sol(n)$	$n(n+1)/2$																		
$Nil(n)$	$n(n-1)/2$																		
$A(p,q)$	pq																		
gilmore-lie-groups_ F08545	62	0.837	$p^2+q^2+r^2+p\ q+p\ r+q\ r$	$p^2 + q^2 + r^2 + pq + pr + qr$	<table><thead><tr><th>Group</th><th>Dimension</th></tr></thead><tbody><tr><td>$UT(p,q)$</td><td>$p^2 + q^2 + pq$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2 + q^2 + r^2 + pq + pr + qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$pq$</td></tr></tbody></table> ens to these dimensions if the matrix groups are over the	Group	Dimension	$UT(p,q)$	$p^2 + q^2 + pq$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2 + q^2 + r^2 + pq + pr + qr$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	pq
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gilmore-lie-groups_ F08546	62	0.837	$n(n+1) / 2$	$n(n+1)/2$	<table><thead><tr><th>Group</th><th>Dimension</th></tr></thead><tbody><tr><td>$UT(p,q)$</td><td>$p^2 + q^2 + pq$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2 + q^2 + r^2 + pq + pr + qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$pq$</td></tr></tbody></table> ens to these dimensions if the matrix groups are over the	Group	Dimension	$UT(p,q)$	$p^2 + q^2 + pq$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2 + q^2 + r^2 + pq + pr + qr$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	pq
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$Nil(n)$	$n(n-1)/2$																		
$A(p,q)$	pq																		
gilmore-lie-groups_ F08547	62	0.837	$n(n-1) / 2$	$n(n-1)/2$	<table><thead><tr><th>Group</th><th>Dimension</th></tr></thead><tbody><tr><td>$UT(p,q)$</td><td>$p^2 + q^2 + pq$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2 + q^2 + r^2 + pq + pr + qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$pq$</td></tr></tbody></table> ens to these dimensions if the matrix groups are over the	Group	Dimension	$UT(p,q)$	$p^2 + q^2 + pq$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2 + q^2 + r^2 + pq + pr + qr$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	pq
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$A(p,q)$	pq																		
gilmore-lie-groups_ F08548	62	0.837	$p\ q$	pq	<table><thead><tr><th>Group</th><th>Dimension</th></tr></thead><tbody><tr><td>$UT(p,q)$</td><td>$p^2 + q^2 + pq$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2 + q^2 + r^2 + pq + pr + qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$pq$</td></tr></tbody></table> ens to these dimensions if the matrix groups are over the	Group	Dimension	$UT(p,q)$	$p^2 + q^2 + pq$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2 + q^2 + r^2 + pq + pr + qr$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	pq
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$Sol(n)$	$n(n+1)/2$																		
$Nil(n)$	$n(n-1)/2$																		
$A(p,q)$	pq																		
gilmore-lie-groups_ F08549	62	1.000	$\mathbf{F}=\mathbf{d}\ \mathbf{p}\ /\ \mathbf{d}\ \mathbf{t}$	$\mathbf{F} = d\mathbf{p}/dt$	ions can be written in the form $\frac{d\mathbf{p}}{dt} = \mathbf{F}(\mathbf{p}, \mathbf{q}, t)$														

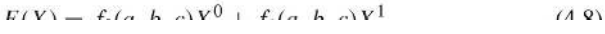
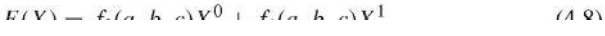
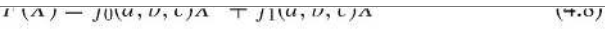
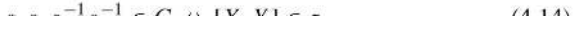
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups-F08550	62	■ 1.000	$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) A_\mu = -\frac{4\pi}{c} j_\mu$		Poincaré not invariant invariant ow do you reconcile these results? ow that the group of 2×2 matrices $SU(2)$ is parameterized by two compl
gilmore-lie-groups-F08551	62	■ 0.814	$SU(2)$	$SU(2)$	$SU(2) = \left \begin{matrix} -c_5^* & c_1^* \end{matrix} \right $
gilmore-lie-groups-F08552	62	■ 1.000	$c_1 = a_1 + ib_1$	$c_1 = a_1 + ib_1$	$\text{matrix for the addition } \vec{c}_2 = \vec{c}_1 + \vec{c}_2 = \vec{c}_2$
gilmore-lie-groups-F08553	62	■ 1.000	$c_2 = a_2 + ib_2$	$c_2 = a_2 + ib_2$	$\text{matrix for the addition } \vec{c}_2 = \vec{c}_1 + \vec{c}_2 = \vec{c}_2$
gilmore-lie-groups-F08554	62	■ 1.000	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$	11. The group of 2×2 matrices $SU(1, 1)$ is parameterized by two complex numbers $a_1 = a_1 + ib_1$ and $a_2 = a_2 + ib_2$, so that
gilmore-lie-groups-F08555	62	■ 0.991	$S^3 \subset R^4$	$S^3 \subset R^4$	$SU(1, 1) = \begin{bmatrix} c_1 & c_2 \\ \pm & \pm \end{bmatrix}$
gilmore-lie-groups-F08556	62	■ 1.000	$a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$	$a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$	
gilmore-lie-groups-F08557	62	■ 1.000	$\left[\begin{matrix} m_{11} & x \\ m_{21} & m_{22} \end{matrix}\right]$	$\left[\begin{matrix} m_{11} & x \\ m_{21} & m_{22} \end{matrix}\right]$	
gilmore-lie-groups-F08558	62	■ 1.000	$m_{11}^2 + x^2 = 1$	$m_{11}^2 + x^2 = 1$	
gilmore-lie-groups-F08559	62	■ 1.000	$m_{11} = \pm \sqrt{1 - x^2}$	$m_{11} = \pm \sqrt{1 - x^2}$	
gilmore-lie-groups-F08560	63	■ 1.000	$m_{21}m_{11} + m_{22}x = 0$	$m_{21}m_{11} + m_{22}x = 0$	is orthogonal to the first, so that $m_{21}m_{11} + m_{22}x = 0$. As a result, we find
gilmore-lie-groups-F08561	63	■ 0.994	$m_{ij}i \geq j$	$m_{ij}i \geq j$	An alternative parameterization of $SO(3)$ is given by
gilmore-lie-groups-F08562	63	■ 0.495	Z_2	Z_2	are necessary to cover $SU(3)$ completely?
gilmore-lie-groups-F08563	63	■ 0.495	Z_1	Z_1	are necessary to cover $SU(3)$ completely?
gilmore-lie-groups-F08564	63	■ 0.495	(x, y)	(x, y)	are necessary to cover $SU(3)$ completely?
gilmore-lie-groups-F08565	63	■ 0.854	$M \in GL(n; \mathbb{Z})$	$M \in GL(n; \mathbb{Z})$	17 Estimate the Fibonacci number $F(n)$ from the eigen
gilmore-lie-groups-F08566	63	■ 0.854	$\operatorname{det}(M)$	$\det(M)$	17 Estimate the Fibonacci number $F(n)$ from the eigen
gilmore-lie-groups-F08567	63	■ 1.000	$O(n; \mathbb{Z}) \supset S_n \supset A_n$	$O(n; \mathbb{Z}) \supset S_n \supset A_n$	generating matrix (3.33). What happens to this sequence if different initial (other than $F(0) = 0, F(1) = 1$) are introduced?
gilmore-lie-groups-F08568	63	■ 1.000	$2^n \times n!, n!, \frac{1}{2}n!$	$2^n \times n!, n!, \frac{1}{2}n!$	generating matrix (3.33). What happens to this sequence if different initial (other than $F(0) = 0, F(1) = 1$) are introduced?
gilmore-lie-groups-F08569	63	■ 1.000	$\lambda_\pm = \frac{1}{2}(1 \pm \sqrt{5})$	$\lambda_\pm = \frac{1}{2}(1 \pm \sqrt{5})$	18. Derive other Fibonacci-type series using other symmetric generating matrices in
gilmore-lie-groups-F08570	63	■ 0.999	$F(0)=0, F(1)=1$	$F(0) = 0, F(1) = 1$	
gilmore-lie-groups-F08571	63	■ 1.000	$\left[\begin{matrix} 2 & -1 \\ -1 & 1 \end{matrix}\right]$	$\left[\begin{matrix} 2 & -1 \\ -1 & 1 \end{matrix}\right]$	quantum number n have the same energy $E(nlm) = -E_0/n^2$ ($E_0 =$
gilmore-lie-groups-F08572	63	■ 1.000	$ n \ l \ m\rangle$	$ nlm\rangle$	potential, the overall symmetry reduces to that of the rotation group; there is a
gilmore-lie-groups-F08573	63	■ 1.000	n^2	n^2	potential, the overall symmetry reduces to that of the rotation group; there is a

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gilmore-lie-groups_ F08574	63	■ 1.000	$E(n\ l\ m)=-E_{\{0\}} / n^{\{2\}}\backslashleft(E_{\{0\}}=13.6\ \mathrm{eV}\backslashright)$	$E(nlm)=-E_0/n^2\ (E_0=13.6\mathrm{eV})$	the description of the hydrogen atom bound states are indexed by the principal
gilmore-lie-groups_ F08575	63	■ 1.000	$S\ 0(4)\ \downarrow S\ 0(3)$	$SO(4)\downarrow SO(3)$	number l , $l=0,1,2,\ldots,n-1$, with $\sum_{l=0}^{n-1}(2l+1)=n^2$. The $SO(5)$ multiplet with quantum number l is $2l+1$ fold degenerate. As associated Hamiltonian with
gilmore-lie-groups_ F08576	63	■ 0.993	$n(n=1,2,3,\ \ldots)$	$n(n=1,2,3,\ldots)$	$SO(4)\times SO(3)$ broken symmetry that describes the filling order. When electrons are introduced into a Coulomb potential established by a central charge $+Ze$ can be
gilmore-lie-groups_ F08577	63	■ 0.980	$l,\ l=0,1,2\ \ldots,\ n-1$	$l,l=0,1,2\ldots,n-1$	
gilmore-lie-groups_ F08578	63	■ 0.980	$\sum_{l=0}^{l=n-1} (2\ l+1)=n^{\{2\}}$	$\sum_{l=0}^{l=n-1} (2l+1)=n^2$	
gilmore-lie-groups_ F08579	63	■ 1.000	l	l	
gilmore-lie-groups_ F08580	63	■ 1.000	$2\ l+1$	$2l+1$	
gilmore-lie-groups_ F08581	63	■ 1.000	$+Z\ e$	$+Ze$	
gilmore-lie-groups_ F08582	64	■ 1.000	$\backslashdelta=0.28$	$\delta=0.28$	periodic table of the chemical elements: $(n,\ l)\rightarrow 1s:\ 2s,\ 2p:\ 3s,\ 3p:\ 4s,\ 3d,\ 4p:$
gilmore-lie-groups_ F08583	64	■ 0.608	$(n,\ l)\ \rightarrowtail 1\ s$	$(n,l)\rightarrow 1s$	$5s,\ 4d,\ 5p;\ 6s,\ 4f,\ 5d,\ 6p;\ 7s,\ 5f,\ 6d,\ 7p;\ 8s,\ 6f,\ 7d,\ 8p;\ldots$
gilmore-lie-groups_ F08584	64	■ 0.608	$2\ s$	$2s$	$5s,\ 4d,\ 5p;\ 6s,\ 4f,\ 5d,\ 6p;\ 7s,\ 5f,\ 6d,\ 7p;\ 8s,\ 6f,\ 7d,\ 8p;\ldots$
gilmore-lie-groups_ F08585	64	■ 0.608	$2\ p$	$2p$	$5s,\ 4d,\ 5p;\ 6s,\ 4f,\ 5d,\ 6p;\ 7s,\ 5f,\ 6d,\ 7p;\ 8s,\ 6f,\ 7d,\ 8p;\ldots$
gilmore-lie-groups_ F08586	64	■ 0.608	$3\ s$	$3s$	$5s,\ 4d,\ 5p;\ 6s,\ 4f,\ 5d,\ 6p;\ 7s,\ 5f,\ 6d,\ 7p;\ 8s,\ 6f,\ 7d,\ 8p;\ldots$
gilmore-lie-groups_ F08587	64	■ 0.608	$3\ p$	$3p$	$5s,\ 4d,\ 5p;\ 6s,\ 4f,\ 5d,\ 6p;\ 7s,\ 5f,\ 6d,\ 7p;\ 8s,\ 6f,\ 7d,\ 8p;\ldots$
gilmore-lie-groups_ F08588	64	■ 0.608	$4\ s$	$4s$	$5s,\ 4d,\ 5p;\ 6s,\ 4f,\ 5d,\ 6p;\ 7s,\ 5f,\ 6d,\ 7p;\ 8s,\ 6f,\ 7d,\ 8p;\ldots$
gilmore-lie-groups_ F08589	64	■ 0.608	$3\ d$	$3d$	$5s,\ 4d,\ 5p;\ 6s,\ 4f,\ 5d,\ 6p;\ 7s,\ 5f,\ 6d,\ 7p;\ 8s,\ 6f,\ 7d,\ 8p;\ldots$
gilmore-lie-groups_ F08590	64	■ 0.608	$4\ p$	$4p$	$5s,\ 4d,\ 5p;\ 6s,\ 4f,\ 5d,\ 6p;\ 7s,\ 5f,\ 6d,\ 7p;\ 8s,\ 6f,\ 7d,\ 8p;\ldots$
gilmore-lie-groups_ F08591	64	■ 0.957	$5\ s,\ 4\ d,\ 5\ p\ ;\ 6\ s,\ 4\ f,\ 5\ d,\ 6\ p\ ;\ 7\ s,\ 5\ f,\ 6\ d,\ 7\ p\ ;\ 8\ s,\ 6\ f,\ 7\ d,\ 8\ p\ ;\ \ldots$	$5s,4d,5p;6s,4f,5d,6p;7s,5f,6d,7p;8s,6f,7d,8p;\ldots$	no summation shown the following summation.....
gilmore-lie-groups_ F08592	64	■ 1.000	$G_{\{2\}}$	G_2	
gilmore-lie-groups_ F08593	64	■ 1.000	$2\ n$	$2n$	
gilmore-lie-groups_ F08594	64	■ 1.000	$M^{\{t\}}\ G_{\{i\}}\ M=G_{\{i\}}$	$M^tG_iM=G_i$	$G_1\qquad G_2\qquad G_1\text{ and }G_2$
gilmore-lie-groups_ F08595	64	■ 1.000	$y^{\{j\}}=x^{\{i\}}\ M_{\{i\}}^{\{j\}}$	$y^j=x^iM_i^j$	$\frac{\sigma}{\partial v^i}=\frac{\sigma y^j}{\partial v^i}\frac{\sigma}{\partial v^j}=M_i^j\frac{\sigma}{\partial v^j}$
gilmore-lie-groups_ F08596	64	■ 0.990	$f(q,\ p)$	$f(q,p)$	$\overline{\overline{\overline{k}}}\frac{\sigma q_k}{\sigma q_k}\frac{\sigma p_k}{\sigma p_k}\frac{\sigma q_k}{\sigma q_k}\frac{\sigma p_k}{\sigma p_k}$
gilmore-lie-groups_ F08597	64	■ 0.990	$g(q,\ p)$	$g(q,p)$	$\overline{\overline{\overline{k}}}\frac{\sigma q_k}{\sigma q_k}\frac{\sigma p_k}{\sigma p_k}\frac{\sigma q_k}{\sigma q_k}\frac{\sigma p_k}{\sigma p_k}$
gilmore-lie-groups_ F08598	65	■ 0.666	$(Q,\ P)$	(Q,P)	b. Introduce a new coordinate system (Q,P) , related to the original by a linear
gilmore-lie-groups_ F08599	65	■ 1.000	$A^{\{t\}}\ C$	A^tC	. Show that the same conditions hold for linear transformations and the quantumm
gilmore-lie-groups_ F08600	65	■ 1.000	$B^{\{t\}}\ D$	B^tD	. Show that the same conditions hold for linear transformations and the quantumm

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gilmore-lie-groups_F00601	65	■ 1.000	$A^t D - B^t C = I_n$	$A^t D - B^t C = I_n$	. Show that the same conditions hold for linear transformations and the quantum
gilmore-lie-groups_F00602	65	■ 0.999	$\left[q_j, q_k \right] = \left[p_j, p_k \right] = 0$	$[q_j, q_k] = [p_j, p_k] = 0$	ote: The transformation from classical mechanics to quantum mechanics is made
gilmore-lie-groups_F00603	65	■ 0.999	$\left[q_j, p_k \right] = i \hbar \delta_{jk}$	$[q_j, p_k] = i\hbar\delta_{jk}$	ote: The transformation from classical mechanics to quantum mechanics is made
gilmore-lie-groups_F00604	65	■ 1.000	$e^{+i k x}$	e^{+ikx}	energy $E = (\hbar k)^2/2m$. The complex number A_L is the probability amplitude for finding a particle moving to the right with this momentum. The expected value of the
gilmore-lie-groups_F00605	65	■ 0.644	m	m	momentum in the left-hand region is $\langle \hat{p} \rangle = (A_L ^2 - B_L ^2)\hbar k$, where the operator
gilmore-lie-groups_F00606	65	■ 0.644	$(+)$	$(+)$	momentum in the left-hand region is $\langle \hat{p} \rangle = (A_L ^2 - B_L ^2)\hbar k$, where the operator
gilmore-lie-groups_F00607	65	■ 0.644	$\hbar k$	$\hbar k$	momentum in the left-hand region is $\langle \hat{p} \rangle = (A_L ^2 - B_L ^2)\hbar k$, where the operator
gilmore-lie-groups_F00608	65	■ 1.000	$E = (\hbar k)^2 / 2 m$	$E = (\hbar k)^2/2m$	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$.
gilmore-lie-groups_F00609	65	■ 1.000	$A_{\{L\}}$	A_L	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$.
gilmore-lie-groups_F00610	65	■ 1.000	$\langle \hat{p} \rangle = \left(A_L ^2 - B_L ^2 \right) \hbar k$	$\langle \hat{p} \rangle = \left(A_L ^2 - B_L ^2 \right) \hbar k$	$ A_L ^2 - B_L ^2 = A_R ^2 - B_R ^2$
gilmore-lie-groups_F00611	65	■ 1.000	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$	
gilmore-lie-groups_F00612	65	■ 1.000	$V_{\{L\}} = V_{\{R\}}$	$V_L = V_R$	
gilmore-lie-groups_F00613	65	■ 1.000	$\psi_{\sigma}(x) = A_{\sigma} e^{+i k x} + B_{\sigma} e^{-i k x}$	$\psi_{\sigma}(x) = A_{\sigma} e^{+ikx} + B_{\sigma} e^{-ikx}$	
gilmore-lie-groups_F00614	65	■ 0.969	$\sigma = L, R$	$\sigma = L, R$	
gilmore-lie-groups_F00615	66	■ 1.000	$A_{\{L\}}, A_{\{R\}}$	A_L, A_R	b. Since the Schrödinger equation is second order the four amplitudes $A_L, A_R,$
gilmore-lie-groups_F00616	66	■ 1.000	$B_{\{L\}}, B_{\{R\}}$	B_L, B_R	B_L, B_R are not independent. Only two are independent. Two linear relations exist
gilmore-lie-groups_F00617	66	■ 1.000	E	E	c. Show that $T \in SU(1, 1)$ by appropria
gilmore-lie-groups_F00618	66	■ 1.000	$T(E) \in U(1, 1)$	$T(E) \in U(1, 1)$	c. Show that $T \in SU(1, 1)$ by appropria
gilmore-lie-groups_F00619	66	■ 1.000	$T \in SU(1, 1)$	$T \in SU(1, 1)$	25. Crossing symmetry: A transfer matrix T for a one-dimens
gilmore-lie-groups_F00620	66	■ 1.000	$S \in U(2)$	$S \in U(2)$	$\begin{pmatrix} r_1 & t_{12} \\ t_{21} & r_2 \end{pmatrix}$
gilmore-lie-groups_F00621	66	■ 1.000	$S(E)$	$S(E)$	ray time $1/j\hbar$.
gilmore-lie-groups_F00622	66	■ 1.000	$T(E)$	$T(E)$	ray time $1/j\hbar$.
gilmore-lie-groups_F00623	66	■ 1.000	$t_{\{11\}}(E)$	$t_{11}(E)$	ray time $1/j\hbar$.
gilmore-lie-groups_F00624	66	■ 1.000	$r_j / \left[\left(E - E_j \right) + i \left(\Gamma_j / 2 \right) \right]$	$r_j / [(E - E_j) + i(\Gamma_j/2)]$	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00625	66	■ 1.000	$E_{\{j\}}$	E_j	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00626	66	■ 1.000	$\Gamma_j / \hbar$	$\Gamma_j/\hbar$	are inputs to
gilmore-lie-groups_F00627	66	■ 1.000	$V_{\{1\}}$	V_1	$\begin{pmatrix} i_2\rangle \\ o_1\rangle \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} o_1\rangle \\ i_2\rangle \end{pmatrix}$

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gilmore-lie-groups_F00628	66	■ 1.000	$V_{\{2\}}$	V_2	
gilmore-lie-groups_F00629	66	■ 1.000	$T_{\{1\}}$	T_1	
gilmore-lie-groups_F00630	66	■ 1.000	$T_{\{2\}}$	T_2	
gilmore-lie-groups_F00631	66	■ 1.000	$S_{\{1\}}$	S_1	
gilmore-lie-groups_F00632	67	■ 1.000	$S_{\{\text{\texttt{Tot}}\}}$	S_{Tot}	
gilmore-lie-groups_F00633	67	■ 1.000	$1/\left(1-s_{\{12\}}s_{\{43\}}\right)$	$1/\left(1-s_{12}s_{43}\right)$	
gilmore-lie-groups_F00634	67	■ 1.000	$V_{\{1\}}^{\prime}$	V_1'	
gilmore-lie-groups_F00635	67	■ 1.000	$V_{\{2\}}^{\prime}$	V_2'	
gilmore-lie-groups_F00636	67	■ 1.000	$T_{\{i\}}(E) \rightarrow T_{\{i\}}^{\prime}(E)$	$T_i(E) \rightarrow T_i'(E)$	
gilmore-lie-groups_F00637	67	■ 1.000	$S_{\{i\}}(E) \rightarrow S_{\{i\}}^{\prime}(E)$	$S_i(E) \rightarrow S_i'(E)$	
gilmore-lie-groups_F00638	67	■ 1.000	$i=1,2$	$i=1,2$	
gilmore-lie-groups_F00639	67	■ 1.000	$E, S_{\{\text{\texttt{Tot}}\}}^{\prime}(E)=S_{\{\text{\texttt{Tot}}\}}(E)$	$E, S_{\text{Tot}}'(E) = S_{\text{Tot}}(E)$	
gilmore-lie-groups_F00640	67	■ 1.000	$S_{\{1\}}^{\prime}(E)$	$S_1'(E)$	
gilmore-lie-groups_F00641	67	■ 1.000	$S_{\{2\}}^{\prime}(E)$	$S_2'(E)$	
gilmore-lie-groups_F00642	67	■ 0.998	$S_{\{\text{\texttt{Tot}}\}}(E)$	$S_{\text{Tot}}(E)$	
gilmore-lie-groups_F00643	67	■ 0.999	$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$	$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$	
gilmore-lie-groups_F00644	67	■ 0.997	$T_{\{1\}}(E) T_{\{2\}}(E)=T_{\{\text{\texttt{Tot}}\}}(E)=T_{\{1\}}^{\prime}(E) T_{\{2\}}^{\prime}(E)$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$	
gilmore-lie-groups_F00645	67	■ 0.997	$T_{\{1\}}^{\prime}(E)=T_{\{1\}}(E) R$	$T_1'(E) = T_1(E)R$	
gilmore-lie-groups_F00646	67	■ 0.997	$T_{\{2\}}^{\prime}(E)=$	$T_2'(E) =$	
gilmore-lie-groups_F00647	67	■ 1.000	$R^{\{-1\}} T_{\{2\}}(E)$	$R^{-1}T_2(E)$	
gilmore-lie-groups_F00648	67	■ 1.000	$R \in U(1,1)$	$R \in U(1,1)$	
gilmore-lie-groups_F00649	67	■ 1.000	$U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$	$U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$	
gilmore-lie-groups_F00650	67	■ 0.932	$T_{\{\text{\texttt{Tot}}\}}(E)$	$T_{\text{Tot}}(E)$	
gilmore-lie-groups_F00651	67	■ 0.932	$\left(T_{\{1\}}(E) R, R^{\{-1\}} T_{\{2\}}(E)\right)$	$(T_1(E)R, R^{-1}T_2(E))$	
gilmore-lie-groups_F00652	67	■ 0.850	$\left(S_{\{1\}}^{\prime}(E), S_{\{2\}}^{\prime}(E)\right)$	$(S_1'(E), S_2'(E))$	
gilmore-lie-groups_F00653	68	■ 0.987	$n_{\{1\}}, n_{\{2\}}, \ldots, n_{\{k\}}$	$n_1, n_2, \dots, n_k$	
gilmore-lie-groups_F00654	68	■ 0.718	$k S$	kS	

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gilmore-lie-groups_F00655	68	0.718	$n_{\{j\}} \times n_{\{j\}} (j=1,2, \ldots, k)$	$n_j \times n_j (j = 1, 2, \dots, k)$	permutation of the rows and columns of this direct sum of S matrices the input-output
gilmore-lie-groups_F00656	68	1.000	Γ	Γ	connectivity, of the network. a. Show that the S matrix that describes the network, defined by $S_{ij} = \delta_{ij} - \Gamma_{ij}$
gilmore-lie-groups_F00657	68	0.999	$\left[o_e\right]=S_{\text {Network }}\left[i_e\right]$	$[o_e] = S_{\text{Network}} [i_e]$	Figure 3.4. A network circuit.
gilmore-lie-groups_F00658	68	1.000	$S_{\text {\text {Newtork}}}$	S_{Newtork}	$S_{\text{Network}} = D + C\Gamma B + C\Gamma A\Gamma B + C\Gamma A\Gamma A\Gamma B + C\Gamma A\Gamma A\Gamma A$
gilmore-lie-groups_F00659	68	1.000	$S_{\text {\text {Newtork}}}^{\dag}=S_{\text {\text {Newtork}}}, S_{\text {\text {Newtork}}} \subset U(d)$	$S_{\text{Newtork}}^{\dagger} = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$	$S_{\text{Network}} = D + C\Gamma B + C\Gamma A\Gamma B + C\Gamma A\Gamma A\Gamma B + C\Gamma A\Gamma A\Gamma A$
gilmore-lie-groups_F00660	68	1.000	$U\left(\sum_{j=1}^k n_{\{j\}}\right) \supset \Pi_{j=1}^k \otimes U\left(n_{\{j\}}\right)$	$U\left(\sum_{j=1}^k n_j\right) \supset \Pi_{j=1}^k \otimes U\left(n_j\right)$	each group operation in $U(d)$.
gilmore-lie-groups_F00661	68	1.000	$U(d)$	$U(d)$	30 All the matrices in this problem are square $n \times n$ with: H hermitian; U unitary; A
gilmore-lie-groups_F00662	68	1.000	d	d	30 All the matrices in this problem are square $n \times n$ with: H hermitian; U unitary; A
gilmore-lie-groups_F00663	68	0.917	$\Pi_{j=1}^k \otimes U\left(n_{\{j\}}\right)$	$\Pi_{j=1}^k \otimes U\left(n_j\right)$	antihermitian. Show the right-hand column follows from the definition in the left-hand
gilmore-lie-groups_F00664	68	1.000	U	U	$H_2 = \frac{H_1 + I_n}{2} \quad [H_1, H_2] = 0 \quad H_1 = \frac{H_2 + I_n}{2}$
gilmore-lie-groups_F00665	70	1.000	b	b	its neighborhood, on the left by group element ba^{-1} (or on the right by $a^{-1}b$). This maps a into b and points near a into points near b .
gilmore-lie-groups_F00666	70	1.000	$b\,a^{-1}$	ba^{-1}	maps a into b and points near a into points near b . It is therefore necessary to study the neighborhood of only one group operation
gilmore-lie-groups_F00667	70	1.000	$a^{-1}\,b$	$a^{-1}b$	maps a into b and points near a into points near b . It is therefore necessary to study the neighborhood of only one group operation
gilmore-lie-groups_F00668	71	0.953	$(a, \, b, \, c)$	(a, b, c)	done by knowing the parameters (a, b, c) to become infinitesimals and expanding the group operation in terms of these infinitesimals to first order
gilmore-lie-groups_F00669	71	1.000	$\phi(x, \, y) \rightarrow \phi(x, \, \theta + \delta y)$	$\phi(x, y) \rightarrow \phi(x, \theta + \delta y)$	the linearization of the group in the neighborhood of the identity is so powerful.
gilmore-lie-groups_F00670	72	1.000	ϵ	ϵ	given in (4.4). Then if ϵ is a small real number, $I + \epsilon X$ represents an element in
gilmore-lie-groups_F00671	72	1.000	$I+\epsilon X$	$I + \epsilon X$	given in (4.4). Then if ϵ is a small real number, $I + \epsilon X$ represents an element in
gilmore-lie-groups_F00672	72	0.744	$X_{\{a\}}$	X_a	given in (7.7)
gilmore-lie-groups_F00673	72	0.744	$X_{\{b\}}$	X_b	given in (7.7)
gilmore-lie-groups_F00674	72	0.744	$X_{\{c\}}$	X_c	given in (7.7)
gilmore-lie-groups_F00675	72	1.000	$X^{\{n\}}$	X^n	must be proportional to X ($= \theta^{-1} X$), X^2 is proportional to the identity, and so on. Eisenstein involves use of the Cayley–Hamilton theorem, that every matrix satisfies
gilmore-lie-groups_F00676	72	1.000	$X^{\{0\}}=I_{\{2\}}, \, X^{\{1\}}=X$	$X^0 = I_2, X^1 = X$	Eisenstein involves use of the Cayley–Hamilton theorem, that every matrix satisfies its secular equation. This means that a 2×2 matrix must satisfy a polynomial
gilmore-lie-groups_F00677	72	1.000	$X^{\{2\}}=\theta^2 I_{\{2\}}$	$X^2 = \theta^2 I_2$	its secular equation. This means that a 2×2 matrix must satisfy a polynomial equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.
gilmore-lie-groups_F00678	72	1.000	$X^{\{2\}}$	X^2	its secular equation. This means that a 2×2 matrix must satisfy a polynomial equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.
gilmore-lie-groups_F00679	72	1.000	$X^{\{3\}}=X^{\{2\}} \, X^{\{1\}}$	$X^3 = X^2 X^1$	its secular equation. This means that a 2×2 matrix must satisfy a polynomial equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.
gilmore-lie-groups_F00680	72	0.991	$X\left(=\theta^2 X\right), \, X^{\{4\}}$	$X \left(= \theta^2 X\right), X^4$	Similarly, X^3 can be replaced by a linear combination of X^2 and X , and then X^2

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gilmore-lie-groups-F00681	72	■ 0.995	$X^{\{0\}}=I_{\{2\}}$	$X^0 = I_2$	
gilmore-lie-groups-F00682	72	■ 0.995	$X^{\{1\}}=X$	$X^1 = X$	
gilmore-lie-groups-F00683	72	■ 1.000	$X^{\{3\}}$	X^3	
gilmore-lie-groups-F00684	72	■ 0.759	$I_{\{2\}}$	I_2	
gilmore-lie-groups-F00685	73	■ 1.000	$f_{\{0\}}, f_{\{1\}}$	f_0, f_1	Furthermore, the functions f_0, f_1 are not arbitrary functions of the three parameters
gilmore-lie-groups-F00686	73	■ 1.000	$\theta^2=a^2+b\ c$	$\theta^2 = a^2 + bc$	X is $\theta^2 = a^2 + bc$. As a result, we know from general and simple considerations
gilmore-lie-groups-F00687	73	■ 1.000	$f_{\{0\}}\left(\theta\right)=1+\theta^2/2!+\theta^4/4!+\theta^6/6!+\cdots=\cosh \theta$	$f_0\left(\theta^2\right)=1+\theta^2/2!+\theta^4/4!+\theta^6/6!+\cdots=\cosh \theta$	$f_1(\theta^2)=1+\theta^2/3!+\theta^4/5!+\theta^6/7!+\cdots=\sinh(\theta)/\theta$. These arguments are ap-
gilmore-lie-groups-F00688	73	■ 1.000	$f_{\{1\}}\left(\theta\right)=1+\theta^2/3!+\theta^4/5!+\theta^6/7!+\cdots=\sinh (\theta) / \theta$	$f_1\left(\theta^2\right)=1+\theta^2/3!+\theta^4/5!+\theta^6/7!+\cdots=\sinh (\theta) / \theta$	plicable to the exponential of any matrix Lie algebra.
gilmore-lie-groups-F00689	73	■ 1.000	Y	Y	
gilmore-lie-groups-F00690	73	■ 1.000	$g_{\{1\}}=I+\epsilon X$	$g_1 = I + \epsilon X$	
gilmore-lie-groups-F00691	73	■ 1.000	$I+\epsilon X$	$I + \epsilon X$	
gilmore-lie-groups-F00692	74	■ 1.000	α	α	for any real number α . We can form the produ
gilmore-lie-groups-F00693	74	■ 1.000	ϵ, δ	ϵ, δ	$\text{EXP}(\epsilon X)$ and $g_1(\epsilon)^{-1} = \text{EXP}(-\epsilon X)$ are group elements near the identity, as are
gilmore-lie-groups-F00694	74	■ 1.000	$g_{\{1\}}(\epsilon)=$	$g_1(\epsilon) =$	$\text{EXP}(\epsilon X)$ and $g_1(\epsilon)^{-1} = \text{EXP}(-\epsilon X)$ are group elements near the identity, as are
gilmore-lie-groups-F00695	74	■ 1.000	$\operatorname{EXP}(\epsilon X)$	$\text{EXP}(\epsilon X)$	$g_2(\delta)^{\pm 1} = \text{EXP}(\pm \delta Y)$. Expanding the commutator to lowest order nonvanishing
gilmore-lie-groups-F00696	74	■ 1.000	$g_{\{1\}}(\epsilon)^{-1}=\operatorname{EXP}(-\epsilon X)$	$g_1(\epsilon)^{-1} = \text{EXP}(-\epsilon X)$	$g_2(\delta)^{\pm 1} = \text{EXP}(\pm \delta Y)$. Expanding the commutator to lowest order nonvanishing
gilmore-lie-groups-F00697	74	■ 1.000	$g_{\{2\}}(\delta)^{\pm 1}=\operatorname{EXP}(\pm \delta Y)$	$g_2(\delta)^{\pm 1} = \text{EXP}(\pm \delta Y)$	terms, we find
gilmore-lie-groups-F00698	74	■ 0.996	$g_{\{1\}}(\epsilon)=\operatorname{EXP}(\epsilon X)$	$g_1(\epsilon) = \text{EXP}(\epsilon X)$	$Y, [X, Y] = (XY - YX)$, to be in its Lie algebra $\mathfrak{g}$
gilmore-lie-groups-F00699	74	■ 0.996	$g_{\{2\}}(\delta)=$	$g_2(\delta) =$	$Y, [X, Y] = (XY - YX)$, to be in its Lie algebra $\mathfrak{g}$
gilmore-lie-groups-F00700	74	■ 1.000	$\operatorname{EXP}(\delta Y)$	$\text{EXP}(\delta Y)$	
gilmore-lie-groups-F00701	74	■ 1.000	$Y,[X,Y]=(XY-YX)$	$Y,[X,Y] = (XY - YX)$	
gilmore-lie-groups-F00702	74	■ 1.000	$\mathfrak{g}$	$\mathfrak{g}$	
gilmore-lie-groups-F00703	74	■ 1.000	$g_{\{1\}}H\,g_{\{1\}}^{-1}\subset H$	$g_1Hg_1^{-1} \subset H$	$g_1Hg_1^{-1} \in H \Rightarrow [X, Y] \in \text{Lie algebra of } H$ (4.10)
gilmore-lie-groups-F00704	74	■ 1.000	X,Y,Z	X,Y,Z	This identity involves the cyclic permutation of the operators in a double commutator. For matrices this identity can be proved by opening up the con
gilmore-lie-groups-F00705	74	■ 0.996	$([X,Y]=X\,Y-Y\,X)$	$([X,Y] = XY - YX)$	
gilmore-lie-groups-F00706	74	■ 1.000	$X\,Y$	XY	

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gilmore-lie-groups_ F00707	75	■ 1.000	$X \in \mathfrak{g}$	$X \in \mathfrak{g}$	$Y \in \mathfrak{g}$ then every linear combination of X and Y is in $\mathfrak{g}$:
gilmore-lie-groups_ F00708	75	■ 1.000	$Y \in \mathfrak{g}$	$Y \in \mathfrak{g}$	
gilmore-lie-groups_ F00709	75	■ 1.000	$X \in \mathfrak{g}, Y \in \mathfrak{g}$	$X \in \mathfrak{g}, Y \in \mathfrak{g}$	$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$
gilmore-lie-groups_ F00710	75	■ 1.000	$Z \in \mathfrak{g}$	$Z \in \mathfrak{g}$	$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$
gilmore-lie-groups_ F00711	75	■ 0.996	$X_1, X_2, \dots, X_n$	$X_1, X_2, \dots, X_n$	any two basis vectors can be expressed as a linear superposition of basis vectors
gilmore-lie-groups_ F00712	75	■ 1.000	(X_i, X_j)	(X_i, X_j)	
gilmore-lie-groups_ F00713	76	■ 1.000	C_{ij}^k	C_{ij}^k	The coefficients C_{ij}^k in this expansion are called structure constants . The structure
gilmore-lie-groups_ F00714	76	■ 0.865	X_a, X_b, X_c	X_a, X_b, X_c	
gilmore-lie-groups_ F00715	76	■ 0.865	$\mathfrak{sl}(2; R)$	$\mathfrak{sl}(2; R)$	
gilmore-lie-groups_ F00716	76	■ 1.000	Z	Z	Under the change of basis (4.21) we find
gilmore-lie-groups_ F00717	76	■ 1.000	$R(Z)$	$R(Z)$	
gilmore-lie-groups_ F00718	77	■ 1.000	$\mathfrak{sl}(2; \mathbb{R})$	$\mathfrak{sl}(2; \mathbb{R})$	structed, since the structure constants have been given in (4.23)
gilmore-lie-groups_ F00719	77	■ 0.844	a, b	a, b	
gilmore-lie-groups_ F00720	77	■ 0.844	c	c	
gilmore-lie-groups_ F00721	78	■ 0.899	X_a, X_b	X_a, X_b	the three generators X_a, X_b , and X_c of $\mathfrak{sl}(2; \mathbb{R})$ to either the nonsemisimple or the
gilmore-lie-groups_ F00722	78	■ 0.997	$E(2)$	$E(2)$	$E(2)$ with surt
gilmore-lie-groups_ F00723	78	■ 1.000	P_x, P_y	P_x, P_y	Lie algebra. The algebra, and the original group, are therefore nonsemisimple.
gilmore-lie-groups_ F00724	78	■ 1.000	L_z	L_z	There is a beautiful structure theory for simple and semisimple Lie algebras. This will be discussed in Chapter 9. A structure theory exists for representations.
gilmore-lie-groups_ F00725	78	■ 0.985	A, B	A, B	This inner product is positive-definite. that is
gilmore-lie-groups_ F00726	78	■ 0.985	$p \times q$	$p \times q$	This inner product is positive-definite. that is
gilmore-lie-groups_ F00727	79	■ 0.998	X_i	X_i	and also on all generators that are not representable as the commutator of two
gilmore-lie-groups_ F00728	79	■ 0.998	$C_{i*}^* = 0$	$C_{i*}^* = 0$	and also on all generators that are not representable as the commutator of two
gilmore-lie-groups_ F00729	79	■ 1.000	$C_{**}^i = 0$	$C_{**}^i = 0$	The Hilbert–Schmidt inner proc
gilmore-lie-groups_ F00730	79	■ 1.000	V_0	V_0	The subspace V_0 is also a subalgebra of $\mathfrak{g}$, consisting of compact (or periodicity) operators. That is to say, the exponential of this subspace is a subset of
gilmore-lie-groups_ F00731	79	■ 1.000	V_-	V_-	Finally, the subspace V_+ is not a subalgebra of $\mathfrak{g}$. It consists of noncompact
gilmore-lie-groups_ F00732	79	■ 1.000	V_+	V_+	

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gilmore-lie-groups_ F00733	80	■ 1.000	$X_{\{\pm\}}=X_{\{b\}}\pm X_{\{c\}}$	$X_{\pm}=X_b\pm X_c$	
gilmore-lie-groups_ F00735	80	■ 1.000	$X_{\{a\}}, X_{\{+\}}$	X_a, X_+	
gilmore-lie-groups_ F00736	81	■ 0.644	$\alpha^1, \alpha^2, \ldots, \alpha^n$	$\alpha^1, \alpha^2, \ldots, \alpha^n$	
gilmore-lie-groups_ F00737	81	■ 0.840	$\alpha^1+d\alpha^1, \alpha^2+d\alpha^2, \ldots, \alpha^n+d\alpha^n$	$\alpha^1+d\alpha^1, \alpha^2+d\alpha^2, \ldots, \alpha^n+d\alpha^n$	
gilmore-lie-groups_ F00738	81	■ 0.866	$d\alpha^n$	$d\alpha^n$	
gilmore-lie-groups_ F00739	81	■ 0.866	$\left(x^1, x^2, \ldots, x^n\right)$	$\left(x^1, x^2, \ldots, x^n\right)$	
gilmore-lie-groups_ F00740	81	■ 0.831	$\alpha^1+d\alpha^1, \alpha^2+d\alpha^2, \ldots, \alpha^n+d\alpha^n$	$\alpha^1+d\alpha^1, \alpha^2+d\alpha^2, \ldots, \alpha^n+d\alpha^n$	
gilmore-lie-groups_ F00741	81	■ 0.831	$x^1+d x^1, x^2+$	$x^1+d x^1, x^2+$	
gilmore-lie-groups_ F00742	81	■ 1.000	$d x^2, \ldots, x^n+d x^n$	$dx^2, \ldots, x^n+d x^n$	
gilmore-lie-groups_ F00743	81	■ 0.997	$d x$	dx	
gilmore-lie-groups_ F00744	81	■ 0.997	$d \alpha$	$d\alpha$	
gilmore-lie-groups_ F00745	81	■ 1.000	$d s$	ds	
gilmore-lie-groups_ F00746	81	■ 1.000	$\alpha^i+d\alpha^i$	$\alpha^i+d\alpha^i$	
gilmore-lie-groups_ F00747	81	■ 0.671	$x, g_{rs}(x)$	$x, g_{rs}(x)$	
gilmore-lie-groups_ F00748	81	■ 1.000	$x, g(x)$	$x, g(x)$	
gilmore-lie-groups_ F00749	82	■ 0.976	$\left(d\alpha^1, d\alpha^2, d\alpha^3\right)$	$\left(d\alpha^1, d\alpha^2, d\alpha^3\right)$	
gilmore-lie-groups_ F00750	82	■ 0.999	$(d x, d y, d z)$	(dx, dy, dz)	
gilmore-lie-groups_ F00751	83	■ 1.000	$[X,Y]=(X\,Y-Y\,X)$	$[X,Y]=(XY-YX)$	
gilmore-lie-groups_ F00752	83	■ 1.000	$g_1=(I+\epsilon X), g_1^{-1}=(I+\epsilon X)^{-1}=I-$	$g_1=(I+\epsilon X), g_1^{-1}=(I+\epsilon X)^{-1}=I-$	
gilmore-lie-groups_ F00753	83	■ 1.000	$\epsilon X+\epsilon^2 X^2-\cdots$	$\epsilon X+\epsilon^2 X^2-\cdots$	
gilmore-lie-groups_ F00754	83	■ 1.000	g_2	g_2	
gilmore-lie-groups_ F00755	83	■ 1.000	$X^{\{\prime\}}$	X'	
gilmore-lie-groups_ F00756	83	■ 1.000	$(X,X)=2\left(a^2+b\,c\right)$	$(X,X)=2\left(a^2+bc\right)$	
gilmore-lie-groups_ F00757	83	■ 1.000	$Y=e^X$	$Y=e^X$	
gilmore-lie-groups_ F00758	83	■ 0.570	$\lambda _i$	$\lambda _i$	

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gilmore-lie-groups_ F00759	83	0.570	$\theta<\lambda_i<+2$	$0<\lambda_i<+2$	
gilmore-lie-groups_ F00760	83	0.570	$Y\in\operatorname{SL}(2;\mathbb{R})$	$Y\in\operatorname{SL}(2;\mathbb{R})$	
gilmore-lie-groups_ F00761	83	0.570	$\operatorname{tr}Y<-2$	$\operatorname{tr}Y<-2$	
gilmore-lie-groups_ F00762	84	1.000	$X\in\mathfrak{sl}(2;\mathbb{R})$	$X\in\mathfrak{sl}(2;\mathbb{R})$	expansion does not converge. That is, there is no 2×2 matrix $X\in\mathfrak{sl}(2;\mathbb{R})$ with the
gilmore-lie-groups_ F00763	84	0.637	$\operatorname{tr}e^X<-2$	$\operatorname{tr}e^X<-2$	property $\operatorname{tr}e^X<$
gilmore-lie-groups_ F00764	84	0.952	$\mathbf{L} =$	$\mathbf{L} =$	4. The Lie algebra of $SO(3)$ is spanned by three 3×3 antisymmetric matrices $\mathbf{L} =$
gilmore-lie-groups_ F00765	84	1.000	$\left(L_{\{1\}},L_{\{2\}},L_{\{3\}}\right)=\left(X_{\{23\}},X_{\{31\}},X_{\{12\}}\right)$	$(L_1,L_2,L_3)=(X_{23},X_{31},X_{12})$	$(L_1,L_2,L_3)=(A_{23},A_{31},A_{12}),$ w
gilmore-lie-groups_ F00766	84	1.000	$\mathbf{X}=\theta\cdot\mathbf{L}$	$\mathbf{X}=\theta\cdot\mathbf{L}$	$\boldsymbol{\sigma}=\sigma_1+\sigma_2+\sigma_3$. Show
gilmore-lie-groups_ F00767	84	1.000	$\theta^2=\theta_1^2+\theta_2^2+\theta_3^2$	$\theta^2=\theta_1^2+\theta_2^2+\theta_3^2$	$f_0(\theta)$
gilmore-lie-groups_ F00768	84	1.000	$SO(n)$	$SO(n)$	$X_{ij}=-X_{ji}$ ($1\leq i\neq j\leq n$) with matrix elements $(X_{ij})_{\alpha\beta}=\delta_{i\alpha}\delta_{j\beta}-\delta_{i\beta}\delta_{j\alpha}$.
gilmore-lie-groups_ F00769	84	1.000	$X_{\{i\ j\}}=-X_{\{j\ i\}}(1\leq i\neq j\leq n)$	$X_{ij}=-X_{ji}(1\leq i\neq j\leq n)$	$[X_{ii},X_{rs}]=X_{is}\delta_{ir}+X_{ir}\delta_{is}-X_{ir}\delta_{is}-X_{is}\delta_{ir}\tag{4}$
gilmore-lie-groups_ F00770	84	1.000	$\left(X_{\{i\ j\}}\right)_{\alpha\beta}=\delta_{i\alpha}\delta_{j\beta}-\delta_{i\beta}\delta_{j\alpha}$	$(X_{ij})_{\alpha\beta}=\delta_{i\alpha}\delta_{j\beta}-\delta_{i\beta}\delta_{j\alpha}$	$[X_{ii},X_{rs}]=X_{is}\delta_{ir}+X_{ir}\delta_{is}-X_{ir}\delta_{is}-X_{is}\delta_{ir}\tag{4}$
gilmore-lie-groups_ F00771	84	1.000	$\mathcal{X}_{\{i\ j\}}=x^{\{i\}}\partial_{\{j\}}-x^{\{j\}}\partial_{\{i\}}$	$\mathcal{X}_{ij}=x^i\partial_j-x^j\partial_i$	$b_i^\dagger b_j-b_j^\dagger b_i$ ($1\leq i\neq j\leq n$) satisfy isomorphic commutation relations.
gilmore-lie-groups_ F00772	84	1.000	$\mathcal{B}_{\{i\ j\}} =$	$\mathcal{B}_{ij} =$	$f_i^\dagger f_j-f_j^\dagger f_i$ ($1\leq i\neq j\leq n$) satisfy isomorphic commutation relations.
gilmore-lie-groups_ F00773	84	1.000	$b_{\{i\}}^\dagger b_{\{j\}}-b_{\{j\}}^\dagger b_{\{i\}}(1\leq i\neq j\leq n)$	$b_i^\dagger b_j-b_j^\dagger b_i(1\leq i\neq j\leq n)$	6. The Jacobi identity for operators D,Y,Z (replace X by D in Eq. (4.17)).
gilmore-lie-groups_ F00774	84	0.963	$\mathcal{F}_{\{i\ j\}} =$	$\mathcal{F}_{ij} =$	rewritten in the form
gilmore-lie-groups_ F00775	84	1.000	$f_{\{i\}}^\dagger f_{\{j\}}-f_{\{j\}}^\dagger f_{\{i\}}(1\leq i\neq j\leq n)$	$f_i^\dagger f_j-f_j^\dagger f_i(1\leq i\neq j\leq n)$	$\left[D, \left[Y, Z \right] \right] = \left[\left[D, Y \right], Z \right] + \left[Y, \left[D, Z \right] \right]$
gilmore-lie-groups_ F00776	84	1.000	D,Y,Z	D,Y,Z	Show this. Compare with the expression for the differential operator
gilmore-lie-groups_ F00777	84	0.998	$\mathfrak{so}(4)$	$\mathfrak{so}(4)$	
gilmore-lie-groups_ F00778	84	0.998	4×4	4×4	
gilmore-lie-groups_ F00779	85	1.000	$a_{\{i\ j\}}X_{\{i\ j\}}$	$a_{ij}X_{ij}$	element $a_{ij}X_{ij}$ in the Lie algebra:
gilmore-lie-groups_ F00780	85	1.000	$SO(p,q)$	$SO(p,q)$	between the proportionality constant and the dimensions of the defining and regular
gilmore-lie-groups_ F00781	85	1.000	$\operatorname{EXP}(V_-)$	$\operatorname{EXP}(V_-)$	torus $T=\operatorname{Exp}(Z_1,Z_2)$.
gilmore-lie-groups_ F00782	85	0.546	$SO(3,1)$	$SO(3,1)$	Construct the infinitesimal generators for the group $SO(3)$ using the parameterizations
gilmore-lie-groups_ F00783	85	0.546	$SO(2,2)$	$SO(2,2)$	Construct the infinitesimal generators for the group $SO(3)$ using the parameterizations
gilmore-lie-groups_ F00784	85	0.982	T^2	T^2	proposed in Problems 1

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gilmore-lie-groups-F00785	85	0.752	$e^{\theta \mathbf{L}} \in SO(3)$	$e^{\theta \cdot \mathbf{L}} \in SO(3)$	Compute the determinant of the metric tensor (4.53) on the group $SL(2; \mathbb{R})$. Show
gilmore-lie-groups-F00786	85	0.752	$\rho(\theta)$	$\rho(\theta)$	Compute the determinant of the metric tensor (4.53) on the group $SL(2; \mathbb{R})$. Show
gilmore-lie-groups-F00787	85	1.000	$g_{\mu\nu}(\theta)$	$g_{\mu\nu}(\theta)$	that the square root of the determinant is equal to the measure, in accordance with the
gilmore-lie-groups-F00788	85	1.000	$dV(x) = \ g(x)\ ^{1/2} d^n x$	$dV(x) = \ g(x)\ ^{1/2} d^n x$	It is represented by a real symmetric nonsingular $n \times n$ matrix $g_{rs} = (e_r, e_s)$, where $\mathbf{x} = \mathbf{e}_i x^i$. The inverse matrix, g^{rs} , is well defined.
gilmore-lie-groups-F00789	85	0.623	$(\mathbf{x}, \mathbf{x})$	$(\mathbf{x}, \mathbf{x})$	d. Show the Lie algebra $\mathfrak{g}$ of G satisfies $X + g + gX = 0$. e. Show that the infinitesimal generators of G are $\mathbf{Y} = \alpha \cdot \mathbf{e}_1 \mathbf{A} = \alpha \cdot \mathbf{e}_2 \mathbf{A}$
gilmore-lie-groups-F00790	85	1.000	$g_{rs} = (\mathbf{e}_r, \mathbf{e}_s)$	$g_{rs} = (\mathbf{e}_r, \mathbf{e}_s)$	e. Show that the infinitesimal generators of G are $X_{rs} = g_{rt} \alpha_s - g_{st} \alpha_r$.
gilmore-lie-groups-F00791	85	1.000	$\mathbf{x} = \mathbf{e}_i x^i$	$\mathbf{x} = \mathbf{e}_i x^i$	
gilmore-lie-groups-F00792	85	1.000	g^{rs}	g^{rs}	
gilmore-lie-groups-F00793	85	1.000	$\mathbf{y} = G\mathbf{x}, (\mathbf{y}, \mathbf{y}) = (\mathbf{x}, \mathbf{x})$	$\mathbf{y} = G\mathbf{x}, (\mathbf{y}, \mathbf{y}) = (\mathbf{x}, \mathbf{x})$	
gilmore-lie-groups-F00794	85	1.000	$G^t g G = g$	$G^t g G = g$	
gilmore-lie-groups-F00795	85	1.000	$H^t g + g H = 0$	$H^t g + g H = 0$	
gilmore-lie-groups-F00796	85	1.000	$X_{rs} = g_{rt} x^t \partial_s - g_{st} x^t \partial_r$	$X_{rs} = g_{rt} x^t \partial_s - g_{st} x^t \partial_r$	
gilmore-lie-groups-F00797	86	1.000	X_{rs}	X_{rs}	d. Show that the operators X_{rs} satisfy commutation relations
gilmore-lie-groups-F00798	86	0.887	$M = SO$	$M = SO$	15. Every real unimodular 2×2 matrix M can be written in the form $M = SO$, where S is a real symmetric matrix, O is a real orthogonal matrix.
gilmore-lie-groups-F00799	86	1.000	$S^t = S^{+1}$	$S^t = S^{+1}$	$\begin{array}{ccccc} S^t = S^{+1} & \det(S) = +1 & S = e^\Sigma & \text{Tr } \Sigma = 0 & \Sigma^t = +\Sigma \\ O^t = O^{-1} & \det(O) = +1 & O = e^A & \text{Tr } A = 0 & A^t = -A \end{array}$
gilmore-lie-groups-F00800	86	1.000	$\operatorname{det}(S) = +1$	$\det(S) = +1$	$\begin{array}{ccccc} S^t = S^{+1} & \det(S) = +1 & S = e^\Sigma & \text{Tr } \Sigma = 0 & \Sigma^t = +\Sigma \\ O^t = O^{-1} & \det(O) = +1 & O = e^A & \text{Tr } A = 0 & A^t = -A \end{array}$
gilmore-lie-groups-F00801	86	1.000	$S = e^\Sigma$	$S = e^\Sigma$	$\begin{array}{ccccc} S^t = S^{+1} & \det(S) = +1 & S = e^\Sigma & \text{Tr } \Sigma = 0 & \Sigma^t = +\Sigma \\ O^t = O^{-1} & \det(O) = +1 & O = e^A & \text{Tr } A = 0 & A^t = -A \end{array}$
gilmore-lie-groups-F00802	86	1.000	$\operatorname{Tr} \Sigma = 0$	$\text{Tr } \Sigma = 0$	$\begin{array}{ccccc} S^t = S^{+1} & \det(S) = +1 & S = e^\Sigma & \text{Tr } \Sigma = 0 & \Sigma^t = +\Sigma \\ O^t = O^{-1} & \det(O) = +1 & O = e^A & \text{Tr } A = 0 & A^t = -A \end{array}$
gilmore-lie-groups-F00803	86	1.000	$\Sigma^t = +\Sigma$	$\Sigma^t = +\Sigma$	$\begin{array}{ccccc} S^t = S^{+1} & \det(S) = +1 & S = e^\Sigma & \text{Tr } \Sigma = 0 & \Sigma^t = +\Sigma \\ O^t = O^{-1} & \det(O) = +1 & O = e^A & \text{Tr } A = 0 & A^t = -A \end{array}$
gilmore-lie-groups-F00804	86	1.000	$O^t = O^{-1}$	$O^t = O^{-1}$	$\begin{array}{ccccc} S^t = S^{+1} & \det(S) = +1 & S = e^\Sigma & \text{Tr } \Sigma = 0 & \Sigma^t = +\Sigma \\ O^t = O^{-1} & \det(O) = +1 & O = e^A & \text{Tr } A = 0 & A^t = -A \end{array}$

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gilmore-lie-groups_F08805	86	■ 1.000	$\operatorname{det}(0)=+1$	$\det(O) = +1$	$\begin{array}{ccccc} S' = S^{+1} & \det(S) = +1 & S = e^{\Sigma} & \operatorname{Tr} \Sigma = 0 & \Sigma' = +\Sigma \\ O' = O^{-1} & \det(O) = +1 & O = e^A & \operatorname{Tr} A = 0 & A' = -A \end{array}$
gilmore-lie-groups_F08806	86	■ 1.000	$0=e^{\{A\}}$	$O = e^A$	$\begin{array}{ccccc} S' = S^{+1} & \det(S) = +1 & S = e^{\Sigma} & \operatorname{Tr} \Sigma = 0 & \Sigma' = +\Sigma \\ O' = O^{-1} & \det(O) = +1 & O = e^A & \operatorname{Tr} A = 0 & A' = -A \end{array}$
gilmore-lie-groups_F08807	86	■ 1.000	$\operatorname{Tr} A=0$	$\operatorname{Tr} A = 0$	$\begin{array}{ccccc} S' = S^{+1} & \det(S) = +1 & S = e^{\Sigma} & \operatorname{Tr} \Sigma = 0 & \Sigma' = +\Sigma \\ O' = O^{-1} & \det(O) = +1 & O = e^A & \operatorname{Tr} A = 0 & A' = -A \end{array}$
gilmore-lie-groups_F08808	86	■ 1.000	$A^{\{t\}}=-A$	$A^t = -A$	$\begin{array}{ccccc} S' = S^{+1} & \det(S) = +1 & S = e^{\Sigma} & \operatorname{Tr} \Sigma = 0 & \Sigma' = +\Sigma \\ O' = O^{-1} & \det(O) = +1 & O = e^A & \operatorname{Tr} A = 0 & A' = -A \end{array}$
gilmore-lie-groups_F08809	86	■ 0.999	$M\ M^{\{t\}}=S^{\{2\}}=e^{\{2\}\ \Sigma\}$	$MM^t = S^2 = e^{2\Sigma}$	b. Show $O = S^{-1}M = e^{-\Sigma}$.
gilmore-lie-groups_F08810	86	■ 1.000	$0=S^{\{-1\}}\ M=e^{\{-\Sigma\}}\ M$	$O = S^{-1}M = e^{-\Sigma}M$	c. Write S as a power series
gilmore-lie-groups_F08811	86	■ 1.000	Σ	Σ	d. Write S as a power series expansion in Σ . Under what conditions are these expansions valid?
gilmore-lie-groups_F08812	86	■ 1.000	$S-I_{\{2\}}$	$S - I_2$	Under what conditions are these expansions valid?
gilmore-lie-groups_F08813	86	■ 1.000	$M=H\ U$	$M = HU$	15. Transfer matrices have been described in Chapter 3, Problem 24. Is any discussion
gilmore-lie-groups_F08814	86	■ 1.000	$H^{\{\dagger\}}=H^{\{+1\}}$	$H^\dagger = H^{+1}$	15. Transfer matrices have been described in Chapter 3, Problem 24. In one of the transfer matrix for a scattering potential with free particles incident from
gilmore-lie-groups_F08815	86	■ 1.000	$U^{\{\dagger\}}=U^{\{-1\}}$	$U^\dagger = U^{-1}$	15. Transfer matrices have been described in Chapter 3, Problem 24. In one of the transfer matrix for a scattering potential with free particles incident from
gilmore-lie-groups_F08816	86	■ 1.000	$\hbar k_{\{L\}}$	$\hbar k_L$	$\left \begin{array}{cc} \beta_R - i\beta_I & \alpha_R - i\alpha_I \\ \beta_R - i\beta_I & \alpha_R - i\alpha_I \end{array} \right $
gilmore-lie-groups_F08817	86	■ 1.000	$-\hbar k_{\{R\}}$	$-\hbar k_R$	$\left \begin{array}{cc} \beta_R - i\beta_I & \alpha_R - i\alpha_I \\ \beta_R - i\beta_I & \alpha_R - i\alpha_I \end{array} \right $
gilmore-lie-groups_F08818	86	■ 1.000	$m_{\{i\ j\}}$	m_{ij}	$\begin{array}{ccccccccc} - & - & - & - & - & - & - & - & - \end{array}$
gilmore-lie-groups_F08819	86	■ 1.000	$\hbar k_{\{L\}}=\hbar k_{\{R\}}$	$\hbar k_L = \hbar k_R$	$\left \begin{array}{cc} i & 0 \\ 0 & i \end{array} \right \quad \left \begin{array}{cc} 0 & i \\ i & 0 \end{array} \right \quad \left \begin{array}{cc} 0 & -1 \\ -1 & 0 \end{array} \right \quad (4.61c)$
gilmore-lie-groups_F08820	86	■ 0.679	$\operatorname{SU}(1,1)$	$\operatorname{SU}(1,1)$	$\left[\begin{array}{cc} \mu_1 - \mu_2 & \alpha_1 - \alpha_2 \end{array} \right]$
gilmore-lie-groups_F08821	87	■ 1.000	$\kappa_{\{R\}}$	κ_R	wavefunction in the asymptotic left- and right-hand regions of the potential. The real
gilmore-lie-groups_F08822	87	■ 1.000	$\kappa_{\{L\}}$	κ_L	wavefunction in the asymptotic left- and right-hand regions of the potential. The real
gilmore-lie-groups_F08823	87	■ 1.000	$\kappa_{\{L\}}=\kappa_{\{R\}}$	$\kappa_L = \kappa_R$	construct the infinitesimal generators for the group of the transfer matrix for bound
gilmore-lie-groups_F08824	87	■ 1.000	$k_{\{*\}}=\sqrt{2\ m\left(E-V_{\{*\}}\right)\ } / \ \hbar^{\{2\}}$	$k_* = \sqrt{2m(E - V_*)/\hbar^2}$	
gilmore-lie-groups_F08825	87	■ 1.000	$E>V_{\{*\}}$	$E > V_*$	
gilmore-lie-groups_F08826	87	■ 1.000	$\kappa_{\{*\}}=\sqrt{2\ m\left(V_{\{*\}}-E\right)\ } / \ \hbar^{\{2\}}$	$\kappa_* = \sqrt{2m(V_* - E)/\hbar^2}$	

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gilmore-lie-groups_ F08827	87	■ 1.000	$E<V_{*}$	$E < V_*$	
gilmore-lie-groups_ F08828	87	■ 0.978	$*\!=L, \, R$	$* = L, R$	
gilmore-lie-groups_ F08829	88	■ 0.998	$\operatorname{det}(M) \neq 0$	$\det(M) \neq 0$	(11) are defined by bilinear and quadratic constraints that are linearized by applying the constraint method in Gaussian elimination to the addition $I + \epsilon A$. The solutions
gilmore-lie-groups_ F08830	88	■ 1.000	$I+\epsilon A$	$I + \epsilon A$	$(I + \epsilon A)^t G (I + \epsilon A) = G$
gilmore-lie-groups_ F08831	88	■ 1.000	$\mathfrak{gl}(n\,;\,\mathbb{F})$	$\mathfrak{gl}(n;\mathbb{F})$	
gilmore-lie-groups_ F08832	89	■ 1.000	$\mathfrak{ut}(p,\,q)$	$\mathfrak{ut}(p,q)$	$m_{iq} = 0 \qquad \qquad \qquad \frac{P^{i+1} + \frac{1}{2} \frac{P^i}{n} - \frac{P^{i+1} + q}{1} < n < n} \qquad (5.3)$
gilmore-lie-groups_ F08833	89	■ 0.998	$\mathfrak{ht}(p,\,q)$	$\mathfrak{ht}(p,q)$	
gilmore-lie-groups_ F08834	90	■ 1.000	$(a,\,b)=(1,0)$	$(a,b) = (1,0)$	The identity is at $(a,b) = (1,0)$. Its algebra is spanned by the two operators
gilmore-lie-groups_ F08835	90	■ 0.999	$\left[X_{\scriptscriptstyle{a}},\,X_{\scriptscriptstyle{b}}\right]\!=\!X_{\scriptscriptstyle{b}}$	$[X_a,X_b] = X_b$	4 $\mathfrak{ut}(n\, \,a\, \,r)$. This matrix algebra is identical in structure
gilmore-lie-groups_ F08836	90	■ 1.000	$\mathfrak{ut}(p,\,q,\,r)$	$\mathfrak{ut}(p,q,r)$	Example A very useful six-parameter subalgebra of $\mathfrak{ut}(1,2,1)$ is given by
gilmore-lie-groups_ F08837	90	■ 0.999	$\mathfrak{ut}(1,2,1)$	$\mathfrak{ut}(1,2,1)$	$\begin{bmatrix} 0 & l & r & -2\delta \end{bmatrix}$
gilmore-lie-groups_ F08838	90	■ 1.000	j	j	$ X_{\scriptscriptstyle{u}}\,X_{\scriptscriptstyle{v}}\,X_{\scriptscriptstyle{r}}\,X_{\scriptscriptstyle{u}}\,X_{\scriptscriptstyle{r}}\,X_{\scriptscriptstyle{s}}$
gilmore-lie-groups_ F08839	90	■ 1.000	$\left[X_{\scriptscriptstyle{i}},\,X_{\scriptscriptstyle{j}}\right]$	$[X_i,X_j]$	
gilmore-lie-groups_ F08840	90	■ 1.000	X_{η}	X_{η}	$\left. \begin{array}{l} X_R \\ X_L \\ X_r \\ X_l \\ X_{\delta} \end{array} \right \begin{array}{ccccc} 0 & -4\Lambda_{\eta} & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_{\delta} & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array}$ $\frac{\hat{n}+\frac{1}{2}I}{\hat{n}+\frac{1}{2}I}\left \begin{array}{ccccc}\hat{n}+\frac{1}{2}I&a^{\dagger}a^{\dagger}&aa&a^{\dagger}&a\\0&2a^{\dagger}a^{\dagger}&-2aa&a^{\dagger}&-a\end{array}\right.$
gilmore-lie-groups_ F08841	90	■ 1.000	$X_{\scriptscriptstyle{R}}$	X_R	$\left. \begin{array}{l} X_R \\ X_L \\ X_r \\ X_l \\ X_{\delta} \end{array} \right \begin{array}{ccccc} 0 & -4\Lambda_{\eta} & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_{\delta} & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array}$ $\frac{\hat{n}+\frac{1}{2}I}{\hat{n}+\frac{1}{2}I}\left \begin{array}{ccccc}\hat{n}+\frac{1}{2}I&a^{\dagger}a^{\dagger}&aa&a^{\dagger}&a\\0&2a^{\dagger}a^{\dagger}&-2aa&a^{\dagger}&-a\end{array}\right.$
gilmore-lie-groups_ F08842	90	■ 1.000	$X_{\scriptscriptstyle{L}}$	X_L	$\left. \begin{array}{l} X_R \\ X_L \\ X_r \\ X_l \\ X_{\delta} \end{array} \right \begin{array}{ccccc} 0 & -4\Lambda_{\eta} & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_{\delta} & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array}$ $\frac{\hat{n}+\frac{1}{2}I}{\hat{n}+\frac{1}{2}I}\left \begin{array}{ccccc}\hat{n}+\frac{1}{2}I&a^{\dagger}a^{\dagger}&aa&a^{\dagger}&a\\0&2a^{\dagger}a^{\dagger}&-2aa&a^{\dagger}&-a\end{array}\right.$

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gilmore-lie-groups_ F08843	90	■ 1.000	$X_{\{r\}}$	X_r	$\begin{array}{c} X_R \\ X_L \\ X_r \\ X_l \\ X_\delta \end{array} \left \begin{array}{ccccc} 0 & -4\Lambda_\eta & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_\delta & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array} \right.$ $\frac{\left \hat{n} + \frac{1}{2}I \right.}{\hat{n} + \frac{1}{2}I} \left \begin{array}{ccccc} a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array} \right.$
gilmore-lie-groups_ F08844	90	■ 1.000	$X_{\{l\}}$	X_l	$\begin{array}{c} X_R \\ X_L \\ X_r \\ X_l \\ X_\delta \end{array} \left \begin{array}{ccccc} 0 & -4\Lambda_\eta & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_\delta & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array} \right.$ $\frac{\left \hat{n} + \frac{1}{2}I \right.}{\hat{n} + \frac{1}{2}I} \left \begin{array}{ccccc} a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array} \right.$
gilmore-lie-groups_ F08845	90	■ 1.000	$X_{\{\delta\}}$	X_δ	$\begin{array}{c} X_R \\ X_L \\ X_r \\ X_l \\ X_\delta \end{array} \left \begin{array}{ccccc} 0 & -4\Lambda_\eta & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_\delta & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array} \right.$ $\frac{\left \hat{n} + \frac{1}{2}I \right.}{\hat{n} + \frac{1}{2}I} \left \begin{array}{ccccc} a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array} \right.$
gilmore-lie-groups_ F08846	90	■ 1.000	$2 X_{\{R\}}$	$2X_R$	$\begin{array}{c} X_R \\ X_L \\ X_r \\ X_l \\ X_\delta \end{array} \left \begin{array}{ccccc} 0 & -4\Lambda_\eta & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_\delta & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array} \right.$ $\frac{\left \hat{n} + \frac{1}{2}I \right.}{\hat{n} + \frac{1}{2}I} \left \begin{array}{ccccc} a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array} \right.$
gilmore-lie-groups_ F08847	90	■ 1.000	$-2 X_{\{L\}}$	$-2X_L$	$\begin{array}{c} X_R \\ X_L \\ X_r \\ X_l \\ X_\delta \end{array} \left \begin{array}{ccccc} 0 & -4\Lambda_\eta & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_\delta & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array} \right.$ $\frac{\left \hat{n} + \frac{1}{2}I \right.}{\hat{n} + \frac{1}{2}I} \left \begin{array}{ccccc} a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array} \right.$
gilmore-lie-groups_ F08848	90	■ 1.000	$-X_{\{l\}}$	$-X_l$	$\begin{array}{c} X_R \\ X_L \\ X_r \\ X_l \\ X_\delta \end{array} \left \begin{array}{ccccc} 0 & -4\Lambda_\eta & 0 & -2\Lambda_r & 0 \\ & 0 & 2X_l & 0 & 0 \\ & & 0 & -X_\delta & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{array} \right.$ $\frac{\left \hat{n} + \frac{1}{2}I \right.}{\hat{n} + \frac{1}{2}I} \left \begin{array}{ccccc} a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array} \right.$

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gilmore-lie-groups_ F08849	90	■ 1.000	-4 X_{\eta}	$-4X_\eta$	$\begin{array}{c ccccc} X_R & 0 & -4X_\eta & 0 & -2X_r & 0 \\ X_L & & 0 & 2X_l & 0 & 0 \\ X_r & & & 0 & -X_\delta & 0 \\ X_l & & & & 0 & 0 \\ X_\delta & & & & & 0 \end{array}$ $\frac{\hat{n} + \frac{1}{2}I}{\hat{n} + \frac{1}{2}I} \begin{array}{c ccccc} \hat{n} + \frac{1}{2}I & a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array}$
gilmore-lie-groups_ F08850	90	■ 1.000	-2 X_{r}	$-2X_r$	$\begin{array}{c ccccc} X_R & 0 & -4X_\eta & 0 & -2X_r & 0 \\ X_L & & 0 & 2X_l & 0 & 0 \\ X_r & & & 0 & -X_\delta & 0 \\ X_l & & & & 0 & 0 \\ X_\delta & & & & & 0 \end{array}$ $\frac{\hat{n} + \frac{1}{2}I}{\hat{n} + \frac{1}{2}I} \begin{array}{c ccccc} \hat{n} + \frac{1}{2}I & a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array}$
gilmore-lie-groups_ F08851	90	■ 1.000	2 X_{l}	$2X_l$	$\begin{array}{c ccccc} X_R & 0 & -4X_\eta & 0 & -2X_r & 0 \\ X_L & & 0 & 2X_l & 0 & 0 \\ X_r & & & 0 & -X_\delta & 0 \\ X_l & & & & 0 & 0 \\ X_\delta & & & & & 0 \end{array}$ $\frac{\hat{n} + \frac{1}{2}I}{\hat{n} + \frac{1}{2}I} \begin{array}{c ccccc} \hat{n} + \frac{1}{2}I & a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array}$
gilmore-lie-groups_ F08852	90	■ 1.000	-X_{\delta}	$-X_\delta$	$\begin{array}{c ccccc} X_R & 0 & -4X_\eta & 0 & -2X_r & 0 \\ X_L & & 0 & 2X_l & 0 & 0 \\ X_r & & & 0 & -X_\delta & 0 \\ X_l & & & & 0 & 0 \\ X_\delta & & & & & 0 \end{array}$ $\frac{\hat{n} + \frac{1}{2}I}{\hat{n} + \frac{1}{2}I} \begin{array}{c ccccc} \hat{n} + \frac{1}{2}I & a^\dagger a^\dagger & aa & a^\dagger & a \\ 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a \end{array}$
gilmore-lie-groups_ F08853	90	■ 0.600	$\hat{n} + \frac{1}{2} I$	$\hat{n} + \frac{1}{2}I$	$\begin{array}{c ccccc} aa & & 0 & 2a & 0 & 0 \\ a^\dagger & & & 0 & -I & 0 \\ a & & & & 0 & 0 \\ I & & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one b ted. It is clear that there is an isomorphism between this ma e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc
gilmore-lie-groups_ F08854	90	■ 0.600	$a^{\dagger} a^{\dagger}$	$a^\dagger a^\dagger$	$\begin{array}{c ccccc} aa & & 0 & 2a & 0 & 0 \\ a^\dagger & & & 0 & -I & 0 \\ a & & & & 0 & 0 \\ I & & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one b ted. It is clear that there is an isomorphism between this ma e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F08855	90	■ 0.600	$a \ a$	aa	$\begin{array}{c ccc} aa & 0 & 2a & 0 & 0 \\ a^\dagger & & 0 & -I & 0 \\ a & & & 0 & 0 \\ I & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one h ted. It is clear that there is an isomorphism between this m e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc
gilmore-lie-groups_ F08856	90	■ 0.600	$2 \ a^{\dagger} a^{\dagger}$	$2a^\dagger a^\dagger$	$\begin{array}{c ccc} aa & 0 & 2a & 0 & 0 \\ a^\dagger & & 0 & -I & 0 \\ a & & & 0 & 0 \\ I & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one h ted. It is clear that there is an isomorphism between this m e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc
gilmore-lie-groups_ F08857	90	■ 0.600	$-2 \ a \ a$	$-2aa$	$\begin{array}{c ccc} aa & 0 & 2a & 0 & 0 \\ a^\dagger & & 0 & -I & 0 \\ a & & & 0 & 0 \\ I & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one h ted. It is clear that there is an isomorphism between this m e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc
gilmore-lie-groups_ F08858	90	■ 0.600	$-a$	$-a$	$\begin{array}{c ccc} aa & 0 & 2a & 0 & 0 \\ a^\dagger & & 0 & -I & 0 \\ a & & & 0 & 0 \\ I & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one h ted. It is clear that there is an isomorphism between this m e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc
gilmore-lie-groups_ F08859	90	■ 0.600	$-4\left(\hat{n}+\frac{1}{2} \ I\right)$	$-4\left(\hat{n}+\frac{1}{2} I\right)$	$\begin{array}{c ccc} aa & 0 & 2a & 0 & 0 \\ a^\dagger & & 0 & -I & 0 \\ a & & & 0 & 0 \\ I & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one h ted. It is clear that there is an isomorphism between this m e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc
gilmore-lie-groups_ F08860	90	■ 0.600	$-2 \ a^{\dagger}$	$-2a^\dagger$	$\begin{array}{c ccc} aa & 0 & 2a & 0 & 0 \\ a^\dagger & & 0 & -I & 0 \\ a & & & 0 & 0 \\ I & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one h ted. It is clear that there is an isomorphism between this m e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00861	90	0.600	$2a$	$2a$	$\begin{array}{c ccc} aa & 0 & 2a & 0 & 0 \\ a^\dagger & & 0 & -I & 0 \\ a & & & 0 & 0 \\ I & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one b ted. It is clear that there is an isomorphism between this ma e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc
gilmore-lie-groups_F00862	90	0.600	$-I$	$-I$	$\begin{array}{c ccc} aa & 0 & 2a & 0 & 0 \\ a^\dagger & & 0 & -I & 0 \\ a & & & 0 & 0 \\ I & & & & 0 \end{array}$ ble inherits the antisymmetry of the commutator, so only one b ted. It is clear that there is an isomorphism between this ma e algebra of the photon energy operator $\hat{n} = a^\dagger a + \frac{1}{2}$, two-phc
gilmore-lie-groups_F00863	90	1.000	$\hat{n}=a^{\dagger}a+\frac{1}{2}$	$\hat{n} = a^\dagger a + \frac{1}{2}$	
gilmore-lie-groups_F00864	91	0.996	$I=\left[a,\,a^{\dagger}\right]$	$I = [a, a^\dagger]$	operators $a^\dagger$ and a , and the identity operator $I = [a, a^\dagger]$. We observe that the 4×4
gilmore-lie-groups_F00865	91	0.887	$\frac{\mathfrak{so}(n)}{\mathfrak{ut}(1,1,1,\ldots,1)}$	$\mathfrak{so}(n) = \mathfrak{ut}(1,1,1,\dots,1)$	er. $\mathfrak{so}(n) = \mathfrak{ut}(1,1,1,\dots,1)$. This matrix algebra is also identical in structure to its parent group. A very useful four-parameter subalgebra of $\mathfrak{ut}(1,1,1,1)$ is given by
gilmore-lie-groups_F00866	91	1.000	$\frac{\mathfrak{ut}(1,1,1)}{\mathfrak{nil}(n)}$	$\mathfrak{ut}(1,1,1)$	matrices of the following form
gilmore-lie-groups_F00867	91	0.936	$\frac{\mathfrak{nil}(n)}{\eta=0}$	$\mathfrak{nil}(n)$	erators of the algebra of nilpotent 5×5 matrices have structure and commutation relations
gilmore-lie-groups_F00868	91	1.000	$\eta=0$	$\eta = 0$	ter 3 (see (3.13)) consists of matrices having the form shown in Fig. 5.1.
gilmore-lie-groups_F00869	91	1.000	$\mathfrak{a}(p,q)$	$\mathfrak{a}(p,q)$	
gilmore-lie-groups_F00870	92	0.998	$O(3)$	$O(3)$	$\Gamma \begin{bmatrix} 0 & \theta_3 & -\theta_5 \end{bmatrix}$
gilmore-lie-groups_F00871	92	0.998	$U(2)$	$U(2)$	$\Gamma \begin{bmatrix} 0 & \theta_3 & -\theta_5 \end{bmatrix}$
gilmore-lie-groups_F00872	92	0.728	σ_μ	σ_μ	groups satisfy
gilmore-lie-groups_F00873	92	1.000	$\frac{\mathfrak{so}(2,1)}{\mathfrak{so}(3,1)}$	$\mathfrak{so}(2,1)$	
gilmore-lie-groups_F00874	93	1.000	$\frac{\mathfrak{so}(3,1)}{\mathfrak{so}(3,1)}$	$\mathfrak{so}(3,1)$	By an identical argument the matrix Lie algebra for the Lorentz group $\mathfrak{so}(3,1)$ is
gilmore-lie-groups_F00875	93	1.000	$M=-G^{-1}M^{\dagger}G$	$M = -G^{-1}M^\dagger G$	$\mathrm{tr} \, M = -\mathrm{tr} \, G^{-1}M^\dagger G = -\mathrm{tr} \, M^\dagger$
gilmore-lie-groups_F00876	93	1.000	$(p+q)\times(p+q)$	$(p+q) \times (p+q)$	$\left \begin{array}{cc} C & D \end{array} \right \left \begin{array}{cc} \tilde{0} & 0 \end{array} \right + \left \begin{array}{cc} \tilde{0} & 0 \end{array} \right \left \begin{array}{cc} B^\dagger & D^\dagger \end{array} \right = \left \begin{array}{cc} 0 & 0 \end{array} \right $
gilmore-lie-groups_F00877	93	0.983	$\left[\begin{array}{l} g \ \& \ 0 \\ 0 \ \& \ 0 \end{array}\right]$	$\left[\begin{array}{cc} g & 0 \\ 0 & 0 \end{array}\right]$	$\Rightarrow A\sigma + \sigma A^\dagger = 0 \quad C = C^\dagger = 0 \quad R = 0$
gilmore-lie-groups_F00878	93	0.983	$\operatorname{det}(g) \neq 0$	$\det(g) \neq 0$	$\Rightarrow A\sigma + \sigma A^\dagger = 0 \quad C = C^\dagger = 0 \quad R = 0$
gilmore-lie-groups_F00879	93	1.000	$\operatorname{diag}(1,1,1,0)$	$\operatorname{diag}(1,1,1,0)$	$\left \begin{array}{c} \tilde{0} \end{array} \right $

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gilmore-lie-groups_F00880	94	1.000	$\backslash\theta_{i}$	θ_i	Here the parameters θ_i describe rotations about the i th coordinate axis and the
gilmore-lie-groups_F00881	94	1.000	$s_{\{4\}}$	s_4	parameter s_4 describes “scaling” of the time axis: $t' = e^{s_4}t$. If s_4 is set to zero
gilmore-lie-groups_F00882	94	1.000	$t^{\backslash\prime}=e^{\{s_{\{4\}}\}}\,t$	$t' = e^{s_4}t$	parameter s_4 describes “scaling” of the time axis: $t' = e^{s_4}t$. If s_4 is set to zero
gilmore-lie-groups_F00883	94	1.000	$E(3)=I\,S\,O(3)$	$E(3) = ISO(3)$	Euclidean group $E(3) = ISO(3)$ (inhomogeneous rotation group in $\mathbb{R}^3$). The matrix Lie algebra for the Poincaré group $ISO(3,1)$ (3.26) is obtained
gilmore-lie-groups_F00884	94	0.981	$I\,S\,O(3,1)(3.26)$	$ISO(3,1)(3.26)$	The matrix Lie algebra for the Poincaré group $ISO(3,1)$ (3.26) is obtained by similar arguments using a singular 5×5 metric $G = \text{diag}(1,1,1,-1,0)$. The Lie
gilmore-lie-groups_F00885	94	0.837	$5\,\times\,5$	5×5	algebra is (setting the trace equal to zero):
gilmore-lie-groups_F00886	94	0.837	$G=\operatorname{diag}\{1,1,1,-1,0\}$	$G = \text{diag}(1,1,1,-1,0)$	algebra is (setting the trace equal to zero):
gilmore-lie-groups_F00887	94	0.988	$I\,S\,O(3,1)$	$ISO(3,1)$	$\left \begin{array}{ccccc}0&\theta_3&-\theta_2&v_1&t_1\\-\hat{\theta}_3&0&\theta_1&v_2&t_2\\0&\hat{\theta}_3&0&v_3&t_3\\0&0&0&0&0\\0&0&0&0&0\end{array}\right $
gilmore-lie-groups_F00888	94	1.000	$\backslash\mathfrak{s}\,\mathfrak{u}(n)$	$\mathfrak{su}(n)$	
gilmore-lie-groups_F00889	94	1.000	$\backslash\mathfrak{u}(n)$	$\mathfrak{u}(n)$	
gilmore-lie-groups_F00890	94	1.000	$\backslash\mathfrak{sl}(n\,;\,\backslash\mathbb{C})$	$\mathfrak{sl}(n;\mathbb{C})$	
gilmore-lie-groups_F00891	95	1.000	$M_{\{i\,j\}}(i<j)(M\,\backslash\in\,\backslash\mathfrak{u}(n))$	$M_{ij}(i < j)(M \in \mathfrak{u}(n))$	diagonal of M by a 2×2 real matrix, and each of the diagonal matrix elements
gilmore-lie-groups_F00892	95	0.588	$M_{\{i\,i\}}$	M_{ii}	b arbitrary in Eq. (3.3)). The matrix elements M_{ij} below the diagonal ($i > j$)
gilmore-lie-groups_F00893	95	0.761	$M_{\{i\,j\}}$	M_{ij}	are obtained from the antihermiticity condition. The result is a real antisymmetric
gilmore-lie-groups_F00894	95	0.761	$(i>j)$	$(i > j)$	are obtained from the antihermiticity condition. The result is a real antisymmetric
gilmore-lie-groups_F00895	95	0.999	$\backslash\mathfrak{u}(n)\,\rightarrow\,\mathfrak{o}\,\mathfrak{u}(2\,n)\,\subset\,\mathfrak{o}(2\,n)$	$\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$	is the dimension of $\mathfrak{u}(n)$: $2 \times n(n-1)/2 + 1 \times n = n^2$, that is, the dimension of $\mathfrak{su}(n)$ consists of n^2 commutation relations M_i and $\mathfrak{u}(1)$.
gilmore-lie-groups_F00896	95	0.999	$\backslash\mathfrak{o}\,\backslash\mathfrak{u}(2\,n)$	$\mathfrak{ou}(2n)$	is the dimension of $\mathfrak{u}(n)$: $2 \times n(n-1)/2 + 1 \times n = n^2$, that is, the dimension of $\mathfrak{su}(n)$ consists of n^2 commutation relations M_i and $\mathfrak{u}(1)$.
gilmore-lie-groups_F00897	95	0.640	$\backslash\mathcal{u}(n):2\,\times\,n(n-1)\,/\,2+1\,\times\,n=n^{\{2\}}$	$\square(n) : 2 \times n(n-1)/2 + 1 \times n = n^2$	The Lie algebra of $sp(n)$ consists of $n \times n$ antihermitian 1
gilmore-lie-groups_F00898	95	0.914	$S\,\mathfrak{p}(n)$	$Sp(n)$	$M \in \mathfrak{sp}(n) \Rightarrow (M^\dagger)_{ij} = -M_{ji}^* \qquad (5.31)$
gilmore-lie-groups_F00899	95	0.791	$\operatorname{Sp}(n)$	$\mathbf{Sp}(n)$	plex matrix, and each of the diagonal matrix elements M_{ii} by a complex 2×2
gilmore-lie-groups_F00900	95	0.712	$M_{\{i\,j\}}(i<j)(M\,\backslash\in\,\backslash\mathfrak{sp}(n))$	$M_{ij}(i < j)(M \in \mathfrak{sp}(n))$	matrix representing an imaginary quaternion ($q_0 = 0, q_i$ arbitrary in Eq. (3.4)). The
gilmore-lie-groups_F00901	95	0.998	$q_{\{0\}}=0,\,q_{\{i\}}$	$q_0 = 0, q_i$	ity condition. The result is a real antihermitian $2n \times 2n$ matrix with the property $\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$.
gilmore-lie-groups_F00902	95	0.615	$i>j$	$i > j$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$, $4 \times n(n-1)/2 + n \times 3 = 2n(2n+1)/2$.
gilmore-lie-groups_F00903	95	1.000	$\backslash\mathfrak{sp}(n)\,\rightarrow\,\backslash\mathfrak{usp}(2\,n)\,\subset\,\mathfrak{su}(2\,n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$	5.8 Modular groups
gilmore-lie-groups_F00904	95	1.000	$\backslash\mathfrak{u}\,\mathfrak{sp}(2\,n)$	$\mathfrak{usp}(2n)$	5.8 Modular groups
gilmore-lie-groups_F00905	95	1.000	$\backslash\mathfrak{sp}(n)$	$\mathfrak{sp}(n)$	5.8 Modular groups

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gilmore-lie-groups_F00906	95	■ 1.000	$4 \times n(n-1) / 2 + n \times 3 = 2n(2n+1) / 2$	$4 \times n(n-1)/2 + n \times 3 = 2n(2n+1)/2$	The modular group $GL(n; \mathbb{Z})$ has no Lie al.
gilmore-lie-groups_F00907	96	■ 0.999	$(X, X)=0 \Rightarrow X=$	$(X, X) = 0 \Rightarrow X =$	is usually very useful. This inner product is positive-definite: $(X, X) = 0 \Rightarrow X =$
gilmore-lie-groups_F00908	96	■ 0.949	$X, X^{\prime}$	X, X'	0. For example, for the algebra $\mathfrak{so}(2, 1)$ (Eq. (5.18)), if X, X' are two 3×3 matrices
gilmore-lie-groups_F00909	96	■ 0.811	(X, Y)	(X, Y)	many instances it is very convenient to use the defining matrix representation of
gilmore-lie-groups_F00910	97	■ 1.000	$\exp (\theta X)$	$\exp (\theta X)$	In the compact case, the group element $\exp (\theta X)$ periodically returns to the identity
gilmore-lie-groups_F00911	97	■ 1.000	$-\pi \leq \theta \leq \pi$	$-\pi \leq \theta \leq +\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00912	97	■ 1.000	$-\pi$	$-\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00913	97	■ 1.000	$+\pi$	$+\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00914	97	■ 1.000	$-\infty < \theta < +\infty$	$-\infty < \theta < +\infty$	the noncompact case the group is parameterized by the entire line $-\infty < \theta < +\infty$. The underlying manifolds for the two groups are the circle S^1 and the line $\mathbb{R}$.
gilmore-lie-groups_F00915	97	■ 1.000	S^1	S^1	The underlying manifolds for the two groups are the circle S^1 and the line $\mathbb{R}$. In the compact case the simplification of parameterizing the group with a bounded
gilmore-lie-groups_F00916	97	■ 1.000	R^k	R^k	generators, to parameterize this piece of the group. This piece of the group is
gilmore-lie-groups_F00917	97	■ 1.000	$G=I_{\{n\}}, I_{\{p, q\}}$	$G=I_n, I_{p,q}$	
gilmore-lie-groups_F00918	97	■ 0.989	$(A, A)=$	$(A, A) =$	$\begin{bmatrix} 0 & 0 \\ -b & 1 \end{bmatrix} X_a$
gilmore-lie-groups_F00919	97	■ 1.000	$\operatorname{tr}(A)^2=a^2+c^2$	$\operatorname{tr}(A)^2=a^2+c^2$	
gilmore-lie-groups_F00920	97	■ 1.000	$\mathfrak{ut}(1,1)$	$\mathfrak{ut}(1,1)$	$\begin{bmatrix} 0 & 0 \\ a-c & 1 \end{bmatrix} X_b$
gilmore-lie-groups_F00921	98	■ 1.000	$(A, A)=\operatorname{tr} R(A)^2=(a-c)^2$	$(A, A)=\operatorname{tr} R(A)^2=(a-c)^2$	inner product in the regular representation is $(A, A)=\operatorname{tr} R(A)^2=(a-c)^2$. The inner
gilmore-lie-groups_F00922	98	■ 0.998	$X_{\{a\}}+X_{\{c\}}$	X_a+X_c	product in the regular representation suggests that the linear combination X_a+X_c
gilmore-lie-groups_F00923	98	■ 1.000	$\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$	$\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$	5. Write down the algebra inclusions $\mathfrak{gl}(1, \mathbb{R}) \subset \mathfrak{gl}(1, \mathbb{C}) \subset \mathfrak{gl}(1, \mathbb{Q})$ explicitly in terms of the 2×2 complex matrices as defined in (3.2) and (3.4).
gilmore-lie-groups_F00924	98	■ 1.000	$X_{\{\eta\}}, X_{\{r\}}, X_{\{l\}}, X_{\{\delta\}}$	$X_\eta, X_r, X_l, X_\delta$	isomorphic. The photon number operator is $\hat{n}=a^\dagger a$ and the photon energy operator
gilmore-lie-groups_F00925	98	■ 1.000	$\hat{E}=\left(a^{\dagger} a+\frac{1}{2}\right) \hbar \omega \rightarrow a^{\dagger} a+\frac{1}{2}$	$\hat{E}=\left(a^\dagger a+\frac{1}{2}\right) \hbar \omega \rightarrow a^\dagger a+\frac{1}{2}$	5. Cartan decomposition Assume a matrix I
gilmore-lie-groups_F00926	98	■ 1.000	$\hbar \omega=1$	$\hbar \omega=1$	5. Cartan decomposition Assume a matrix I
gilmore-lie-groups_F00927	98	■ 1.000	$X, X^{\prime} \in \mathfrak{h}$	$X, X' \in \mathfrak{h}$	6. Show that an inner product for the Cartan decomposition given in the previous problem
gilmore-lie-groups_F00928	98	■ 1.000	$Y, Y^{\prime} \in \mathfrak{p}$	$Y, Y' \in \mathfrak{p}$	6. Show that an inner product for the Cartan decomposition given in the previous problem
gilmore-lie-groups_F00929	98	■ 1.000	$\left[X, X^{\prime}\right] \in \mathfrak{h},[X, Y] \in \mathfrak{p},\left[Y, Y^{\prime}\right] \in \mathfrak{h}$	$[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$	6. Show that an inner product for the Cartan decomposition given in the previous problem
gilmore-lie-groups_F00930	98	■ 1.000	$\mathfrak{h}$	$\mathfrak{h}$	6. Show that an inner product for the Cartan decomposition given in the previous problem is
gilmore-lie-groups_F00931	98	■ 1.000	$\mathfrak{p}$	$\mathfrak{p}$	6. Show that an inner product for the Cartan decomposition given in the previous problem is

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gilmore-lie-groups_F00932	98	■ 0.995	$X=\left[\begin{array}{cc}A&\theta\\0&D\end{array}\right]\in\mathfrak{h}$	$X=\begin{bmatrix}A&0\\0&D\end{bmatrix}\in\mathfrak{h}$	show that A and D are orthogonal matrices $\Rightarrow$ $\begin{bmatrix}A&0\\0&D\end{bmatrix}^t = -\begin{bmatrix}A&0\\0&D\end{bmatrix}$
gilmore-lie-groups_F00933	98	■ 0.995	$Y=\left[\begin{array}{cc}0&B\\C&0\end{array}\right]\in\mathfrak{p}$	$Y=\begin{bmatrix}0&B\\C&0\end{bmatrix}\in\mathfrak{p}$	show that A and C are orthogonal matrices $\Rightarrow$ $\begin{bmatrix}0&B\\C&0\end{bmatrix}^t = -\begin{bmatrix}0&B\\C&0\end{bmatrix}$
gilmore-lie-groups_F00934	98	■ 1.000	$(X,Y)=0$	$(X,Y)=0$	$(X,X)\leq 0, \quad (X,X)=0\Rightarrow X=0$
gilmore-lie-groups_F00935	98	■ 0.970	$\mathfrak{so}(p,q)$	$\mathfrak{so}(p,q)$	$(X,X)\leq 0, \quad (X,X)=0\Rightarrow X=0$
gilmore-lie-groups_F00936	98	■ 0.970	$\left[\begin{array}{cc}A&B\\B^t&C\end{array}\right]$	$\begin{bmatrix}A&B\\B^t&C\end{bmatrix}$	$(X,X)\leq 0, \quad (X,X)=0\Rightarrow X=0$
gilmore-lie-groups_F00937	98	■ 0.970	$p\times p$	$p\times p$	$(X,X)\leq 0, \quad (X,X)=0\Rightarrow X=0$
gilmore-lie-groups_F00938	98	■ 0.970	$q\times q$	$q\times q$	$(X,X)\leq 0, \quad (X,X)=0\Rightarrow X=0$
gilmore-lie-groups_F00939	98	■ 0.986	$C^t=-C$	$C^t=-C$	
gilmore-lie-groups_F00940	98	■ 0.986	$X=\left[\begin{array}{cc}A&\theta\\0&C\end{array}\right]\in\mathfrak{h}$	$X=\begin{bmatrix}A&0\\0&C\end{bmatrix}\in\mathfrak{h}$	
gilmore-lie-groups_F00941	98	■ 0.986	$Y=\left[\begin{array}{cc}0&B\\B^t&0\end{array}\right]\in\mathfrak{p}$	$Y=\begin{bmatrix}0&B\\B^t&0\end{bmatrix}\in\mathfrak{p}$	
gilmore-lie-groups_F00942	98	■ 0.988	$(X,X)\leq 0, \quad \text{quad}(X,X)=0\Rightarrow X=0$	$(X,X)\leq 0, \quad (X,X)=0\Rightarrow X=0$	
gilmore-lie-groups_F00943	98	■ 1.000	$(Y,Y)\geq 0, \quad \text{quad}(Y,Y)=0\Rightarrow Y=0$	$(Y,Y)\geq 0, \quad (Y,Y)=0\Rightarrow Y=0$	
gilmore-lie-groups_F00944	99	■ 0.818	$\mathfrak{su}(p,q)$	$\mathfrak{su}(p,q)$	6. The Lie algebra $\mathfrak{su}(p,q)$ has the structure $\begin{bmatrix}B^t&C\end{bmatrix}$ where the $p\times p$ and $q\times q$ matrices A and C satisfy $A^\dagger=-A$, $C^\dagger=-C$, and $\text{tr}(A+C)=0$. If $X=\begin{bmatrix}A&0\\0&0\end{bmatrix}\in\mathfrak{h}$ and
gilmore-lie-groups_F00945	99	■ 0.818	$\left[\begin{array}{c}A\\B^t\\C\end{array}\right]$	$\begin{bmatrix}A\\B^t\\C\end{bmatrix}$	6. The Lie algebra $\mathfrak{su}(p,q)$ has the structure $\begin{bmatrix}B^t&C\end{bmatrix}$ where the $p\times p$ and $q\times q$ matrices A and C satisfy $A^\dagger=-A$, $C^\dagger=-C$, and $\text{tr}(A+C)=0$. If $X=\begin{bmatrix}A&0\\0&0\end{bmatrix}\in\mathfrak{h}$ and
gilmore-lie-groups_F00946	99	■ 0.999	$A^\dagger=-A, \quad C^\dagger=-C$	$A^\dagger=-A, C^\dagger=-C$	X and C satisfy $A^\dagger=-A$, $C^\dagger=-C$, and $\text{tr}(A+C)=0$. If $X=\begin{bmatrix}A&0\\0&0\end{bmatrix}\in\mathfrak{h}$ and $Y=\begin{bmatrix}0&B\\B^t&0\end{bmatrix}\in\mathfrak{p}$, then show once again that
gilmore-lie-groups_F00947	99	■ 0.999	$\text{tr}(A+C)=0$	$\text{tr}(A+C)=0$	X and C satisfy $A^\dagger=-A$, $C^\dagger=-C$, and $\text{tr}(A+C)=0$. If $X=\begin{bmatrix}A&0\\0&0\end{bmatrix}\in\mathfrak{h}$ and $Y=\begin{bmatrix}0&B\\B^t&0\end{bmatrix}\in\mathfrak{p}$, then show once again that
gilmore-lie-groups_F00948	99	■ 1.000	$Y=\left[\begin{array}{cc}0&B\\B^t&0\end{array}\right]\in\mathfrak{p}$	$Y=\begin{bmatrix}0&B\\B^t&0\end{bmatrix}\in\mathfrak{p}$	$\begin{bmatrix}B^t&0\\0&0\end{bmatrix}^t = -\begin{bmatrix}B^t&0\\0&0\end{bmatrix}$ and $\begin{bmatrix}0&B\\0&0\end{bmatrix}^t = -\begin{bmatrix}0&B\\0&0\end{bmatrix}$
gilmore-lie-groups_F00949	99	■ 1.000	$(X,X)=0\Rightarrow X=0$	$(X,X)=0\Rightarrow X=0$	The Lie algebra for $\mathfrak{sl}(n;\mathbb{R})$ has a decomposition in terms of real
gilmore-lie-groups_F00950	99	■ 1.000	$(Y,Y)=0\Rightarrow Y=0$	$(Y,Y)=0\Rightarrow Y=0$	The Lie algebra for $\mathfrak{sl}(n;\mathbb{R})$ has a decomposition in terms of real
gilmore-lie-groups_F00951	99	■ 1.000	$\mathfrak{sl}(n;\mathbb{R})$	$\mathfrak{sl}(n;\mathbb{R})$	traceless symmetric matrices $A=A^t$ and $B=B^t$ such that $A+B=0$.
gilmore-lie-groups_F00952	99	■ 1.000	$B^t=B$	$B^t=B$	$\mathfrak{sl}(n;\mathbb{R})=A+B$
gilmore-lie-groups_F00953	99	■ 1.000	$\text{tr} B=0$	$\text{tr} B=0$	$\mathfrak{sl}(n;\mathbb{R})=A+B$
gilmore-lie-groups_F00954	99	■ 1.000	$(A,A)=0\Rightarrow A=0$	$(A,A)=0\Rightarrow A=0$	matrices $A^\dagger=-A$ and traceless hermitian matrices $H^\dagger=H$.
gilmore-lie-groups_F00955	99	■ 1.000	$(B,B)=0\Rightarrow B=0$	$(B,B)=0\Rightarrow B=0$	matrices $A^\dagger=-A$ and traceless hermitian matrices $H^\dagger=H$.
gilmore-lie-groups_F00956	99	■ 1.000	$A^\dagger=-A$	$A^\dagger=-A$	$\mathfrak{g}=\mathfrak{g}^+\oplus\mathfrak{g}^-\oplus\mathfrak{h}$

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gilmore-lie-groups_ F08957	99	■ 1.000	$H^{\dagger}=H$	$H^{\dagger}=H$	
gilmore-lie-groups_ F08958	99	■ 1.000	$(H, H)=0 \Rightarrow H=0$	$(H, H)=0 \Rightarrow H=0$	
gilmore-lie-groups_ F08959	99	■ 1.000	$\mathfrak{g}=\mathfrak{h}+\mathfrak{p}$	$\mathfrak{g}=\mathfrak{h}+\mathfrak{p}$	
gilmore-lie-groups_ F08960	100	■ 1.000	$\mathfrak{g}^{\prime}=\mathfrak{h}+i\mathfrak{p}=\mathfrak{h}+\mathfrak{p}^{\prime}$	$\mathfrak{g}^{\prime}=\mathfrak{h}+i\mathfrak{p}=\mathfrak{h}+\mathfrak{p}^{\prime}$	
gilmore-lie-groups_ F08961	100	■ 1.000	$\mathfrak{g}=\mathfrak{sl}(n;\mathbb{Q})$	$\mathfrak{g}=\mathfrak{sl}(n;\mathbb{Q})$	
gilmore-lie-groups_ F08962	100	■ 1.000	$\mathfrak{h}=\mathfrak{sl}(n;\mathbb{C})$	$\mathfrak{h}=\mathfrak{sl}(n;\mathbb{C})$	
gilmore-lie-groups_ F08963	100	■ 0.817	μ	μ	
gilmore-lie-groups_ F08964	100	■ 0.817	σ	σ	
gilmore-lie-groups_ F08965	100	■ 0.817	$\mathfrak{so}(3,2)$	$\mathfrak{so}(3,2)$	
gilmore-lie-groups_ F08966	100	■ 0.817	$\mathfrak{so}(4,1)$	$\mathfrak{so}(4,1)$	
gilmore-lie-groups_ F08967	100	■ 0.817	$\mathfrak{so}(5)$	$\mathfrak{so}(5)$	
gilmore-lie-groups_ F08968	100	■ 0.817	t_{μ}	t_{μ}	

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gilmore-lie-groups_F00969	100	■ 1.000	$\mathfrak{g}=A$	$\mathfrak{g} = A$	values of A are rationally related ($\lambda_i = \gamma n_i$, n_i are integers, $\gamma \neq 0$ is rational or irrational), $\text{EXP}(tA)$ returns periodically to I_n . What is t_0 , the minimum period in t ?
gilmore-lie-groups_F00970	100	■ 1.000	$\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$	$\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$	irrational), $\text{EXP}(tA)$ returns periodically to I_n . What is t_0 , the minimum period in t ?
gilmore-lie-groups_F00971	100	■ 1.000	$\operatorname{EXP}(t\,A)$	$\text{EXP}(tA)$	irrational), $\text{EXP}(tA)$ returns periodically to I_n . What is t_0 , the minimum period in t ?
gilmore-lie-groups_F00972	100	■ 1.000	t	t	15. Use the parameterization of $\mathfrak{so}(3)$ given in Problem 3.14. Show that the differentials
gilmore-lie-groups_F00973	100	■ 1.000	$\lambda_i=\gamma\,n_i, \, n_i$	$\lambda_i = \gamma n_i, n_i$	(dx, dy, dz) of a point in the neighborhood of (x, y, z) are related to the displacements
gilmore-lie-groups_F00974	100	■ 1.000	$\gamma \neq 0$	$\gamma \neq 0$	(dx, dy, dz) of a point in the neighborhood of (x, y, z) are related to the displacements
gilmore-lie-groups_F00975	100	■ 1.000	$t_{\{0\}}$	t_0	$(\delta x, \delta y, \delta z)$ in the neighborhood of the identity by
gilmore-lie-groups_F00976	100	■ 0.979	$\mathfrak{so}(3)$	$\mathfrak{so}(3)$	$\left \begin{matrix} \omega_x \\ \delta v \end{matrix} \right = \left \begin{matrix} \omega_{x1} & \omega_{x2} & \omega_{x3} \\ n & m & v \end{matrix} \right \left \begin{matrix} \omega_x \\ \delta v \end{matrix} \right $
gilmore-lie-groups_F00977	100	■ 1.000	$(\delta x, \delta y, \delta z)$	$(\delta x, \delta y, \delta z)$	Use the parameterization of $\mathfrak{so}(3)$ given in Problem 3.14. Show that the differentials
gilmore-lie-groups_F00978	100	■ 1.000	$g(x, y, z)$	$g(x, y, z)$	
gilmore-lie-groups_F00979	100	■ 1.000	$d\,\mu(x, y, z)$	$d\mu(x, y, z)$	
gilmore-lie-groups_F00980	101	■ 1.000	$(\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p})$	$(\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p})$	exist in the algebra ($\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p}$),
gilmore-lie-groups_F00981	102	■ 0.987	$b_i^{\dagger}$	$b_i^{\dagger}$	$[b_i, b_i^{\dagger}] = I \delta_{ii}$ (6.1)
gilmore-lie-groups_F00982	102	■ 0.987	$b_{\{j\}}$	b_j	$[b_i, b_i^{\dagger}] = I \delta_{ii}$ (6.1)
gilmore-lie-groups_F00983	102	■ 0.971	$\left[b_{\{i\}}, b_{\{j\}}\right], \left[b_{\{i\}}^{\dagger}, b_{\{j\}}^{\dagger}\right], \left[b_{\{i\}}, I\right], \left[b_{\{j\}}^{\dagger}, I\right]$	$[b_i, b_j], \left[b_i^{\dagger}, b_j^{\dagger}\right], [b_i, I], \left[b_j^{\dagger}, I\right]$	operators:
gilmore-lie-groups_F00984	102	■ 1.000	$\mathcal{A}$	$\mathcal{A}$	$\mathcal{A} \cong \mathcal{A} = \mathcal{O}(\mathcal{A}) = \sum_i \sum_j u_i \alpha_{ij} u_j$ (0.2)
gilmore-lie-groups_F00985	103	■ 1.000	$[A, B]=C$	$[A, B] = C$	where $[A, B] = C$. This argument is invertible. An algebra of operators bilinear in
gilmore-lie-groups_F00986	103	■ 0.999	$2\,n+1$	$2n + 1$	Remark The $2n+1$ operators $b_i, b_j^{\dagger}$, ($i = 1, j = n$) span the Heisenberg algebra.
gilmore-lie-groups_F00987	103	■ 0.999	$b_{\{i\}}, b_{\{j\}}^{\dagger}, I(1 \leq i, j \leq n)$	$b_i, b_j^{\dagger}, I(1 \leq i, j \leq n)$	Remark The $2n+1$ operators $b_i, b_j^{\dagger}$, ($i = 1, j = n$) span the Heisenberg algebra.
gilmore-lie-groups_F00988	103	■ 1.000	$X_{\{i\,j\}}$	X_{ij}	$[X_{ij}, X_{rs}] = X_{is} \delta_{jr} - X_{rj} \delta_{si}$ (6
gilmore-lie-groups_F00989	103	■ 1.000	$b_{\{i\}}^{\dagger} b_{\{j\}}$	$b_i^{\dagger} b_j$	$\mathcal{A} \cong \mathcal{A} = \sum_{ij} X_{ij} \mathcal{A}_{ij}$
gilmore-lie-groups_F00990	103	■ 0.970	$f_{\{i\}}^{\dagger}$	$f_i^{\dagger}$	$\{f, f^{\dagger}\} = f f^{\dagger} + f^{\dagger} f = I \otimes$ (6.9)
gilmore-lie-groups_F00991	103	■ 0.970	$f_{\{j\}}$	f_j	$\{f, f^{\dagger}\} = f f^{\dagger} + f^{\dagger} f = I \otimes$ (6.9)
gilmore-lie-groups_F00992	103	■ 1.000	$\left\{f_i, f_j\right\}, \left\{f_i^{\dagger}, f_j^{\dagger}\right\}$	$\{f_i, f_j\}, \left\{f_i^{\dagger}, f_j^{\dagger}\right\}$	$\left[f_i^{\dagger} f_j, f_r^{\dagger} f_s\right] = f_i^{\dagger} f_j f_r^{\dagger} f_s - f_r^{\dagger} f_s f_i^{\dagger} f_j$
gilmore-lie-groups_F00993	104	■ 1.000	$S\,U(n)$	$SU(n)$	construction of the symmetric representations of $U(n)$. The mapping to a fermion

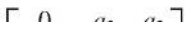
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00994	104	1.000	$\mathcal{L}_1 = x_2 \partial_3 - x_3 \partial_2, \mathcal{L}_2 = x_3 \partial_1 - x_1 \partial_3, \mathcal{L}_3 = x_1 \partial_2 - x_2 \partial_1$	$\mathcal{L}_1 = x_2 \partial_3 - x_3 \partial_2, \mathcal{L}_2 = x_3 \partial_1 - x_1 \partial_3, \mathcal{L}_3 = x_1 \partial_2 - x_2 \partial_1$	where L_i are 3×3 matrices and $\mathcal{L}_i$ are first order differential operators.
gilmore-lie-groups_F00995	104	0.705	$L_{\{i\}}$	L_i	
gilmore-lie-groups_F00996	104	0.705	$\mathcal{L}_{\{i\}}$	$\mathcal{L}_i$	
gilmore-lie-groups_F00997	105	1.000	$E(2) = \text{ISO}(2)$	$E(2) = \text{ISO}(2)$	As another example, we treat the Lie algebra for the group $E(2) = \text{ISO}(2)$ of
gilmore-lie-groups_F00998	105	1.000	$x-y$	$x - y$	rigid motions (translations and rotations) in the x - y plane, whose matrix algebra
gilmore-lie-groups_F00999	105	1.000	$t_{\{1\}}$	t_1	operator algebra is
gilmore-lie-groups_F01000	105	1.000	$t_{\{2\}}$	t_2	operator algebra is
gilmore-lie-groups_F01001	105	1.000	$\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$	$\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$	Differential operator realizations of Lie algebras come about in a natural way.
gilmore-lie-groups_F01002	105	1.000	$\mathcal{T}_i = \partial_i$	$\mathcal{T}_i = \partial_i$	Differential operator realizations of Lie algebras come about in a natural way.
gilmore-lie-groups_F01003	105	0.867	(a, b)	(a, b)	$(a, b) \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
gilmore-lie-groups_F01004	105	1.000	$x(p)$	$x(p)$	$J(p)$
gilmore-lie-groups_F01005	105	1.000	$S^{\{\prime\}}$	S'	group transformation
gilmore-lie-groups_F01006	105	1.000	$p, x^{\{\prime\}}(p)$	$p, x'(p)$	
gilmore-lie-groups_F01007	105	1.000	$f_{\{S\}}$	f_S	$\begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} 1+aa & av \\ a & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}$
gilmore-lie-groups_F01008	105	1.000	$f_{\{S^{\{\prime\}}\}}$	$f_{S'}$	$\begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} 1+aa & av \\ a & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}$
gilmore-lie-groups_F01009	105	1.000	$x^{\{\prime\}}(p)$	$x'(p)$	
gilmore-lie-groups_F01010	106	1.000	$G(x, y)$	$G(x, y)$	
gilmore-lie-groups_F01011	106	1.000	$\left(x^{\{\prime\}}, y^{\{\prime\}}\right)$	(x', y')	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} =$
gilmore-lie-groups_F01012	107	1.000	$\mathcal{A} = \sum_i \sum_j A_{ij} X_{ij}$	$\mathcal{A} = \sum_i \sum_j A_{ij} X_{ij}$	tion relations of the matrix Lie algebra and the operator algebra are isomorphic: $[A, B] = C \Leftrightarrow [A, B] = C$ Under these conditions, complicated commutators in
gilmore-lie-groups_F01013	107	1.000	$[A, B] = C \Leftrightarrow [\mathcal{A}, \mathcal{B}] = C$	$[A, B] = C \Leftrightarrow [\mathcal{A}, \mathcal{B}] = C$	these results extend to the respective Lie groups: products of exponentials of op- erators can be replaced by products of exponentials of the corresponding matrices
gilmore-lie-groups_F01014	107	1.000	$e^{\mathcal{A}} e^{\mathcal{B}} = e^{\mathcal{D}} \Leftrightarrow e^{\mathcal{A}} e^{\mathcal{B}} = e^{\mathcal{D}}$	$e^{\mathcal{A}} e^{\mathcal{B}} = e^{\mathcal{D}} \Leftrightarrow e^{\mathcal{A}} e^{\mathcal{B}} = e^{\mathcal{D}}$	1. Bilinear products involving one creation and on
gilmore-lie-groups_F01015	107	1.000	$a_{\{i\}}^{\{\dagger\}} a_{\{j\}}, 1 \leq i, j \leq 2$	$a_i^{\dagger} a_j, 1 \leq i, j \leq 2$	commutation relations.
gilmore-lie-groups_F01016	107	1.000	$\hat{n} = a_1^{\{\dagger\}} a_{\{1\}} + a_2^{\{\dagger\}} a_{\{2\}}$	$\hat{n} = a_1^{\dagger} a_1 + a_2^{\dagger} a_2$	c. These calculations simplify considerably under the operator to matrix mapping
gilmore-lie-groups_F01017	107	1.000	$\hat{n}$	$\hat{n}$	$a_1^{\dagger} a_1 + a_2^{\dagger} a_2 \quad a_1^{\dagger} a_1 - a_2^{\dagger} a_2 \quad a_1^{\dagger} a_2 \quad a_2^{\dagger} a_1$
gilmore-lie-groups_F01018	107	1.000	$a_{\{1\}}^{\{\dagger\}} a_{\{1\}} - a_{\{2\}}^{\{\dagger\}} a_{\{2\}}, a_{\{1\}}^{\{\dagger\}} a_{\{2\}}$	$a_1^{\dagger} a_1 - a_2^{\dagger} a_2, a_1^{\dagger} a_2$	$\begin{matrix} a_1^{\dagger} a_1 + a_2^{\dagger} a_2 & a_1^{\dagger} a_1 - a_2^{\dagger} a_2 & a_1^{\dagger} a_2 & a_2^{\dagger} a_1 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{matrix}$

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gilmore-lie-groups_ F01019	107	■ 1.000	$a_2^{\dagger} a_1$	$a_2^{\dagger} a_1$	
gilmore-lie-groups_ F01020	108	■ 0.999	$\frac{1}{2} \left(a_1^{\dagger} a_1 - a_2^{\dagger} a_2 \right) a_1^{\dagger} a_2$	$\frac{1}{2} \left(a_1^{\dagger} a_1 - a_2^{\dagger} a_2 \right) a_1^{\dagger} a_2$	d. Show that the three operators $\frac{1}{2}(a_1^{\dagger} a_1 - a_2^{\dagger} a_2)$, $a_1^{\dagger} a_2$, and $a_2^{\dagger} a_1$ satisfy commutation
gilmore-lie-groups_ F01021	108	■ 1.000	$J_z, J_{\pm}$	$J_z, J_{\pm}$	$J_z, J_{\pm}$. In particular, s
gilmore-lie-groups_ F01022	108	■ 1.000	$J^2 = J_x^2 + J_y^2 + J_z^2$	$J^2 = J_x^2 + J_y^2 + J_z^2$	and show
gilmore-lie-groups_ F01023	108	■ 1.000	$i(i=1,2)$	$i(i=1,2)$	operators $a_i^{\dagger}$ and a_i act on state $ n_i\rangle$ in the usual wav:
gilmore-lie-groups_ F01024	108	■ 1.000	$\left n_i \right\rangle$	$ n_i\rangle$	operators $a_i^{\dagger}$ and a_i act on state $ n_i\rangle$ in the usual wav:
gilmore-lie-groups_ F01025	108	■ 1.000	n_i	n_i	operators $a_i^{\dagger}$ and a_i act on state $ n_i\rangle$ in the usual wav:
gilmore-lie-groups_ F01026	108	■ 1.000	$a_i^{\dagger}$	$a_i^{\dagger}$	$a_i^{\dagger} n_i\rangle = \sqrt{n_i + 1} n_i + 1\rangle$ $a_i n_i\rangle = \sqrt{n_i} n_i - 1\rangle$
gilmore-lie-groups_ F01027	108	■ 1.000	a_i	a_i	$a_i^{\dagger} n_i\rangle = \sqrt{n_i + 1} n_i + 1\rangle$ $a_i n_i\rangle = \sqrt{n_i} n_i - 1\rangle$
gilmore-lie-groups_ F01028	108	■ 1.000	$\left n_1 \right\rangle \otimes \left n_2 \right\rangle = \left n_1, n_2 \right\rangle$	$ n_1\rangle \otimes n_2\rangle = n_1, n_2\rangle$	$ n_1, n_2\rangle = n_1\rangle \otimes n_2\rangle$ $J = \frac{1}{2}(n_1 + n_2)$, $m = \frac{1}{2}(n_1 - n_2)$
gilmore-lie-groups_ F01029	108	■ 1.000	$\left n_1, n_2 \right\rangle = j, m\rangle$	$ n_1, n_2\rangle = jm\rangle$	b. Show that the four operators $a_i^{\dagger} a_j$ leave invariant the sum $n_1 + n_2$.
gilmore-lie-groups_ F01030	108	■ 0.994	$\frac{1}{2} \left(a_1^{\dagger} a_1 - a_2^{\dagger} a_2 \right)$	$\frac{1}{2} \left(a_1^{\dagger} a_1 - a_2^{\dagger} a_2 \right)$	c. $J^2 jm\rangle = j(j+1) jm\rangle$.
gilmore-lie-groups_ F01031	108	■ 0.994	J_z	J_z	c. $J^2 jm\rangle = j(j+1) jm\rangle$.
gilmore-lie-groups_ F01032	108	■ 0.994	$a_1^{\dagger} a_2, a_2^{\dagger} a_1$	$a_1^{\dagger} a_2, a_2^{\dagger} a_1$	c. $J^2 jm\rangle = j(j+1) jm\rangle$.
gilmore-lie-groups_ F01033	108	■ 1.000	J_+	J_+	d. $J_z jm\rangle =$
gilmore-lie-groups_ F01034	108	■ 1.000	J_-	J_-	d. $J_z jm\rangle =$
gilmore-lie-groups_ F01035	108	■ 1.000	$a_i^{\dagger} a_j$	$a_i^{\dagger} a_j$	e. $J_+ jm\rangle = a_1^{\dagger} a_2 n_1, n_2\rangle = \sqrt{n_1 + 1} \sqrt{n_2} n_1 + 1, n_2 - 1\rangle =$
gilmore-lie-groups_ F01036	108	■ 1.000	$n_1 + n_2$	$n_1 + n_2$	e. $J_+ jm\rangle = a_1^{\dagger} a_2 n_1, n_2\rangle = \sqrt{n_1 + 1} \sqrt{n_2} n_1 + 1, n_2 - 1\rangle =$
gilmore-lie-groups_ F01037	108	■ 1.000	$J^2 jm\rangle = j(j+1) jm\rangle$	$J^2 jm\rangle = j(j+1) jm\rangle$	e. $J^2 jm\rangle = j(j+1) jm\rangle$
gilmore-lie-groups_ F01038	108	■ 1.000	$J_z jm\rangle = m jm\rangle$	$J_z jm\rangle = m jm\rangle$	e. $J_z jm\rangle = m jm\rangle$
gilmore-lie-groups_ F01039	108	■ 1.000	$J_+ jm\rangle = a_1^{\dagger} a_2 n_1, n_2\rangle = \sqrt{n_1 + 1} \sqrt{n_2} n_1 + 1, n_2 - 1\rangle =$	$J_+ jm\rangle = a_1^{\dagger} a_2 n_1, n_2\rangle = \sqrt{n_1 + 1} \sqrt{n_2} n_1 + 1, n_2 - 1\rangle =$	g. $J_+ jm\rangle = i, m \pm 1\rangle \sqrt{(j \pm m + 1)(j \mp m)}$. Note that $J_+ i, m\rangle =$
gilmore-lie-groups_ F01040	108	■ 0.999	$ j, m+1\rangle \sqrt{j+m+1} \sqrt{j-m}$	$ j, m+1\rangle \sqrt{j+m+1} \sqrt{j-m}$	0.
gilmore-lie-groups_ F01041	108	■ 1.000	$J_- jm\rangle = a_2^{\dagger} a_1 n_1, n_2\rangle = \sqrt{n_1} \sqrt{n_2 + 1} n_1 - 1, n_2 + 1\rangle =$	$J_- jm\rangle = a_2^{\dagger} a_1 n_1, n_2\rangle = \sqrt{n_1} \sqrt{n_2 + 1} n_1 - 1, n_2 + 1\rangle =$	h. $\langle j' m' J_{\pm} j m \rangle = \sqrt{(j' \pm m')(j' \mp m)} \delta_{j' j} \delta_{m' m \pm 1}$.
gilmore-lie-groups_ F01042	108	■ 0.999	$ j, m-1\rangle \sqrt{j+m} \sqrt{j-m+1}$	$ j, m-1\rangle \sqrt{j+m} \sqrt{j-m+1}$	2 Basic vectors in the Lie algebra
gilmore-lie-groups_ F01043	108	■ 1.000	$J_{\pm} jm\rangle = j, m \pm 1\rangle \sqrt{(j \pm m + 1)(j \mp m)}$	$J_{\pm} jm\rangle = j, m \pm 1\rangle \sqrt{(j \pm m + 1)(j \mp m)}$	that are isomorphic to the commutation relations of the nine boson operators $a_i^{\dagger} a_i$.

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gilmore-lie-groups_F01044	108	1.000	$J_{+} j,j\rangle=0, J_{-} j,-j\rangle=$	$J_{+} j,j\rangle=0, J_{-} j,-j\rangle=$	that are isomorphic to the commutation relations of the nine boson operators $a_i^{\dagger}a_i$.
gilmore-lie-groups_F01045	108	1.000	$\left\langle j^{\prime} m^{\prime} \right J_{\pm} \left j m \right\rangle = \sqrt{\left(j^{\prime} \pm m^{\prime} \right) \left(j \mp m \right)} \delta_{j^{\prime} j} \delta_{m^{\prime}, m \pm 1}$	$\langle j' m' J_{\pm} j m \rangle = \sqrt{(j' \pm m')(j \mp m)} \delta_{j' j} \delta_{m', m \pm 1}$	the form $ n_1, n_2, n_3\rangle = n_1\rangle \otimes n_2\rangle \otimes n_3\rangle$, where f_c
gilmore-lie-groups_F01046	108	0.999	$\mathfrak{u}(3)$	$\mathfrak{u}(3)$	
gilmore-lie-groups_F01047	108	0.999	$\mathrm{U}(3)$	$U(3)$	
gilmore-lie-groups_F01048	108	1.000	$1 \leq i, j \leq 3$	$1 \leq i, j \leq 3$	
gilmore-lie-groups_F01049	108	1.000	$\left n_1, n_2, n_3 \right\rangle = \left n_1 \right\rangle \otimes \left n_2 \right\rangle \otimes \left n_3 \right\rangle$	$ n_1, n_2, n_3\rangle = n_1\rangle \otimes n_2\rangle \otimes n_3\rangle$	
gilmore-lie-groups_F01050	108	1.000	$b_i \left n_i \right\rangle = \left n_i - 1 \right\rangle \sqrt{n_i}$	$b_i n_i\rangle = n_i - 1\rangle \sqrt{n_i}$	
gilmore-lie-groups_F01051	109	1.000	$N=\sum_{i=1}^3 n_i$	$N = \sum_{i=1}^3 n_i$	b. Show that the dimension, D , of this representation is $D = (N + 3 - 1)!/N!(3 - 1)!$. This is the number of ways three nonnegative integers can be chosen whose
gilmore-lie-groups_F01052	109	1.000	$D=(N+3-1)!/N!(3-$	$D = (N + 3 - 1)!/N!(3-$	n . D is also the number of monomials of degree N in the Taylor series expansion
gilmore-lie-groups_F01053	109	0.953	N	N	. Compute the matrix elements of all operators $b_i^{\dagger}b_j$ in this representation:
gilmore-lie-groups_F01054	109	0.953	(n)	(n)	. Compute the matrix elements of all operators $b_i^{\dagger}b_j$ in this representation:
gilmore-lie-groups_F01055	109	0.852	$n \cdot D$	$n.D$	
gilmore-lie-groups_F01056	109	0.999	$f\left(x_1,x_2,\ldots,x_n\right)$	$f(x_1, x_2, \ldots, x_n)$	
gilmore-lie-groups_F01057	109	1.000	$I_{\mathcal{D}}$	I_D	What is $\mathcal{O}$?
gilmore-lie-groups_F01058	109	1.000	$\mathcal{O}$	$\mathcal{O}$	onstruct oper
gilmore-lie-groups_F01059	109	1.000	$f_i^{\dagger}f_j$	$f_i^{\dagger}f_j$	$\left \begin{array}{cc} a & b \\ c & d \end{array} \right = \left \begin{array}{cc} A & B \\ C & D \end{array} \right \left \begin{array}{c} a \\ a^{\dagger} \end{array} \right $
gilmore-lie-groups_F01060	109	1.000	$d, d^{\dagger}$	$d, d^{\dagger}$	
gilmore-lie-groups_F01061	109	0.996	$a, a^{\dagger}$	$a, a^{\dagger}$	
gilmore-lie-groups_F01062	110	0.996	$\left[d, d^{\dagger} \right] = 1$	$[d, d^{\dagger}] = 1$	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD =$
gilmore-lie-groups_F01063	110	0.996	$[d, d]=\left[d^{\dagger}, d^{\dagger} \right]=0$	$[d, d] = [d^{\dagger}, d^{\dagger}] = 0$	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD =$
gilmore-lie-groups_F01064	110	0.996	$A \, D$	AD	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD =$
gilmore-lie-groups_F01065	110	1.000	$B \, C=1$	$BC = 1$	$B\{$
gilmore-lie-groups_F01066	110	0.998	$\operatorname{Sp}(2;\mathbb{R})=$	$\operatorname{Sp}(2;\mathbb{R}) =$	b. Argue that the commutation relations are invariant under the group $\operatorname{Sp}(2;\mathbb{R}) =$
gilmore-lie-groups_F01067	110	0.982	$\operatorname{Sp}(2;\mathbb{R})$	$\operatorname{Sp}(2;\mathbb{R})$	c. Show that under $\operatorname{Sp}(2;\mathbb{R})$, linear combinations of the coordinate and differential operators $x, \hat{p}$ preserve the commutation relations. In particular, show that
gilmore-lie-groups_F01068	110	1.000	x, ∂	x, ∂	

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gilmore-lie-groups_F01069	110	■ 0.946	$a_{\{1\}}, a_{\{2\}}, \ldots, a_{\{n\}}$	$a_1, a_2, \dots, a_n$	some linear transformation as given above, with A, B, C, D now $n \times n$ matrices. Determine the conditions on these matrices under which the structure of the
gilmore-lie-groups_F01070	110	■ 1.000	$\left n_{\{1\}}, n_{\{2\}}, \ldots, n_{\{N\}}\right\rangle$	$ n_1, n_2, \dots, n_N\rangle$	counting problem is exact
gilmore-lie-groups_F01071	110	■ 1.000	$\hbar \omega(n+$	$\hbar \omega(n+$	Taylor series expansion of a function of N variables: $f(x_1, x_2, \dots, x_N)$.
gilmore-lie-groups_F01072	110	■ 0.531	$\frac{N}{2}$	$\frac{N}{2}$	b. Show that the symmetry group of this hamiltonian has Lie algebra spanned by the
gilmore-lie-groups_F01073	110	■ 0.531	$\operatorname{deg}(N, n)=(n+N-1)! / n!(N-1)!$	$\deg(N, n) = (n + N - 1)! / n!(N - 1)!$	b. Show that the symmetry group of this hamiltonian has Lie algebra spanned by the
gilmore-lie-groups_F01074	110	■ 0.999	$f\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{N\}}\right)$	$f\left(x_1, x_2, \dots, x_N\right)$	this algebra is a direct sum of a simple Lie algebra, $\mathfrak{su}(N)$, plus the one-unit
gilmore-lie-groups_F01075	110	■ 1.000	N^2	N^2	c. If the generators $a_i^\dagger a_j$ that span the invariance algebra are supplemented with the single creation and annihilation operators $a_i^\dagger$ and a_j , as well as their commutator I ,
gilmore-lie-groups_F01076	110	■ 1.000	$\mathfrak{u}(N)$	$\mathfrak{u}(N)$	c. If the generators $a_i^\dagger a_j$ that span the invariance algebra are supplemented with the single creation and annihilation operators $a_i^\dagger$ and a_j , as well as their commutator I ,
gilmore-lie-groups_F01077	110	■ 1.000	$\left[\mathcal{H}, a_i^\dagger a_j\right]=0$	$\left[\mathcal{H}, a_i^\dagger a_j\right]=0$	c. If the generators $a_i^\dagger a_j$ that span the invariance algebra are supplemented with the single creation and annihilation operators $a_i^\dagger$ and a_j , as well as their commutator I ,
gilmore-lie-groups_F01078	110	■ 1.000	$\mathfrak{su}(N)$	$\mathfrak{su}(N)$	the resulting set of operators closes to form an $(N+1)^2$ dimensional Lie algebra
gilmore-lie-groups_F01079	110	■ 1.000	$\mathcal{H}$	$\mathcal{H}$	that is nonsemisi
gilmore-lie-groups_F01080	110	■ 1.000	$a_{\{j\}}$	a_j	from this algebra that transform any state in a multiplet with n excitations to any
gilmore-lie-groups_F01081	110	■ 1.000	$(N+1)^2$	$(N+1)^2$	state in a multiplet with n' excitations.
gilmore-lie-groups_F01082	110	■ 1.000	$n^{\prime}$	n'	
gilmore-lie-groups_F01083	110	■ 1.000	$R, S, T, U, \ldots$	$R, S, T, U, \dots$	
gilmore-lie-groups_F01084	110	■ 1.000	$a^{\dagger}=$	$a^\dagger=$	
gilmore-lie-groups_F01085	110	■ 0.999	$a_{\{1\}}^{\dagger}, a_{\{2\}}^{\dagger}, \ldots, a_{\{n\}}^{\dagger}$	$a_1^\dagger, a_2^\dagger, \dots, a_n^\dagger$	
gilmore-lie-groups_F01086	111	■ 1.000	$\mathcal{R}=a^{\dagger} R a=a_{\{i\}}^{\dagger} R_{i j} a_{\{j\}}$	$\mathcal{R}=a^\dagger R a=a_i^\dagger R_{ij} a_j$	adjoint, a column vector of annihilation operators. Define $\mathcal{R}=a^\dagger R a=a_i^\dagger R_{ij} a_j$, and
gilmore-lie-groups_F01087	111	■ 1.000	$\mathcal{S}, T, U, \ldots$	$S, T, U, \dots$	similarly for $S, T, U, \dots$
gilmore-lie-groups_F01088	111	■ 1.000	$[R, S]=T \Leftrightarrow[\mathcal{R}, S]=\mathcal{T}$	$[R, S]=T \Leftrightarrow[\mathcal{R}, S]=\mathcal{T}$	a. $[R, S]=T \Leftrightarrow[\mathcal{R}, S]=\mathcal{T}$
gilmore-lie-groups_F01089	111	■ 1.000	$e^{\mathcal{R}} e^{\mathcal{S}}=e^{\mathcal{U}} \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}}=e^{\mathcal{U}}$	$e^{\mathcal{R}} e^{\mathcal{S}}=e^{\mathcal{U}} \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}}=e^{\mathcal{U}}$	b. $e^{\mathcal{R}} e^{\mathcal{S}}=e^{\mathcal{U}} \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}}=e^{\mathcal{U}}$
gilmore-lie-groups_F01090	111	■ 0.994	$\left[\frac{d}{d x}, e^{-x^2 / 2}\right]=-x e^{-x^2 / 2}$	$\left[\frac{d}{d x}, e^{-x^2 / 2}\right]=-x e^{-x^2 / 2}$	b. Use this result to show
gilmore-lie-groups_F01091	111	■ 1.000	$a=\frac{1}{\sqrt{2}}\left(x+\frac{d}{d x}\right)$	$a=\frac{1}{\sqrt{2}}\left(x+\frac{d}{d x}\right)$	$\langle x 0\rangle=e^{-x^2 / 4 \sqrt{\pi}}$. e. Introduce the creation operator $a^\dagger=\frac{1}{\sqrt{2}}\left(x-\frac{d}{d x}\right)$ and show
gilmore-lie-groups_F01092	111	■ 1.000	$\langle x \mid 0\rangle$	$\langle x \mid 0\rangle$	$\langle x \mid 0\rangle=\frac{1}{\sqrt{2 \sqrt{\pi}}} \int_{-\infty}^{\infty} d x e^{-x^2 / 4} e^{i x x}=\frac{1}{\sqrt{2 \sqrt{\pi}}} \int_{-\infty}^{\infty} d x e^{-x^2 / 4}$
gilmore-lie-groups_F01093	111	■ 1.000	$a\langle x \mid 0\rangle=0$	$a\langle x \mid 0\rangle=0$	$\langle x \mid 0\rangle=\frac{1}{\sqrt{2 \sqrt{\pi}}} \int_{-\infty}^{\infty} d x e^{-x^2 / 4} e^{i x x}=\frac{1}{\sqrt{2 \sqrt{\pi}}} \int_{-\infty}^{\infty} d x e^{-x^2 / 4}$
gilmore-lie-groups_F01094	111	■ 0.990	$\langle x \mid 0\rangle=e^{-x^2 / 2} / \sqrt{1 \sqrt{\pi}}$	$\langle x \mid 0\rangle=e^{-x^2 / 2} / \sqrt{1 \sqrt{\pi}}$	$\langle x \mid 0\rangle=e^{-x^2 / 2} / \sqrt{1 \sqrt{\pi}}$

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gilmore-lie-groups_F01095	111	1.000	$a^{\dagger}=\frac{1}{\sqrt{2}}\left(x-\frac{d}{dx}\right)$	$a^{\dagger}=\frac{1}{\sqrt{2}}\left(x-\frac{d}{dx}\right)$	$\langle x n\rangle=\frac{1}{\sqrt{2^n n!}}\langle x 0\rangle=\frac{1}{\sqrt{2^n n!}\sqrt{\pi}}=\psi_n(x)$
gilmore-lie-groups_F01096	111	1.000	$\psi_n(x)$	$\psi_n(x)$	$L_{ij}=a_i^{\dagger}a_j-a_j^{\dagger}a_i$ and a quadrupole interaction ($Q_{ij}=a_i^{\dagger}a_j+a_j^{\dagger}a_i$).
gilmore-lie-groups_F01097	111	1.000	$\langle x\mid n\rangle=$	$\langle x\mid n\rangle=$	$L_{ij}=a_i^{\dagger}a_j-a_j^{\dagger}a_i$ and a quadrupole interaction ($Q_{ij}=a_i^{\dagger}a_j+a_j^{\dagger}a_i$).
gilmore-lie-groups_F01098	111	0.999	$\frac{\left(a^{\dagger}\right)^n}{\sqrt{n!}}\langle x\mid 0\rangle$	$\frac{\left(a^{\dagger}\right)^n}{\sqrt{n!}}\langle x\mid 0\rangle$	Show that
gilmore-lie-groups_F01099	111	0.749	$L_{ij}=a_i^{\dagger}a_j-a_j^{\dagger}a_i$	$L_{ij}=a_i^{\dagger}a_j-a_j^{\dagger}a_i$	$H=\sum_{i=1}^n\hbar\omega_i\left(u_iu_i+\frac{1}{2}\right)+\sum_{i<j}\hbar\omega_{ij}\left(u_iu_j-u_ju_i\right)+\sum_{i\leq j}q_{ij}\left(u_iu_j+u_ju_i\right)$
gilmore-lie-groups_F01100	111	0.749	$Q_{ij}=a_i^{\dagger}a_j+a_j^{\dagger}a_i$	$Q_{ij}=a_i^{\dagger}a_j+a_j^{\dagger}a_i$	$H=\sum_{i=1}^n\hbar\omega_i\left(u_iu_i+\frac{1}{2}\right)+\sum_{i<j}\hbar\omega_{ij}\left(u_iu_j-u_ju_i\right)+\sum_{i\leq j}q_{ij}\left(u_iu_j+u_ju_i\right)$
gilmore-lie-groups_F01101	111	1.000	$i\leq j$	$i\leq j$	
gilmore-lie-groups_F01102	111	0.999	$\Gamma_{ji}^*=\Gamma_{ij}$	$\Gamma_{ji}^*=\Gamma_{ij}$	
gilmore-lie-groups_F01103	112	0.992	$b_i=m_{ij}a_j:H=\sum_{i=1}^n\hbar\omega_i\left(b_i^{\dagger}b_i+\frac{1}{2}\right)+$	$b_i=m_{ij}a_j:H=\sum_{i=1}^n\hbar\omega_i\left(b_i^{\dagger}b_i+\frac{1}{2}\right)+$	annihilation operators $b_i=m_{ij}a_j$; $H=\sum_{i=1}^n\hbar\omega_i(b_i^{\dagger}b_i+\frac{1}{2})$ + constant. Express
gilmore-lie-groups_F01104	112	1.000	$\Gamma(H)$	$\Gamma(H)$	the amplitudes m_{ij} in terms of the eigenvectors of $\Gamma(H)$ d. Compute the shift in the zero point energy ("constant")
gilmore-lie-groups_F01105	114	1.000	$\operatorname{EXP}(X)$	$\operatorname{EXP}(X)$	We consider the Lie group $SL(2;\mathbb{R})$ with Lie algebra $\mathfrak{sl}(2;\mathbb{R})$:
gilmore-lie-groups_F01106	114	1.000	$\operatorname{Tr}X=0$	$\operatorname{Tr}X=0$	$\operatorname{tr}e^{X}=\frac{1}{2}\left(e^{\lambda_1}+e^{\lambda_2}\right)\longrightarrow\begin{cases}2\cos\theta&\geq-2\\<-2\end{cases}$ imaginary eigenvalues (7.6)
gilmore-lie-groups_F01107	114	1.000	$\pm\theta$	$\pm\theta$	
gilmore-lie-groups_F01108	114	1.000	$\pm i\theta$	$\pm i\theta$	
gilmore-lie-groups_F01109	116	0.997	$a=b=0$	$a=b=0$	To see this most easily set $a=b=0$. Points on the c -axis map onto
gilmore-lie-groups_F01110	116	1.000	$2\pi n$	$2\pi n$	and therefore points separated by $2\pi n$ along the c -axis map onto the same group operation in $SO(2)\subset SL(2;\mathbb{R})$. The complementary subenace ($a=b=0$) maps onto
gilmore-lie-groups_F01111	116	0.651	$SO(2)\subset SL(2;\mathbb{R})$	$SO(2)\subset SL(2;\mathbb{R})$	noncompact group operations in $SL(2;\mathbb{R})$
gilmore-lie-groups_F01112	116	0.651	$a,b,0$	$a,b,0$	noncompact group operations in $SL(2;\mathbb{R})$
gilmore-lie-groups_F01113	116	1.000	$z=\cosh r\geq 1$	$z=\cosh r\geq 1$	which is just the upper sheet of the two-sheeted hyperboloid H^2_+ , shown in
gilmore-lie-groups_F01114	116	1.000	$(x,y)=(b,a)\sinh(r)/r,r^2=a^2+$	$(x,y)=(b,a)\sinh(r)/r,r^2=a^2+$	which is just the upper sheet of the two-sheeted hyperboloid H^2_+ , shown in
gilmore-lie-groups_F01115	116	1.000	b^2	b^2	Fig. 7.2(a). T
gilmore-lie-groups_F01116	116	1.000	H_{2+}^2	H_{2+}^2	
gilmore-lie-groups_F01117	116	1.000	$-\pi\leq c\leq+\pi$	$-\pi\leq c\leq+\pi$	
gilmore-lie-groups_F01118	117	0.999	$SO(1,1)$	$SO(1,1)$	
gilmore-lie-groups_F01119	117	1.000	H_1^2	H_1^2	algebra into the product of the associated group operations (cf. (7.11) and (7.13)).
gilmore-lie-groups_F01120	118	1.000	$c,0\leq c<2\pi$	$c,0\leq c<2\pi$	and coset representatives are usually interesting spaces. From this discussion we conclude that the group $SL(2;\mathbb{D})$ can be viewed in two

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gilmore-lie-groups_ F01121	118	■ 0.999	$S L(2 ; \mathbb{R}) / S O(2)$	$SL(2;\mathbb{R})/SO(2)$	with a circle. Since the upper sheet of a two-sheeted hyperboloid is topologically
gilmore-lie-groups_ F01122	118	■ 0.996	$R^2 \times S^1$	$R^2 \times S^1$	manifold underlying $SL(2;\mathbb{R})$ is the direct product of the single-sheeted hyperboloid (equivalent to $p1 \times c1$) with $p1$. This product is once again $p2 \times c1$.
gilmore-lie-groups_ F01123	118	■ 1.000	$[S L(2 ; \mathbb{R}) / S O(1,1)] \times S O(1,1)\left(S O(1,1) \simeq R^1\right)$	$[SL(2;\mathbb{R})/SO(1,1)] \times SO(1,1) (SO(1,1) \simeq R^1)$	boloid (equivalent to $R^1 \times S^1$) with R^1 . This product is once again $R^1 \times S^1$. Since matrix Lie groups are defined by algebraic constraints, so are their sub-
gilmore-lie-groups_ F01124	118	■ 1.000	$R^1 \times S^1$	$R^1 \times S^1$	matrix Lie group is an algebraic manifold. For example, for subgroups of $GL(n$
gilmore-lie-groups_ F01125	118	■ 1.000	$R^N, N=n^2$	$R^N, N = n^2$	We conclude this discussion of the covering problem by stating a theorem due to
gilmore-lie-groups_ F01126	119	■ 1.000	R^m	R^m	generators is R^m , for suitable m (m is the number of noncompact generators). The
gilmore-lie-groups_ F01127	119	■ 0.943	$S O(2,1)$	$SO(2,1)$	
gilmore-lie-groups_ F01128	119	■ 0.993	$S O(2,1) / S O(2)$	$SO(2,1)/SO(2)$	2π does not change the 3×3 rotation matrix in (7.18). The 2:1 correspondence
gilmore-lie-groups_ F01129	119	■ 0.993	$S U(1,1) / U(1)$	$SU(1,1)/U(1)$	2π does not change the 3×3 rotation matrix in (7.18). The 2:1 correspondence
gilmore-lie-groups_ F01130	119	■ 0.969	$U(1)$	$U(1)$	can be seen in a better and simpler way. One can ask: how far along a straight line
gilmore-lie-groups_ F01131	119	■ 0.969	$2: 1$	$2: 1$	can be seen in a better and simpler way. One can ask: how far along a straight line
gilmore-lie-groups_ F01132	119	■ 0.936	b_3	b_3	through the origin does one have to go to return to the identity? For the subgroup
gilmore-lie-groups_ F01133	119	■ 1.000	$U\left(b_3+2 \pi\right)=-U\left(b_3\right)$	$U\left(b_3+2 \pi\right)=-U\left(b_3\right)$	
gilmore-lie-groups_ F01134	119	■ 1.000	a_3	a_3	
gilmore-lie-groups_ F01135	120	■ 0.984	$U(1) \subset S U(1,1)$	$U(1) \subset SU(1,1)$	$U(1) \subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2,1)$ the result is
gilmore-lie-groups_ F01136	120	■ 0.984	4π	4π	$U(1) \subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2,1)$ the result is
gilmore-lie-groups_ F01137	120	■ 0.984	$S O(2) \subset S O(2,1)$	$SO(2) \subset SO(2,1)$	$U(1) \subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2,1)$ the result is
gilmore-lie-groups_ F01138	120	■ 0.811	$\operatorname{SO}(2,1)$	$SO(2,1)$	2π . Therefore, $SU(1,1)$ is “twice as large” as $SO(2,1)$. More formally, there is a
gilmore-lie-groups_ F01139	120	■ 0.992	$2 \rightarrow 1$	$2 \rightarrow 1$	$2 \rightarrow 1$ homomorphism of $SU(1,1)$ onto $SO(2,1)$
gilmore-lie-groups_ F01140	120	■ 1.000	$\sqrt{a_1^2+a_2^2+a_3^2} \leq$	$\sqrt{a_1^2+a_2^2+a_3^2} \leq$	π) map onto $SO(3)$ provided antipodal points at $ \mathbf{a} =\pi$ are identified
gilmore-lie-groups_ F01141	120	■ 0.994	$ \mathbf{a} =\pi$	$ \mathbf{a} =\pi$	$\pi(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \sim-\pi(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos$
gilmore-lie-groups_ F01142	120	■ 0.999	$\operatorname{SU}(2)$	$SU(2)$	$SU(2)$ and an points at a radius of 2π are mapped onto $-I_2$. There is an easier way
gilmore-lie-groups_ F01143	120	■ 0.998	$2 \pi\left(\sqrt{b_1^2+b_2^2+b_3^2}<2 \pi\right)$	$2 \pi\left(\sqrt{b_1^2+b_2^2+b_3^2}<2 \pi\right)$	to verify the $2 \rightarrow 1$ nature of the map $SU(2)$ to $SO(3)$. All straight lines through the origin of the Lie algebra are equivalent (since the algebra has rank 1, see Chapter
gilmore-lie-groups_ F01144	120	■ 0.986	$-I_2$	$-I_2$	8). Therefore, we can compare how a convenient line (z -axis) maps onto the two
gilmore-lie-groups_ F01145	121	■ 0.652	$x^2+y^2+z^2=1$	$x^2+y^2+z^2=1$	numbers (x', y', z') subject to $x'^2+y'^2+z'^2=1$, are in 1:1 correspondence with
gilmore-lie-groups_ F01146	121	■ 0.652	$S U(2) / U(1)$	$SU(2)/U(1)$	numbers (x', y', z') subject to $x'^2+y'^2+z'^2=1$, are in 1:1 correspondence with

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gilmore-lie-groups_ F01147	121	■ 0.989	$x^{\{\prime\}}, y^{\{\prime\}}, z^{\{\prime\}}$	x', y', z'	points in the same geometric space – a sphere in this case. As a result, the 2:1 nature of the mapping $SU(2) \rightarrow SO(3)$ can be seen from the 2:1 nature of the mapping
gilmore-lie-groups_ F01148	121	■ 0.989	$x^{\{\prime\prime\}}+y^{\{\prime\prime\}}+z^{\{\prime\prime\}}=1$	$x'^2 + y'^2 + z'^2 = 1$	points in the same geometric space – a sphere in this case. As a result, the 2:1 nature of the mapping $SU(2) \rightarrow SO(3)$ can be seen from the 2:1 nature of the mapping
gilmore-lie-groups_ F01149	121	■ 0.604	$SU(2) \rightarrow SO(3)$	$SU(2) \rightarrow SO(3)$	around the z axis. Yet another result of Cartan establishes a unique connection between Lie groups
gilmore-lie-groups_ F01150	121	■ 1.000	$\bar{G}$	$\bar{G}$	identity. Moreover, every other Lie group with this Lie algebra is either identical to
gilmore-lie-groups_ F01151	121	■ 1.000	$\bar{G} / D$	$\bar{G}/D$	for $a_i \in D$ and $g \in G$. If G is compact it is useful to determine the largest such subgroup, D_{MAX} , of $\bar{G}$. Then all compact Lie groups with the same Lie algebra as
gilmore-lie-groups_ F01152	121	■ 1.000	$\bar{G}: g d_i = d_i g$	$\bar{G}: g d_i = d_i g$	$\bar{G}$ are obtained by “dividing” $\bar{G}$ by all possible subgroups of D_{MAX} , as shown in
gilmore-lie-groups_ F01153	121	■ 1.000	$d_i \in D$	$d_i \in D$	Fig. 7.3.
gilmore-lie-groups_ F01154	121	■ 0.863	$D_{\{\text{MAX}\}}$	D_{MAX}	For simple matrix Lie groups G , computation of the discrete invariant subgroup
gilmore-lie-groups_ F01155	121	■ 0.979	$D_{\{\text{MAX}\}}$	D_{MAX}	D is a simple matter. The only discrete group operations d_i that commute with all
gilmore-lie-groups_ F01156	121	■ 1.000	d_i	d_i	$g \in G, \quad a_i \in D, \quad G \text{ simple}, \quad g a_i = a_i g \Rightarrow a_i = \lambda I_n \quad (7.25)$
gilmore-lie-groups_ F01157	121	■ 1.000	$G_1 = \bar{G} / D_1$	$G_1 = \bar{G}/D_1$	
gilmore-lie-groups_ F01158	121	■ 0.848	$\bar{G} / D_{\{\text{MAX}\}}$	$\bar{G}/D_{\text{MAX}}$	
gilmore-lie-groups_ F01159	122	■ 1.000	$SG = \bar{G}$	$SG = \bar{G}$	discrete invariant subgroups D_i of $\bar{G}$.
gilmore-lie-groups_ F01160	122	■ 1.000	D_i	D_i	
gilmore-lie-groups_ F01161	122	■ 1.000	λI_2	λI_2	with $\lambda = +1$. As a result $SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$. For each group operation in $SO(3)$ there are two matrices in $SU(2)$ that represent it.
gilmore-lie-groups_ F01162	122	■ 1.000	$\lambda^* \lambda = 1$	$\lambda^* \lambda = 1$	with $\lambda = +1$. As a result $SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$. For each group operation in $SO(3)$ there are two matrices in $SU(2)$ that represent it.
gilmore-lie-groups_ F01163	122	■ 1.000	$\det(\lambda I_2) = +1$	$\det(\lambda I_2) = +1$	with $\lambda = +1$. As a result $SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$. For each group operation in $SO(3)$ there are two matrices in $SU(2)$ that represent it.
gilmore-lie-groups_ F01164	122	■ 1.000	$\lambda = \pm 1 \cdot D$	$\lambda = \pm 1 \cdot D$	with $\lambda = +1$. As a result $SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$. For each group operation in $SO(3)$ there are two matrices in $SU(2)$ that represent it.
gilmore-lie-groups_ F01165	122	■ 1.000	$D = \{I_2, -I_2\}$	$D = \{I_2, -I_2\}$	operation in $SO(3)$ there are two matrices in $SU(2)$ that differ in sign.
gilmore-lie-groups_ F01166	122	■ 1.000	$SO(3), D = \lambda I_3$	$SO(3), D = \lambda I_3$	operation in $SO(3)$ there are two matrices in $SU(2)$ that differ in sign.
gilmore-lie-groups_ F01167	122	■ 0.874	$\lambda = +1$	$\lambda = +1$	Remark The maximal compact subgroups $SO(2)$ of $SO(2, 1)$ and $U(1)$ of
gilmore-lie-groups_ F01168	122	■ 0.874	$SU(2) / \{I_2, -I_2\} = SO(3) / I_3 = SO(3)$	$SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$	Remark The maximal compact subgroups $SO(2)$ of $SO(2, 1)$ and $U(1)$ of
gilmore-lie-groups_ F01169	122	■ 1.000	$\overline{SO(2, 1)} =$	$\overline{SO(2, 1)} =$	$SO(2) = SU(1, 1)/U(1) \times U(1) = R^2 \times R^1$. It is the only group we will encounter
gilmore-lie-groups_ F01170	122	■ 1.000	$\overline{SU(1, 1)}$	$\overline{SU(1, 1)}$	in this book that is not a matrix group. The covering group $SO(2, 1) = SU(1, 1)$ has
gilmore-lie-groups_ F01171	122	■ 1.000	$\overline{SO(2, 1) / SO(2)} \times \overline{SO(2)} = \overline{SU(1, 1) / U(1)} \times \overline{U(1)} = [SO(2, 1) / SO(2)] \times$	$\overline{SO(2, 1) / SO(2)} \times \overline{SO(2)} = \overline{SU(1, 1) / U(1)} \times \overline{U(1)} = [SO(2, 1) / SO(2)] \times$	many discrete invariant subgroups but does not have a maximal discrete invariant

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gilmore-lie-groups_F01172	122	■ 1.000	$\overline{SO(2)}=SU(1,1)/U(1)\times \overline{U(1)}=R^2\times R^1$		subgroup.
gilmore-lie-groups_F01173	122	■ 1.000	$\overline{SO(2,1)}=\overline{SU(1,1)}$	$\overline{SO(2,1)}=\overline{SU(1,1)}$	
gilmore-lie-groups_F01174	123	■ 1.000	R_{+}^2	R_+^2	right half plane R_+ into the group operator
gilmore-lie-groups_F01175	123	■ 0.974	(w, z)	(w, z)	$\begin{bmatrix} m-\tau \\ \rho^w-(\rho^w-1)\tau/m \end{bmatrix}$
gilmore-lie-groups_F01176	123	■ 1.000	$R_{+}^2(x>0, y)$	$R_+^2(x>0, y)$	from them, equivalent? The transformation between these two parameterizations is
gilmore-lie-groups_F01177	123	■ 1.000	$R^2(w, z)$	$R^2(w, z)$	from them, equivalent? The transformation between these two parameterizations is
gilmore-lie-groups_F01178	123	■ 0.995	$x>0$	$x>0$	$\begin{vmatrix} 0 & 0 & 1 \\ \alpha & \alpha & \alpha \end{vmatrix}=-\alpha\alpha+\alpha^2$
gilmore-lie-groups_F01179	124	■ 0.981	$a, a^{\dagger}, I$	$a, a^{\dagger}, I$	The single-mode photon operators $a, a^{\dagger}, I$ obey isomorphic commutation relations
gilmore-lie-groups_F01180	124	■ 1.000	$l=l^{\prime}=l^{\prime\prime}, r=r^{\prime}=r^{\prime\prime}, \delta^{\prime}=\delta+\frac{1}{2}lr=\delta^{\prime\prime}+l^{\prime\prime}r^{\prime\prime}$	$l=l^{\prime}=l^{\prime\prime}, r=r^{\prime}=r^{\prime\prime}, \delta^{\prime}=\delta+\frac{1}{2}lr=\delta^{\prime\prime}+l^{\prime\prime}r^{\prime\prime}$	As a third example we treat the four-parameter Lie group of solvable 3×3
gilmore-lie-groups_F01181	124	■ 1.000	$\delta=0$	$\delta=0$	matrices with Lie algebra
gilmore-lie-groups_F01182	125	■ 1.000	$\hat{n}=a^{\dagger}a, a, a^{\dagger}, I$	$\hat{n}=a^{\dagger}a, a, a^{\dagger}, I$	photon operators $\hat{n}=a^{\dagger}a, a, a^{\dagger}, I$ under the identification
gilmore-lie-groups_F01183	125	■ 1.000	$\operatorname{EXP}\!\left(\eta\,a^{\dagger}a+r\,a^{\dagger}+l\,a\right)$	$\operatorname{EXP}\left(\eta a^{\dagger}a+ra^{\dagger}+la\right)$	veniently ordered form $\operatorname{EXP}(r'a^{\dagger})\operatorname{EXP}(\eta'a^{\dagger}a+\delta'I)\operatorname{EXP}(l'a)$, then the reparam-
gilmore-lie-groups_F01184	125	■ 1.000	$\operatorname{EXP}\!\left(r^{\prime}a^{\dagger}a^{\dagger}\right)\operatorname{EXP}\!\left(\eta^{\prime}a^{\dagger}a+\delta^{\prime}I\right)\operatorname{EXP}\!\left(l^{\prime}a\right)$	$\operatorname{EXP}\left(r'a^{\dagger}\right)\operatorname{EXP}\left(\eta'a^{\dagger}a+\delta'I\right)\operatorname{EXP}\left(l'a\right)$	terization computation could be carried out in the 3×3 matrix representation
gilmore-lie-groups_F01185	125	■ 0.948	$e^{l^{\prime}a^{\dagger}a} 0\rangle\!\rangle=\left 0\right\rangle\!\rangle, \langle 0 e^{r^{\prime}a^{\dagger}}=\langle 0 $	$e^{l'a^{\dagger}} 0\rangle= 0\rangle, \langle 0 e^{r'a^{\dagger}}=\langle 0 $	$\backslash \qquad \qquad \qquad /$
gilmore-lie-groups_F01186	125	■ 0.948	$e^{\eta^{\prime}a^{\dagger}a^{\dagger}a} 0\rangle\!\rangle=\left 0\right\rangle\!\rangle$	$e^{\eta'a^{\dagger}a} 0\rangle= 0\rangle$	$\backslash \qquad \qquad \qquad /$
gilmore-lie-groups_F01187	125	■ 1.000	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	$\left\{ \begin{array}{l} J_z \\ \left e^{i\theta\cdot\mathbf{J}} \right \\ J_z \end{array} \right\} \qquad (7.42)$
gilmore-lie-groups_F01188	125	■ 1.000	$(j,-j\rangle)$	$(j,-j\rangle)$	
gilmore-lie-groups_F01189	126	■ 1.000	$\theta_z^{\prime}$	θ_z'	we carry out the operator disentangling calculations in the familiar 2×2 matrix representation $\mathbf{I}\rightarrow\frac{1}{2}\boldsymbol{\sigma}$ where $\boldsymbol{\sigma}$ are the Pauli spin matrices (5.14).
gilmore-lie-groups_F01190	126	■ 1.000	$\mathbf{J}\rightarrow\frac{1}{2}\boldsymbol{\sigma}$	$\mathbf{J}\rightarrow\frac{1}{2}\boldsymbol{\sigma}$	$e^{i\theta\cdot\mathbf{J}}\rightarrow\operatorname{EXP}\frac{i}{2}\begin{bmatrix} \theta_z&\theta_x-i\theta_y \\ \theta_x+i\theta_y&\theta_z \end{bmatrix}$
gilmore-lie-groups_F01191	126	■ 1.000	$\theta_{\pm}=\theta_1\pm i\theta_2$	$\theta_{\pm}=\theta_1\pm i\theta_2$	As a result, we find
gilmore-lie-groups_F01192	127	■ 1.000	$e^{\alpha J_+}e^{\beta J_-}$	$e^{\alpha J_+}e^{\beta J_-}$	$\operatorname{EXP}(n'J_z)\operatorname{EXP}(\alpha'J_+)$ and the parameters α',β',n' computed. We find
gilmore-lie-groups_F01193	127	■ 1.000	$\operatorname{EXP}\!\left(\beta^{\prime}J_{-}\right)$	$\operatorname{EXP}\left(\beta'J_{-}\right)$	$\operatorname{EXP}(n'J_z)\operatorname{EXP}(\alpha'J_+)$ and the parameters α',β',n' computed. We find
gilmore-lie-groups_F01194	127	■ 0.999	$\operatorname{EXP}\!\left(n^{\prime}J_z\right)\operatorname{EXP}\!\left(\alpha^{\prime}J_{+}\right)$	$\operatorname{EXP}\left(n'J_z\right)\operatorname{EXP}\left(\alpha'J_{+}\right)$	$\left i\right \left \sigma_1-\sigma_2-\sigma_3\right i\left \begin{array}{l} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{array} \right $
gilmore-lie-groups_F01195	127	■ 0.999	$\alpha^{\prime},\beta^{\prime},n^{\prime}$	α',β',n'	$\left i\right \left \sigma_1-\sigma_2-\sigma_3\right i\left \begin{array}{l} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{array} \right $

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gilmore-lie-groups_ F01196	127	■ 1.000	$(1+\alpha \beta)^{2j}$	$(1+\alpha \beta)^{2j}$	$\left\langle j\left J^rJ^s\right j\right\rangle=\frac{(2j)!r!}{s!}\delta_{rs}$
gilmore-lie-groups_ F01197	127	■ 1.000	$\mathcal{A},\mathcal{B}$	$\mathcal{A},\mathcal{B}$	(i) finding a faithful matrix representation of the operator algebra
gilmore-lie-groups_ F01198	127	■ 0.995	$e^{\mathcal{A}}e^{\mathcal{B}}$	$e^{\mathcal{A}}e^{\mathcal{B}}$	(i) finding a faithful matrix representation of the operator algebra, (ii) identifying the operators $\mathcal{A},\mathcal{B}$ with matrices A,B .
gilmore-lie-groups_ F01199	127	■ 0.999	$e^{\mathcal{B}^{\prime}}e^{\mathcal{A}^{\prime}}\left(\mathcal{A}^{\prime},\mathcal{B}^{\prime}\right)$	$e^{\mathcal{B}^{\prime}}e^{\mathcal{A}^{\prime}}\left(\mathcal{A}^{\prime},\mathcal{B}^{\prime}\right)$	(ii) identifying the operators $\mathcal{A},\mathcal{B}$ with matrices A,B , (iii) Carrying out the matrix calculations e^Ae^B and $e^{B^{\prime}}e^{A^{\prime}}$,
gilmore-lie-groups_ F01200	127	■ 0.999	$\left(\mathcal{A},\mathcal{B}\right)$	$\mathcal{A},\mathcal{B}$	(ii) identifying the operators $\mathcal{A},\mathcal{B}$ with matrices A,B , (iii) Carrying out the matrix calculations e^Ae^B and $e^{B^{\prime}}e^{A^{\prime}}$,
gilmore-lie-groups_ F01201	127	■ 1.000	$e^{\mathcal{A}}e^{\mathcal{B}}$	$e^{\mathcal{A}}e^{\mathcal{B}}$	This procedure will produce a local analytic repa
gilmore-lie-groups_ F01202	127	■ 1.000	$e^{\mathcal{B}^{\prime}}e^{\mathcal{A}^{\prime}}$	$e^{\mathcal{B}^{\prime}}e^{\mathcal{A}^{\prime}}$	This procedure will produce a local analytic repa
gilmore-lie-groups_ F01203	127	■ 1.000	$\mathcal{A}^{\prime},\mathcal{B}^{\prime}$	$\mathcal{A}^{\prime},\mathcal{B}^{\prime}$	$(\mathcal{A},\mathcal{B})$. If the matrix group used to construct this reparameterization
gilmore-lie-groups_ F01204	127	■ 1.000	$\mathcal{A}^{\prime}\leftrightarrow\mathcal{A}^{\prime}\mathcal{B}^{\prime}\leftrightarrow\mathcal{B}^{\prime}$	$\mathcal{A}^{\prime}\leftrightarrow\mathcal{A}^{\prime}\mathcal{B}^{\prime}\leftrightarrow\mathcal{B}^{\prime}$	connected (the covering group) the analytic rep
gilmore-lie-groups_ F01205	127	■ 1.000	$(\mathcal{A},\mathcal{B})\leftrightarrow$	$(\mathcal{A},\mathcal{B})\leftrightarrow$	groups of the operator group and the matrix group. When the operators $\mathcal{A},\mathcal{B},\dots$ are
gilmore-lie-groups_ F01206	127	■ 0.998	$\left(\mathcal{A}^{\prime},\mathcal{B}^{\prime}\right)$	$(\mathcal{A}^{\prime},\mathcal{B}^{\prime})$	
gilmore-lie-groups_ F01207	127	■ 1.000	$\mathcal{A},\mathcal{B},\ldots$	$\mathcal{A},\mathcal{B},\dots$	
gilmore-lie-groups_ F01208	128	■ 1.000	$A,B,\ldots$	$A,B,\dots$	related to matrices $A,B,\dots$ by a matrix–operator mapping (see Chapter 6) $\mathcal{A}\leftrightarrow A$,
gilmore-lie-groups_ F01209	128	■ 1.000	$\mathcal{A}\leftrightarrow A$	$\mathcal{A}\leftrightarrow A$	related to matrices $A,B,\dots$ by a matrix–operator mapping (see Chapter 6) $\mathcal{A}\leftrightarrow A$,
gilmore-lie-groups_ F01210	128	■ 1.000	$t+\delta t$	$t+\delta t$	$ \psi(t+\delta t)\rangle=\left(I-\frac{i}{\hbar}H\delta t\right) \psi(t)\rangle=e^{-\frac{i}{\hbar}H\delta t} \psi(t)\rangle$
gilmore-lie-groups_ F01211	128	■ 1.000	$\left \psi\left(t_f\right)\right\rangle$	$ \psi(t_f)\rangle$	
gilmore-lie-groups_ F01212	128	■ 1.000	t_f	t_f	
gilmore-lie-groups_ F01213	128	■ 1.000	$\left \psi\left(t_f\right)\right\rangle=$	$ \psi(t_f)\rangle=$	
gilmore-lie-groups_ F01214	128	■ 0.996	$U\left(t_f,t_i\right)\left \psi\left(t_i\right)\right\rangle$	$U\left(t_f,t_i\right)\left \psi\left(t_i\right)\right\rangle$	$U\left(t_f,t_i\right)=U\left(t_f,t_f-\delta t\right)\cdots U\left(t_i+2\delta t,t_i+\delta t\right)U\left(t_i+\delta t,t_i\right)$
gilmore-lie-groups_ F01215	128	■ 1.000	$H\left(t^{\prime}\right)$	$H\left(t^{\prime}\right)$	Riemann integral.
gilmore-lie-groups_ F01216	128	■ 1.000	$H(t),t^{\prime}\neq t$	$H(t),t'\neq t$	
gilmore-lie-groups_ F01217	129	■ 1.000	$2j+1$	$2j+1$	three operators $J_z,J_{\pm}$ that obey angular momentum commutation relations. We
gilmore-lie-groups_ F01218	129	■ 1.000	$\left j,m_j\right\rangle$	$ j,m_j\rangle$	is
gilmore-lie-groups_ F01219	129	■ 1.000	$g(t)\in S U(2)$	$g(t)\in SU(2)$	the equation of motion <i>in the group</i> is
gilmore-lie-groups_ F01220	129	■ 1.000	$(2j+1)\times(2j+1)$	$(2j+1)\times(2j+1)$	$d\left[\begin{smallmatrix}a(t)&b(t)\end{smallmatrix}\right]_{-i}\left[\frac{1}{2}\epsilon(t)-\alpha(t)\right]\left[\begin{smallmatrix}a(t)&b(t)\end{smallmatrix}\right]_{+i}\leq 0$
gilmore-lie-groups_ F01221	129	■ 1.000	$a(t)$	$a(t)$	$1,b(t_i)=0$. The final 2×2 unitary matrix is determined by $a(t_f),b(t_f)$. This is

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gilmore-lie-groups_F01248	131	■ 1.000	$\left\langle J_{-} \right\rangle$	$\langle J_{-} \rangle$	
gilmore-lie-groups_F01249	131	■ 1.000	$\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$	$\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$	
gilmore-lie-groups_F01250	131	■ 1.000	$\Lambda = 0$	$\Lambda = 0$	
gilmore-lie-groups_F01251	131	■ 1.000	$\left. \frac{\partial}{\partial \lambda^*} \log (\langle e^\Lambda \rangle) \right _{\Lambda=0}$	$\left. \frac{\partial}{\partial \lambda^*} \log (\langle e^\Lambda \rangle) \right _{\Lambda=0}$	
gilmore-lie-groups_F01252	132	■ 0.999	$\rho = e^{-\beta(\hbar \omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$	$\rho = e^{-\beta(\hbar \omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$	operator expectation values is $\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger a + \lambda a^\dagger + \lambda^* a + d I} / Z = \langle e^\Lambda \rangle$.
gilmore-lie-groups_F01253	132	■ 1.000	$\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger a + \lambda a^\dagger + \lambda^* a + d I} / Z = \langle e^\Lambda \rangle$	$\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger a + \lambda a^\dagger + \lambda^* a + d I} / Z = \langle e^\Lambda \rangle$	The trace is taken in the infinite-dimensional Hilbert space with Fock basis
gilmore-lie-groups_F01254	132	■ 0.967	$ 0\rangle, 1\rangle, 2\rangle, \dots$	$ 0\rangle, 1\rangle, 2\rangle, \dots$	without exploiting opportunities allowed by choice of a smaller, more convenient
gilmore-lie-groups_F01255	132	■ 0.989	H, Λ	H, Λ	(ii) Compute the exponential of each. For example
gilmore-lie-groups_F01256	132	■ 0.999	Z_l	Z_l	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$
gilmore-lie-groups_F01257	132	■ 0.999	Z_r	Z_r	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$
gilmore-lie-groups_F01258	133	■ 1.000	$A < 0$	$A < 0$	(vi) Take the trace. Assuming $A < 0$ the sum converges to
gilmore-lie-groups_F01259	133	■ 1.000	$\lambda_n =$	$\lambda_n =$	
gilmore-lie-groups_F01260	133	■ 1.000	$d = 0$	$d = 0$	
gilmore-lie-groups_F01261	133	■ 1.000	$D = \text{Id}, G$	$D = \text{Id}, G$	each other.
gilmore-lie-groups_F01262	133	■ 1.000	λI_n	λI_n	ful ones typically involve a sequence of exponential mappings of linear vector
gilmore-lie-groups_F01263	134	■ 0.940	$it/\hbar \leftrightarrow 1/k_B T$	$it/\hbar \leftrightarrow 1/k_B T$	mapping, and suitable stepping back and forth through large and small unitary or
gilmore-lie-groups_F01264	134	■ 1.000	$\phi((x_1, y_1), (x_2, y_2))$	$\phi((x_1, y_1), (x_2, y_2))$	2. Show that a straight line through the origin of the parameter space (a, b, c) that is
gilmore-lie-groups_F01265	134	■ 1.000	$\phi((w_1, z_1), (w_2, z_2))$	$\phi((w_1, z_1), (w_2, z_2))$	inside the light cone $a^2 + b^2 - c^2 < 0$ (Eq. (7.7)) maps onto the subgroup $SO(2) \subset$
gilmore-lie-groups_F01266	134	■ 1.000	$a^2 + b^2 - c^2 < 0$	$a^2 + b^2 - c^2 < 0$	$\beta^2 = (a^2 + b^2)/c^2$. Compare this renormalization of periodicity with “time dilation.”
gilmore-lie-groups_F01267	134	■ 1.000	$SO(2) \subset$	$SO(2) \subset$	$\beta^2 = (a^2 + b^2)/c^2$. Compare this renormalization of periodicity with “time dilation.”
gilmore-lie-groups_F01268	134	■ 0.999	c_T	c_T	3. Compute the maximal discrete invariant subgroup D_{MAX} of $SO(3)$ and show that it is $\{I, \pm I, \pm 2I\}$ where $I = a2\pi i/3$. Next, show that $SO(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_F01269	134	■ 0.999	$a^2 + b^2 > 0 \left(\sqrt{a^2 + b^2} = \beta \times c, \beta < 1 \right)$	$a^2 + b^2 > 0 \left(\sqrt{a^2 + b^2} = \beta \times c, \beta < 1 \right)$	3. Compute the maximal discrete invariant subgroup D_{MAX} of $SO(3)$ and show that it is $\{I, \pm I, \pm 2I\}$ where $I = a2\pi i/3$. Next, show that $SO(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_F01270	134	■ 0.999	$2\pi\gamma$	$2\pi\gamma$	is $\{I_3, \kappa I_3, \kappa^{-1} I_3\}$, where $\kappa = e^{\pm i2\pi/3}$. Next, show that $SO(3)/D_{\text{MAX}}$ is isomorphic to the group of real 8×8 matrices $\text{EXP}[\text{Re}n(\text{su}(3))]$ (“eight-fold wav”)
gilmore-lie-groups_F01271	134	■ 0.999	$\gamma = 1/\sqrt{1 - \beta^2}$	$\gamma = 1/\sqrt{1 - \beta^2}$	is $\{I_3, \kappa I_3, \kappa^{-1} I_3\}$, where $\kappa = e^{\pm i2\pi/3}$. Next, show that $SO(3)/D_{\text{MAX}}$ is isomorphic to the group of real 8×8 matrices $\text{EXP}[\text{Re}n(\text{su}(3))]$ (“eight-fold wav”)
gilmore-lie-groups_F01272	134	■ 1.000	$\beta^2 = (a^2 + b^2) / c^2$	$\beta^2 = (a^2 + b^2) / c^2$	the group of real 8×8 matrices $\text{EXP}[\text{Re}n(\text{su}(3))]$ (“eight-fold wav”)

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gilmore-lie-groups_F01273	134	■ 0.947	$SU(3)$	$SU(3)$	and show that it is the cyclic group of order n generated by $\epsilon I_n, \epsilon = e^{2\pi i/n}$. What real
gilmore-lie-groups_F01274	134	■ 0.954	$\left\{I_3, \lambda I_3, \lambda^2 I_3\right\}$	$\left\{I_3, \lambda I_3, \lambda^2 I_3\right\}$	matrix group is $SU(n)/D_{\text{MAX}}$ equivalent to?
gilmore-lie-groups_F01275	134	■ 0.954	$\lambda = e^{2\pi i/3}$	$\lambda = e^{2\pi i/3}$	matrix group is $SU(n)/D_{\text{MAX}}$ equivalent to?
gilmore-lie-groups_F01276	134	■ 0.954	$SU(3)/D_{\text{MAX}}$	$SU(3)/D_{\text{MAX}}$	matrix group is $SU(n)/D_{\text{MAX}}$ equivalent to?
gilmore-lie-groups_F01277	134	■ 0.229	$\Re\left(\frac{su(3)}{3}\right)$	$\Re g(\mathfrak{su}(3))$	
gilmore-lie-groups_F01278	134	■ 1.000	$\epsilon I_n, \epsilon = e^{2\pi i/n}$	$\epsilon I_n, \epsilon = e^{2\pi i/n}$	
gilmore-lie-groups_F01279	134	■ 0.988	$SU(n)/D_{\text{MAX}}$	$SU(n)/D_{\text{MAX}}$	
gilmore-lie-groups_F01280	135	■ 1.000	$a^\dagger, a$	$a^\dagger, a$	6. It is convenient to introduce the creation and annihilation operators $a^\dagger, a$ to study the
gilmore-lie-groups_F01281	135	■ 1.000	$\langle n' x^k n \rangle$	$\langle n' x^k n \rangle$	indicates how.
gilmore-lie-groups_F01282	135	■ 1.000	$e^{\lambda x}$	$e^{\lambda x}$	$d^k = \frac{1}{k!} \frac{d^k}{dx^k} e^{\lambda x} \Big _{x=0}$
gilmore-lie-groups_F01283	135	■ 1.000	x^k	x^k	$d^k = \frac{1}{k!} \frac{d^k}{dx^k} x^k \Big _{x=0}$
gilmore-lie-groups_F01284	135	■ 1.000	$[a, a^\dagger] = I$	$[a, a^\dagger] = I$	$\lambda \bar{x} = \lambda(a^\dagger + a)/\sqrt{2} \Rightarrow \begin{pmatrix} \lambda & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$
gilmore-lie-groups_F01285	135	■ 0.955	$e^{ra^\dagger} e^{\delta I} e^{la}$	$e^{ra^\dagger} e^{\delta I} e^{la}$	$e^{ra^\dagger} e^{\delta I} e^{la} = \begin{vmatrix} 0 & 1 & r \end{vmatrix} \begin{vmatrix} 0 & 1 & 0 \end{vmatrix} \begin{vmatrix} 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 1 & r \end{vmatrix}$
gilmore-lie-groups_F01286	135	■ 0.999	$\langle n' x^4 n \rangle$	$\langle n' x^4 n \rangle$	$\frac{1}{4!} \frac{d^4}{dx^4} (n' (a^\dagger)^p (a)^q n)$
gilmore-lie-groups_F01287	135	■ 1.000	$\langle e^{ikx} \rangle$	$\langle e^{ikx} \rangle$	
gilmore-lie-groups_F01288	136	■ 1.000	$P_n \simeq e^{-n\beta\hbar\omega}$	$P_n \simeq e^{-n\beta\hbar\omega}$	thermal: $P_n \simeq e^{-n\beta\hbar\omega}$. This expectation value can be written in algebraic form as
gilmore-lie-groups_F01289	136	■ 1.000	$k \rightarrow 0$	$k \rightarrow 0$	we concentrate on the numerator, as the denominator is obtained in the limit $k \rightarrow 0$.
gilmore-lie-groups_F01290	136	■ 1.000	$\langle e^{ikx} \rangle = e^\delta$	$\langle e^{ikx} \rangle = e^\delta$	c. Compute σ using relations in the .
gilmore-lie-groups_F01291	136	■ 0.976	δ	δ	from the group element $g = e^{\delta H}$.
gilmore-lie-groups_F01292	136	■ 1.000	$\infty \times \infty$	$\infty \times \infty$	... $\frac{1}{2} \ln \frac{1}{2} = -\ln 2$, ... $\frac{1}{2} \ln \frac{1}{2} = -\ln 2$, ...
gilmore-lie-groups_F01293	136	■ 0.999	$\omega' = \omega$	$\omega' = \omega$	8. A finite set of operators X_i closes under commutation: $[X_i$
gilmore-lie-groups_F01294	136	■ 0.999	α, β	α, β	8. A finite set of operators X_i closes under commutation: $[X_i$
gilmore-lie-groups_F01295	136	■ 0.999	γ	γ	8. A finite set of operators X_i closes under commutation: $[X_i$
gilmore-lie-groups_F01296	136	■ 1.000	$\left[X_i, X_j\right] = \sum_{k=1}^N C_{ij}^k X_k$	$\left[X_i, X_j\right] = \sum_{k=1}^N C_{ij}^k X_k$	κ is hermitian: $(\kappa(a^\dagger X_i))^\dagger = (a^\dagger \kappa(X_i))^\dagger = (a^\dagger)^\dagger \kappa^\dagger(X_i)$. S is faithful: $S(a^\dagger X_i) = 0 \Rightarrow a^\dagger = 0$.
gilmore-lie-groups_F01297	136	■ 1.000	R	R	ble. We require R to be hermitian to make an immediate connection with quantum
gilmore-lie-groups_F01298	136	■ 1.000	$\left(R\left(a^i X_i\right)\right)^\dagger = \left(a^i R\left(X_i\right)\right)^\dagger = \left(a^i\right)^* R^\dagger\left(X_i\right)$	$\left(R\left(a^i X_i\right)\right)^\dagger = \left(a^i R\left(X_i\right)\right)^\dagger = \left(a^i\right)^* R^\dagger\left(X_i\right)$	mechanics.

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gilmore-lie-groups_F01299	136	■ 1.000	$S\left(a^{\{i\}}X_{\{i\}}\right)=0\Rightarrow a^{\{i\}}=0$	$S\left(a^iX_i\right)=0\Rightarrow a^i=0$	a. It happens frequently that $\mathcal{H} = \bar{K}$
gilmore-lie-groups_F01300	136	■ 1.000	$\mathcal{H}=R\left(a^{\{i\}}X_{\{i\}}\right)$	$\mathcal{H}=R\left(a^iX_i\right)$	representation, $\mathrm{Tr}\left(a^{\{i\}}/J_{\alpha\beta}\right)=\mathrm{Tr}\left(a^i/\omega_{\alpha\beta}\right)$.
gilmore-lie-groups_F01301	136	■ 0.999	$H_{\{1\}},H_{\{2\}},\ldots,H_{\{r\}}\in\mathfrak{g}$	$H_1,H_2,\ldots,H_r\in\mathfrak{g}$	
gilmore-lie-groups_F01302	136	■ 1.000	$\left[H_{\{i\}},H_{\{j\}}\right]=0,1\leq i,j\leq r$	$[H_i,H_j]=0,1\leq i,j\leq r$	
gilmore-lie-groups_F01303	136	■ 1.000	$R\left(H_{\{i\}}\right)$	$R\left(H_i\right)$	
gilmore-lie-groups_F01304	136	■ 1.000	$\left[R\left(H_{\{i\}}\right)\right]_{\alpha\beta}=\delta_{\alpha\beta}r_{\alpha}(i)$	$[R\left(H_i\right)]_{\alpha\beta}=\delta_{\alpha\beta}r_{\alpha}(i)$	
gilmore-lie-groups_F01305	137	■ 1.000	$\left[S\left(H_{\{i\}}\right),S\left(H_{\{j\}}\right)\right]=0$	$[S\left(H_i\right),S\left(H_j\right)]=0$	b. Show that $[S(H_i),S(H_j)]=0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_F01306	137	■ 1.000	r	r	b. Show that $[S(H_i),S(H_j)]=0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_F01307	137	■ 1.000	$S\left(H_{\{i\}}\right)$	$S\left(H_i\right)$	b. Show that $[S(H_i),S(H_j)]=0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_F01308	137	■ 1.000	$U(t)=R\left(e^{-\frac{i}{\hbar}t}\mathcal{H}\right)=e^{-\frac{i}{\hbar}R(\mathcal{H})t}$	$U(t)=R\left(e^{-\frac{i}{\hbar}t\mathcal{H}}\right)=e^{-\frac{i}{\hbar}R(\mathcal{H})t}$	$U(t)=R(e^{-\frac{i}{\hbar}t\mathcal{H}})=e^{-\frac{i}{\hbar}R(\mathcal{H})t}$.
gilmore-lie-groups_F01309	137	■ 1.000	$\rho(T)=e^{-\beta\mathcal{H}}/Z$	$\rho(T)=e^{-\beta\mathcal{H}}/Z$	a. Show that the density operator for thermal expectation values is $\rho(T)=e^{-\beta\mathcal{H}}/Z=$ $e^{-\beta R(\mathcal{H})}/Z=e^{-\beta R(\mathcal{H})}/Z$ with $\mathcal{H}=\mathcal{H}$.
gilmore-lie-groups_F01310	137	■ 0.975	$R\left(e^{-\beta\mathcal{H}}\right)/Z=e^{-\beta R(\mathcal{H})}/Z$	$R\left(e^{-\beta\mathcal{H}}\right)/Z=e^{-\beta R(\mathcal{H})}/Z$	$R(e^{-\beta\mathcal{H}})/Z=e^{-\beta R(\mathcal{H})}/Z$. What is Z ? Show that the unitary time evolution on
gilmore-lie-groups_F01311	137	■ 1.000	$U(t)$	$U(t)$	operator $\rho(T)$ are related by a Wick rotation $it/\hbar\leftrightarrow\beta=1/k_BT$.
gilmore-lie-groups_F01312	137	■ 1.000	$\rho(T)$	$\rho(T)$	. A generating function for thermal expectation values has the form
gilmore-lie-groups_F01313	137	■ 1.000	$it/\hbar\leftrightarrow\beta=1/k_BT$	$it/\hbar\leftrightarrow\beta=1/k_BT$	. A generating function for thermal expectation values has the form
gilmore-lie-groups_F01314	137	■ 1.000	$G=e^{\mathfrak{g}}$	$G=e^{\mathfrak{g}}$	tion, this operator product can be transformed to "diagonal" form (i.e., expressed in terms of the operators H_i); the trace can easily be constructed. Show that for
gilmore-lie-groups_F01315	137	■ 1.000	$H_{\{i\}}$	H_i	$S=e^{y^iX_i}$ with the property
gilmore-lie-groups_F01316	137	■ 1.000	$x^{\{i\}}$	x^i	
gilmore-lie-groups_F01317	137	■ 1.000	$S=e^{y^{\{k\}}X_{\{k\}}}$	$S=e^{y^kX_k}$	$Se^{x^jX_j}\epsilon$
gilmore-lie-groups_F01318	137	■ 1.000	$e^{x^{\{i\}}X_{\{i\}}}$	$e^{x^iX_i}$	he unitary transformation has the form U
gilmore-lie-groups_F01319	137	■ 1.000	$U(\alpha)=e^{\left(\alpha a^{\dagger}-\alpha^*a\right)}$	$U(\alpha)=e^{\left(\alpha a^{\dagger}-\alpha^*a\right)}$	$U(\alpha) 0\rangle=e^{(\alpha a^{\dagger}-\alpha^*a)} 0\rangle=e^{\beta a^{\dagger}}e^{\delta I}e^{\beta^*a} 0\rangle\tag{7.80}$
gilmore-lie-groups_F01320	138	■ 0.896	$a^{\dagger} n\rangle=$	$a^{\dagger} n\rangle=$	∞ . . .
gilmore-lie-groups_F01321	138	■ 1.000	$ n+1\rangle/\sqrt{n+1}$	$ n+1\rangle/\sqrt{n+1}$	$ \alpha\rangle$
gilmore-lie-groups_F01322	138	■ 1.000	$\langle\beta \alpha\rangle$	$\langle\beta \alpha\rangle$	8. Show $\langle\alpha \alpha\rangle=\langle\alpha+\alpha\rangle/\sqrt{2}$.
gilmore-lie-groups_F01323	138	■ 1.000	$\langle\alpha \alpha\rangle=1$	$\langle\alpha \alpha\rangle=1$	8. Show $\langle\alpha \alpha\rangle=\langle\alpha+\alpha\rangle/\sqrt{2}$.
gilmore-lie-groups_F01324	138	■ 1.000	$a \alpha\rangle=\alpha \alpha\rangle$	$a \alpha\rangle=\alpha \alpha\rangle$	9. An $SU(2)$ coher
gilmore-lie-groups_F01325	138	■ 1.000	$\langle\alpha x \alpha\rangle=(\alpha^*+\alpha)/\sqrt{2}$	$\langle\alpha x \alpha\rangle=(\alpha^*+\alpha)/\sqrt{2}$	action of an arbitrary $SU(2)$;

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gilmore-lie-groups_F01326	138	0.966	$e^{i\phi} \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle e^{-ij\phi}$		Coherent states parameterized by coset representatives in $SO(2)/U(1)$.
gilmore-lie-groups_F01327	138	1.000	$e^{i(\theta_x J_x + \theta_y J_y)}$	$e^{i(\theta_x J_x + \theta_y J_y)}$	
gilmore-lie-groups_F01328	138	1.000	$e^{i\alpha_+ J_+} e^{i\alpha_z J_z} e^{i\alpha_- J_-}$	$e^{i\alpha_+ J_+} e^{i\alpha_z J_z} e^{i\alpha_- J_-}$	
gilmore-lie-groups_F01329	138	0.859	$e^{i\alpha_- J_-} \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle$	$e^{i\alpha_- J_-} \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle$	
gilmore-lie-groups_F01330	138	0.952	$e^{i\alpha_z J_z} \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle e^{-ij\alpha_z}$	$e^{i\alpha_z J_z} \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle e^{-ij\alpha_z}$	
gilmore-lie-groups_F01331	139	0.815	$J_- \left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$	$J_- \left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$	g. Show that $J_- \left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$ cannot be proportional to $\left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$ because the state $\left \begin{smallmatrix} j \\ +j \end{smallmatrix} \right\rangle$ is not occupied. This is different from the harmonic oscillator (photon operator) case.
gilmore-lie-groups_F01332	139	0.815	$\left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$	$\left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$	g. Show that $J_- \left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$ cannot be proportional to $\left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$ because the state $\left \begin{smallmatrix} j \\ +j \end{smallmatrix} \right\rangle$ is not occupied. This is different from the harmonic oscillator (photon operator) case.
gilmore-lie-groups_F01333	139	0.815	$\left \begin{smallmatrix} j \\ +j \end{smallmatrix} \right\rangle$	$\left \begin{smallmatrix} j \\ +j \end{smallmatrix} \right\rangle$	g. Show that $J_- \left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$ cannot be proportional to $\left \begin{smallmatrix} j \\ \theta_x \theta_y \end{smallmatrix} \right\rangle$ because the state $\left \begin{smallmatrix} j \\ +j \end{smallmatrix} \right\rangle$ is not occupied. This is different from the harmonic oscillator (photon operator) case.
gilmore-lie-groups_F01334	139	1.000	H_4	H_4	with only an infinite-dimensional unitary irreducible representation of interest.
gilmore-lie-groups_F01335	139	1.000	$e^{-i\phi} = (\theta_x - i\theta_y) / \theta$	$e^{-i\phi} = (\theta_x - i\theta_y) / \theta$	A number of important quantum eigenvalue equations can be expressed compactly. A few examples are:
gilmore-lie-groups_F01336	139	1.000	θ'	θ'	A number of important quantum eigenvalue equations can be expressed compactly. A few examples are:
gilmore-lie-groups_F01337	139	1.000	$ u\rangle$	$ u\rangle$	a. Show that a unitary transformation U transforms this equation to the diagonal form
gilmore-lie-groups_F01338	139	1.000	J_3	J_3	$(E' J_3 - Z) v\rangle = 0$, where $E' = \sqrt{E^2 + p^2}$ and $ v\rangle = U u\rangle$.
gilmore-lie-groups_F01339	139	1.000	$(E' J_3 - Z) v\rangle = 0$	$(E' J_3 - Z) v\rangle = 0$	. Under what conditions on j, p, Z are these solutions valid?
gilmore-lie-groups_F01340	139	1.000	$E' = \sqrt{E^2 + p^2}$	$E' = \sqrt{E^2 + p^2}$	. Under what conditions on j, p, Z are these solutions valid?
gilmore-lie-groups_F01341	139	1.000	$ v\rangle = U u\rangle$	$ v\rangle = U u\rangle$	. Under what conditions on j, p, Z are these solutions valid?
gilmore-lie-groups_F01342	139	1.000	$E = \pm \sqrt{(Z/m)^2 - p^2}$	$E = \pm \sqrt{(Z/m)^2 - p^2}$	e. Construct the unitary transformation and show that $H = \hbar \omega (J_+ J_- + 1/2)$. Can you
gilmore-lie-groups_F01343	139	0.871	$(p \rightarrow 0)$	$(p \rightarrow 0)$	and show that $ u\rangle = e^{-ipJ/m} 0\rangle$. Compute $\langle u u\rangle$ for each E .
gilmore-lie-groups_F01344	139	1.000	j, p, Z	j, p, Z	Compute the matrix elements of the rotation matrices in the $2j + 1$ uni
gilmore-lie-groups_F01345	139	0.986	$ u\rangle = e^{i\theta J_2} \left \begin{smallmatrix} j \\ m \end{smallmatrix} \right\rangle$	$ u\rangle = e^{i\theta J_2} \left \begin{smallmatrix} j \\ m \end{smallmatrix} \right\rangle$	$(-1)^{j-m} \Gamma((j-n)!) /$
gilmore-lie-groups_F01346	139	1.000	$z = \cos(\beta)$	$z = \cos(\beta)$	
gilmore-lie-groups_F01347	139	1.000	D_{mn}^j	D_{mn}^j	
gilmore-lie-groups_F01348	139	1.000	$j = l$	$j = l$	
gilmore-lie-groups_F01349	140	1.000	$I_2 \in SU(2)$	$I_2 \in SU(2)$	a. Geodesics through $I_2 \in SU(2)$ focus at $-I_2$ and geodesics through $I_3 \in SO(3)$
gilmore-lie-groups_F01350	140	1.000	$I_3 \in SO(3)$	$I_3 \in SO(3)$	a. Geodesics through $I_2 \in SU(2)$ focus at $-I_2$ and geodesics through $I_3 \in SO(3)$

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gilmore-lie-groups_F01351	140	0.835	$I_{\{3\}}$	I_3	focus at I_3 . Conclude that $SU(2)$ is a two-fold covering group of $SO(3)$.
gilmore-lie-groups_F01352	140	0.999	$SU(2)/U(1)(z=1, x=y=0)$	$SU(2)/U(1)(z=1, x=y=0)$	b. Geodesics through the “north pole” of $SU(2)/U(1)$ ($z=1, x=y=0$) focus at
gilmore-lie-groups_F01353	140	0.793	$z=-1, x=y=0$	$z=-1, x=y=0$	its “south pole” ($z=-1, x=y=0$) and geodesics through the north pole of $SO(3)/SO(2)$ ($z=1, x=y=0$) focus at its south pole ($z=-1, x=y=0$).
gilmore-lie-groups_F01354	140	0.701	$SO(3)/SO(2)(z=1, x=y=0)$	$SO(3)/SO(2)(z=1, x=y=0)$	$SU(2)/U(1)(z=1, x=y=0)$ focus at its south pole ($z=-1, x=y=0$). c. Conclude that $SU(2)/U(1) = \mathbb{S}^2 = SO(3)/SO(2)$ and the $2 \rightarrow 1$ nature of the
gilmore-lie-groups_F01355	140	0.701	$(z=-1, x=y=0)$	$(z=-1, x=y=0)$	$SU(2)/U(1)(z=1, x=y=0)$ focus at its south pole ($z=-1, x=y=0$). c. Conclude that $SU(2)/U(1) = \mathbb{S}^2 = SO(3)/SO(2)$ and the $2 \rightarrow 1$ nature of the
gilmore-lie-groups_F01356	140	1.000	$SU(2)/U(1)=S^2=SO(3)/SO(2)$	$SU(2)/U(1) = S^2 = SO(3)/SO(2)$	c. Conclude that $SU(2)/U(1) = \mathbb{S}^2 = SO(3)/SO(2)$ and the $2 \rightarrow 1$ nature of the covering $SU(2) \downarrow SO(3)$ is contained in the subgroup of rotations about the z -
gilmore-lie-groups_F01357	140	0.986	$SU(2) \searrow SO(3)$	$SU(2) \downarrow SO(3)$	axis $U(1) \downarrow SO(2)$:
gilmore-lie-groups_F01358	140	0.918	$U(1) \searrow SO(2)$	$U(1) \downarrow SO(2)$	
gilmore-lie-groups_F01359	140	1.000	$e^{2\pi i k/r} I_n$	$e^{2\pi i k/r} I_n$	How are the roots related to the group operations of the form $e^{2\pi i k/r} I_n$?
gilmore-lie-groups_F01360	140	1.000	n/r	n/r	How are the roots related to the group operations of the form $e^{2\pi i k/r} I_n$?
gilmore-lie-groups_F01361	140	1.000	$e^{2\pi i k/n} I_n$	$e^{2\pi i k/n} I_n$	14. Show that the matrix $\begin{bmatrix} -\lambda & 0 \\ 0 & -1/\lambda \end{bmatrix}$ in $SL(2; \mathbb{R})$ cannot be reached by $\exp$
gilmore-lie-groups_F01362	140	0.999	$\left[\begin{array}{cc} -\lambda & 0 \\ 0 & -1/\lambda \end{array}\right]$	$\begin{bmatrix} -\lambda & 0 \\ 0 & -1/\lambda \end{bmatrix}$	“broken geodesic” $e^A e^B$. Find matrices A and B that do this. (Hint: do not work too
gilmore-lie-groups_F01363	140	1.000	$\lambda > 1$	$\lambda > 1$	hard.)
gilmore-lie-groups_F01364	140	1.000	$\hbar \omega$	$\hbar \omega$	$\sum_{i=1}^N 2^{\frac{1}{2}(\sigma_z^{(i)} + 1)} = \sqrt{N} 2^{\frac{1}{2}}$
gilmore-lie-groups_F01365	140	1.000	$\sigma_z^{(i)}$	$\sigma_z^{(i)}$	transition is accompanied by the absorption (annihilation) of a single photon. The operator $\sigma_z^{(i)\dagger}$ describes deexcitation of an atom with emission (creation, $a^\dagger$) of a
gilmore-lie-groups_F01366	140	1.000	$a^{\dagger} a$	$a^\dagger a$	transition is accompanied by the absorption (annihilation) of a single photon. The operator $\sigma_z^{(i)\dagger}$ describes deexcitation of an atom with emission (creation, $a^\dagger$) of a
gilmore-lie-groups_F01367	140	0.999	$\sigma_+^{(j)} \left(\sigma_\pm^{(j)} = \frac{1}{2} \left(\sigma_x^{(j)} \pm i \sigma_y^{(j)} \right) \right)$	$\sigma_+^{(j)} \left(\sigma_\pm^{(j)} = \frac{1}{2} \left(\sigma_x^{(j)} \pm i \sigma_y^{(j)} \right) \right)$	operator $\sigma_-^{(j)\dagger} a^\dagger$, describes deexcitation of an atom with emission (creation, $a^\dagger$) of a photon. The strength of interaction of the atom with the electromagnetic field (the
gilmore-lie-groups_F01368	140	1.000	$\sigma_-^{(j)} a^\dagger$	$\sigma_-^{(j)} a^\dagger$	we recover the atomic dipole moment $\vec{d} = q \vec{r}$:
gilmore-lie-groups_F01369	140	1.000	$\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$	$\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$	ical operators $J_z, J_\pm$ can be replaced by their c -number expectation values:
gilmore-lie-groups_F01370	140	0.999	$\sum_{i=1}^N \sigma_\pm^{(i)} \rightarrow J_\pm$	$\sum_{i=1}^N \sigma_\pm^{(i)} \rightarrow J_\pm$	
gilmore-lie-groups_F01371	140	0.999	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	
gilmore-lie-groups_F01372	141	1.000	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$. Show that this semiclassical
gilmore-lie-groups_F01373	141	1.000	$ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$	$ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$	of the electromagnetic field, $\vec{p}(t) = \vec{p}(t)/\omega, \vec{q}(t) = \vec{q}(t)/\omega$. Use the Ehrenfest and Heisenberg theorems to compute the relation between the coherent state parameter $\alpha(t)$
gilmore-lie-groups_F01374	141	1.000	$\alpha(t)$	$\alpha(t)$	and the classical driving fields $\langle J_z(t) \rangle$ and $\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$.

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gilmore-lie-groups_F01375	141	■ 1.000	$\left\langle J_z(t) \right\rangle$	$\langle J_z(t) \rangle$	. Show that if the initial state of the field is not the ground state, but rath
gilmore-lie-groups_F01376	141	■ 1.000	$\left\langle J_+(t) \right\rangle = \left\langle J_-(t) \right\rangle^*$	$\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$	. Show that if the initial state of the field is not the ground state, but rath
gilmore-lie-groups_F01377	141	■ 1.000	$ \beta\rangle$	$ \beta\rangle$	state. How are the parameters β, describing the initial condition, and α, describing
gilmore-lie-groups_F01378	141	■ 1.000	β	β	the unitary evolution of the field, related?
gilmore-lie-groups_F01379	141	■ 1.000	$a^\dagger \rightarrow \langle a(t) \rangle^*$	$a^\dagger \rightarrow \langle a(t) \rangle^*$	. Show that the resulting semiclassical hamiltonian is
gilmore-lie-groups_F01380	141	■ 1.000	$a \rightarrow \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$	
gilmore-lie-groups_F01381	141	■ 1.000	$a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$	$a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$	
gilmore-lie-groups_F01382	141	■ 1.000	$m = -\frac{1}{2}$	$m = -\frac{1}{2}$	parameterized by a point in the coset $SO(2)/U(1)$. . Show that, under the action of this semiclassical Hamiltonian a coherent state will
gilmore-lie-groups_F01383	141	■ 1.000	$M = -J, J = N/2$	$M = -J, J = N/2$	parameterized by a point in the coset $SO(2)/U(1)$. . Show that, under the action of this semiclassical Hamiltonian a coherent state will
gilmore-lie-groups_F01384	141	■ 0.997	$ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$	$ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$	a classical current will produce a coherent state of the electromagnetic field; a
gilmore-lie-groups_F01385	141	■ 0.997	$J = N/2$	$J = N/2$	a classical current will produce a coherent state of the electromagnetic field; a
gilmore-lie-groups_F01386	141	■ 1.000	$\theta(t)$	$\theta(t)$	classical electromagnetic field will produce a coherent atomic state.
gilmore-lie-groups_F01387	141	■ 1.000	$\langle a(t) \rangle$	$\langle a(t) \rangle$	classical electromagnetic field will produce a coherent atomic state.
gilmore-lie-groups_F01388	141	■ 1.000	$\langle a(t) \rangle^*$	$\langle a(t) \rangle^*$	classical electromagnetic field will produce a coherent atomic state.
gilmore-lie-groups_F01389	141	■ 1.000	$\langle a \rangle$	$\langle a \rangle$	classical electromagnetic field will produce a coherent atomic state.
gilmore-lie-groups_F01390	141	■ 1.000	$\langle a \rangle^*$	$\langle a \rangle^*$	classical electromagnetic field will produce a coherent atomic state.
gilmore-lie-groups_F01391	141	■ 1.000	$\langle J_+ \rangle = \langle J_- \rangle^*$	$\langle J_+ \rangle = \langle J_- \rangle^*$	assume N identical atoms interacting with a single field mode are in thermodynamic equilibrium at temperature T ($\beta = 1/k_B T$)
gilmore-lie-groups_F01392	141	■ 1.000	$T(\beta = 1/k_B T)$	$T(\beta = 1/k_B T)$	expectation values for the operators $a, a^\dagger, a, a^\dagger$
gilmore-lie-groups_F01393	141	■ 1.000	$\langle \sigma_+^{(i)} \rangle_T$	$\langle \sigma_+^{(i)} \rangle_T$	
gilmore-lie-groups_F01394	141	■ 1.000	$a^\dagger, a, a^\dagger a$	$a^\dagger, a, a^\dagger a$	
gilmore-lie-groups_F01395	142	■ 1.000	$\sigma_z, \sigma_+, \sigma_-$	$\sigma_z, \sigma_+, \sigma_-$	operators σ_z, σ_+
gilmore-lie-groups_F01396	142	■ 1.000	$\epsilon, \hbar\omega, \lambda, N$	$\epsilon, \hbar\omega, \lambda, N$	e. Show that a thermodynamic phase transition occurs as $\lambda^-/$
gilmore-lie-groups_F01397	142	■ 1.000	$\langle J_+ \rangle_T \neq 0$	$\langle J_+ \rangle_T \neq 0$	e. Show that a thermodynamic phase transition occurs as $\lambda^-/$
gilmore-lie-groups_F01398	142	■ 1.000	$\lambda^2 / \epsilon \hbar \omega$	$\lambda^2 / \epsilon \hbar \omega$	+1. Is this a first or second order phase transition?
gilmore-lie-groups_F01399	142	■ 1.000	$a(t), b(t)$	$a(t), b(t)$	(unitarity condition). This condition simply reflects that the state of the system is given
gilmore-lie-groups_F01400	142	■ 1.000	$a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$	$a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$	by a unit quaternion. As numerical integration proceeds, imprecisions may cause these

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gilmore-lie-groups_F01401	142	0.542	$\left(a_{\{r\}}, a_{\{i\}}, b_{\{r\}}, b_{\{i\}}\right)$	(a_r, a_i, b_r, b_i)	(7.67) and (7.70) simplify considerably if the “diagonal operator” is not include
gilmore-lie-groups_F01402	142	1.000	$\lambda_3 \rightarrow 0$	$\lambda_3 \rightarrow 0$	For many reasons it is less desirable to compute inner product values for <i>symmetric operator products</i> such as $(T - T^\dagger)^\dagger(T - T^\dagger)$ or $(a^\dagger + a)^\dagger(a^\dagger + a)$ than it is to construct
gilmore-lie-groups_F01403	142	1.000	$\lambda_n \rightarrow 0$	$\lambda_n \rightarrow 0$	For many reasons it is less desirable to compute inner product values for <i>symmetric operator products</i> such as $(T - T^\dagger)^\dagger(T - T^\dagger)$ or $(a^\dagger + a)^\dagger(a^\dagger + a)$ than it is to construct
gilmore-lie-groups_F01404	142	1.000	$d=\delta=0$	$d = \delta = 0$	metric operator products
gilmore-lie-groups_F01405	142	1.000	$\left\langle J_+ J_- + J_- J_+ \right\rangle$	$\langle J_+ J_- + J_- J_+ \rangle$	metric operator products in (7.67) and (7.70), or their simplified forms constructed in the previous problem, into generating functions for ordered products of operators.
gilmore-lie-groups_F01406	142	1.000	$\left\langle a a^\dagger + a^\dagger a \right\rangle$	$\langle a a^\dagger + a^\dagger a \rangle$	metric operator products in (7.67) and (7.70), or their simplified forms constructed in the previous problem, into generating functions for ordered products of operators.
gilmore-lie-groups_F01407	142	1.000	$\left\langle J_+ J_- \right\rangle$	$\langle J_+ J_- \rangle$	previous problem, into generating functions for ordered products of operators.
gilmore-lie-groups_F01408	142	1.000	$\left\langle a^\dagger a \right\rangle$	$\langle a^\dagger a \rangle$	previous problem, into generating functions for ordered products of operators.
gilmore-lie-groups_F01409	143	0.749	$X_{\{j\}}=A_{\{j\}}^s Y_{\{s\}}$	$X_j = A_j^s Y_s$	
gilmore-lie-groups_F01410	144	1.000	$N=p \times q$	$N = pq$	operation is equivalent to addition of the $p \times q$ matrices. The regular representation
gilmore-lie-groups_F01411	144	1.000	$\mathfrak{so}(n)$	$\mathfrak{so}(n)$	
gilmore-lie-groups_F01412	145	1.000	$\hat{n} =$	$\hat{n} =$	$\frac{1}{2} \{a, a^\dagger\} = a^\dagger a + \frac{1}{2}, a^\dagger, a, I = [a, a^\dagger]$. This is isomorphic to the algebra of six 4×4 matrices presented in (5.7). The algebra of 4×4 matrices (the “defining” representation) and the regular representation of this algebra are given below:
gilmore-lie-groups_F01413	145	0.993	$\frac{1}{2} \{a, a^\dagger\} = a^\dagger a + \frac{1}{2}, a^\dagger, a, I = [a, a^\dagger]$	$\frac{1}{2} \{a, a^\dagger\} = a^\dagger a + \frac{1}{2}, a^\dagger, a, I = [a, a^\dagger]$	or six 4×4 matrices presented in (5.7). The algebra of 4×4 matrices (the “defining” representation) and the regular representation of this algebra are given below:
gilmore-lie-groups_F01414	145	1.000	$X_{\{i \ j\}}=-X_{\{j \ i\}}, 1 \leq i, j \leq 4$	$X_{ij} = -X_{ji}, 1 \leq i, j \leq 4$	
gilmore-lie-groups_F01415	146	1.000	$\mathfrak{so}(n), n \geq 4$	$\mathfrak{so}(n), n \geq 4$	algebra is simple.
gilmore-lie-groups_F01416	146	0.999	$\mathfrak{su}(n)(n \geq 2), \mathfrak{so}(n)(n \geq 4)$	$\mathfrak{su}(n)(n \geq 2), \mathfrak{so}(n)(n \geq 4)$	
gilmore-lie-groups_F01417	146	0.999	$\mathfrak{sp}(n)(n \geq 1)$	$\mathfrak{sp}(n)(n \geq 1)$	
gilmore-lie-groups_F01418	147	0.999	$[Z, X_i]$	$[Z, X_i]$	subalgebra for each value of $i = 1, 2, \dots, n$ since the subalgebra is mapped onto itself by every element Z in the original algebra. Each subalgebra is an invariant
gilmore-lie-groups_F01419	147	1.000	$X_{\{j\}}$	X_j	subalgebra. The result is shown schematically in Fig. 8.2 and is summarized by
gilmore-lie-groups_F01420	147	1.000	$j \geq i$	$j \geq i$	subalgebra. The result is shown schematically in Fig. 8.2 and is summarized by
gilmore-lie-groups_F01421	147	1.000	$X_{\{i\}}, X_{\{i+1\}}, \dots, X_{\{n\}}$	$X_i, X_{i+1}, \dots, X_n$	subalgebra. The result is shown schematically in Fig. 8.2 and is summarized by
gilmore-lie-groups_F01422	147	0.593	$i=1,2, \dots, n$	$i = 1, 2, \dots, n$	
gilmore-lie-groups_F01423	147	1.000	$V_{\{i\}}$	V_i	
gilmore-lie-groups_F01424	147	1.000	$V_{\{j\}}, i > j$	$V_j, i > j$	
gilmore-lie-groups_F01425	148	1.000	$[V_2, V_2] \subseteq V_2, V_2$	$[V_2, V_2] \subseteq V_2, V_2$	invariant subalgebra. The complementary subspace V_1 is not generally even a
gilmore-lie-groups_F01426	148	1.000	V_2, V_2	V_2, V_2	subalgebra of the original algebra.

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gilmore-lie-groups_ F01427	149	■ 1.000	$\mathfrak{g}^{\{\prime\}}=\mathfrak{g}-V_0$	$\mathfrak{g}' = \mathfrak{g} - V_0$	
gilmore-lie-groups_ F01428	149	■ 1.000	$\mathfrak{g}^{\{\prime\}}$	$\mathfrak{g}'$	) The original Lie algebra has the follow
gilmore-lie-groups_ F01429	150	■ 1.000	$V_0^{\{\prime\}}=0$	$V_0' = 0$	
gilmore-lie-groups_ F01430	150	■ 1.000	$a^{\{\dagger\}2}$	$a^{\dagger 2}$	
gilmore-lie-groups_ F01431	150	■ 1.000	$a^{\{2\}}$	a^2	
gilmore-lie-groups_ F01432	150	■ 1.000	$a^{\{\dagger\}2}+a^{\{2\}}$	$a^{\dagger 2} + a^2$	
gilmore-lie-groups_ F01433	150	■ 1.000	$a^{\{\dagger\}}a+\frac{1}{2}\{2\}$	$a^{\dagger}a + \frac{1}{2}$	8.6 Conclusion
gilmore-lie-groups_ F01434	150	■ 1.000	$a^{\{\dagger\}2}-a^{\{2\}}$	$a^{\dagger 2} - a^2$	
gilmore-lie-groups_ F01435	151	■ 1.000	$\hat{n}, a^{\{\dagger\}}, a, I$	$\hat{n}, a^{\dagger}, a, I$	a. The photon algebra $\hat{n}, a^{\dagger}, a, I$ (Eq. (8.5)).
gilmore-lie-groups_ F01436	151	■ 0.786	$\left[b_{\{i\}}, b_{\{j\}}^{\{\dagger\}}\right]=I\delta_{ij}, 1\leq i,j\leq n$	$\left[b_i, b_j^{\dagger}\right] = I\delta_{ij}, 1 \leq i, j \leq n$	are computed in the usual way.
gilmore-lie-groups_ F01437	151	■ 1.000	$b_{\{i\}}, b_{\{j\}}^{\{\dagger\}}, I$	$b_i, b_j^{\dagger}, I$	b. $b_i^{\dagger}b_j$.
gilmore-lie-groups_ F01438	151	■ 1.000	$b_{\{i\}}^{\{\dagger\}}b_{\{j\}}, b_{\{i\}}, b_{\{j\}}^{\{\dagger\}}, I$	$b_i^{\dagger}b_j, b_i, b_j^{\dagger}, I$	d. $b_i^{\dagger}b_j^{\dagger}, b_i^{\dagger}b_j + \frac{1}{2}\delta_i$
gilmore-lie-groups_ F01439	151	■ 1.000	$b_{\{i\}}^{\{\dagger\}}b_{\{j\}}^{\{\dagger\}}, b_{\{i\}}^{\{\dagger\}}b_{\{j\}}+\frac{1}{2}\delta_{ij}, b_{\{i\}}b_{\{j\}}$	$b_i^{\dagger}b_j^{\dagger}, b_i^{\dagger}b_j + \frac{1}{2}\delta_{ij}, b_i b_j$	e. $\partial_i\partial_j, \partial_i\partial_j + \frac{1}{2}\delta_{ij}, \partial_i\partial_j, \ell$ f. $h-h^{\dagger}h-h^{\dagger}h^{\dagger}h$
gilmore-lie-groups_ F01440	151	■ 0.998	$b_{\{i\}}^{\{\dagger\}}b_{\{j\}}^{\{\dagger\}}, b_{\{i\}}^{\{\dagger\}}b_{\{j\}}+\frac{1}{2}\delta_{ij}, b_{\{i\}}b_{\{j\}}, b_{\{i\}}, b_{\{j\}}^{\{\dagger\}}, I$	$b_i^{\dagger}b_j^{\dagger}, b_i^{\dagger}b_j + \frac{1}{2}\delta_{ij}, b_i b_j, b_i, b_j^{\dagger}, I$	g. $b^{\dagger}, b^{\dagger}b, b^{\dagger}bb$.
gilmore-lie-groups_ F01441	151	■ 1.000	$b, b^{\{\dagger\}}b, b^{\{\dagger\}}b^{\{\dagger\}}b$	$b, b^{\dagger}b, b^{\dagger}b^{\dagger}b$	Fermion creation
gilmore-lie-groups_ F01442	151	■ 0.788	$b^{\{\dagger\}}$	$b^{\dagger}$	$\{f_i, f_i^{\dagger}\} = \delta_{ii}$, but
gilmore-lie-groups_ F01443	151	■ 0.788	$b^{\{\dagger\}}b, b^{\{\dagger\}}b^{\{\dagger\}}b$	$b^{\dagger}b, b^{\dagger}bb$	$\{f_i, f_i^{\dagger}\} = \delta_{ii}$, but
gilmore-lie-groups_ F01444	151	■ 1.000	$\left\{f_{\{i\}}, f_{\{j\}}^{\{\dagger\}}\right\}=\delta_{ij}$	$\left\{f_i, f_j^{\dagger}\right\} = \delta_{ij}$	a. $f_i^{\dagger}f_j$. b. $f_i^{\dagger}f_i^{\dagger}-f_i^{\dagger}f_i\pm 1\delta_{ii}-f_i f_i$.
gilmore-lie-groups_ F01445	151	■ 0.985	$f_{\{i\}}^{\{\dagger\}}f_{\{j\}}^{\{\dagger\}}$	$f_i^{\dagger}f_j^{\dagger}$	of the position (x_i) and moment a. $\mathbf{v}, \mathbf{a}, I$
gilmore-lie-groups_ F01446	151	■ 0.985	$f_{\{i\}}^{\{\dagger\}}f_{\{j\}}+\frac{1}{2}\delta_{ij}$	$f_i^{\dagger}f_j + \frac{1}{2}\delta_{ij}$	of the position (x_i) and moment a. $\mathbf{v}, \mathbf{a}, I$
gilmore-lie-groups_ F01447	151	■ 0.985	$f_{\{i\}}f_{\{j\}}$	$f_i f_j$	of the position (x_i) and moment a. $\mathbf{v}, \mathbf{a}, I$
gilmore-lie-groups_ F01448	151	■ 1.000	$\left(x_{\{i\}}\right)$	(x_i)	c. $x_i\partial_j, x_i, \partial_j, I$.
gilmore-lie-groups_ F01449	151	■ 1.000	$\left(\partial_{\{j\}}\right)$	(∂_j)	c. $x_i\partial_j, x_i, \partial_j, I$.
gilmore-lie-groups_ F01450	151	■ 1.000	$x_{\{i\}}, \partial_{\{j\}}, I$	x_i, ∂_j, I	d. $x_i x_j, x_i \partial_j$.
gilmore-lie-groups_ F01451	151	■ 1.000	$x_{\{i\}}\partial_{\{j\}}$	$x_i \partial_j$	e. $x_i x_j$ a. $\mathbf{v}, \mathbf{a}$

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gilmore-lie-groups_F01452	151	■ 1.000	$x_{\{i\}} \partial_{\{j\}}, x_{\{i\}}, \partial_{\{j\}}, I$	$x_i \partial_j, x_i, \partial_j, I$	
gilmore-lie-groups_F01453	151	■ 0.991	$x_{\{i\}} x_{\{j\}}, x_{\{i\}} \partial_{\{j\}} + \frac{1}{2} I \delta_{ij}, \partial_i \partial_j$	$x_i x_j, x_i \partial_j + \frac{1}{2} I \delta_{ij}, \partial_i \partial_j$	
gilmore-lie-groups_F01454	151	■ 1.000	$x_{\{i\}} x_{\{j\}}, x_{\{i\}} \partial_{\{j\}}, \partial_{\{i\}} \partial_{\{j\}}, x_{\{i\}}, \partial_{\{j\}}, I$	$x_i x_j, x_i \partial_j, \partial_i \partial_j, x_i, \partial_j, I$	What is the relation between the
gilmore-lie-groups_F01455	151	■ 1.000	$\frac{d}{dx}, x \frac{d}{dx}, x^2 \frac{d}{dx}$	$\frac{d}{dx}, x \frac{d}{dx}, x^2 \frac{d}{dx}$	defining matrix repr santantation of the
gilmore-lie-groups_F01456	151	■ 1.000	$x, x \frac{d}{dx}, x \frac{d^2}{dx^2}$	$x, x \frac{d}{dx}, x \frac{d^2}{dx^2}$	representation of a the Lie algebra. Note
gilmore-lie-groups_F01457	151	■ 0.799	$2+1$	$2+1$	
gilmore-lie-groups_F01458	151	■ 0.799	$(x, y$	$(x, y$	
gilmore-lie-groups_F01459	151	■ 0.799	$t)$	$t)$	
gilmore-lie-groups_F01460	151	■ 1.000	$\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array} \right]$	$\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array} \right]$	
gilmore-lie-groups_F01461	151	■ 1.000	$\left[\begin{array}{ccc c} 0 & \theta & v_1 & t_1 \\ -\theta & 0 & v_2 & t_2 \\ \hline v_1 & v_2 & 0 & t_3 \\ 0 & 0 & 0 & 0 \end{array} \right]$	$\left[\begin{array}{ccc c} 0 & \theta & v_1 & t_1 \\ -\theta & 0 & v_2 & t_2 \\ \hline v_1 & v_2 & 0 & t_3 \\ 0 & 0 & 0 & 0 \end{array} \right]$	
gilmore-lie-groups_F01462	151	■ 1.000	$\left[\begin{array}{ccc c} 0 & \theta & v_1 & t_1 \\ -\theta & 0 & v_2 & t_2 \\ \hline 0 & 0 & 0 & t_3 \\ 0 & 0 & 0 & 0 \end{array} \right]$	$\left[\begin{array}{ccc c} 0 & \theta & v_1 & t_1 \\ -\theta & 0 & v_2 & t_2 \\ \hline 0 & 0 & 0 & t_3 \\ 0 & 0 & 0 & 0 \end{array} \right]$	
gilmore-lie-groups_F01463	152	■ 1.000	$x_{\{i\}}, \partial_{\{j\}}$	x_i, ∂_j	d. For each Lie algebra, express the generators in terms of the operators x_i, ∂_j .
gilmore-lie-groups_F01464	152	■ 0.781	$b_{\{i\}}^{\dagger}, b_{\{j\}}$	$b_i^{\dagger}, b_j$	e. For each Lie algebra, express the generators in terms of the boson operators b_i, b_j , $1 \leq i, j \leq 3$.
gilmore-lie-groups_F01465	152	■ 1.000	$g_{\{i\} \{j\}} = C_{\{i\} \{r\}} \{ \}^{\{s\}} C_{\{j\} \{s\}} \{ \}^{\{r\}}$	$g_{ij} = C_{ir}^s C_{js}^r$	therefore the contravariant metric g^{ij} is well defined. Show that the bilinear operator
gilmore-lie-groups_F01466	152	■ 1.000	$g^{\{i\} \{j\}}$	g^{ij}	$\mathcal{C}^2 = g^{ij} X_i X_j$ satisfies $[\mathcal{C}^2, X_k] = 0$. If there is one quadratic Casimir operator, it must
gilmore-lie-groups_F01467	152	■ 0.940	$\mathcal{C}^2 = g^{\{i\} \{j\}} X_{\{i\}} X_{\{j\}}$	$\mathcal{C}^2 = g^{ij} X_i X_j$	therefore be proportional to $\mathcal{C}^2$.
gilmore-lie-groups_F01468	152	■ 0.940	$\left[\mathcal{C}^2, X_k \right] = 0$	$[\mathcal{C}^2, X_k] = 0$	therefore be proportional to $\mathcal{C}^2$.
gilmore-lie-groups_F01469	152	■ 1.000	$\mathcal{C}^2$	$\mathcal{C}^2$	8. Show that $C_{ijk} = C_{jik}$ is
gilmore-lie-groups_F01470	152	■ 1.000	$C_{\{i\} \{j\} \{k\}} = C_{\{i\} \{j\}} \{ \}^{\{r\}} g_{\{r\} \{k\}}$	$C_{ijk} = C_{ij}^r g_{rk}$	$-C_{kji} = -C_{jik} = -C_{ikj}$. (Hint: use the Jacobi identity.)
gilmore-lie-groups_F01471	152	■ 1.000	$C_{\{i\} \{j\} \{k\}} = C_{\{j\} \{k\} \{i\}} = C_{\{k\} \{i\} \{j\}} =$	$C_{ijk} = C_{jki} = C_{kij} =$	$-C_{kji} = -C_{jik} = -C_{ikj}$. (Hint: use the Jacobi identity.)

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gilmore-lie-groups-F01472	152	■ 1.000	$-C_{\{k\ j\ i\}}=-C_{\{j\ i\ k\}}=-C_{\{i\ k\ j\}}$	$-C_{kji} = -C_{jik} = -C_{ikj}$	9. Determine the structure of the Lie algebra defined by
gilmore-lie-groups-F01473	154	■ 0.989	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$ have the same complex extension and have their different commutation relations cast into the same canonical form. We return to this question in Chapter 11.
gilmore-lie-groups-F01474	154	■ 1.000	Z, X	Z, X	“eigenoperator” commutation relations
gilmore-lie-groups-F01475	154	■ 1.000	$X=\sum_{i=1}^N a^{\{i\}} X_{\{i\}}$	$X = \sum_{i=1}^N a^i X_i$	$\left[Z, \sum a^i X_i \right] = \lambda \sum \epsilon$
gilmore-lie-groups-F01476	154	■ 1.000	$a^{\{i\}}$	a^i	$\ R(Z) - \lambda I_N \ = 0 \quad (9.3)$
gilmore-lie-groups-F01477	154	■ 1.000	$\phi_{\{j\}}(Z)$	$\phi_j(Z)$	Example The regular representation for the three-dimensional Lie algebra
gilmore-lie-groups-F01478	154	■ 0.973	$z^{\{i\}}\backslash\left(Z=\sum z^{\{i\}} X_{\{i\}}\right)$	$z^i \left(Z = \sum z^i X_i \right)$	Example The regular representation is spanned by the photon creation operators
gilmore-lie-groups-F01479	155	■ 1.000	$a^{\{\dag\}}, a, I=\left[a, a^{\{\dag\}}\right]$	$a^\dagger, a, I = [a, a^\dagger]$	$a^\dagger, a, I = [a, a^\dagger]$ is
gilmore-lie-groups-F01480	155	■ 1.000	$(-\lambda)^N=0$	$(-\lambda)^N = 0$	a basis can be found in which the matrix has strictly upper (or lower) triangular form. Therefore, there is a permutation transformation of the basis vectors that brings the
gilmore-lie-groups-F01481	155	■ 0.852	$X=\sum a_{\{i\}} X_{\{i\}} \in \mathfrak{su}(2)$	$X = \sum a_i X_i \in \mathfrak{su}(2)$	
gilmore-lie-groups-F01482	155	■ 0.953	$\mathfrak{def}(X)$	$\mathfrak{def}(X)$	$\mathfrak{def}(X) = \frac{1}{\alpha} \left \begin{array}{cc} i\alpha_3 & i(\alpha_1 - i\alpha_2) \\ i\alpha_1 + i\alpha_2 & i\alpha_3 \end{array} \right $
gilmore-lie-groups-F01483	155	■ 0.953	$\mathfrak{Reg}(X)$	$\mathfrak{Reg}(X)$	$\mathfrak{def}(X) = \frac{1}{\alpha} \left \begin{array}{cc} i\alpha_3 & i(\alpha_1 - i\alpha_2) \\ i\alpha_1 + i\alpha_2 & i\alpha_3 \end{array} \right $
gilmore-lie-groups-F01484	155	■ 1.000	$\phi_2(\mathbf{a}) \geq 0$	$\phi_2(\mathbf{a}) \geq 0$	
gilmore-lie-groups-F01485	155	■ 0.760	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	
gilmore-lie-groups-F01486	155	■ 0.760	$\lambda=0, \lambda=\pm i a$	$\lambda = 0, \lambda = \pm i a$	
gilmore-lie-groups-F01487	155	■ 0.999	$a^2=+a_1^2+a_2^2+a_3^2$	$a^2 = +a_1^2 + a_2^2 + a_3^2$	
gilmore-lie-groups-F01488	156	■ 0.861	$Y=\sum b_{\{i\}} Y_{\{i\}} \in \mathfrak{su}(1,1)$	$Y = \sum b_i Y_i \in \mathfrak{su}(1,1)$	Example For $Y = \sum b_i Y_i \in \mathfrak{su}(1,1)$ the defining 2×2 matrix representation
gilmore-lie-groups-F01489	156	■ 0.976	$\mathfrak{def}(Y)$	$\mathfrak{def}(Y)$	$\mathfrak{def}(Y)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(Y)$ are
gilmore-lie-groups-F01490	156	■ 0.976	$\mathfrak{Reg}(Y)$	$\mathfrak{Reg}(Y)$	$\mathfrak{def}(Y)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(Y)$ are
gilmore-lie-groups-F01491	156	■ 0.999	$\left(a_{\{1\}}, a_{\{2\}}, a_{\{3\}}\right) \rightarrow \left(i b_{\{1\}}, i b_{\{2\}}, b_{\{3\}}\right)$	$(a_1, a_2, a_3) \rightarrow (i b_1, i b_2, b_3)$	of coordinates maps the Lie algebra $\mathfrak{su}(2)$ to the Lie algebra $\mathfrak{su}(1,1)$.
gilmore-lie-groups-F01492	156	■ 0.999	$\phi_2(\mathbf{a})=a_1^2+a_2^2+a_3^2$	$\phi_2(\mathbf{a}) = a_1^2 + a_2^2 + a_3^2$	of coordinates maps the Lie algebra $\mathfrak{su}(2)$ to the Lie algebra $\mathfrak{su}(1,1)$.
gilmore-lie-groups-F01493	156	■ 0.953	$\phi_2(\mathbf{b})=-b_1^2-b_2^2+b_3^2$	$\phi_2(\mathbf{b}) = -b_1^2 - b_2^2 + b_3^2$	The secular equation was written down for the regular representation, since it
gilmore-lie-groups-F01494	156	■ 1.000	ϕ_2	ϕ_2	
gilmore-lie-groups-F01495	157	■ 0.515	$\mathfrak{Reg}$	$\mathfrak{Reg}$	equation of its regular representation, $\mathfrak{Reg}$. Since the number of independent roots
gilmore-lie-groups-F01496	157	■ 1.000	$\phi_{\{j\}}\left(z^{\{i\}}\right)$	$\phi_j \left(z^i \right)$	of the secular equation is equal to the number of independent coefficients $\phi_j(z^i)$, the rank is also the number of independent roots of the secular equation. The rank
gilmore-lie-groups-F01497	157	■ 1.000	$X=Z$	$X = Z$	$l^2 \sim N$, so describing commutation relations in terms of rank rather than dimension

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gilmore-lie-groups_ F01498	157	■ 1.000	$l^2 \sim N$	$l^2 \sim N$	effects a big simplification.
gilmore-lie-groups_ F01499	157	■ 1.000	$\mathfrak{m}$	$\mathfrak{m}$	$ \mathbf{v} ^2 = \sum_j u_j \mathbf{v} ^2_j \quad (9.16)$
gilmore-lie-groups_ F01500	158	■ 1.000	$\phi_j(X)$	$\phi_j(X)$	
gilmore-lie-groups_ F01501	158	■ 0.955	$a^r{}_s$	$a^r{}_s$	In this figure the vertical symbol is the Levi-Civita symbol for n dimension $n=3$ $\epsilon_{ijk} = \epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$, $\epsilon_{321} = \epsilon_{213} = \epsilon_{132} = -1$, $\epsilon_{111} = \epsilon_{222} = \epsilon_{333} = \epsilon_{112} = \dots = 0$
gilmore-lie-groups_ F01502	158	■ 0.955	$X^r{}_s$	$X^r{}_s$	In this figure the vertical symbol is the Levi-Civita symbol for n dimension $n=3$ $\epsilon_{ijk} = \epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$, $\epsilon_{321} = \epsilon_{213} = \epsilon_{132} = -1$, $\epsilon_{111} = \epsilon_{222} = \epsilon_{333} = \epsilon_{112} = \dots = 0$
gilmore-lie-groups_ F01503	158	■ 0.955	ϕ_j	ϕ_j	In this figure the vertical symbol is the Levi-Civita symbol for n dimension $n=3$ $\epsilon_{ijk} = \epsilon_{123} = \epsilon_{231} = \epsilon_{312} = +1$, $\epsilon_{321} = \epsilon_{213} = \epsilon_{132} = -1$, $\epsilon_{111} = \epsilon_{222} = \epsilon_{333} = \epsilon_{112} = \dots = 0$
gilmore-lie-groups_ F01504	158	■ 0.836	$X=a_s^{}X^r{}_s$	$X = a_s^r X^r{}_s$	(e.g., in $\mathcal{K}^+$, $\epsilon_{ijk} = +1$ for (ijk) a cyclic permutation
gilmore-lie-groups_ F01505	158	■ 0.751	$R^3,=\epsilon_{ijk}=+1$	$R^3, = \epsilon_{ijk} = +1$	operators depends only on the commutation relations of the Lie algebra. Therefore
gilmore-lie-groups_ F01506	158	■ 0.751	$(i\,j\,k)$	(ijk)	operators depends only on the commutation relations of the Lie algebra. Therefore
gilmore-lie-groups_ F01507	159	■ 1.000	$\phi_j(X^r{}_s)$	$\phi_j(X^r{}_s)$	these invariant operators $\phi_j(X^r{}_s)$ remain invariant when the matrices are replaced
gilmore-lie-groups_ F01508	159	■ 1.000	$O(n)$	$O(n)$	Example The orthonogonal groups $O(n)$ and their subgroups $SO(n)$ have Lie alge-
gilmore-lie-groups_ F01509	159	■ 0.651	$d \times d$	$d \times d$	
gilmore-lie-groups_ F01510	159	■ 0.651	$\mathfrak{so}(n)$	$\mathfrak{so}(n)$	
gilmore-lie-groups_ F01511	159	■ 0.651	$d=n(n-1)/2$	$d = n(n-1)/2$	
gilmore-lie-groups_ F01512	159	■ 1.000	$\mathfrak{def}(X)^t=-\mathfrak{def}(X)$	$\mathfrak{def}(X)^t = -\mathfrak{def}(X)$	
gilmore-lie-groups_ F01513	159	■ 0.986	$[n/2]$	$[n/2]$	Example The second order Casimir invariant operator for $\mathfrak{so}(n)$
gilmore-lie-groups_ F01514	159	■ 0.996	$j=2$	$j = 2$	antisymmetric tensors, as shown in Fig. 9.3.
gilmore-lie-groups_ F01515	159	■ 0.996	$X_{ij}=-X_{ji}$	$X_{ij} = -X_{ji}$	antisymmetric tensors, as shown in Fig. 9.3.
gilmore-lie-groups_ F01516	159	■ 1.000	$\mathfrak{so}(5)$	$\mathfrak{so}(5)$	$\left \begin{array}{c} r_1 \\ \hline \end{array} \right \left \begin{array}{c} X^1_{s_1} \\ \hline \end{array} \right \left \begin{array}{c} s_1 \\ \hline \end{array} \right \quad \left \begin{array}{c} r_1 \\ \hline \end{array} \right \left \begin{array}{c} \mathbf{v} r_1 \\ \hline \end{array} \right \left \begin{array}{c} \mathbf{v} r \\ \hline \end{array} \right \left \begin{array}{c} r_2 \\ \hline \end{array} \right $
gilmore-lie-groups_ F01517	159	■ 1.000	$X^r{}_s=-X^s{}_r$	$X^r{}_s = -X^s{}_r$	
gilmore-lie-groups_ F01518	160	■ 0.940	$v^m=\epsilon^{ijklm}X_{ij}X_{kl}$	$v^m = \epsilon^{ijklm} X_{ij} X_{kl}$	where the components of the five-vector $\mathbf{v}$ are $v^m = \epsilon^{ijklm} X_{ij} X_{kl}$, for example
gilmore-lie-groups_ F01519	160	■ 1.000	$n/2$	$n/2$	independent Casimir operator. The existence of two second-order Casimir operators for $\mathfrak{so}(4)$ is another piece of evidence that this algebra is semisimple rather than
gilmore-lie-groups_ F01520	160	■ 1.000	$Z=0$	$Z = 0$	we do this by choosing a Z for which we:
gilmore-lie-groups_ F01521	160	■ 1.000	$[Z,X]=0X$	$[Z, X] = 0X$	1. maximize the number of nonzer
gilmore-lie-groups_ F01522	160	■ 1.000	$\hat{n}+\frac{1}{2}=\frac{1}{2}\left\{a,a^{\dagger}\right\}$	$\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$	$= \frac{1}{2} \left(\frac{1}{2} + 2 \right) = \frac{5}{2}$
gilmore-lie-groups_ F01523	160	■ 1.000	$a^{\dagger 2}, a^{\dagger}, I=\left[a,a^{\dagger}\right], a, a^2$	$a^{\dagger 2}, a^{\dagger}, I = [a, a^\dagger], a, a^2$	The secular equations for the 6×6 regular repre

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gilmore-lie-groups_ F01524	161	■ 1.000	$\pm 2z_1, \pm z_1, 0, 0$	$\pm 2z_1, \pm z_1, 0, 0$	From this diagram we learn
gilmore-lie-groups_ F01525	161	■ 0.999	$X_{(2,0)}=a^{\dagger 2}, X_{(1,0)}=a^{\dagger}, X_{(0,0)}=\hat{n}+\frac{1}{2}I, X_{(-1,0)}=a, X_{(-2,0)}=a^2$	$X_{(2,0)}=a^{\dagger 2}, X_{(1,0)}=a^{\dagger}, X_{(0,0)}=\hat{n}+\frac{1}{2}I, X_{(-1,0)}=a, X_{(-2,0)}=a^2$	$[A(k,0), A(l,0)] \sim A(k+l,0)$ (9.30)
gilmore-lie-groups_ F01526	161	■ 1.000	$k, l \in \{-2, -1, 0, +1, +2\}$	$k, l \in \{-2, -1, 0, +1, +2\}$	if $k+l$ is in the set $\{-2, -1, 0, +1, +2\}$ an
gilmore-lie-groups_ F01527	161	■ 1.000	$k+l$	$k+l$	
gilmore-lie-groups_ F01528	161	■ 1.000	$\{-2, -1, 0, +1, +2\}$	$\{-2, -1, 0, +1, +2\}$	
gilmore-lie-groups_ F01529	161	■ 1.000	$k+l=0$	$k+l=0$	
gilmore-lie-groups_ F01530	161	■ 0.968	$(0,0): \hat{n}+\frac{1}{2}$	$(0,0): \hat{n}+\frac{1}{2}$	For simple and sem
gilmore-lie-groups_ F01531	162	■ 1.000	$\alpha_1, \alpha_2, \dots, \alpha_l$	$\alpha_1, \alpha_2, \dots, \alpha_l$	pendent roots can be collected together as the components of an l -dimensional vector
gilmore-lie-groups_ F01532	162	■ 0.570	$(\alpha_1, \alpha_2, \dots, \alpha_l)$	$(\alpha_1, \alpha_2, \dots, \alpha_l)$	(iii) the dimension of the subspa
gilmore-lie-groups_ F01533	162	■ 1.000	V_{β}	V_{β}	$V_{\alpha+\beta} = V_{\alpha} \oplus V_{\beta}$
gilmore-lie-groups_ F01534	162	■ 1.000	$V_{\alpha+\beta}$	$V_{\alpha+\beta}$	(iv) the root spaces $V_{\alpha+\beta}, V_{\alpha+2\beta}, \dots, V_{\alpha+l\beta}$ are l -dimensional subspaces of $\mathfrak{g}$.
gilmore-lie-groups_ F01535	162	■ 1.000	$\alpha+\beta$	$\alpha+\beta$	$[H_i, H_j] = 0$
gilmore-lie-groups_ F01536	162	■ 1.000	$H_1, H_2, \dots, H_l$	$H_1, H_2, \dots, H_l$	(iii) Each subsnace V_{α} ($\alpha \neq 0$) is one dimensional. Therefore each subspace V_{α} is span
gilmore-lie-groups_ F01537	162	■ 1.000	$V_{\alpha} (\alpha \neq 0)$	$V_{\alpha} (\alpha \neq 0)$	$[H_i, E_{\alpha}] = \alpha_i E_{\alpha}$
gilmore-lie-groups_ F01538	162	■ 0.995	E_{α}	E_{α}	$E_{\alpha} = \frac{1}{2}(\alpha_1 \alpha_1^{-1} + \alpha_2 \alpha_2^{-1} + \dots + \alpha_l \alpha_l^{-1})$ (9.33)
gilmore-lie-groups_ F01539	162	■ 0.995	$[V_0, V_{\alpha}] \subset V_{\alpha}$	$[V_0, V_{\alpha}] \subset V_{\alpha}$	$E_{\alpha} = \frac{1}{2}(\alpha_1 \alpha_1^{-1} + \alpha_2 \alpha_2^{-1} + \dots + \alpha_l \alpha_l^{-1})$ (9.33)
gilmore-lie-groups_ F01540	162	■ 1.000	$-\alpha$	$-\alpha$	of the H_i :
gilmore-lie-groups_ F01541	162	■ 1.000	$c \alpha$	$c \alpha$	of the H_i :
gilmore-lie-groups_ F01542	162	■ 1.000	$ c =1$	$ c =1$	of the H_i :
gilmore-lie-groups_ F01543	162	■ 1.000	$E_{-\alpha}$	$E_{-\alpha}$	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij} \alpha_j$ can be introduced in this root space
gilmore-lie-groups_ F01544	162	■ 1.000	α^i	α^i	
gilmore-lie-groups_ F01545	162	■ 1.000	α_j	α_j	
gilmore-lie-groups_ F01546	162	■ 1.000	$\alpha^i = h^{ij} \alpha_j$	$\alpha^i = h^{ij} \alpha_j$	
gilmore-lie-groups_ F01547	163	■ 1.000	$h^{ij} = \delta^{ij}$	$h^{ij} = \delta^{ij}$	then $h^{ii} = 0$ and we can identify α^i with α_i , $\alpha^i = \alpha_i$.) It remains to compute
gilmore-lie-groups_ F01548	163	■ 1.000	$\alpha_i: \alpha^i = \alpha_i$	$\alpha_i: \alpha^i = \alpha_i$	then $h^{ii} = 0$ and we can identify α^i with α_i , $\alpha^i = \alpha_i$.) It remains to compute
gilmore-lie-groups_ F01549	163	■ 1.000	$N_{\alpha, \beta}$	$N_{\alpha, \beta}$	

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gilmore-lie-groups_ F01550	163	■ 1.000	$E_{\alpha}, E_{\beta}, E_{\gamma}$	$E_{\alpha}, E_{\beta}, E_{\gamma}$	$[[E_{\alpha}, E_{\beta}], E_{\gamma}] + [[E_{\beta}, E_{\gamma}], E_{\alpha}] + [[E_{\gamma},$
gilmore-lie-groups_ F01551	163	■ 1.000	$\alpha, \beta + k \alpha$	$\alpha, \beta + k \alpha$	
gilmore-lie-groups_ F01552	164	■ 1.000	$\left[E_{\alpha}, E_{\beta}\right]=N_{\alpha, \beta} E_{\alpha+\beta}$	$[E_{\alpha}, E_{\beta}] = N_{\alpha, \beta} E_{\alpha+\beta}$	
gilmore-lie-groups_ F01553	164	■ 1.000	$N_{\alpha, \beta+n \alpha}=0$	$N_{\alpha, \beta+n \alpha} = 0$	$N^2_{\alpha, \beta-(n-k) \alpha} = (n-k+1) \left(\alpha \cdot \beta + \frac{1}{\gamma}(n-m) \alpha \cdot \alpha\right) = 0$
gilmore-lie-groups_ F01554	164	■ 1.000	$N_{-\alpha, \beta-m \alpha}^2=N_{\alpha, \beta-(m+1) \alpha}^2=0$	$N_{-\alpha, \beta-m \alpha}^2 = N_{\alpha, \beta-(m+1) \alpha}^2 = 0$	$N^2_{\alpha, \beta-(m+1) \alpha} = (n+m+1) \left(\alpha \cdot \beta + \frac{1}{\gamma}(n-m) \alpha \cdot \alpha\right) = 0$
gilmore-lie-groups_ F01555	164	■ 1.000	$N_{\alpha, \beta+k \alpha}^2=(n-k)(m+k+1)(\alpha \cdot \alpha) / 2 \geq 0$	$N_{\alpha, \beta+k \alpha}^2 = (n-k)(m+k+1)(\alpha \cdot \alpha) / 2 \geq 0$	(ii) The inner products obey
gilmore-lie-groups_ F01556	164	■ 1.000	$\alpha \cdot \beta > 0$	$\alpha \cdot \beta > 0$	$\alpha \cdot \beta > 0$
gilmore-lie-groups_ F01557	164	■ 1.000	$m -$	$m -$	$\alpha \cdot \beta > 0$
gilmore-lie-groups_ F01558	164	■ 1.000	$n > 0$	$n > 0$	
gilmore-lie-groups_ F01559	164	■ 1.000	$\alpha \cdot \beta < 0$	$\alpha \cdot \beta < 0$	
gilmore-lie-groups_ F01560	164	■ 1.000	$m - n < 0$	$m - n < 0$	
gilmore-lie-groups_ F01561	165	■ 0.998	$A_{\{2\}}, B_{\{2\}}, C_{\{2\}}$	A_2, B_2, C_2	The root space diagram encapsulates in a very convenient way all the structure
gilmore-lie-groups_ F01562	165	■ 0.998	$D_{\{2\}}, G_{\{2\}}$	D_2, G_2	The root space diagram encapsulates in a very convenient way all the structure
gilmore-lie-groups_ F01563	165	■ 0.456	$l(l$	$l(l$	
gilmore-lie-groups_ F01564	165	■ 0.977	$\mathbf{H}=\left(H_{\{1\}}, H_{\{2\}}, \ldots, H_{\{l\}}\right)$	$\mathbf{H} = (H_1, H_2, \dots, H_l)$	$[H_i, H_j] = 0$
gilmore-lie-groups_ F01565	165	■ 0.994	$\alpha=\left(\alpha_{\{1\}}, \alpha_{\{2\}}, \ldots, \alpha_{\{l\}}\right)$	$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_l)$	$[\mathbf{H}, E_{\alpha}] = \alpha E_{\alpha}$
gilmore-lie-groups_ F01566	166	■ 1.000	$C_{\{2\}}$	C_2	
gilmore-lie-groups_ F01567	166	■ 1.000	$\mathbf{e}_{\{1\}}, \mathbf{e}_{\{2\}}$	$\mathbf{e}_1, \mathbf{e}_2$	
gilmore-lie-groups_ F01568	166	■ 1.000	$\alpha_{\{1\}}, \alpha_{\{2\}}$	α_1, α_2	
gilmore-lie-groups_ F01569	166	■ 1.000	$\mathbf{e}_{\{1\}}$	$\mathbf{e}_1$	these roots by $\sum \alpha \cdot \alpha = 2$ so that
gilmore-lie-groups_ F01570	166	■ 1.000	$\mathbf{e}_{\{2\}}$	$\mathbf{e}_2$	these roots by $\sum \alpha \cdot \alpha = 2$ so that
gilmore-lie-groups_ F01571	166	■ 1.000	$\pm 2 \mathbf{e}_{\{1\}}, \pm 2 \mathbf{e}_{\{2\}}, \pm \mathbf{e}_{\{1\}} \pm \mathbf{e}_{\{2\}}$	$\pm 2 \mathbf{e}_1, \pm 2 \mathbf{e}_2, \pm \mathbf{e}_1 \pm \mathbf{e}_2$	
gilmore-lie-groups_ F01572	166	■ 0.986	$H_{\{i\}}, i=1,2$	$H_i, i = 1, 2$	$(\mathbf{e}_i, \mathbf{e}_i) = \frac{1}{\gamma} \delta_{ii}$ (9.48)
gilmore-lie-groups_ F01573	166	■ 1.000	$\sum \alpha \cdot \alpha=2$	$\sum \alpha \cdot \alpha = 2$	

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gilmore-lie-groups_ F01574	167	■ 0.608	$\left[H_1, H_2\right]$	$[H_1, H_2]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01575	167	■ 0.608	$\left[\mathbf{H}, E_{\pm 2\mathbf{e}_1}\right]$	$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

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gilmore-lie-groups_ F01576	167	■ 0.608	$(\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1}$	$(\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6}) E_{\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01577	167	■ 0.608	$\left[\mathbf{H}, E_{\pm 2\mathbf{e}_2}\right]$	$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6}) E_{\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

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gilmore-lie-groups_ F01578	167	■ 0.608	$(0, \sqrt{2}) E_{\pm 2}$	$(0, \pm 2/\sqrt{12}) E_{\pm 2e_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2e_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2e_1} \\ [\mathbf{H}, E_{\pm 2e_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2e_2} \\ [\mathbf{H}, E_{\pm e_1 \pm e_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm e_1 \pm e_2} \\ [E_{+2e_1}, E_{-2e_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2e_2}, E_{-2e_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm e_1 \pm e_2}, E_{-(\pm e_1 \pm e_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2e_1}, E_{-(e_1+e_2)}] &= *(1/\sqrt{6}) E_{e_1-e_2} \\ [E_{-e_1+e_2}, E_{-e_1-e_2}] &= (1/\sqrt{6}) E_{-2e_1} \\ [E_{-e_1-e_2}, E_{+2e_1}] &= (1/\sqrt{6}) E_{+e_1-e_2} \\ [E_{-2e_1}, E_{e_1-e_2}] &= (-1/\sqrt{6}) E_{-e_1-e_2} \\ [E_{+e_1-e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6}) E_{+2e_1} \\ [E_{+e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6}) E_{+e_1-e_2} \\ [E_{+2e_2}, E_{-e_1-e_2}] &= *(1/\sqrt{6}) E_{-e_1+e_2} \\ [E_{-e_1-e_2}, E_{+e_1-e_2}] &= (1/\sqrt{6}) E_{-2e_2} \\ [E_{+e_1-e_2}, E_{+2e_2}] &= (1/\sqrt{6}) E_{+e_1+e_2} \\ [E_{-2e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6}) E_{+e_1-e_2} \\ [E_{+e_1+e_2}, E_{-e_1+e_2}] &= (-1/\sqrt{6}) E_{+2e_2} \\ [E_{-e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6}) E_{-e_1-e_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01579	167	■ 0.608	$\left[\mathbf{H}, E_{\pm 1} \pm \mathbf{e}_2\right]$	$[\mathbf{H}, E_{\pm e_1 \pm e_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2e_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2e_1} \\ [\mathbf{H}, E_{\pm 2e_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2e_2} \\ [\mathbf{H}, E_{\pm e_1 \pm e_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm e_1 \pm e_2} \\ [E_{+2e_1}, E_{-2e_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2e_2}, E_{-2e_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm e_1 \pm e_2}, E_{-(\pm e_1 \pm e_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2e_1}, E_{-(e_1+e_2)}] &= *(1/\sqrt{6}) E_{e_1-e_2} \\ [E_{-e_1+e_2}, E_{-e_1-e_2}] &= (1/\sqrt{6}) E_{-2e_1} \\ [E_{-e_1-e_2}, E_{+2e_1}] &= (1/\sqrt{6}) E_{+e_1-e_2} \\ [E_{-2e_1}, E_{e_1-e_2}] &= (-1/\sqrt{6}) E_{-e_1-e_2} \\ [E_{+e_1-e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6}) E_{+2e_1} \\ [E_{+e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6}) E_{+e_1-e_2} \\ [E_{+2e_2}, E_{-e_1-e_2}] &= *(1/\sqrt{6}) E_{-e_1+e_2} \\ [E_{-e_1-e_2}, E_{+e_1-e_2}] &= (1/\sqrt{6}) E_{-2e_2} \\ [E_{+e_1-e_2}, E_{+2e_2}] &= (1/\sqrt{6}) E_{+e_1+e_2} \\ [E_{-2e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6}) E_{+e_1-e_2} \\ [E_{+e_1+e_2}, E_{-e_1+e_2}] &= (-1/\sqrt{6}) E_{+2e_2} \\ [E_{-e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6}) E_{-e_1-e_2} \end{aligned}$ 9.8 Conclusion

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gilmore-lie-groups_ F01580	167	■ 0.608	$(\pm \sqrt{12}, \pm \sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$	$(\pm \sqrt{12}, \pm \sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01581	167	■ 0.608	$\left[E_{+2 \mathrm{e}_1}, E_{-2 \mathrm{e}_1}\right]$	$[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01582	167	■ 0.608	$(2 / \sqrt{12}) H_1$	$(2/\sqrt{12})H_1$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01583	167	■ 0.608	$\left[E_{+2 \mathrm{e}_{2}}, E_{-2 \mathrm{e}_{2}}\right]$	$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01584	167	■ 0.608	$(2 / \sqrt{12}) H_2$	$(2/\sqrt{12})H_2$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01585	167	■ 0.608	$\left[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}\right], E_{-\left(\pm \mathbf{e}_1 \pm \mathbf{e}_2\right)}$	$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01586	167	■ 0.608	$(1 / \sqrt{12}) \left(\pm H_1 \pm H_2 \right)$	$(1/\sqrt{12}) (\pm H_1 \pm H_2)$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12}) (\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01587	167	■ 0.608	$\left[E_{+2 \mathbf{e}_1}, E_{-\left(\mathbf{e}_1 + \mathbf{e}_2 \right)} \right]$	$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12}) (\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01588	167	■ 0.608	$\ast(1 / \sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2}$	$\ast(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= \ast(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= \ast(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01589	167	■ 0.608	$\left[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2} \right]$	$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= \ast(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= \ast(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01590	167	■ 0.608	$(1 / \sqrt{6}) E_{-2} \mathrm{e}_1$	$(1/\sqrt{6})E_{-2\mathbf{e}_1}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01591	167	■ 0.608	$\left[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}\right]$	$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01592	167	■ 0.608	$(1 / \sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01593	167	■ 0.608	$\left[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}\right]$	$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01594	167	■ 0.608	$(-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2}$	$(-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01595	167	■ 0.608	$\left[E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}\right]$	$[E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01596	167	■ 0.608	$(-1/\sqrt{6})E_{+2}\mathrm{e}_1$	$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01597	167	■ 0.608	$\left[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}\right]$	$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01598	167	■ 0.608	$(-1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2}$	$(-1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6}) E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6}) E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01599	167	■ 0.608	$\left[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2} \right]$	$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12}) E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12}) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12}) H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12}) H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6}) E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6}) E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6}) E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6}) E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01600	167	■ 0.608	$\ast(1 / \sqrt{6}) E_{-\mathbf{e}_1+\mathbf{e}_2}$	$\ast(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= \ast(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= \ast(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01601	167	■ 0.608	$\left[E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}\right]$	$[E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= \ast(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= \ast(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

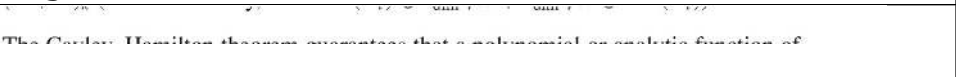
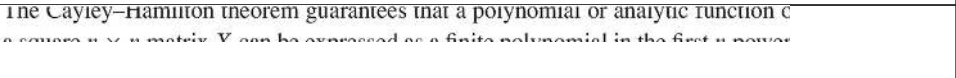
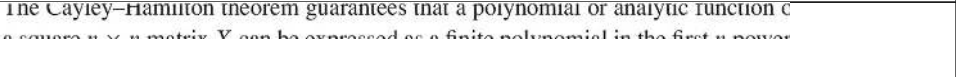
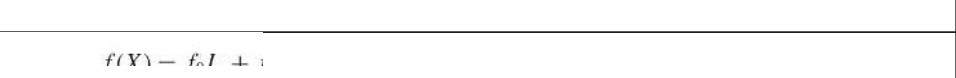
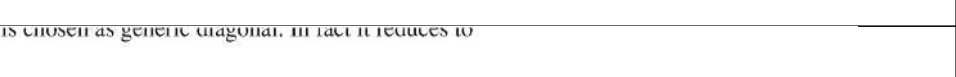
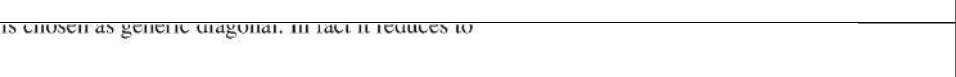
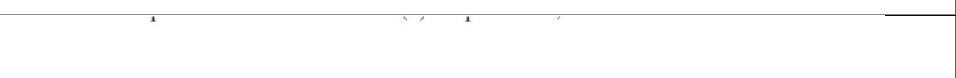
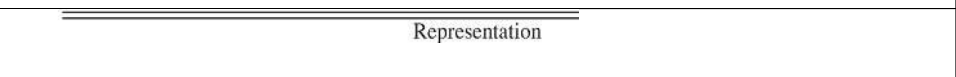
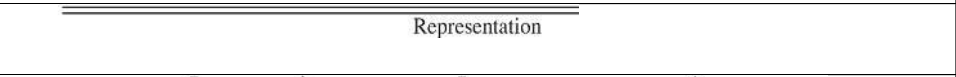
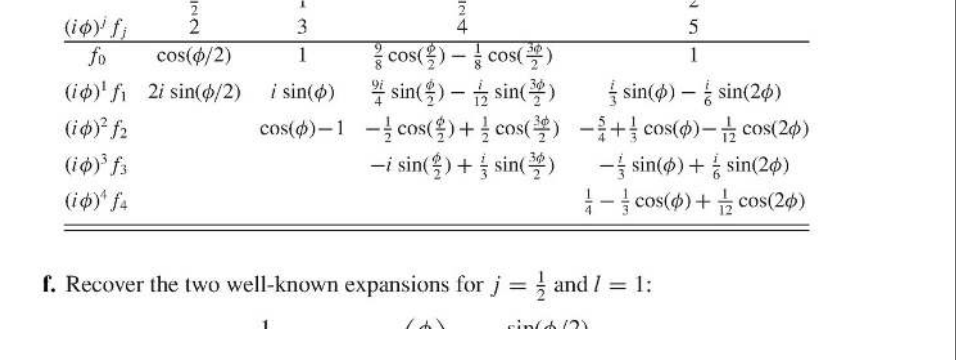
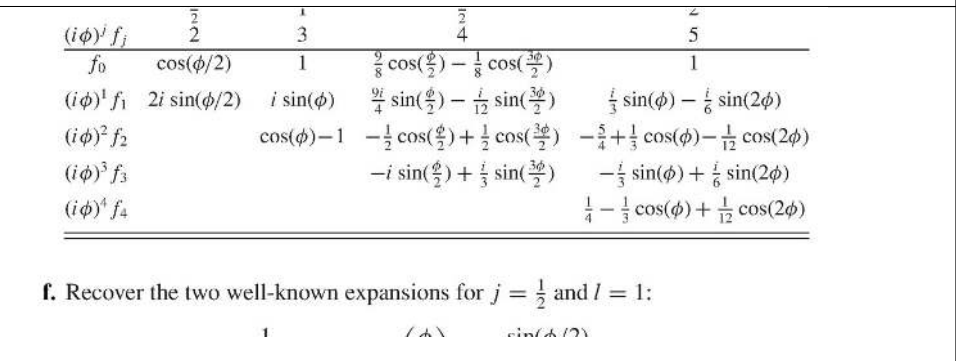
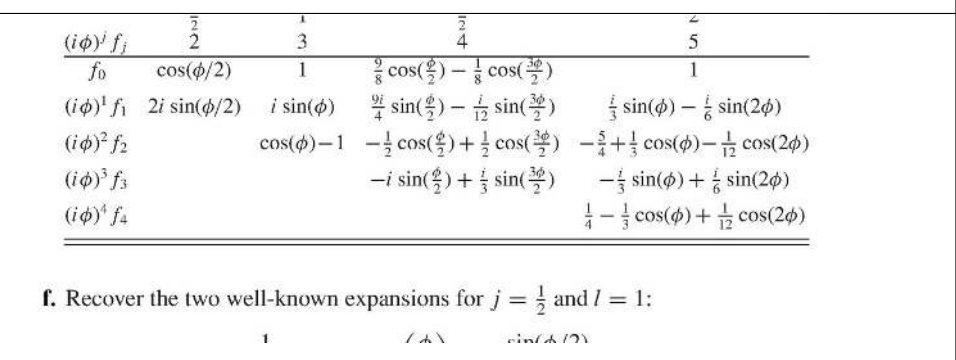
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01602	167	■ 0.608	$(1 / \sqrt{6}) E_{-2} \mathrm{e}_{\{2\}}$	$(1/\sqrt{6})E_{-2\mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01603	167	■ 0.608	$\left[E_{+\mathbf{e}_{\{1\}} - \mathbf{e}_{\{2\}}}, E_{+2\mathbf{e}_{\{2\}}}\right]$	$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01604	167	■ 0.608	$(1 / \sqrt{6}) E_{+\mathbf{e}_1+\mathbf{e}_2}$	$(1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01605	167	■ 0.608	$\left[E_{-2 \mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}+\mathbf{e}_2\right]$	$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01606	167	■ 0.608	$\left[E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}\right]$	$[E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion
gilmore-lie-groups_ F01607	167	■ 0.608	$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$	$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$ 9.8 Conclusion

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01608	167	0.608	$\left[E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}\right]$	$[E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	$\begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned}$
gilmore-lie-groups_ F01609	168	1.000	$\mathfrak{sl}(2; \mathbb{C})$	$\mathfrak{sl}(2; \mathbb{C})$	extension $\mathfrak{sl}(2, \mathbb{C})$, root spaces easily summarize relations of these complex Lie algebras. Root spaces also summarize the commutation relations for the various
gilmore-lie-groups_ F01610	168	1.000	α_i	α_i	will be imaginary. However, determining the real subalgebras of a complex Lie
gilmore-lie-groups_ F01611	168	1.000	$\frac{1}{2}\{a^\dagger, a\}$	$\frac{1}{2}\{a^\dagger, a\}$	$a^\dagger, a^\dagger, I, a, a^\dagger$. Determine the secular equation for this matrix. Determine the rank of this Lie algebra.
gilmore-lie-groups_ F01612	168	1.000	$a^{\dagger^2}, a^\dagger, I, a, a^2$	$a^{\dagger^2}, a^\dagger, I, a, a^2$	of this Lie algebra.
gilmore-lie-groups_ F01613	168	1.000	$X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$	$X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$	3. The Lie algebra $\mathfrak{su}(4)$ has a 4×4 defining matrix representation and a 15×15 regular
gilmore-lie-groups_ F01614	168	1.000	$\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$	$\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$	3. The Lie algebra $\mathfrak{su}(4)$ has a 4×4 defining matrix representation and a 15×15 regular
gilmore-lie-groups_ F01615	168	0.980	$\mathcal{C}'_2 = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$	$\mathcal{C}'_2 = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$	matrix representation. Show th
gilmore-lie-groups_ F01616	168	0.994	$\mathfrak{su}(4)$	$\mathfrak{su}(4)$	between the secular equation for the regular representation and the secular equation
gilmore-lie-groups_ F01617	168	1.000	$\mathfrak{so}(2n+1)$	$\mathfrak{so}(2n+1)$	Compare your results with Fig. 9.3.
gilmore-lie-groups_ F01618	168	1.000	$2, 4, \dots, 2n$	$2, 4, \dots, 2n$	Compare your results with Fig. 9.3.
gilmore-lie-groups_ F01619	168	1.000	$\mathfrak{so}(2n)$	$\mathfrak{so}(2n)$	5 In Chapter 11 we will show that $\mathfrak{su}(4) = \mathfrak{so}(6)$. Both Lie algebras have invariant
gilmore-lie-groups_ F01620	168	0.600	$\mathcal{C}'_n = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$	$\mathcal{C}'_n = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$	operators of degree $2n$, it is sufficient to consider the isomorphism between these Lie algebras and their invariant operators.
gilmore-lie-groups_ F01621	168	0.600	$\mathcal{C}'_2$	$\mathcal{C}'_2$	operators of degree $2n$, it is sufficient to consider the isomorphism between these Lie algebras and their invariant operators.
gilmore-lie-groups_ F01622	168	0.600	$\mathcal{C}'_3$	$\mathcal{C}'_3$	operators of degree $2n$, it is sufficient to consider the isomorphism between these Lie algebras and their invariant operators.
gilmore-lie-groups_ F01623	168	0.600	$\mathfrak{so}(6)$	$\mathfrak{so}(6)$	operators of degree $2n$, it is sufficient to consider the isomorphism between these Lie algebras and their invariant operators.
gilmore-lie-groups_ F01624	168	1.000	$\mathfrak{su}(4) = \mathfrak{so}(6)$	$\mathfrak{su}(4) = \mathfrak{so}(6)$	two modes. This algebra is ten dimensional. It contains the four operators $a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$

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gilmore-lie-groups_F01648	169	■ 0.919	$2^{\{n-1\}} n!$	$2^{n-1}n!$	
gilmore-lie-groups_F01649	169	■ 0.919	$B_{\{n\}}$	B_n	
gilmore-lie-groups_F01650	169	■ 0.919	$C_{\{n\}}$	C_n	
gilmore-lie-groups_F01651	169	■ 0.919	$2^{\{n\}} n!$	$2^n n!$	
gilmore-lie-groups_F01652	170	■ 1.000	$f_{\{r\}}(\backslash Gamma)$	$f_r(\Gamma)$	
gilmore-lie-groups_F01653	170	■ 1.000	$\backslash mathcal{C}^{\{r\}}$	$\mathcal{C}^r$	
gilmore-lie-groups_F01654	170	■ 1.000	$\backslash mathfrak{s o}(2 n), \backslash mathfrak{s o}(2 n+1), \backslash mathfrak{s p}(2 n)$	$\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$	reality restrictions are imposed on the coefficients $a^{\pm}$. a. What are the conditions on a^{ij} for each matrix Lie algebra?
gilmore-lie-groups_F01655	170	■ 1.000	$\backslash sum_{i j} a^{\{i j\}} M_{i j}$	$\sum_{ij} a^{ij} M_{ij}$	reality restrictions are imposed on the coefficients $a^{\pm}$. a. What are the conditions on a^{ij} for each matrix Lie algebra?
gilmore-lie-groups_F01656	170	■ 1.000	$\backslash mathfrak{s o}(2 n), \backslash mathfrak{s p}(2 n)$	$\mathfrak{so}(2n), \mathfrak{sp}(2n)$	these matrix Lie algebras.
gilmore-lie-groups_F01657	170	■ 1.000	$(2 n+1) \backslash times(2 n+1)$	$(2n+1) \times (2n+1)$	these matrix Lie algebras.
gilmore-lie-groups_F01658	170	■ 1.000	$a^{\{i j\}}$	a^{ij}	c. Show that all odd coefficients $\phi_l(a^{ij})$ vanish for ea
gilmore-lie-groups_F01659	170	■ 1.000	$\backslash phi_{\{r\}}\backslash left(a^{\{i j\}}\backslash right)$	$\phi_r\left(a^{ij}\right)$	$(l=2n, 2n+1, 2n)$.
gilmore-lie-groups_F01660	170	■ 0.999	$\backslash epsilon_{\{i_1\} i_2\} \backslash dots i_{\{l\}}\}$	$\epsilon_{i_1 i_2 \cdots i_l}$	g. Show that $\phi_{2n}(a^{\pm})$ is a perfect square for $\mathfrak{so}(2n)$, write down its square root, show that it is of degree n
gilmore-lie-groups_F01661	170	■ 1.000	$(l=2 n, 2 n+1,2 n)$	$(l=2n, 2n+1, 2n)$	what is its degree?
gilmore-lie-groups_F01662	170	■ 1.000	$\backslash phi_{\{2 n\}}\backslash left(a^{\{i j\}}\backslash right)$	$\phi_{2n}\left(a^{ij}\right)$	
gilmore-lie-groups_F01663	171	■ 0.999	$\backslash operatorname{tr} M^{\{2\}}=-2 \backslash theta \backslash cdot \backslash theta$	$\mathrm{tr} M^2=-2\theta \cdot \theta$	b. Show $\mathrm{tr} \mathcal{M}^2=-2\mathbf{J} \cdot \mathbf{J}$.
gilmore-lie-groups_F01664	171	■ 0.987	$\backslash operatorname{tr} \backslash mathcal{M}^{\{2\}}=-2 \backslash mathbf{J} \backslash cdot \backslash mathbf{J}$	$\mathrm{tr} \mathcal{M}^2=-2\mathbf{J} \cdot \mathbf{J}$	c. Show $[\mathbf{J}, \mathrm{tr} \mathcal{M}^2]=0$.
gilmore-lie-groups_F01665	171	■ 1.000	$\backslash left[\backslash mathbf{J}, \backslash operatorname{tr} \backslash mathcal{M}^{\{2\}} \backslash right]=0$	$[\mathbf{J}, \mathrm{tr} \mathcal{M}^2]=0$	d. Show $\mathrm{tr} \mathcal{M}^{2n+1}=0$ and
gilmore-lie-groups_F01666	171	■ 0.999	$\backslash operatorname{tr} \backslash mathcal{M}^{\{2 n+1\}}=0$	$\mathrm{tr} \mathcal{M}^{2n+1}=0$	Casimir covariants A semisimple Lie algebra has ba
gilmore-lie-groups_F01667	171	■ 0.999	$\backslash operatorname{tr} \backslash mathcal{M}^{\{2 n\}}=(-2)^{\{n\}}(\backslash mathbf{J} \backslash cdot \backslash mathbf{J})^{\{n\}}$	$\mathrm{tr} \mathcal{M}^{2n}=(-2)^n(\mathbf{J} \cdot \mathbf{J})^n$	Casimir covariants A semisimple Lie algebra has ba
gilmore-lie-groups_F01668	171	■ 1.000	$\backslash left[X_{\{i\}}, X_{\{j\}}\backslash right]=C_{\{i j\}}^{\{k\}} X_{\{k\}}$	$[X_i, X_j]=C_{ij}^{\ k} X_k$	and $V^{(1)}$, that carry irreducible representations of this Lie algebra, $X_i \rightarrow \Gamma^{(1)}(X_i)=Y_i$ and $X_i \rightarrow \Gamma^{(2)}(X_i)=Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_F01669	171	■ 1.000	$V^{\{1\}}$	$V^{(1)}$	and $V^{(1)}$, that carry irreducible representations of this Lie algebra, $X_i \rightarrow \Gamma^{(1)}(X_i)=Y_i$ and $X_i \rightarrow \Gamma^{(2)}(X_i)=Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_F01670	171	■ 1.000	$V^{\{2\}}$	$V^{(2)}$	X_i and $X_j \rightarrow X_{\{ij\}}=Z_{\{ij\}}$ Show that the Casimir covariant $g_{\{ij\}}^{\{kl\}}$ commutes with $(Y+Z)_k$ (more accurately, with $\Gamma^{(1)}(X_i) \otimes I_{\dim V(2)}+I_{\dim V(1)} \otimes \Gamma^{(2)}(X_i)$).
gilmore-lie-groups_F01671	171	■ 1.000	$X_{\{i\}} \backslash rightharpoonup \backslash Gamma^{\{1\}}\backslash left(X_{\{i\}}\backslash right)=$	$X_i \rightarrow \Gamma^{(1)}(X_i)=$	X_i and $X_j \rightarrow X_{\{ij\}}=Z_{\{ij\}}$ Show that the Casimir covariant $g_{\{ij\}}^{\{kl\}}$ commutes with $(Y+Z)_k$ (more accurately, with $\Gamma^{(1)}(X_i) \otimes I_{\dim V(2)}+I_{\dim V(1)} \otimes \Gamma^{(2)}(X_i)$).
gilmore-lie-groups_F01672	171	■ 1.000	$Y_{\{i\}}$	Y_i	$X_i \rightarrow \Gamma^{(1)}(X_i)=Y_i$ and $X_i \rightarrow \Gamma^{(2)}(X_i)=Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_F01673	171	■ 1.000	$X_{\{i\}} \backslash rightharpoonup \backslash Gamma^{\{2\}}\backslash left(X_{\{i\}}\backslash right)=Z_{\{i\}}$	$X_i \rightarrow \Gamma^{(2)}(X_i)=Z_i$	The Cartan–Hamilton theorem guarantees that a polynomial or analytic function of

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gilmore-lie-groups_ F01674	171	■ 1.000	$g^{\{i\ j\}} Y_{\{i\}} Z_{\{j\}}$	$g^{ij} Y_i Z_j$	
gilmore-lie-groups_ F01675	171	■ 0.857	$(Y+Z)_{\{k\}}$	$(Y + Z)_k$	
gilmore-lie-groups_ F01676	171	■ 0.857	$\Gamma^{\{1\}}(X_{\{i\}}) \otimes I_{\operatorname{dim} V^{\{2\}}} + I_{\operatorname{dim} V^{\{1\}}} \otimes \Gamma^{\{2\}}(X_i)$	$\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$	
gilmore-lie-groups_ F01677	171	■ 1.000	$X^{\{0\}}=I_{\{n\}}$	$X^0 = I_n$	
gilmore-lie-groups_ F01678	171	■ 1.000	$\lambda_{\{i\}}(X)$	$\lambda_i(X)$	
gilmore-lie-groups_ F01679	171	■ 1.000	$\phi_{\{i\}}(X)$	$\phi_i(X)$	
gilmore-lie-groups_ F01680	171	■ 0.967	$e^{\{i\ \phi_{\ J\{z\}}\}}$	$e^{i\phi J_z}$	
gilmore-lie-groups_ F01681	171	■ 0.999	$j=\frac{1}{2},\ 1,\ \frac{3}{2},\ 2$	$j = \frac{1}{2}, 1, \frac{3}{2}, 2$	
gilmore-lie-groups_ F01682	172	■ 1.000	$f_{\{j\}}(\phi)$	$f_j(\phi)$	
gilmore-lie-groups_ F01683	172	■ 1.000	$e^{\{X\}}$	e^X	
gilmore-lie-groups_ F01684	172	■ 1.000	$X \in \mathfrak{su}(2)$	$X \in \mathfrak{su}(2)$	
gilmore-lie-groups_ F01685	172	■ 1.000	$(i\ \phi)^{\{j\}}\ f_{\{j\}}$	$(i\phi)^j f_j$	
gilmore-lie-groups_ F01686	172	■ 1.000	$\frac{1}{2}$	$\frac{1}{2}$	
gilmore-lie-groups_ F01687	172	■ 1.000	$\frac{3}{2}$	$\frac{3}{2}$	

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gilmore-lie-groups_ F01716	175	■0.995	$\backslash \cos ^{2}(\backslash \alpha , \backslash \beta)$	$\cos ^2(\alpha , \beta)$	<table><tr><th>$\cos ^2(\alpha , \beta)$</th><th>$\theta (\alpha , \beta)$</th><th>$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$</th><th>$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$</th><th>$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi }{2} \pm \frac{\pi }{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{4}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{6}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{6}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>0</td><td>$\frac{\pi }{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos ^2(\alpha , \beta)$	$\theta (\alpha , \beta)$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$	1	$\frac{\pi }{2} \pm \frac{\pi }{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{4}$	± 1	± 3	3^{+1}	$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 1	± 2	2^{+1}	0	$\frac{\pi }{2}$	0	0	–
$\cos ^2(\alpha , \beta)$	$\theta (\alpha , \beta)$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$																																				
1	$\frac{\pi }{2} \pm \frac{\pi }{2}$	± 2	± 2	1																																				
$\frac{3}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{3}$	± 3	± 1	3^{-1}																																				
$\frac{2}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{4}$	± 1	± 3	3^{+1}																																				
$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 2	± 1	2^{-1}																																				
$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 1	± 2	2^{+1}																																				
0	$\frac{\pi }{2}$	0	0	–																																				
gilmore-lie-groups_ F01717	175	■0.995	$\backslash \theta (\backslash \alpha , \backslash \beta)$	$\theta (\alpha , \beta)$	<table><tr><th>$\cos ^2(\alpha , \beta)$</th><th>$\theta (\alpha , \beta)$</th><th>$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$</th><th>$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$</th><th>$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi }{2} \pm \frac{\pi }{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{4}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{6}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{6}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>0</td><td>$\frac{\pi }{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos ^2(\alpha , \beta)$	$\theta (\alpha , \beta)$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$	1	$\frac{\pi }{2} \pm \frac{\pi }{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{4}$	± 1	± 3	3^{+1}	$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 1	± 2	2^{+1}	0	$\frac{\pi }{2}$	0	0	–
$\cos ^2(\alpha , \beta)$	$\theta (\alpha , \beta)$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$																																				
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gilmore-lie-groups_ F01718	175	■0.995	$n=\frac{2 \ \alpha \ \backslash \mathrm{cdot} \ \backslash \beta \mathrm{a}}{\backslash \alpha \ \backslash \mathrm{cdot} \ \backslash \alpha \mathrm{a}}$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	<table><tr><th>$\cos ^2(\alpha , \beta)$</th><th>$\theta (\alpha , \beta)$</th><th>$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$</th><th>$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$</th><th>$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi }{2} \pm \frac{\pi }{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{4}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{6}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{6}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>0</td><td>$\frac{\pi }{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos ^2(\alpha , \beta)$	$\theta (\alpha , \beta)$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$	1	$\frac{\pi }{2} \pm \frac{\pi }{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{4}$	± 1	± 3	3^{+1}	$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 1	± 2	2^{+1}	0	$\frac{\pi }{2}$	0	0	–
$\cos ^2(\alpha , \beta)$	$\theta (\alpha , \beta)$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$																																				
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gilmore-lie-groups_ F01719	175	■0.995	$n^{\backslash \mathrm{prime}}=\frac{2 \ \alpha \ \backslash \mathrm{cdot} \ \backslash \beta \mathrm{a}}{\backslash \beta \mathrm{a} \ \backslash \mathrm{cdot} \ \backslash \beta \mathrm{a}}$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	<table><tr><th>$\cos ^2(\alpha , \beta)$</th><th>$\theta (\alpha , \beta)$</th><th>$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$</th><th>$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$</th><th>$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi }{2} \pm \frac{\pi }{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{4}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{6}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi }{2} \pm \frac{\pi }{6}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>0</td><td>$\frac{\pi }{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos ^2(\alpha , \beta)$	$\theta (\alpha , \beta)$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$	1	$\frac{\pi }{2} \pm \frac{\pi }{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{4}$	± 1	± 3	3^{+1}	$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi }{2} \pm \frac{\pi }{6}$	± 1	± 2	2^{+1}	0	$\frac{\pi }{2}$	0	0	–
$\cos ^2(\alpha , \beta)$	$\theta (\alpha , \beta)$	$n = \frac{2\alpha \cdot \beta }{\alpha \cdot \alpha }$	$n' = \frac{2\alpha \cdot \beta }{\beta \cdot \beta }$	$\frac{\alpha \cdot \alpha }{\beta \cdot \beta } = \frac{n'}{n}$																																				
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01720	175	■ 0.995	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n^{\prime}}{n}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n^{\prime}}{n}$	
gilmore-lie-groups_ F01721	175	■ 0.995	$\frac{\pi}{2} \pm \frac{\pi}{2}$	$\frac{\pi}{2} \pm \frac{\pi}{2}$	
gilmore-lie-groups_ F01722	175	■ 0.995	± 2	± 2	
gilmore-lie-groups_ F01723	175	■ 0.995	$\frac{\pi}{2} \pm \frac{\pi}{3}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																			
gilmore-lie-groups_ F01724	175	■ 0.995	\pm 3	±3	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>±2</td><td>±2</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>±3</td><td>±1</td><td>3^{-1}</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>±1</td><td>±3</td><td>3^{+1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>±2</td><td>±1</td><td>2^{-1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>±1</td><td>±2</td><td>2^{+1}</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	±2	±2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	±3	±1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	±1	±3	3^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	±2	±1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	±1	±2	2^{+1}	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																				
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gilmore-lie-groups_ F01725	175	■ 0.995	3^{-1}	3^{-1}	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>±2</td><td>±2</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>±3</td><td>±1</td><td>3^{-1}</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>±1</td><td>±3</td><td>3^{+1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>±2</td><td>±1</td><td>2^{-1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>±1</td><td>±2</td><td>2^{+1}</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	±2	±2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	±3	±1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	±1	±3	3^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	±2	±1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	±1	±2	2^{+1}	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																				
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gilmore-lie-groups_ F01726	175	■ 0.995	3^{+1}	3^{+1}	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>±2</td><td>±2</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>±3</td><td>±1</td><td>3^{-1}</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>±1</td><td>±3</td><td>3^{+1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>±2</td><td>±1</td><td>2^{-1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>±1</td><td>±2</td><td>2^{+1}</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	±2	±2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	±3	±1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	±1	±3	3^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	±2	±1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	±1	±2	2^{+1}	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																				
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gilmore-lie-groups_ F01727	175	■ 0.995	\frac{2}{4}	$\frac{2}{4}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>±2</td><td>±2</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>±3</td><td>±1</td><td>3^{-1}</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>±1</td><td>±3</td><td>3^{+1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>±2</td><td>±1</td><td>2^{-1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>±1</td><td>±2</td><td>2^{+1}</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	±2	±2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	±3	±1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	±1	±3	3^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	±2	±1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	±1	±2	2^{+1}	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																				
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gilmore-lie-groups_ F01728	175	■ 0.995	$\frac{\pi}{2} \pm \frac{\pi}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																									
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gilmore-lie-groups_ F01729	175	■ 0.995	2^{-1}	2^{-1}	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																									
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gilmore-lie-groups_ F01730	175	■ 0.995	2^{+1}	2^{+1}	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
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gilmore-lie-groups_ F01731	175	■ 0.995	$\frac{1}{4}$	$\frac{1}{4}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																									
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gilmore-lie-groups_ F01732	175	■ 0.995	$\frac{\pi}{2} \pm \frac{\pi}{6}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	$ \begin{array}{ccccc} \cos^2(\alpha, \beta) & \theta(\alpha, \beta) & n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha} & n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta} & \frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n} \\ \hline 1 & \frac{\pi}{2} \pm \frac{\pi}{2} & \pm 2 & \pm 2 & 1 \\ \frac{3}{4} & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 1 & \pm 3 & 3^{+1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 2 & \pm 1 & 2^{-1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 2 & 2^{+1} \\ 0 & \frac{\pi}{2} & 0 & 0 & - \\ \hline \hline \end{array} $
gilmore-lie-groups_ F01733	175	■ 0.995	$\frac{\pi}{2}$	$\frac{\pi}{2}$	$ \begin{array}{ccccc} \cos^2(\alpha, \beta) & \theta(\alpha, \beta) & n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha} & n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta} & \frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n} \\ \hline 1 & \frac{\pi}{2} \pm \frac{\pi}{2} & \pm 2 & \pm 2 & 1 \\ \frac{3}{4} & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 1 & \pm 3 & 3^{+1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 2 & \pm 1 & 2^{-1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 2 & 2^{+1} \\ 0 & \frac{\pi}{2} & 0 & 0 & - \\ \hline \hline \end{array} $
gilmore-lie-groups_ F01734	175	■ 0.998	$(l-1)$	$(l-1)$	$A_{l-1} + \mathbf{e}_i - \mathbf{e}_j \quad 1 \leq i \neq j \leq l \quad l-1 \geq 1$
gilmore-lie-groups_ F01735	175	■ 0.998	$D_{\{l\}}$	D_l	$A_1 = B_1 = C_1$
gilmore-lie-groups_ F01736	175	■ 0.998	$\pm(\mathbf{e}_i + \mathbf{e}_j)$	$\pm(\mathbf{e}_i + \mathbf{e}_j)$	$A_1 = B_1 = C_1$
gilmore-lie-groups_ F01737	175	■ 0.998	$A_{\{l-1\}}$	A_{l-1}	$A_1 = B_1 = C_1$
gilmore-lie-groups_ F01738	175	■ 0.998	$B_{\{l\}}, C_{\{l\}}$	B_l, C_l	$A_1 = B_1 = C_1$
gilmore-lie-groups_ F01739	175	■ 1.000	$\pm \mathbf{e}_i, \pm 2\mathbf{e}_i$	$\pm \mathbf{e}_i, \pm 2\mathbf{e}_i$	$B_2 = C_2 \quad (10.7)$
gilmore-lie-groups_ F01740	175	■ 1.000	$A_{\{l-1\}}, D_{\{l\}}, B_{\{l\}}, C_{\{l\}}$	A_{l-1}, D_l, B_l, C_l	$B_2 = C_2 \quad (10.7)$
gilmore-lie-groups_ F01741	176	■ 1.000	$A_{\{3\}}, D_{\{3\}}$	A_3, D_3	
gilmore-lie-groups_ F01742	176	■ 1.000	$B_{\{3\}}, C_{\{3\}}$	B_3, C_3	
gilmore-lie-groups_ F01743	178	■ 1.000	$A_{\{1\}}$	A_1	
gilmore-lie-groups_ F01744	178	■ 1.000	$SL(l; \mathbb{C})$	$SL(l; \mathbb{C})$	Remark 2. The root space D_2 consists of two orthogonal sets of roots :
gilmore-lie-groups_ F01745	178	■ 0.997	$\pm(\mathbf{e}_1 - \mathbf{e}_2)$	$\pm(\mathbf{e}_1 - \mathbf{e}_2)$	describe semisimple Lie algebras. Root subspaces that do not have an orthogonal
gilmore-lie-groups_ F01746	178	■ 1.000	$\pm(\mathbf{e}_1 + \mathbf{e}_2)$	$\pm(\mathbf{e}_1 + \mathbf{e}_2)$	decomposition describe simple Lie algebras. Complete reducibility of the regu-

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gilmore-lie-groups_ F01747	178	■ 0.964	$B_{\{2\}}$	B_2	quaternion by a complex 2×2 matrix. Therefore we should expect $SO(5)$ to
gilmore-lie-groups_ F01748	178	■ 0.513	$S_{O(5)}$	$SO(5)$	have a four-dimensional “spinor” representation based on $U(2; \mathbb{Q})$ in the same way
gilmore-lie-groups_ F01749	178	■ 0.513	$Sp(2) = U(2; \mathbb{Q})$	$Sp(2) = U(2; \mathbb{Q})$	have a four-dimensional “spinor” representation based on $U(2; \mathbb{Q})$ in the same way
gilmore-lie-groups_ F01750	178	■ 1.000	$U(2; \mathbb{Q})$	$U(2; \mathbb{Q})$	Remark 4. In the building-up construction the roots in each root space diagram are
gilmore-lie-groups_ F01751	178	■ 0.975	$S_{O(3)} \left(B_{\{1\}} \right)$	$SO(3) (B_1)$	explicitly constructed. What is not immediately obvious is that there are no more
gilmore-lie-groups_ F01752	178	■ 0.975	$U(1; \mathbb{Q})$	$U(1; \mathbb{Q})$	explicitly constructed. What is not immediately obvious is that there are no more
gilmore-lie-groups_ F01753	178	■ 0.995	$\operatorname{SU}(2) \left(A_{\{1\}} \right)$	$SU(2) (A_1)$	simpl
gilmore-lie-groups_ F01754	179	■ 1.000	$\alpha_{\{1\}} = \mathbf{e}_{\{1\}} - \mathbf{e}_{\{2\}}$	$\alpha_1 = \mathbf{e}_1 - \mathbf{e}_2$	
gilmore-lie-groups_ F01755	179	■ 1.000	$\alpha_{\{2\}} =$	$\alpha_2 =$	
gilmore-lie-groups_ F01756	179	■ 1.000	$-\mathbf{e}_{\{1\}} + 2\mathbf{e}_{\{2\}} - \mathbf{e}_{\{3\}}$	$-\mathbf{e}_1 + 2\mathbf{e}_2 - \mathbf{e}_3$	
gilmore-lie-groups_ F01757	179	■ 1.000	$\mathbf{R} = \mathbf{e}_{\{1\}} + \mathbf{e}_{\{2\}} + \mathbf{e}_{\{3\}}$	$\mathbf{R} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$	
gilmore-lie-groups_ F01758	180	■ 1.000	$G_{\{2\}} \oplus B_{\{3\}}$	$G_2 \oplus B_3$	
gilmore-lie-groups_ F01759	180	■ 0.508	$\alpha_{\{1\}}$	α_1	$\begin{array}{ccccccc} A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \end{array}$ Inner products among the fundamental roots are summarized conv
gilmore-lie-groups_ F01760	180	■ 0.508	$\alpha_{\{2\}}$	α_2	$\begin{array}{ccccccc} A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \end{array}$ Inner products among the fundamental roots are summarized conv
gilmore-lie-groups_ F01761	180	■ 0.508	$\alpha_{\{3\}}$	α_3	$\begin{array}{ccccccc} A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \end{array}$ Inner products among the fundamental roots are summarized conv
gilmore-lie-groups_ F01762	180	■ 0.508	$\alpha_{\{l-1\}}$	α_{l-1}	$\begin{array}{ccccccc} A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \end{array}$ Inner products among the fundamental roots are summarized conv

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups-F01771	180	0.508	$C_{\{l\}}$	C_l	$\begin{array}{ccccccc} A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \end{array}$ Inner products among the fundamental roots are summarized conv
gilmore-lie-groups-F01772	180	0.508	$2 \mathbf{e}_{\{l\}}$	$2\mathbf{e}_l$	$\begin{array}{ccccccc} A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \end{array}$ Inner products among the fundamental roots are summarized conv
gilmore-lie-groups-F01773	180	1.000	$n_{\{i \ j\}}$	n_{ij}	is called a Dynkin diagram . The Dynkin diagram for the semisimple Lie algebra represented by orthogonal root spaces $G_2 \oplus B_2$ is shown in Fig. 10.5
gilmore-lie-groups-F01774	180	1.000	$0, 1, 2$	$0, 1, 2$	is called a Dynkin diagram . The Dynkin diagram for the semisimple Lie algebra represented by orthogonal root spaces $G_2 \oplus B_2$ is shown in Fig. 10.5
gilmore-lie-groups-F01775	180	1.000	$G_{\{2\}} + B_{\{3\}}$	$G_2 + B_3$	fundamental roots (3, 1 for G_2) are indicated over the root symbol, by
gilmore-lie-groups-F01776	180	0.665	$\mathbf{v}_{\{i\}}$	$\mathbf{v}_i$	These two observations are now used to list a set of properties that constrain the
gilmore-lie-groups-F01777	180	0.665	$\mathbf{u}$	$\mathbf{u}$	These two observations are now used to list a set of properties that constrain the
gilmore-lie-groups-F01778	180	1.000	$\mathbf{u} \cdot \mathbf{v}_{\{i\}}$	$\mathbf{u} \cdot \mathbf{v}_i$	allowed Dynkin diagrams ever more tightly.
gilmore-lie-groups-F01779	181	1.000	$\mathbf{u}_{\{i\}} = \alpha_{\{i\}} / \alpha_{\{i\}} $	$\mathbf{u}_i = \alpha_i / \alpha_i $	
gilmore-lie-groups-F01780	181	1.000	$2 \mathbf{u}_{\{i\}} \cdot \mathbf{u}_{\{j\}} \leq -1$	$2\mathbf{u}_i \cdot \mathbf{u}_j \leq -1$	$\sum (\mathbf{u} \cdot \mathbf{v}_i)^2 = \sum n_i / 4 < 1$
gilmore-lie-groups-F01781	181	0.979	(B, C, F)	(B, C, F)	$i=1 \qquad \qquad \qquad j=1$
gilmore-lie-groups-F01782	182	1.000	$\mathbf{u}_{\{i\}}, \mathbf{v}_{\{j\}}$	$\mathbf{u}_i, \mathbf{v}_j$	where $\mathbf{u}_i, \mathbf{v}_j$ are unit vectors $\mathbf{u}_i = \alpha_i / \alpha_i $ and $\mathbf{v}_i = \alpha_j / \alpha_j $, can be transformed to
gilmore-lie-groups-F01783	182	1.000	$\mathbf{v}_{\{i\}} = \alpha_{\{j\}} / \alpha_{\{j\}} $	$\mathbf{v}_i = \alpha_j / \alpha_j $	where $\mathbf{u}_i, \mathbf{v}_j$ are unit vectors $\mathbf{u}_i = \alpha_i / \alpha_i $ and $\mathbf{v}_i = \alpha_j / \alpha_j $, can be transformed to
gilmore-lie-groups-F01784	182	0.985	$p \geq q$	$p \geq q$	This has the following solutions with $p \geq$
gilmore-lie-groups-F01785	182	0.657	(D, E)	(D, E)	as in Eq. (10.16) leads to the inequality
gilmore-lie-groups-F01786	182	0.997	$p \geq q \geq r \geq 2$	$p \geq q \geq r \geq 2$	$\begin{array}{ccc} p & q & r \\ \hline \end{array}$ Root space
gilmore-lie-groups-F01787	182	0.999	$D_{\{p+2\}}$	D_{p+2}	$\begin{array}{ccc} 3 & 3 & 2 \\ 4 & 3 & 2 \\ 5 & 3 & 2 \end{array} \quad \begin{array}{c} E_6 \\ E_7 \\ E_8 \end{array}$ grams are summarized in 1 : root spaces. Each root spa
gilmore-lie-groups-F01788	182	0.999	$E_{\{6\}}$	E_6	$\begin{array}{ccc} 3 & 3 & 2 \\ 4 & 3 & 2 \\ 5 & 3 & 2 \end{array} \quad \begin{array}{c} E_6 \\ E_7 \\ E_8 \end{array}$ grams are summarized in 1 : root spaces. Each root spa

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01789	182	■ 0.999	$E_{\{7\}}$	E_7	$\begin{array}{ccc} 3 & 3 & 2 \\ 4 & 3 & 2 \\ 5 & 3 & 2 \end{array} \quad \begin{array}{c} E_6 \\ E_7 \\ E_8 \end{array}$ grams are summarized in 1 : root spaces. Each root spa
gilmore-lie-groups_ F01790	182	■ 0.999	$E_{\{8\}}$	E_8	$\begin{array}{ccc} 3 & 3 & 2 \\ 4 & 3 & 2 \\ 5 & 3 & 2 \end{array} \quad \begin{array}{c} E_6 \\ E_7 \\ E_8 \end{array}$ grams are summarized in 1 : root spaces. Each root spa
gilmore-lie-groups_ F01791	182	■ 1.000	$l-1$	$l-1$	
gilmore-lie-groups_ F01792	184	■ 1.000	$f_{\{1\}}^{\dagger} f_{\{1\}}^{\dagger} =$	$f_1^{\dagger} f_1^{\dagger} =$	
gilmore-lie-groups_ F01793	184	■ 1.000	$U(l)$	$U(l)$	
gilmore-lie-groups_ F01794	184	■ 1.000	$b_{\{i\}}^{\dagger} b_{\{i\}}$	$b_i^{\dagger} b_i$	$\begin{array}{ccc} H_i & b_i^{\dagger} b_i & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \end{array} \quad \begin{array}{c} f_i^{\dagger} f_i \\ f_i^{\dagger} f_j \end{array}$ that the following identifications are appropriate for the eigenopera paces C_l and D_l:
gilmore-lie-groups_ F01795	184	■ 1.000	$x^{\{i\}} \partial_{\{i\}}$	$x^i \partial_i$	$\begin{array}{ccc} H_i & b_i^{\dagger} b_i & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \end{array} \quad \begin{array}{c} f_i^{\dagger} f_i \\ f_i^{\dagger} f_j \end{array}$ that the following identifications are appropriate for the eigenopera paces C_l and D_l:
gilmore-lie-groups_ F01796	184	■ 1.000	$f_{\{i\}}^{\dagger} f_{\{i\}}$	$f_i^{\dagger} f_i$	$\begin{array}{ccc} H_i & b_i^{\dagger} b_i & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \end{array} \quad \begin{array}{c} f_i^{\dagger} f_i \\ f_i^{\dagger} f_j \end{array}$ that the following identifications are appropriate for the eigenopera paces C_l and D_l:
gilmore-lie-groups_ F01797	184	■ 1.000	$E_{\{+\mathbf{e}_{\{i\}} - \mathbf{e}_{\{j\}}\}}$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$\begin{array}{ccc} H_i & b_i^{\dagger} b_i & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \end{array} \quad \begin{array}{c} f_i^{\dagger} f_i \\ f_i^{\dagger} f_j \end{array}$ that the following identifications are appropriate for the eigenopera paces C_l and D_l:
gilmore-lie-groups_ F01798	184	■ 1.000	$x^{\{i\}} \partial_{\{j\}}$	$x^i \partial_j$	$\begin{array}{ccc} H_i & b_i^{\dagger} b_i & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \end{array} \quad \begin{array}{c} f_i^{\dagger} f_i \\ f_i^{\dagger} f_j \end{array}$ that the following identifications are appropriate for the eigenopera paces C_l and D_l:

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01799	184	■ 0.960	$b_{i}^{\dagger} b_{i} + \frac{1}{2}$	$b_i^\dagger b_i + \frac{1}{2}$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{ll} f_i^\dagger f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} \\ f_i^\dagger f_j & x^i \partial_j \\ f_i^\dagger f_j^\dagger & x^i x^j \\ f_i f_j & \partial_i \partial_j \end{array} $
gilmore-lie-groups_ F01800	184	■ 0.960	$f_{i}^{\dagger} f_{i} + \frac{1}{2}$	$f_i^\dagger f_i + \frac{1}{2}$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{ll} f_i^\dagger f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} \\ f_i^\dagger f_j & x^i \partial_j \\ f_i^\dagger f_j^\dagger & x^i x^j \\ f_i f_j & \partial_i \partial_j \end{array} $
gilmore-lie-groups_ F01801	184	■ 0.960	$x^{\mathbf{i}} \partial_{\mathbf{i}} + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{ll} f_i^\dagger f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} \\ f_i^\dagger f_j & x^i \partial_j \\ f_i^\dagger f_j^\dagger & x^i x^j \\ f_i f_j & \partial_i \partial_j \end{array} $
gilmore-lie-groups_ F01802	184	■ 0.960	$E_{+\mathbf{e}_i + \mathbf{e}_j}$	$E_{+\mathbf{e}_i + \mathbf{e}_j}$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{ll} f_i^\dagger f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} \\ f_i^\dagger f_j & x^i \partial_j \\ f_i^\dagger f_j^\dagger & x^i x^j \\ f_i f_j & \partial_i \partial_j \end{array} $

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gilmore-lie-groups_ F01803	184	■ 0.960	$b_{\{i\}^{\dagger}} b_{\{j\}^{\dagger}}$	$b_i^{\dagger} b_j^{\dagger}$	$ \begin{array}{lll} H_i & b_i^{\dagger} b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^{\dagger} b_j^{\dagger} & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^{\dagger} b_i^{\dagger} & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{lll} f_i^{\dagger} f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} & \\ f_i^{\dagger} f_j & x^i \partial_j & \\ f_i^{\dagger} f_j^{\dagger} & x^i x^j & \\ f_i f_j & \partial_i \partial_j & \end{array} $
gilmore-lie-groups_ F01804	184	■ 0.960	$x^{\{i\}} x^{\{j\}}$	$x^i x^j$	$ \begin{array}{lll} H_i & b_i^{\dagger} b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^{\dagger} b_j^{\dagger} & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^{\dagger} b_i^{\dagger} & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{lll} f_i^{\dagger} f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} & \\ f_i^{\dagger} f_j & x^i \partial_j & \\ f_i^{\dagger} f_j^{\dagger} & x^i x^j & \\ f_i f_j & \partial_i \partial_j & \end{array} $
gilmore-lie-groups_ F01805	184	■ 0.960	$E_{\{-\mathbf{e}_i - \mathbf{e}_j\}}$	$E_{-\mathbf{e}_i - \mathbf{e}_j}$	$ \begin{array}{lll} H_i & b_i^{\dagger} b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^{\dagger} b_j^{\dagger} & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^{\dagger} b_i^{\dagger} & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{lll} f_i^{\dagger} f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} & \\ f_i^{\dagger} f_j & x^i \partial_j & \\ f_i^{\dagger} f_j^{\dagger} & x^i x^j & \\ f_i f_j & \partial_i \partial_j & \end{array} $
gilmore-lie-groups_ F01806	184	■ 0.960	$b_{\{i\}} b_{\{j\}}$	$b_i b_j$	$ \begin{array}{lll} H_i & b_i^{\dagger} b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^{\dagger} b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^{\dagger} b_j^{\dagger} & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^{\dagger} b_i^{\dagger} & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{lll} f_i^{\dagger} f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} & \\ f_i^{\dagger} f_j & x^i \partial_j & \\ f_i^{\dagger} f_j^{\dagger} & x^i x^j & \\ f_i f_j & \partial_i \partial_j & \end{array} $

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gilmore-lie-groups_ F01807	184	■ 0.960	$\partial_i \partial_j$	$\partial_i \partial_j$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $
gilmore-lie-groups_ F01808	184	■ 0.960	$E_{+2} \mathrm{e}_i$	$E_{+2\mathbf{e}_i}$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $
gilmore-lie-groups_ F01809	184	■ 0.960	$b_i^{\dagger\dagger} b_i^{\dagger\dagger}$	$b_i^\dagger b_i^\dagger$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $
gilmore-lie-groups_ F01810	184	■ 0.960	$x^i x^i$	$x^i x^i$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $

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gilmore-lie-groups_ F01811	184	■ 0.960	$E_{-2} \ \mathbf{e}_i$	$E_{-2\mathbf{e}_i}$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{lll} f_i^\dagger f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} \\ f_i^\dagger f_j & x^i \partial_j \\ f_i^\dagger f_j^\dagger & x^i x^j \\ f_i f_j & \partial_i \partial_j \end{array} $
gilmore-lie-groups_ F01812	184	■ 0.960	$b_{\{i} \ b_{\{i}$	$b_i b_i$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{lll} f_i^\dagger f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} \\ f_i^\dagger f_j & x^i \partial_j \\ f_i^\dagger f_j^\dagger & x^i x^j \\ f_i f_j & \partial_i \partial_j \end{array} $
gilmore-lie-groups_ F01813	184	■ 0.960	$\partial_{\{i} \ \partial_{\{i}$	$\partial_i \partial_i$	$ \begin{array}{lll} H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j \\ E_{+\mathbf{e}_i + \mathbf{e}_j} & b_i^\dagger b_j^\dagger & x^i x^j \\ E_{-\mathbf{e}_i - \mathbf{e}_j} & b_i b_j & \partial_i \partial_j \\ E_{+2\mathbf{e}_i} & b_i^\dagger b_i^\dagger & x^i x^i \\ E_{-2\mathbf{e}_i} & b_i b_i & \partial_i \partial_i \end{array} $ $ \begin{array}{lll} f_i^\dagger f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} \\ f_i^\dagger f_j & x^i \partial_j \\ f_i^\dagger f_j^\dagger & x^i x^j \\ f_i f_j & \partial_i \partial_j \end{array} $
gilmore-lie-groups_ F01814	185	■ 0.852	$D, \ E$	D, E	Dynkin diagrams of type $(D, \ E)$ to obtain the inequality of Eq. (10.19).
gilmore-lie-groups_ F01815	185	■ 1.000	$1 \leq i, \ j \leq n$	$1 \leq i, j \leq n$	8. A Lie algebra is spanned by n^2 operators of the form $a_i a_j$, with $1 \leq i, j \leq n$. Show that the linear vector space for this algebra can be written as the direct sum of two
gilmore-lie-groups_ F01816	185	■ 1.000	$n=3$	$n = 3$	tensor. Show that the commutation relations are
gilmore-lie-groups_ F01817	185	■ 1.000	$\hat{\mathbf{Q}}$	$\hat{\mathbf{Q}}$	$\hat{N} = \sum a_i^\dagger a_i$
gilmore-lie-groups_ F01818	185	■ 1.000	$\hat{N}$	$\hat{N}$	mathematical terms (commutative invariant subalgebra $\mathcal{W}$, Cartan decomposition of a simple Lie algebra $\mathbf{L} + \hat{\mathbf{O}}$).
gilmore-lie-groups_ F01819	185	■ 0.998	$\mathbf{L} + \hat{\mathbf{Q}}$	$\mathbf{L} + \hat{\mathbf{Q}}$	the discussion of Problem 8 i
gilmore-lie-groups_ F01820	185	■ 0.623	$\mathbf{L}(3 \rightarrow n(n-1) / 2)$	$\mathbf{L}(3 \rightarrow n(n-1)/2)$	$(6 \rightarrow n(n+1)/2)$ and the d_k
gilmore-lie-groups_ F01821	185	■ 0.623	$\mathbf{Q}$	$\mathbf{Q}$	
gilmore-lie-groups_ F01822	185	■ 1.000	$(6 \rightarrow n(n+1) / 2)$	$(6 \rightarrow n(n+1)/2)$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01823	185	■ 1.000	$\hat{N}(3 \rightarrow n)$	$\hat{N}(3 \rightarrow n)$	
gilmore-lie-groups_ F01824	186	■ 1.000	$Z=r^i X_i$	$Z = r^i X_i$	general unless the field is extended from the real to the complex numbers. Allowing
gilmore-lie-groups_ F01825	186	■ 1.000	r^i	r^i	general unless the field is extended from the real to the complex numbers. Allowing
gilmore-lie-groups_ F01826	186	■ 1.000	$\sum \sum r^i \left[R_i{}^j(Z) - \lambda \delta_i{}^j \right] X_j = 0$	$\sum \sum r^i [R_i{}^j(Z) - \lambda \delta_i{}^j] X_j = 0$	Lie algebras. The general operator in such algebras has the form
gilmore-lie-groups_ F01827	186	■ 1.000	h^i	h^i	
gilmore-lie-groups_ F01828	186	■ 1.000	e^{α}	e^{α}	
gilmore-lie-groups_ F01829	187	■ 1.000	A_{n-1}, D_n, B_n, C_n	A_{n-1}, D_n, B_n, C_n	submatrix decomposition: subfield restriction: and field embeddings.
gilmore-lie-groups_ F01830	187	■ 1.000	a_1, a_2, a_3	a_1, a_2, a_3	matrices – the Lie algebra for $SL(2; \mathbb{C})$. This relation is shown in Fig. 11.1.
gilmore-lie-groups_ F01831	187	■ 1.000	$\left(a_1, a_2, a_3 \right)$	(a_1, a_2, a_3)	The complex extension Lie algebra has root space A_1 describing canonical com-
gilmore-lie-groups_ F01832	187	■ 1.000	$\left(b_1, b_2, b_3 \right)$	(b_1, b_2, b_3)	The complex extension Lie algebra has root space A_1 describing canonical com-
gilmore-lie-groups_ F01833	188	■ 0.998	H_i, E_{α}	H_i, E_{α}	$\left[\begin{array}{c} 1 \end{array} \right] H_1$
gilmore-lie-groups_ F01834	188	■ 1.000	$\frac{1}{\sqrt{2}} \left(E_{\alpha} \pm E_{-\alpha} \right)$	$\frac{1}{\sqrt{2}} (E_{\alpha} \pm E_{-\alpha})$	$\left \begin{array}{c} 1 \\ \end{array} \right $
gilmore-lie-groups_ F01835	189	■ 0.826	$H_i, \frac{1}{\sqrt{2}} \left(E_{\alpha} + E_{-\alpha} \right)$	$H_i, \frac{1}{\sqrt{2}} (E_{\alpha} + E_{-\alpha})$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_{\alpha} + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_{\alpha} - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01836	189	■ 0.826	$\frac{1}{\sqrt{2}} \left(E_{\alpha} - E_{-\alpha} \right)$	$\frac{1}{\sqrt{2}} (E_{\alpha} - E_{-\alpha})$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_{\alpha} + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_{\alpha} - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01837	189	■ 0.826	$\alpha \neq 0$	$\alpha \neq 0$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_{\alpha} + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_{\alpha} - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01838	189	■ 1.000	$\frac{1}{\sqrt{2}} \left(E_{\alpha} + E_{-\alpha} \right)$	$\frac{1}{\sqrt{2}} (E_{\alpha} + E_{-\alpha})$	generators H_i and $\frac{1}{\sqrt{2}}(E_{\alpha} + E_{-\alpha})$ span a noncompact subspace. On the other hand, if we restrict the coefficients of H_i and $\frac{1}{\sqrt{2}}(E_{\alpha} - E_{-\alpha})$
gilmore-lie-groups_ F01839	189	■ 0.999	$A_1, \mathfrak{sl}(2; \mathbb{R})$	$A_1, \mathfrak{sl}(2; \mathbb{R})$	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right] \left[\begin{array}{cc} 0 & 1 \\ 0 & 1 \end{array} \right]$
gilmore-lie-groups_ F01840	189	■ 1.000	$h_r, h_i; a_r, a_i; b_r, b_i$	$h_r, h_i; a_r, a_i; b_r, b_i$	
gilmore-lie-groups_ F01841	190	■ 0.850	n_+, n_-	n_+, n_-	It is useful to specify how compact a real form is by specifying its index (n_+, n_-) ,
gilmore-lie-groups_ F01842	190	■ 1.000	n_+	n_+	where n_+ is the dimension of the subspace on which the nonsingular Cartan–
gilmore-lie-groups_ F01843	190	■ 1.000	n_-	n_-	Killing inner product is positive-definite and n_- is the dimension of the subspace
gilmore-lie-groups_ F01844	190	■ 1.000	$\chi = n_+ - n_-$	$\chi = n_+ - n_-$	be abbreviated to a single integer, the character $\chi = n_+ - n_-$, to describe a real form. This is the trace of the normalized Cartan–Killing form. Inspection of (11.3)
gilmore-lie-groups_ F01845	190	■ 1.000	$i H_i, i \frac{1}{\sqrt{2}} \left(E_{\alpha} + E_{-\alpha} \right)$	$i H_i, i \frac{1}{\sqrt{2}} (E_{\alpha} + E_{-\alpha})$	and $\frac{1}{\sqrt{2}}(E_{\alpha} - E_{-\alpha})$. In general, for all real forms the character satisfies the bounds
gilmore-lie-groups_ F01846	190	■ 0.917	$g_{\mu, \nu} = \operatorname{diag}(1, 1, 1, -1)$	$g_{\mu, \nu} = \operatorname{diag}(1, 1, 1, -1)$	$\operatorname{diag}(1, 1, 1, 1)$.
gilmore-lie-groups_ F01847	190	■ 0.870	$x, y, z, i c t$	$x, y, z, i c t$	Since the compact real form can always be constructed easily for a simple Lie algebra (see Eq. (11.6)) it is useful to begin with that form. The compact Lie

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gilmore-lie-groups_ F01848	190	0.870	$g_{\mu,\nu}=$	$g_{\mu,\nu} =$	Since the compact real form can always be constructed easily for a simple Lie algebra (see Ex. (11.6)) it is useful to begin with that form. The compact Lie
gilmore-lie-groups_ F01849	190	0.672	$\operatorname{diag}(1,1,1,1)$	$\operatorname{diag}(1,1,1,1)$	algebra (algebra $\mathfrak{a}$
gilmore-lie-groups_ F01850	190	1.000	$\frac{p}{} \rightarrow \frac{p^{\prime}}{}=i\frac{p}{}$	$p \rightarrow p' = ip$	
gilmore-lie-groups_ F01851	191	1.000	$\frac{g^{\prime}}{}, \frac{h}{}$	g', h	generators. The character of this algebra is
gilmore-lie-groups_ F01852	191	1.000	$\frac{p^{\prime}}{}$	p'	generators. The character of this algebra is
gilmore-lie-groups_ F01853	191	1.000	$T^2=I$	$T^2 = I$	All of the real forms of all of the simple classical (matrix) Lie alge
gilmore-lie-groups_ F01854	192	1.000	$\frac{u}{}(n,\mathbb{F})$	$u(n,\mathbb{F})$	$\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$
gilmore-lie-groups_ F01855	192	0.949	$A_{\{p\}}=-A_{\{p\}}^{\dagger}, A_{\{q\}}=-A_{\{q\}}^{\dagger}$	$A_p = -A_p^\dagger, A_q = -A_q^\dagger$	procedure described in the previous section the off-diagonal block is multiplied by ± 1 . This is equivalent to changing the matrix T to $-T$, that is, changing the real form to the
gilmore-lie-groups_ F01856	192	0.999	$I_{\{p+q\}}$	I_{p+q}	metric $\kappa(p,q)$ that is preserved by $\mathfrak{so}(p,q;\mathbb{R})$, where $p+q=n$. The factor ϵ can be absorbed into the $n \times a$ off-diagonal blocks, so that the noncompact algebra has
gilmore-lie-groups_ F01857	192	0.999	$\frac{u}{}(n;\mathbb{F})$	$u(n;\mathbb{F})$	metric $\kappa(p,q)$ that is preserved by $\mathfrak{so}(p,q;\mathbb{R})$, where $p+q=n$. The factor ϵ can be absorbed into the $n \times a$ off-diagonal blocks, so that the noncompact algebra has
gilmore-lie-groups_ F01858	192	1.000	$I_{\{p,q\}}$	$I_{p,q}$	matrix form
gilmore-lie-groups_ F01859	192	1.000	$\frac{u}{}(p,q;\mathbb{F})$	$u(p,q;\mathbb{F})$	matrix form
gilmore-lie-groups_ F01860	192	0.995	$\mathbb{F}=\mathbb{R},\mathbb{C},\mathbb{Q}$	$\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	$\mathbb{R} \quad \mathfrak{so}(p,q) \quad D, B$
gilmore-lie-groups_ F01861	192	0.995	$(D,B),A,C$	$(D,B), A, C$	$\mathbb{R} \quad \mathfrak{so}(p,q) \quad D, B$
gilmore-lie-groups_ F01862	192	0.998	$\frac{\mathfrak{sp}}{\phantom{\mathfrak{sp}}}(2n;\mathbb{R})$	$\mathfrak{sp}(2n;\mathbb{R})$	
gilmore-lie-groups_ F01863	193	1.000	$\frac{\mathfrak{sp}}{\phantom{\mathfrak{sp}}}(n)=\frac{u}{}(n;\mathbb{Q})$	$\mathfrak{sp}(n) = u(n;\mathbb{Q})$	To make the mapping more comprehensible, it is useful to recall the mappings of
gilmore-lie-groups_ F01864	193	1.000	α,β,γ	α, β, γ	these matrices $\mathfrak{ou}(2n)$, since they form an orthogonal representation of the unitary
gilmore-lie-groups_ F01865	193	0.999	$\frac{\mathfrak{ou}}{\phantom{\mathfrak{ou}}}(2n)$	$\mathfrak{ou}(2n)$	matrices for $\mathfrak{sp}(n)$ is replaced by a set of $2n \times 2n$ complex matrices $\mathfrak{asp}(2n)$, the unitary representation of the symplectic algebra of $2n \times 2n$ matrices:
gilmore-lie-groups_ F01866	193	1.000	$\underline{2n} \times 2n$	$\underline{2n} \times 2n$	unitary representation of the symplectic algebra of $\underline{2n} \times 2n$ matrices.
gilmore-lie-groups_ F01867	194	1.000	R^{2n}	R^{2n}	metric metric in κ^{sym} , it is possible to choose coordinates $p_1, q_1, p_2, q_2, \dots, p_n, q_n$ in this space so that the inner product between two vectors $v^i \in T_x M$ is
gilmore-lie-groups_ F01868	194	1.000	$p_{\{1\}}, q_{\{1\}}, p_{\{2\}}, q_{\{2\}}, \ldots, p_{\{n\}}, q_{\{n\}}$	$p_1, q_1, p_2, q_2, \dots, p_n, q_n$	metric metric in κ^{sym} , it is possible to choose coordinates $p_1, q_1, p_2, q_2, \dots, p_n, q_n$ in this space so that the inner product between two vectors $v^i \in T_x M$ is
gilmore-lie-groups_ F01869	194	1.000	$v_{\{i\}}^{\prime} G_{ij} v_{\{j\}}$	$v_i' G_{ij} v_j$	in this space so that the inner product between two vectors $v_i \in T_x M$ is
gilmore-lie-groups_ F01870	194	0.967	$M \in \operatorname{Sp}(2n;\mathbb{R})$	$M \in \operatorname{Sp}(2n;\mathbb{R})$	of the hamiltonian equations of motion in classical mechanics,
gilmore-lie-groups_ F01871	194	1.000	$M^t G M=G$	$M^t G M = G$	
gilmore-lie-groups_ F01872	194	1.000	$\frac{p}{} \rightarrow i \frac{p}{}$	$p \rightarrow ip$	
gilmore-lie-groups_ F01873	194	1.000	$\frac{\mathfrak{so}}{\phantom{\mathfrak{so}}}^*(2n)$	$\mathfrak{so}^*(2n)$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																											
gilmore-lie-groups_ F01885	195	0.899	$\mathfrak{s} \mathfrak{u}(p, q)$	$\mathfrak{su}(p, q)$	<table><tr><td>Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td></td><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td></td><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td></td><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td>Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td></td><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td>Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td></td><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$		$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$		$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$		$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}			$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n			$\mathfrak{su}^*(2n)$	A_{2n-1}																																												
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gilmore-lie-groups_ F01886	195	0.899	$\mathfrak{s} \mathfrak{p}(p, q)$	$\mathfrak{sp}(p, q)$	<table><tr><td>Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td></td><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td></td><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td></td><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td>Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td></td><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td>Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td></td><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$		$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$		$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$		$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}			$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n			$\mathfrak{su}^*(2n)$	A_{2n-1}																																												
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gilmore-lie-groups_ F01887	195	0.899	$\mathfrak{s} \mathfrak{p}(2n; \mathbb{R})$	$\mathfrak{sp}(2n; \mathbb{R})$	<table><tr><td>Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td></td><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td></td><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td></td><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td>Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td></td><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td>Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td></td><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$		$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$		$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$		$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}			$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n			$\mathfrak{su}^*(2n)$	A_{2n-1}																																												
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gilmore-lie-groups_ F01888	195	0.500	B_1	B_1	<table><tr><td>D_2</td><td>$=$</td><td>A_1</td><td>$+$</td><td>A_1</td></tr><tr><td>$\mathfrak{so}(4)$</td><td>$=$</td><td>$\mathfrak{so}(3)$</td><td>$+$</td><td>$\mathfrak{so}(3)$</td></tr><tr><td>$\mathfrak{so}^*(4)$</td><td>$\sim$</td><td>$\mathfrak{so}(3)$</td><td>$+$</td><td>$\mathfrak{so}(2, 1)$</td></tr><tr><td>$\mathfrak{so}(3, 1)$</td><td>$\sim$</td><td>$\mathfrak{sl}(2; \mathbb{C})$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(2, 2)$</td><td>$\sim$</td><td>$\mathfrak{so}(2, 1)$</td><td>$+$</td><td>$\mathfrak{so}(2, 1)$</td></tr><tr><td>B_2</td><td>$=$</td><td>C_2</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5)$</td><td>$\sim$</td><td>$\mathfrak{sp}(2) = \mathfrak{usp}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4, 1)$</td><td>$\sim$</td><td>$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3, 2)$</td><td>$\sim$</td><td>$\mathfrak{sp}(4; \mathbb{R})$</td><td></td><td></td></tr><tr><td>D_3</td><td>$=$</td><td>A_3</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(6)$</td><td>$\sim$</td><td>$\mathfrak{su}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5, 1)$</td><td>$\sim$</td><td>$\mathfrak{su}^*(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}^*(6)$</td><td>$\sim$</td><td>$\mathfrak{su}(3, 1)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4, 2)$</td><td>$\sim$</td><td>$\mathfrak{su}(2, 2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3, 3)$</td><td>$\sim$</td><td>$\mathfrak{sl}(4; \mathbb{R})$</td><td></td><td></td></tr></table>	D_2	$=$	A_1	$+$	A_1	$\mathfrak{so}(4)$	$=$	$\mathfrak{so}(3)$	$+$	$\mathfrak{so}(3)$	$\mathfrak{so}^*(4)$	$\sim$	$\mathfrak{so}(3)$	$+$	$\mathfrak{so}(2, 1)$	$\mathfrak{so}(3, 1)$	$\sim$	$\mathfrak{sl}(2; \mathbb{C})$			$\mathfrak{so}(2, 2)$	$\sim$	$\mathfrak{so}(2, 1)$	$+$	$\mathfrak{so}(2, 1)$	B_2	$=$	C_2			$\mathfrak{so}(5)$	$\sim$	$\mathfrak{sp}(2) = \mathfrak{usp}(4)$			$\mathfrak{so}(4, 1)$	$\sim$	$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$			$\mathfrak{so}(3, 2)$	$\sim$	$\mathfrak{sp}(4; \mathbb{R})$			D_3	$=$	A_3			$\mathfrak{so}(6)$	$\sim$	$\mathfrak{su}(4)$			$\mathfrak{so}(5, 1)$	$\sim$	$\mathfrak{su}^*(4)$			$\mathfrak{so}^*(6)$	$\sim$	$\mathfrak{su}(3, 1)$			$\mathfrak{so}(4, 2)$	$\sim$	$\mathfrak{su}(2, 2)$			$\mathfrak{so}(3, 3)$	$\sim$	$\mathfrak{sl}(4; \mathbb{R})$		
D_2	$=$	A_1	$+$	A_1																																																																												
$\mathfrak{so}(4)$	$=$	$\mathfrak{so}(3)$	$+$	$\mathfrak{so}(3)$																																																																												
$\mathfrak{so}^*(4)$	$\sim$	$\mathfrak{so}(3)$	$+$	$\mathfrak{so}(2, 1)$																																																																												
$\mathfrak{so}(3, 1)$	$\sim$	$\mathfrak{sl}(2; \mathbb{C})$																																																																														
$\mathfrak{so}(2, 2)$	$\sim$	$\mathfrak{so}(2, 1)$	$+$	$\mathfrak{so}(2, 1)$																																																																												
B_2	$=$	C_2																																																																														
$\mathfrak{so}(5)$	$\sim$	$\mathfrak{sp}(2) = \mathfrak{usp}(4)$																																																																														
$\mathfrak{so}(4, 1)$	$\sim$	$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$																																																																														
$\mathfrak{so}(3, 2)$	$\sim$	$\mathfrak{sp}(4; \mathbb{R})$																																																																														
D_3	$=$	A_3																																																																														
$\mathfrak{so}(6)$	$\sim$	$\mathfrak{su}(4)$																																																																														
$\mathfrak{so}(5, 1)$	$\sim$	$\mathfrak{su}^*(4)$																																																																														
$\mathfrak{so}^*(6)$	$\sim$	$\mathfrak{su}(3, 1)$																																																																														
$\mathfrak{so}(4, 2)$	$\sim$	$\mathfrak{su}(2, 2)$																																																																														
$\mathfrak{so}(3, 3)$	$\sim$	$\mathfrak{sl}(4; \mathbb{R})$																																																																														

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01889	195	■ 0.500	$C_{\{1\}}$	C_1	$ \begin{array}{llll} D_2 & = & A_1 & + \quad A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \quad \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \quad \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \quad \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01890	195	■ 0.500	$\mathfrak{so}(3)$	$\mathfrak{so}(3)$	$ \begin{array}{llll} D_2 & = & A_1 & + \quad A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \quad \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \quad \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \quad \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01891	195	■ 0.500	$\mathfrak{sp}(1) = \mathfrak{usp}(2)$	$\mathfrak{sp}(1) = \mathfrak{usp}(2)$	$ \begin{array}{llll} D_2 & = & A_1 & + \quad A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \quad \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \quad \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \quad \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01892	195	■ 0.500	$\mathfrak{su}(1,1)=\mathfrak{sl}(2;\mathbb{R})$	$\mathfrak{su}(1,1) = \mathfrak{sl}(2;\mathbb{R})$	$\begin{array}{llll} D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2;\mathbb{C}) & & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + & \mathfrak{so}(2,1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4;R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4;\mathbb{R}) & & \end{array}$
gilmore-lie-groups_ F01893	195	■ 0.500	$\mathfrak{so}(2,1)$	$\mathfrak{so}(2,1)$	$\begin{array}{llll} D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2;\mathbb{C}) & & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + & \mathfrak{so}(2,1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4;R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4;\mathbb{R}) & & \end{array}$
gilmore-lie-groups_ F01894	195	■ 0.500	$\mathfrak{sp}(2;\mathbb{R})$	$\mathfrak{sp}(2;\mathbb{R})$	$\begin{array}{llll} D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2;\mathbb{C}) & & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + & \mathfrak{so}(2,1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4;R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4;\mathbb{R}) & & \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01895	195	■ 0.500	$\mathfrak{s} \ \mathfrak{o}(4)$	$\mathfrak{so}(4)$	$ \begin{array}{llll} D_2 & = & A_1 & + \ A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \ \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \ \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \ \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01896	195	■ 0.500	$\mathfrak{s} \ \mathfrak{o}^*(4)$	$\mathfrak{so}^*(4)$	$ \begin{array}{llll} D_2 & = & A_1 & + \ A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \ \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \ \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \ \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01897	195	■ 0.500	$\mathfrak{s} \ \mathfrak{o}(3, 1)$	$\mathfrak{so}(3, 1)$	$ \begin{array}{llll} D_2 & = & A_1 & + \ A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \ \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \ \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \ \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01898	195	■ 0.500	$\mathfrak{sl}(2; \mathbb{C})$	$\mathfrak{sl}(2; \mathbb{C})$	$ \begin{array}{llll} D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $
gilmore-lie-groups_ F01899	195	■ 0.500	$\mathfrak{so}(2, 2)$	$\mathfrak{so}(2, 2)$	$ \begin{array}{llll} D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $
gilmore-lie-groups_ F01900	195	■ 0.500	$\mathfrak{sp}(2) = \mathfrak{usp}(4)$	$\mathfrak{sp}(2) = \mathfrak{usp}(4)$	$ \begin{array}{llll} D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01981	195	■ 0.500	$\frac{s}{p}(1,1)=\frac{u}{s}(2,2)$	$\mathfrak{sp}(1,1) = \mathfrak{usp}(2,2)$	
gilmore-lie-groups_ F01982	195	■ 0.500	$\frac{s}{p}(4 ; R)$	$\mathfrak{sp}(4; R)$	
gilmore-lie-groups_ F01983	195	■ 0.500	$D_{\{3\}}$	D_3	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01984	195	■ 0.500	$\mathfrak{s} \mathfrak{o}(6)$	$\mathfrak{so}(6)$	$ \begin{array}{llll} D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01985	195	■ 0.500	$\mathfrak{s} \mathfrak{u}(4)$	$\mathfrak{su}(4)$	$ \begin{array}{llll} D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01986	195	■ 0.500	$\mathfrak{s} \mathfrak{o}(5, 1)$	$\mathfrak{so}(5, 1)$	$ \begin{array}{llll} D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01907	195	■ 0.500	$\frac{s}{\mathfrak{u}^*}(4)$	$\mathfrak{su}^*(4)$	$ \begin{array}{llll} D_2 & = & A_1 & + \quad A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \quad \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \quad \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \quad \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01908	195	■ 0.500	$\frac{s}{\mathfrak{o}^*}(6)$	$\mathfrak{so}^*(6)$	$ \begin{array}{llll} D_2 & = & A_1 & + \quad A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \quad \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \quad \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \quad \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01909	195	■ 0.500	$\frac{s}{\mathfrak{u}}(3, 1)$	$\mathfrak{su}(3, 1)$	$ \begin{array}{llll} D_2 & = & A_1 & + \quad A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \quad \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \quad \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \quad \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01910	195	■ 0.500	$\mathfrak{s} \mathfrak{o}(4,2)$	$\mathfrak{so}(4,2)$	$ \begin{array}{llll} D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + \mathfrak{so}(2,1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4; R) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01911	195	■ 0.500	$\mathfrak{s} \mathfrak{u}(2,2)$	$\mathfrak{su}(2,2)$	$ \begin{array}{llll} D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + \mathfrak{so}(2,1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4; R) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01912	195	■ 0.500	$\mathfrak{s} \mathfrak{o}(3,3)$	$\mathfrak{so}(3,3)$	$ \begin{array}{llll} D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + \mathfrak{so}(2,1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4; R) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01913	195	■ 0.500	$\mathfrak{sl}(4; \mathbb{R})$	$\mathfrak{sl}(4; \mathbb{R})$	D_2 $\mathfrak{so}(4)$ $\mathfrak{so}^*(4)$ $\mathfrak{so}(3, 1)$ $\mathfrak{so}(2, 2)$ B_2 $\mathfrak{so}(5)$ $\mathfrak{so}(4, 1)$ $\mathfrak{so}(3, 2)$ D_3 $\mathfrak{so}(6)$ $\mathfrak{so}(5, 1)$ $\mathfrak{so}^*(6)$ $\mathfrak{so}(4, 2)$ $\mathfrak{so}(3, 3)$ $=$ $=$ $\sim$ $\$

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																														
gilmore-lie-groups_ F01915	196	1.000	$G_{2(-14)}$	$G_{2(-14)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
					Root space			Class _{rank(character)}	Maximal compact subgroup																																																										
						Root space	Dimension																																																												
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	$E_{8(-24)}$	$E_7 + A_1$	136																																																																
	$E_{8(+8)}$	D_8	120																																																																
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gilmore-lie-groups_ F01916	196	1.000	$G_{2(+2)}$	$G_{2(+2)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
					Root space			Class _{rank(character)}	Maximal compact subgroup																																																										
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						$E_{6(-26)}$	F_4	52																																																											
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$E_{6(+2)}$	$A_5 + A_1$	38																																																																	
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	$E_{8(-24)}$	$E_7 + A_1$	136																																																																
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gilmore-lie-groups_01917	196	1.000	$A_{\{1\}}+A_{\{1\}}$	$A_1 + A_1$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01928	196	1.000	$E_{\{6(+2)\}}$	$E_{6(+2)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01942	196	1.000	$E_{8(+8)}$	$E_{8(+8)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_F01943	196	■ 1.000	$D_{\{8\}}$	D_8	completeness we list the real forms for the exceptional Lie																																																														
gilmore-lie-groups_F01944	196	■ 1.000	$T(\frac{\mathfrak{p}}{ })=-\frac{\mathfrak{p}}{ }$	$T(\mathfrak{p}) = -\mathfrak{p}$	decomposition; subfield restriction; and field embedding.																																																														
gilmore-lie-groups_F01945	196	■ 1.000	$T(\frac{\mathfrak{h}}{ })=+\frac{\mathfrak{h}}{ }$	$T(\mathfrak{h}) = +\mathfrak{h}$																																																															
gilmore-lie-groups_F01946	196	■ 1.000	$\frac{\mathfrak{g}}{ }^{\prime}: \frac{\mathfrak{g}}{ }=\frac{\mathfrak{h}}{ }+\frac{\mathfrak{p}}{ } \rightarrow \frac{\mathfrak{g}}{ }^{\prime}=\frac{\mathfrak{h}}{ }+i\frac{\mathfrak{p}}{ }^{\prime}$	$\mathfrak{g}' : \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \rightarrow \mathfrak{g}' = \mathfrak{h} + i\mathfrak{p}'$																																																															
gilmore-lie-groups_F01947	197	■ 1.000	$1 \leq i$	$1 \leq i$	1. Four operators $a_i^\dagger a_j$ can be constructed from boson operators for two modes $1 \leq i,$																																																														
gilmore-lie-groups_F01948	197	■ 1.000	$j \leq 2$	$j \leq 2$	$j \leq 2$. These operators close under commutation.																																																														
gilmore-lie-groups_F01949	197	■ 1.000	$a_{\{1\}}^{\dag} a_{\{1\}}+a_{\{2\}}^{\dag} a_{\{2\}}$	$a_1^\dagger a_1 + a_2^\dagger a_2$	and $a_1 a_2 + a_2 a_1$ are noncompact. Identify the simple three-dimensional subalgebra as $\mathfrak{sl}(2, \mathbb{R})$ or $\mathfrak{su}(1, 1)$. Show that																																																														
gilmore-lie-groups_F01950	197	■ 1.000	$a_{\{1\}}^{\dag} a_{\{2\}}-$	$a_1^\dagger a_2 -$	and $a_1 a_2 + a_2 a_1$ are noncompact. Identify the simple three-dimensional subalgebra as $\mathfrak{sl}(2, \mathbb{R})$ or $\mathfrak{su}(1, 1)$. Show that																																																														
gilmore-lie-groups_F01951	197	■ 1.000	$a_{\{1\}}^{\dag} a_{\{1\}}-a_{\{2\}}^{\dag} a_{\{2\}}$	$a_1^\dagger a_1 - a_2^\dagger a_2$	Identify the simple three-dimensional subalgebra as $\mathfrak{sl}(2, \mathbb{R})$ or $\mathfrak{su}(1, 1)$. Show that the compact real form is obtained by multiplying the two noncompact generators																																																														
gilmore-lie-groups_F01952	197	■ 1.000	$a_{\{1\}}^{\dag} a_{\{2\}}+a_{\{2\}}^{\dag} a_{\{1\}}$	$a_1^\dagger a_2 + a_2^\dagger a_1$	the compact real form is obtained by multiplying by i .																																																														
gilmore-lie-groups_F01953	197	■ 1.000	$i \sigma_{\{j\}}$	$i\sigma_j$	nages of their universal covering groups $\overline{SO(n)} = Spin(n)$, for some integer k .																																																														
gilmore-lie-groups_F01954	197	■ 1.000	$\sigma_{\{j\}}$	σ_j	nages of their universal covering groups $\overline{SO(n)} = Spin(n)$, for some integer k .																																																														
gilmore-lie-groups_F01955	197	■ 0.982	$k \rightarrow 1$	$k \rightarrow 1$	$\mathfrak{so}(2, 1) \oplus \mathfrak{su}(1, 1) = \mathfrak{sl}(2, \mathbb{C})$																																																														
gilmore-lie-groups_F01956	197	■ 0.844	$\overline{SO(n)}=Spin(n)$	$\overline{SO(n)} = Spin(n)$	$Spin(3) = SU(2)$ $Spin(4) = SU(2) \otimes SU(2)$																																																														
gilmore-lie-groups_F01957	197	■ 1.000	$\operatorname{Spin}(n)$	$Spin(n)$	$Spin(3) = SU(2) \otimes SU(2)$ $Spin(5) = U/SU(4)$																																																														
gilmore-lie-groups_F01958	197	■ 1.000	$n=3,4,5,6$	$n = 3, 4, 5, 6$	$Spin(3) = SU(2) \otimes SU(2)$ $Spin(5) = U/SU(4)$																																																														
gilmore-lie-groups_F01959	197	■ 1.000	$n>6$	$n > 6$																																																															

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gilmore-lie-groups_F01960	198	■ 0.968	$H_{\{i\}}:\left n_{\{1\}},n_{\{2\}},\ldots,n_{\{r\}}\right\rangle\rightarrow$	$H_i: n_1,n_2,\dots,n_r\rangle$	
gilmore-lie-groups_F01961	198	■ 1.000	$H_{\{j\}}\rightarrow$	$H_j\rightarrow$	... iH_j , etc.), choose a Hilbert space containing the state $ l,0,\dots,0\rangle$ and show that
gilmore-lie-groups_F01962	198	■ 1.000	$iH_{\{j\}}$	iH_j	the value of $\mathcal{C}^-$ on every vector (i.e., apply shift operators E_{α^-} until no new states
gilmore-lie-groups_F01963	198	■ 1.000	$ l,0,\ldots,0\rangle$	$ l,0,\dots,0\rangle$	the value of $\mathcal{C}^-$ on every vector (i.e., apply shift operators E_{α^-} until no new states
gilmore-lie-groups_F01964	198	■ 0.989	$-m^{\{2\}}$	$-m^2$	4. Master analytic representation for A_1 The complex Lie algebra with root space
gilmore-lie-groups_F01965	198	■ 0.989	$e^{\{i\}m\phi}$	$e^{im\phi}$	4. Master analytic representation for A_1 The complex Lie algebra with root space
gilmore-lie-groups_F01966	198	■ 0.989	$-l(l+1)$	$-l(l+1)$	4. Master analytic representation for A_1 The complex Lie algebra with root space
gilmore-lie-groups_F01967	198	■ 0.989	$Y_{\{m\}}^{\{l\}}(\theta,\phi)$	$Y_m^l(\theta,\phi)$	4. Master analytic representation for A_1 The complex Lie algebra with root space
gilmore-lie-groups_F01968	198	■ 1.000	$J_{\{i\}},i=1,2,3$	$J_i,i=1,2,3$	$(2J+1)$ and are obtained in this way. In this problem we will review the construction
gilmore-lie-groups_F01969	198	■ 0.926	$\operatorname{EXP}\left(i\,r^{\{k\}}J_{\{k\}}\right)$	$\operatorname{EXP}\left(ir^kJ_k\right)$	of the UIK (unitary irreducible representations) of the compact group $SU(2)$ and will
gilmore-lie-groups_F01970	198	■ 0.926	$r^{\{k\}}$	r^k	of the UIK (unitary irreducible representations) of the compact group $SU(2)$ and will
gilmore-lie-groups_F01971	198	■ 1.000	$\mathfrak{su}\left(\mathfrak{s}\right)\mathfrak{su}\left(\mathfrak{u}\right)\left(1,1\right)$	$\mathfrak{su}(1,1)$	
gilmore-lie-groups_F01972	199	■ 0.826	$K_{\{3\}}$	K_3	$SU(2)$ with J_3 diagonal, or in any matrix representation of $SU(1,1)$ with K_3
gilmore-lie-groups_F01973	199	■ 1.000	$e^{\{i\}m\phi}\,\delta_{m'\,m}$	$e^{im\phi}\delta_{m'm}$	diagonal, the matrix is diagonal with matrix elements $e^{im\phi}\delta_{m'm}$. Show that the
gilmore-lie-groups_F01974	199	■ 1.000	$\phi\rightarrow\phi+4\pi$	$\phi\rightarrow\phi+4\pi$	single-valuedness condition under $\phi\rightarrow\phi+4\pi$ requires that $m=\frac{1}{2}(n_1-n_2)$ is
gilmore-lie-groups_F01975	199	■ 1.000	$m=\frac{1}{2}\left(n_{\{1\}}-n_{\{2\}}\right)$	$m=\frac{1}{2}(n_1-n_2)$	single-valuedness condition under $\phi\rightarrow\phi+4\pi$ requires that $m=\frac{1}{2}(n_1-n_2)$ is
gilmore-lie-groups_F01976	199	■ 1.000	$J_{\pm}\,\mathfrak{p}\mathfrak{m}$	$J_{\pm}$	integer or half-integer Show that the shift operators $J_{\pm}$ and $K_{\pm}$ require that all m
gilmore-lie-groups_F01977	199	■ 1.000	$K_{\pm}\,\mathfrak{p}\mathfrak{m}$	$K_{\pm}$	integer or half-integer Show that the shift operators $J_{\pm}$ and $K_{\pm}$ require that all m
gilmore-lie-groups_F01978	199	■ 1.000	$n_{\{1\}},n_{\{2\}}$	n_1,n_2	$ n_2\rangle$ must be nonnegative integers. Construct the matrix elements of the operators
gilmore-lie-groups_F01979	199	■ 1.000	$\left n_{\{1\}},n_{\{2\}}\right\rangle=\left n_{\{1\}}\right\rangle\otimes$	$ n_1,n_2\rangle= n_1\rangle\otimes$	$ n_2\rangle$ must be nonnegative integers. Construct the matrix elements of the operators
gilmore-lie-groups_F01980	199	■ 1.000	$\left n_{\{2\}}\right\rangle$	$ n_2\rangle$	$J_3,J_{\pm},K_3,K_{\pm}$ under this relaxed assumption. Show that all commutation relations
gilmore-lie-groups_F01981	199	■ 0.999	$J_{\{3\}},J_{\pm}\,\mathfrak{p}\mathfrak{m},K_{\{3\}},K_{\pm}\,\mathfrak{p}\mathfrak{m}$	$J_3,J_{\pm},K_3,K_{\pm}$	are satisfied in the representation afforded by this set of basis states.
gilmore-lie-groups_F01982	199	■ 1.000	$J_{\{x\}}$	J_x	$\langle n_1+1,n_2-1 J_{\pm} n_1,n_2\rangle=\sqrt{(n_1+1)n_2}$
gilmore-lie-groups_F01983	199	■ 1.000	$J_{\{y\}}$	J_y	$\langle n_1+1,n_2-1 J_{\pm} n_1,n_2\rangle=\sqrt{(n_1+1)n_2}$
gilmore-lie-groups_F01984	199	■ 0.749	$(0,0)$	$(0,0)$	. With the identification $ j\rangle= n_1,n_2\rangle$, $j=\frac{1}{2}(n_1+n_2)$, $m=\frac{1}{2}(n_1-n_2)$, show
gilmore-lie-groups_F01985	199	■ 0.975	$\left \left\{\frac{1}{2}\right\}_{\{m\}}^{\{j\}}\right\rangle=\left n_{\{1\}},n_{\{2\}}\right\rangle,\mathfrak{j}=\frac{1}{2}\left(n_{\{1\}}+n_{\{2\}}\right),\mathfrak{m}=\frac{1}{2}\left(n_{\{1\}}-n_{\{2\}}\right)$	$ j_m\rangle= n_1,n_2\rangle,j=\frac{1}{2}(n_1+n_2),m=\frac{1}{2}(n_1-n_2)$	$ j-m\rangle= r+j\backslash\sqrt{r^2+m^2+1},m-1\rangle,\sqrt{(r+j+1)^2-(m-1)^2}=\sqrt{(r+j+1)^2-(m-1)^2}$

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gilmore-lie-groups-F01986	199	■ 1.000	$n_{\{1\}}$	n_1	
gilmore-lie-groups-F01987	199	■ 1.000	$n_{\{2\}}$	n_2	
gilmore-lie-groups-F01988	199	■ 1.000	$n_{\{2\}}=0$	$n_2 = 0$	elements for the shift operators $K_{\pm}$ in $\mathfrak{su}(1,1)$ $i\sqrt{(j \mp m)(j \pm m + 1)} = \sqrt{\left(m \pm \frac{1}{2}\right)^2 - \frac{1}{4}}$
gilmore-lie-groups-F01989	199	■ 1.000	$n_{\{1\}}=0$	$n_1 = 0$	elements for the shift operators $K_{\pm}$ in $\mathfrak{su}(1,1)$ $i\sqrt{(j \mp m)(j \pm m + 1)} = \sqrt{\left(m \pm \frac{1}{2}\right)^2 - \frac{1}{4}}$
gilmore-lie-groups-F01990	199	■ 1.000	$n_{\{1\}}=-1$	$n_1 = -1$	elements for the shift operators $K_{\pm}$ in $\mathfrak{su}(1,1)$ $i\sqrt{(j \mp m)(j \pm m + 1)} = \sqrt{\left(m \pm \frac{1}{2}\right)^2 - \frac{1}{4}}$
gilmore-lie-groups-F01991	199	■ 1.000	$n_{\{2\}}=-1$	$n_2 = -1$	elements for the shift operators $K_{\pm}$ in $\mathfrak{su}(1,1)$ $i\sqrt{(j \mp m)(j \pm m + 1)} = \sqrt{\left(m \pm \frac{1}{2}\right)^2 - \frac{1}{4}}$
gilmore-lie-groups-F01992	200	■ 0.982	$n_{\{1\}}+n_{\{2\}}=$	$n_1 + n_2 =$	
gilmore-lie-groups-F01993	200	■ 0.872	$-1 < n_{\{1\}} < 0$	$-1 < n_1 < 0$	
gilmore-lie-groups-F01994	200	■ 1.000	$-1 < n_{\{2\}} < 0$	$-1 < n_2 < 0$	
gilmore-lie-groups-F01995	200	■ 0.997	$n_{\{1\}}+n_{\{2\}}=-\frac{1}{2}+i\beta$	$n_1 + n_2 = -\frac{1}{2} + i\beta$	
gilmore-lie-groups-F01996	200	■ 0.998	$\left n_{\{1\}}+k, n_{\{2\}}-k\right\rangle$	$ n_1 + k, n_2 - k\rangle$	
gilmore-lie-groups-F01997	200	■ 1.000	$\left n_{\{1\}}, n_{\{2\}}=-1\right\rangle$	$ n_1, n_2 = -1\rangle$	
gilmore-lie-groups-F01998	200	■ 1.000	$n_{\{1\}}=0, 1, 2, \dots$	$n_1 = 0, 1, 2, \dots$	
gilmore-lie-groups-F01999	200	■ 0.995	$\left \left\{ \right\}_m^j\right\rangle=\left n_{\{1\}}, n_{\{2\}}\right\rangle$	$ ^j_m\rangle = n_1, n_2\rangle$	
gilmore-lie-groups-F02000	200	■ 0.995	$j+\frac{1}{2}=$	$j + \frac{1}{2} =$	
gilmore-lie-groups-F02001	200	■ 0.993	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	
gilmore-lie-groups-F02002	200	■ 0.993	$2j+1=0, 1, 2, 3, \dots$	$2j + 1 = 0, 1, 2, 3, \dots$	

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gilmore-lie-groups_ F02003	200	■ 0.993	m=j+1, j+2, \ldots	$m = j + 1, j + 2, \dots$	(highest m) state must be $ n_1 = -1, n_2\rangle$ for $n_2 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_ F02004	200	■ 1.000	\mathcal{D}_{+}^{+}\{j\}	$\mathcal{D}_{+}^j$	this bounded discrete series of representations are $ \begin{smallmatrix} j \\ m \end{smallmatrix} \rangle = \begin{smallmatrix} j \\ 0 \end{smallmatrix} \rangle, \begin{smallmatrix} j \\ 1 \end{smallmatrix} \rangle, \begin{smallmatrix} j \\ 3 \end{smallmatrix} \rangle, \dots, \begin{smallmatrix} j \\ -1 \end{smallmatrix} \rangle, \begin{smallmatrix} j \\ -3 \end{smallmatrix} \rangle, \dots$ and $m = \begin{smallmatrix} j \\ 1 \end{smallmatrix}$
gilmore-lie-groups_ F02005	200	■ 0.831	\mathbf{j}	j	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ of $j \neq \frac{1}{2} = 0, 1, \frac{3}{2}, \dots$ and $m = -j = -1, -\frac{3}{2}, \dots$. This is the discrete series of representations that is bounded above: $\mathcal{D}_{+}^j$
gilmore-lie-groups_ F02006	200	■ 1.000	\left n_{1}=-1, n_{2}\right\rangle	$ n_1 = -1, n_2\rangle$	
gilmore-lie-groups_ F02007	200	■ 1.000	n_{2}=0,1,2, \ldots	$n_2 = 0, 1, 2, \dots$	
gilmore-lie-groups_ F02008	200	■ 1.000	m=-j-1,-j-2, \ldots	$m = -j - 1, -j - 2, \dots$	
gilmore-lie-groups_ F02009	200	■ 0.999	\mathcal{D}_{-}^{-}\{j\}	$\mathcal{D}_{-}^j$	
gilmore-lie-groups_ F02010	201	■ 0.998	\left(\mathcal{D}_{+}^{+}\{j\}\right)	$\left(\mathcal{D}_{+}^j\right)$	$(\mathcal{D}_{+}^j)$ or from QII $(\mathcal{D}_{-}^j)$.
gilmore-lie-groups_ F02011	201	■ 0.998	\left(\mathcal{D}_{-}^{-}\{j\}\right)	$\left(\mathcal{D}_{-}^j\right)$	$(\mathcal{D}_{+}^j)$ or from QII $(\mathcal{D}_{-}^j)$.
gilmore-lie-groups_ F02012	201	■ 0.999	\mid n_{1}+k, n_{2}-	$ n_1 + k, n_2 -$	I. Now relax the condition that n_1 and n_2 are integers. The set of states $ n_1 + k, n_2 -$
gilmore-lie-groups_ F02013	201	■ 1.000	k\rangle(k=\ldots,-2,-1,0,+1,+2, \ldots)	$k\rangle(k = \dots, -2, -1, 0, +1, +2, \dots)$	$k\rangle(k = \dots, -2, -1, 0, +1, +2, \dots)$ connected by $K_{\pm}$ carries a hermitian repre-
gilmore-lie-groups_ F02014	201	■ 1.000	p, q\rangle	$ p, q\rangle$	vertices of the four quadrants. If this state is $ p, q\rangle$, with $-1 \leq p, q \leq 0$ then the single-valuedness condition requires $\frac{1}{2}(p - q) = \text{integer or half-integer}$. In the lat-
gilmore-lie-groups_ F02015	201	■ 1.000	-1 \leq p, q \leq 0	$-1 \leq p, q \leq 0$	vertices of the four quadrants. If this state is $ p, q\rangle$, with $-1 \leq p, q \leq 0$ then the single-valuedness condition requires $\frac{1}{2}(p - q) = \text{integer or half-integer}$. In the lat-
gilmore-lie-groups_ F02016	201	■ 1.000	\frac{1}{2}(p-q)=	$\frac{1}{2}(p - q) =$	single-valuedness condition requires $\frac{1}{2}(p - q) = \text{integer or half-integer}$. In the lat-
gilmore-lie-groups_ F02017	201	■ 0.891	p-q=0	$p - q = 0$	$ \begin{smallmatrix} j \\ m \end{smallmatrix} \rangle$ with m integer, j real and $-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_ F02018	201	■ 0.891	-1 \leq p=q \leq 0	$-1 \leq p = q \leq 0$	$ \begin{smallmatrix} j \\ m \end{smallmatrix} \rangle$ with m integer, j real and $-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_ F02019	201	■ 0.679	\left.\left.\right _{m} ^{j}\right\rangle	$ \begin{smallmatrix} j \\ m \end{smallmatrix} \rangle$	complementary series of UIR for $SU(1, 1)$.
gilmore-lie-groups_ F02020	201	■ 0.679	-\frac{1}{2} \leq j+\frac{1}{2} \leq+\frac{1}{2}	$-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$	complementary series of UIR for $SU(1, 1)$.
gilmore-lie-groups_ F02021	201	■ 0.679	\mathcal{D}^p	$\mathcal{D}^p$	complementary series of UIR for $SU(1, 1)$.
gilmore-lie-groups_ F02022	201	■ 1.000	j+\frac{1}{2}=i \beta	$j + \frac{1}{2} = i\beta$	$\sqrt{\frac{1}{2} + \beta^2}$
gilmore-lie-groups_ F02023	201	■ 1.000	j^{\prime}<-\frac{1}{2}	$j' < -\frac{1}{2}$	indices is $j + \frac{1}{2} = -(j' + \frac{1}{2})$ and the relation among states is $ \begin{smallmatrix} j' \\ m' \end{smallmatrix} \rangle \simeq \begin{smallmatrix} j \\ m=m' \end{smallmatrix} \rangle$.
gilmore-lie-groups_ F02024	201	■ 0.693	\left(n_{1}, n_{2}\right)=\left(-\frac{1}{2},-\frac{1}{2}\right)	$(n_1, n_2) = \left(-\frac{1}{2}, -\frac{1}{2}\right)$	Show that the following equivalences occur among representations of these four series:
gilmore-lie-groups_ F02025	201	■ 1.000	j>-\frac{1}{2}	$j > -\frac{1}{2}$	series:
gilmore-lie-groups_ F02026	201	■ 0.311	j+\frac{1}{2}=-\left(j^{\prime}+\frac{1}{2}\right)	$j + \frac{1}{2} = -\left(j' + \frac{1}{2}\right)$	principal series $j' = -\frac{1}{2} - i \beta \Leftrightarrow j = j'^* = -\frac{1}{2} + i \beta $
gilmore-lie-groups_ F02027	201	■ 0.311	\left k_{m^{\prime}}^{\prime}\right\rangle \simeq\left j_{m=m^{\prime}}\right\rangle	$ \begin{smallmatrix} j' \\ m' \end{smallmatrix} \rangle \simeq \begin{smallmatrix} j \\ m=m' \end{smallmatrix} \rangle$	principal series $j' = -\frac{1}{2} - i \beta \Leftrightarrow j = j'^* = -\frac{1}{2} + i \beta $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F02028	201	■ 0.996	$X=h^{\{i\}}H_{\{i\}}+e^{\{\alpha\}}E_{\{\alpha\}}$	$X=h^iH_i+e^\alpha E_\alpha$	
gilmore-lie-groups_ F02029	201	■ 0.996	$h^{\{i\}},e^{\{\alpha\}}$	h^i,e^α	
gilmore-lie-groups_ F02030	201	■ 1.000	$\frac{1}{2}(n-l)$	$\frac{1}{2}(n-l)$	
gilmore-lie-groups_ F02031	201	■ 1.000	$\left(E_{\{\alpha\}}-E_{\{-\alpha\}}\right)/\sqrt{2}$	$(E_\alpha-E_{-\alpha})/\sqrt{2}$	
gilmore-lie-groups_ F02032	202	■ 1.000	$\mathfrak{sp}(p,q)$	$\mathfrak{sp}(p,q)$	6. The noncompact real form $\mathfrak{sp}(p,q)$ of the symplectic algebra was constructed from
gilmore-lie-groups_ F02033	202	■ 1.000	$\mathfrak{sp}(p+q)$	$\mathfrak{sp}(p+q)$	the compact real form $\mathfrak{sp}(p+q)$ by “Minkowski’s trick,” or analytic continuation.
gilmore-lie-groups_ F02034	203	■ 0.999	$\mathfrak{p},i\mathfrak{p}$	$\mathfrak{p},i\mathfrak{p}$	subspaces exponentiate onto algebraic manifolds on which the invariant
gilmore-lie-groups_ F02035	203	■ 1.000	$i\mathfrak{p}$	$i\mathfrak{p}$	metric g_{ij} is definite, either negative or positive. Manifolds with a definite
gilmore-lie-groups_ F02036	203	■ 1.000	$g_{\{i\}j}$	g_{ij}	ric in the sense that every point looks like every other point – because
gilmore-lie-groups_ F02037	203	■ 1.000	$\operatorname{EXP}(\mathfrak{p})$	$\operatorname{EXP}(\mathfrak{p})$	Riemannian globally symmetric spaces in this chapter.
gilmore-lie-groups_ F02038	203	■ 1.000	$\operatorname{EXP}(i\mathfrak{p})$	$\operatorname{EXP}(i\mathfrak{p})$	Riemannian globally symmetric spaces in this chapter.
gilmore-lie-groups_ F02039	203	■ 0.997	$S^2\sim S U(2)/U(1)$	$S^2\sim SU(2)/U(1)$	sheeted hyperboloid. These spaces are shown in Fig. 12.1.
gilmore-lie-groups_ F02040	203	■ 0.997	$H_{2+}^2=S L(2;\mathbb{R})/$	$H_{2+}^2=SL(2;\mathbb{R})/$	sheeted hyperboloid. These spaces are shown in Fig. 12.1.
gilmore-lie-groups_ F02041	203	■ 1.000	$S O(2)=S U(1,1)/U(1)$	$SO(2)=SU(1,1)/U(1)$	The Cartan–Killing inner product in the linear vector subspace $\mathfrak{su}(2)-\mathfrak{u}(1)$ is
gilmore-lie-groups_ F02042	203	■ 1.000	$H_1^2=S L(2;\mathbb{R})/S O(1,1)$	$H_1^2=SL(2;\mathbb{R})/SO(1,1)$	negative definite. This is mapped, under the EXPonential function, to the Cartan–
gilmore-lie-groups_ F02043	203	■ 0.943	$\mathfrak{su}(2)-\mathfrak{u}(1)$	$\mathfrak{su}(2)-\mathfrak{u}(1)$	metric is negative-definite. We may just as well take it as positive-definite. Under this metric the sphere becomes a Riemannian manifold since there is a metric on it
gilmore-lie-groups_ F02044	203	■ 0.999	$S U(2)/U(1)\sim S^2$	$SU(2)/U(1)\sim S^2$	The Cartan–Killing inner product on $\mathfrak{su}(1,1)-\mathfrak{u}(1)\simeq\mathfrak{sl}(2;\mathbb{R})-\mathfrak{so}(2)$ is
gilmore-lie-groups_ F02045	203	■ 0.991	$\mathfrak{su}(1,1)-\mathfrak{u}(1)\simeq\mathfrak{sl}(2;\mathbb{R})-\mathfrak{so}(2)$	$\mathfrak{su}(1,1)-\mathfrak{u}(1)\simeq\mathfrak{sl}(2;\mathbb{R})-\mathfrak{so}(2)$	via its Cartan–Killing metric and the curvature tensor derived from it.
gilmore-lie-groups_ F02046	203	■ 1.000	$H_{2+}^2=S U(1,1)/S O(2)$	$H_{2+}^2=SU(1,1)/SO(2)$	tion
gilmore-lie-groups_ F02047	204	■ 1.000	$S^2=S O(3)/S O(2)=S U(2)/U(1),\quad H_{2+}^2=S O(2,1)/S O(2)=$	$S^2=SO(3)/SO(2)=SU(2)/U(1),\quad H_{2+}^2=SO(2,1)/SO(2)=$	space.
gilmore-lie-groups_ F02048	204	■ 1.000	$S U(1,1)/U(1),H_1^2=S O(2,1)/S O(1,1)=S L(2;\mathbb{R})/S O(1,1)$	$SU(1,1)/U(1),H_1^2=SO(2,1)/SO(1,1)=SL(2;\mathbb{R})/SO(1,1)$	
gilmore-lie-groups_ F02049	204	■ 0.998	$\mathfrak{su}(1,1)-\mathfrak{so}(1,1)$	$\mathfrak{su}(1,1)-\mathfrak{so}(1,1)$	The space is a pseudo-Riemannian manifold. In addition it is multiply connected.
gilmore-lie-groups_ F02050	204	■ 0.966	$\operatorname{EXP}[\mathfrak{su}(1,1)-\mathfrak{so}(1,1)]=S U(1,1)/S O(1,1)$	$\operatorname{EXP}[\mathfrak{su}(1,1)-\mathfrak{so}(1,1)]=SU(1,1)/SO(1,1)$	12.2 Globally symmetric spaces
gilmore-lie-groups_ F02051	204	■ 0.952	$\mathfrak{so}(n),\mathfrak{su}(n),\mathfrak{sp}(n)$	$\mathfrak{so}(n),\mathfrak{su}(n),\mathfrak{sp}(n)$	
gilmore-lie-groups_ F02052	205	■ 1.000	$\mathfrak{p}(i\mathfrak{p})$	$\mathfrak{p}(i\mathfrak{p})$	On the linear vector space $\mathfrak{p}$ ($i\mathfrak{p}$) the Cartan–Killing inner product is negative
gilmore-lie-groups_ F02053	205	■ 1.000	$P=G/H=\operatorname{EXP}(\mathfrak{p})$	$P=G/H=\operatorname{EXP}(\mathfrak{p})$	straight line through the origin in $\mathfrak{p}$ returns periodically to the neighborhood of the

Identifier	Page	Conf.	LaTeX source	Rendered	Image																								
gilmore-lie-groups_ F02054	205	■ 1.000	P	P	When a straight line (geodesic) through the origin never returns to the origin, the space P may be simply connected or multiply connected.																								
gilmore-lie-groups_ F02055	205	■ 0.999	$P^{\prime} = G^{\prime} / H = \operatorname{EXP}(i \frac{p}{\mathfrak{p}})$	$P' = G' / H = \operatorname{EXP}(ip)$	Simply goes away from this point without ever returning. The space $P' = \operatorname{EXP}(ip)$ is																								
gilmore-lie-groups_ F02056	205	■ 1.000	$P^{\prime} = \operatorname{EXP}(i \frac{p}{\mathfrak{p}})$	$P' = \operatorname{EXP}(ip)$	it is not Euclidean since it has nonzero curvature. This space is simply connected.																								
gilmore-lie-groups_ F02057	205	■ 1.000	$n = \operatorname{dim} i \frac{p}{\mathfrak{p}}$	$n = \dim ip$	The Riemannian spaces $P = \operatorname{EXP}(\mathfrak{p})$ and $P' = \operatorname{EXP}(ip)$ are symmetric but not																								
gilmore-lie-groups_ F02058	205	■ 1.000	$P = \operatorname{EXP}(\frac{p}{\mathfrak{p}})$	$P = \operatorname{EXP}(\mathfrak{p})$	$\mathfrak{g}$ is simple with a Cartan decomposition of the form $\mathfrak{g} = \mathfrak{k} + \mathfrak{p}$, with standard commutation relations $[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{k}$, $[\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}$, and $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$; the quotient space $D =$																								
gilmore-lie-groups_ F02059	205	■ 1.000	$\frac{\mathfrak{g}}{\mathfrak{p}} = \frac{\mathfrak{k}}{\mathfrak{p}} + \frac{\mathfrak{p}}{\mathfrak{p}}$	$\mathfrak{g} = \mathfrak{k} + \mathfrak{p}$																									
gilmore-lie-groups_ F02060	205	■ 1.000	$[\frac{\mathfrak{k}}{\mathfrak{p}}, \frac{\mathfrak{k}}{\mathfrak{p}}] \subseteq \frac{\mathfrak{k}}{\mathfrak{p}}, [\frac{\mathfrak{k}}{\mathfrak{p}}, \frac{\mathfrak{p}}{\mathfrak{p}}] \subseteq \frac{\mathfrak{p}}{\mathfrak{p}}, [\frac{\mathfrak{p}}{\mathfrak{p}}, \frac{\mathfrak{p}}{\mathfrak{p}}] \subseteq \frac{\mathfrak{k}}{\mathfrak{p}}$	$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$																									
gilmore-lie-groups_ F02061	205	■ 1.000	$[\frac{\mathfrak{p}}{\mathfrak{p}}, \frac{\mathfrak{p}}{\mathfrak{p}}] \subseteq \frac{\mathfrak{k}}{\mathfrak{p}}$	$[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$																									
gilmore-lie-groups_ F02062	205	■ 1.000	$P =$	$P =$																									
gilmore-lie-groups_ F02063	205	■ 1.000	G / K	G / K	12.3 RANK																								
gilmore-lie-groups_ F02064	205	■ 1.000	$P = G / H$	$P = G / H$	$\ \Re \mathfrak{e} \mathfrak{a}(\mathfrak{h} + \mathfrak{p}) - \lambda I_n \ = \sum_{i=1}^n (-\lambda)^{n-j} \phi_i(\mathfrak{h}, \mathfrak{p}) \quad (12.3)$																								
gilmore-lie-groups_ F02065	205	■ 1.000	$P^{\prime} = G^{\prime} / H$	$P' = G' / H$	$\ \Re \mathfrak{e} \mathfrak{a}(\mathfrak{h} + \mathfrak{p}) - \lambda I_n \ = \sum_{i=1}^n (-\lambda)^{n-j} \phi_i(\mathfrak{h}, \mathfrak{p}) \quad (12.3)$																								
gilmore-lie-groups_ F02066	205	■ 0.993	$S O(p, q) / S O(p) \times S O(q)$	$SO(p, q) / SO(p) \times SO(q)$																									
gilmore-lie-groups_ F02067	206	■ 1.000	$B^{\dagger} B$	$B^{\dagger} B$	It is easy to check that the function ψ_j depends on the $q \times q$ matrix $B^{\dagger} B$ or the $n \times n$ matrix $RR^{\dagger}$ whichever is smaller. The rank of these spaces is $\min(n, n)$.																								
gilmore-lie-groups_ F02068	206	■ 1.000	$B B^{\dagger}$	$BB^{\dagger}$	$p \times p$ matrix $BB^{\dagger}$, whenever it is smaller. The rank of these spaces is $\min(p, q)$. For Riemannian globally symmetric spaces the rank is (cf. Section 10.1):																								
gilmore-lie-groups_ F02069	206	■ 1.000	$\min(p, q)$	$\min(p, q)$	$p \times p$ matrix $BB^{\dagger}$, whenever it is smaller. The rank of these spaces is $\min(p, q)$. For Riemannian globally symmetric spaces the rank is (cf. Section 10.1):																								
gilmore-lie-groups_ F02070	206	■ 1.000	$P \left(P^{\prime} \right)$	$P(P')$	(v) the dimension of a positive-definite root space that can be used to define diagrammatically the projection of these spaces (Aub1). (Catala post discussion).																								
gilmore-lie-groups_ F02071	206	■ 0.999	$G^{\prime} / H$	G' / H	<table><tr><th>Root space</th><th>Quotient G/H</th><th>Dimension D</th><th>Rank D</th></tr></table>	Root space	Quotient G/H	Dimension D	Rank D																				
Root space	Quotient G/H	Dimension D	Rank D																										
gilmore-lie-groups_ F02072	206	■ 0.999	$G^{\prime}$	G'	<table><tr><th>Root space</th><th>Quotient G/H</th><th>Dimension D</th><th>Rank D</th></tr></table>	Root space	Quotient G/H	Dimension D	Rank D																				
Root space	Quotient G/H	Dimension D	Rank D																										
gilmore-lie-groups_ F02073	206	■ 0.877	A_{p+q-1}	A_{p+q-1}	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n) / USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p, q) / SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q) / SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n) / U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q) / USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R}) / U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n) / USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q) / SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q) / SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n) / U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q) / USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R}) / U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n) / USp(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p, q) / SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_{p+q}	$SO(p, q) / SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_n	$SO^*(2n) / U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$USp(2p, 2q) / USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$																										
C_n	$Sp(2n; \mathbb{R}) / U(n)$	$n(n+1)$	n																										

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gilmore-lie-groups_F02074	206	0.877	$S(U(p, q) / S[U(p) \otimes U(q)]$	$SU(p, q)/S[U(p) \otimes U(q)]$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$																										
C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_F02075	206	0.877	$2pq$	$2pq$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$																										
C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_F02076	206	0.877	$SL(n; \mathbb{R}) / SO(n)$	$SL(n; \mathbb{R})/SO(n)$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$																										
C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_F02077	206	0.877	$\frac{1}{2}(n+2)(n-1)$	$\frac{1}{2}(n+2)(n-1)$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$																										
C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_F02078	206	0.877	$n-1$	$n-1$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$																										
C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n																										

Identifier	Page	Conf.	LaTeX source	Rendered	Image																								
gilmore-lie-groups_ F02679	206	0.877	$SU^*(2n)/U\operatorname{Sp}(2n)$	$SU^*(2n)/U\operatorname{Sp}(2n)$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U\operatorname{Sp}(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_ F02680	206	0.877	$(2n+1)(n-1)$	$(2n+1)(n-1)$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U\operatorname{Sp}(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_ F02681	206	0.877	$B_{\{p+q\}}$	B_{p+q}	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U\operatorname{Sp}(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_ F02682	206	0.877	$SO(p,q)/SO(p)\otimes SO(q)$	$SO(p,q)/SO(p)\otimes SO(q)$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U\operatorname{Sp}(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_ F02683	206	0.877	$D_{\{p+q\}}$	D_{p+q}	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U\operatorname{Sp}(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$U\operatorname{Sp}(2p,2q)/U\operatorname{Sp}(2p)\otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										

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gilmore-lie-groups_F02084	206	0.877	$S O^{\{*\}}(2 n) / U(n)$	$SO^*(2n)/U(n)$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U Sp(2n)$</td><td>$(\tilde{2}n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_F02085	206	0.877	$n(n-1)$	$n(n-1)$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U Sp(2n)$</td><td>$(\tilde{2}n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_F02086	206	0.877	$C_{\{p+q\}}$	C_{p+q}	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U Sp(2n)$</td><td>$(\tilde{2}n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$																										
C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_F02087	206	0.877	$U S p(2 p, 2 q) / U S p(2 p) \otimes U S p(2 q)$	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U Sp(2n)$</td><td>$(\tilde{2}n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
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C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$																										
C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										
gilmore-lie-groups_F02088	206	0.877	$4 p q$	$4pq$	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/U Sp(2n)$</td><td>$(\tilde{2}n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U Sp(2p,2q)/U Sp(2p)\otimes U Sp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
A_{2n-1}	$SU^*(2n)/U Sp(2n)$	$(\tilde{2}n+1)(n-1)$	$n-1$																										
B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$																										
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C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																										

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gilmore-lie-groups_ F02689	206	0.877	$S p(2 n ; \mathbb{R}) / U(n)$	$Sp(2n;\mathbb{R})/U(n)$	A_{2n-1} B_{p+q} D_{p+q} D_n C_{p+q} C_n $SU^*(2n)/USp(2n)$ $SO(p,q)/SO(p)\otimes SO(q)$ $SO(p,q)/SO(p)\otimes SO(q)$ $SO^*(2n)/U(n)$ $USp(2p,2q)/USp(2p)\otimes USp(2q)$ $Sp(2n;\mathbb{R})/U(n)$ $(2n+1)(n-1)$ pq pq $n(n-1)$ $4pq$ $n(n+1)$ $n-1$ $\min(p,q)$ $\min(p,q)$ $[n/2]$ $\min(p,q)$ n

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gilmore-lie-groups_ F02093	207	■ 0.979	$\backslash\operatorname{Dim} P$	$\operatorname{Dim} P$	<table> <tr> <th>Root space</th><th>G'/H</th><th>$\operatorname{Dim} G'$</th><th>$\operatorname{Dim} H$</th><th>$\operatorname{Dim} P$</th><th>$\operatorname{Rank} P$</th></tr> <tr> <td>G_2</td><td>$G_{2(+2)}/(A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr> <tr> <td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr> <tr> <td></td><td>$F_{4(+4)}/(C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr> <tr> <td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr> <tr> <td></td><td>$E_{6(-14)}/(D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr> <tr> <td></td><td>$E_{6(+2)}/(A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr> <tr> <td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr> <tr> <td>E_7</td><td>$E_{7(-25)}/(E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr> <tr> <td></td><td>$E_{7(-5)}/(D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr> <tr> <td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr> <tr> <td>E_8</td><td>$E_{8(-24)}/(E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr> <tr> <td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr> </table> subgroup in G'. To each there is a compact real form under $G'/H \rightrightarrows G/H$. For	Root space	G'/H	$\operatorname{Dim} G'$	$\operatorname{Dim} H$	$\operatorname{Dim} P$	$\operatorname{Rank} P$	G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
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gilmore-lie-groups_ F02094	207	■ 0.979	$G_{2(+2)}/\left(A_{11} \oplus A_{11}\right)$	$G_{2(+2)}/(A_1 \oplus A_1)$	<table> <tr> <th>Root space</th><th>G'/H</th><th>$\operatorname{Dim} G'$</th><th>$\operatorname{Dim} H$</th><th>$\operatorname{Dim} P$</th><th>$\operatorname{Rank} P$</th></tr> <tr> <td>G_2</td><td>$G_{2(+2)}/(A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr> <tr> <td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr> <tr> <td></td><td>$F_{4(+4)}/(C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr> <tr> <td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr> <tr> <td></td><td>$E_{6(-14)}/(D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr> <tr> <td></td><td>$E_{6(+2)}/(A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr> <tr> <td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr> <tr> <td>E_7</td><td>$E_{7(-25)}/(E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr> <tr> <td></td><td>$E_{7(-5)}/(D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr> <tr> <td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr> <tr> <td>E_8</td><td>$E_{8(-24)}/(E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr> <tr> <td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr> </table> subgroup in G'. To each there is a compact real form under $G'/H \rightrightarrows G/H$. For	Root space	G'/H	$\operatorname{Dim} G'$	$\operatorname{Dim} H$	$\operatorname{Dim} P$	$\operatorname{Rank} P$	G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
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gilmore-lie-groups_ F02095	207	■ 0.979	$F_{4(-20)}/B_{44}$	$F_{4(-20)}/B_4$	<table> <tr> <th>Root space</th><th>G'/H</th><th>$\operatorname{Dim} G'$</th><th>$\operatorname{Dim} H$</th><th>$\operatorname{Dim} P$</th><th>$\operatorname{Rank} P$</th></tr> <tr> <td>G_2</td><td>$G_{2(+2)}/(A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr> <tr> <td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr> <tr> <td></td><td>$F_{4(+4)}/(C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr> <tr> <td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr> <tr> <td></td><td>$E_{6(-14)}/(D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr> <tr> <td></td><td>$E_{6(+2)}/(A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr> <tr> <td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr> <tr> <td>E_7</td><td>$E_{7(-25)}/(E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr> <tr> <td></td><td>$E_{7(-5)}/(D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr> <tr> <td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr> <tr> <td>E_8</td><td>$E_{8(-24)}/(E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr> <tr> <td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr> </table> subgroup in G'. To each there is a compact real form under $G'/H \rightrightarrows G/H$. For	Root space	G'/H	$\operatorname{Dim} G'$	$\operatorname{Dim} H$	$\operatorname{Dim} P$	$\operatorname{Rank} P$	G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
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E_8	$E_{8(-24)} / (E_7 \oplus A_1)$	248	136	112	4																																																																														
	$E_{8(+8)} / D_8$	248	120	128	8																																																																														

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gilmore-lie-groups_ F02105	207	■ 0.979	$E_{8(+8)} / D_8$	$E_{8(+8)}/D_8$	Root space G'/H Dim G' Dim H Dim P Rank P <table> <tr> <td>G_2</td><td>$G_{2(+2)}/(A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr> <tr> <td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr> <tr> <td></td><td>$F_{4(+4)}/(C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr> <tr> <td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr> <tr> <td></td><td>$E_{6(-14)}/(D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr> <tr> <td></td><td>$E_{6(+2)}/(A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr> <tr> <td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr> <tr> <td>E_7</td><td>$E_{7(-25)}/(E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr> <tr> <td></td><td>$E_{7(-5)}/(D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr> <tr> <td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr> <tr> <td>E_8</td><td>$E_{8(-24)}/(E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr> <tr> <td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr> </table> subgroup in G'. To each there is a compact real form under $G'/H \hookrightarrow G/H$. For	G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2																																																																								
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gilmore-lie-groups_ F02106	207	■ 1.000	$G^{\prime} / H \rightarrow G / H$	$G'/H \rightarrow G/H$	These spaces are classical because they involve the classical series of Lie groups:																																																																								
gilmore-lie-groups_ F02107	207	■ 1.000	$S O(p+q) / S O(p) \otimes S O(q)$	$SO(p+q)/SO(p) \otimes SO(q)$	the orthogonal, the unitary, and the symplectic.																																																																								
gilmore-lie-groups_ F02108	207	■ 1.000	$P, P^{\prime}$	P, P'	If $dx(\mathrm{Id})$ are infinitesimal displacements at the identity that are translated to																																																																								
gilmore-lie-groups_ F02109	207	■ 0.987	$d x(\mathrm{Id})$	$dx(\mathrm{Id})$	$dx^i(\mathrm{Id}) = M^i \quad dx^\mu(n) \quad (17.5)$																																																																								
gilmore-lie-groups_ F02110	207	■ 1.000	$d x(p)$	$dx(p)$																																																																									
gilmore-lie-groups_ F02111	208	■ 1.000	$M^{\mathrm{i}}_{\mu}(p)$	$M^i_{\mu}(p)$	The matrix $M^i_{\mu}(p)$ is not easy to compute in general. For the rank-one spaces																																																																								
gilmore-lie-groups_ F02112	208	■ 0.998	$S O(n, 1) / S O(n), S U(n, 1) / U(n), S p(n, 1) / S p(n) \otimes S p(1)$	$SO(n,1)/SO(n), SU(n,1)/U(n), Sp(n,1)/Sp(n) \times Sp(1)$	$SO(n,1)/SO(n), SU(n,1)/U(n), Sp(n,1)/Sp(n) \times Sp(1)$ defined by																																																																								
gilmore-lie-groups_ F02113	208	■ 1.000	$M^{\mathrm{i}}_{\mu}(X)$	$M^i_{\mu}(X)$																																																																									
gilmore-lie-groups_ F02114	208	■ 1.000	W^{-1}	W^{-1}	at the identity, we find																																																																								
gilmore-lie-groups_ F02115	208	■ 1.000	$H_2^2=S O(2,1) / S O(2)$	$H_2^2 = SO(2,1)/SO(2)$																																																																									
gilmore-lie-groups_ F02116	209	■ 0.999	$S^2=S O(3) / S O(2)$	$S^2 = SO(3)/SO(2)$	$S^2 = SO(3)/SO(2)$. The metric tensors in the two cases are the 2×2 matrices																																																																								
gilmore-lie-groups_ F02117	209	■ 1.000	$S O(2+1) / S O(2)$	$SO(2+1)/SO(2)$	Since the (intrinsic) properties of the Riemannian symmetric space are entirely																																																																								
gilmore-lie-groups_ F02118	209	■ 0.999	$Y^2=1-\left(x^2+y^2\right) \geq 0$	$Y^2 = 1 - \left(x^2 + y^2\right) \geq 0$	Since the (intrinsic) pro encoded in its metric tens																																																																								
gilmore-lie-groups_ F02119	209	■ 1.000	x, y	x, y	the neighborhood of the origin (identity in the coset) only up to quadratic terms,																																																																								
gilmore-lie-groups_ F02120	209	■ 1.000	$x^{\mathrm{i}}, \mathrm{i}=1,2, \ldots, N$	$x^i, i = 1, 2, \dots, N$																																																																									
gilmore-lie-groups_ F02121	209	■ 1.000	$N=2$	$N = 2$																																																																									
gilmore-lie-groups_ F02122	210	■ 1.000	$g^{\mathrm{i} j} \simeq \delta^{\mathrm{i} j}-x^{\mathrm{i}} x^{\mathrm{j}}$	$g^{ij} \simeq \delta^{ij} - x^i x^j$	The inverse (contravariant metric) is $g^{ij} \simeq \delta^{ij} - x^i x^j$, but we will not need this																																																																								
gilmore-lie-groups_ F02123	210	■ 0.900	$g^{\mathrm{i} j} \rightarrow \delta^{\mathrm{i} j}$	$g^{ij} \rightarrow \delta^{ij}$	result, in the neighborhood of the identity ($g^{ij} \rightarrow \delta^{ij}$)																																																																								
gilmore-lie-groups_ F02124	210	■ 1.000	$R_{\mu \sigma, \alpha \beta} =$	$R_{\mu\sigma,\alpha\beta} =$	The Ricci tensor is obtained by contraction																																																																								

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gilmore-lie-groups_ F02125	210	1.000	$\delta_{\alpha\mu}\delta_{\beta\sigma}-\delta_{\alpha\sigma}\delta_{\beta\mu}$	$\delta_{\alpha\mu}\delta_{\beta\sigma}-\delta_{\alpha\sigma}\delta_{\beta\mu}$	$R_{\sigma\beta}=R_{\sigma,\mu\beta}^{\mu}=\delta_{\sigma\beta}\delta_{\mu\mu}-\delta_{\sigma\mu}\delta_{\beta\mu}=N\delta_{\sigma\beta}-\delta_{\sigma\beta}$
gilmore-lie-groups_ F02126	210	0.874	R=2	$R=2$	the origin undergoes the ch
gilmore-lie-groups_ F02127	210	1.000	H_2^2	H_2^2	$SO(2+1)/SO(2) \quad SO(2,1)/SO(2)$ (12.19)
gilmore-lie-groups_ F02128	210	0.848	$\Gamma_{\mu\nu}^{\sigma}\rightarrow-\delta_{\mu\nu}x^{\sigma}$	$\Gamma_{\mu\nu}^{\sigma}\rightarrow-\delta_{\mu\nu}x^{\sigma}$	
gilmore-lie-groups_ F02129	210	0.848	R=-2	$R=-2$	
gilmore-lie-groups_ F02130	211	0.998	$ds^2=g_{\mu\nu}dx^{\mu}dx^{\nu}>0$	$ds^2=g_{\mu\nu}dx^{\mu}dx^{\nu}>0$	positive-definite ($ds^2=g_{\mu\nu}dx^{\mu}dx^{\nu}>0$, equality $\Rightarrow dx=0$) nor negative-
gilmore-lie-groups_ F02131	211	0.998	$\Rightarrow dx=0$	$\Rightarrow dx=0$	positive-definite ($ds^2=g_{\mu\nu}dx^{\mu}dx^{\nu}>0$, equality $\Rightarrow dx=0$) nor negative-
gilmore-lie-groups_ F02132	211	0.958	$(ds^2<0)$	$(ds^2<0)$	dennite ($ds^2<0$), but which is nonsingular ($\ g\ \neq 0$) are called pseudo-Riemannian spaces. Pseudo-Riemannian spaces that are globally symmetric are
gilmore-lie-groups_ F02133	211	0.958	$\ g\ \neq 0$	$\ g\ \neq 0$	dennite ($ds^2<0$), but which is nonsingular ($\ g\ \neq 0$) are called pseudo-Riemannian spaces. Pseudo-Riemannian spaces that are globally symmetric are
gilmore-lie-groups_ F02134	211	1.000	$\mathfrak{g}^{\prime}$	$\mathfrak{g}''$	ias a decomposition
gilmore-lie-groups_ F02135	211	1.000	$\mathfrak{h}^{\prime}$	$\mathfrak{h}''$	$\mathfrak{p}''=\mathbf{EXP}(\mathfrak{n}'')=\mathfrak{G}''/H''$ (12.21)
gilmore-lie-groups_ F02136	211	1.000	$\mathfrak{p}^{\prime}$	$\mathfrak{p}''$	$\mathfrak{p}''=\mathbf{EXP}(\mathfrak{n}'')=\mathfrak{G}''/H''$ (12.21)
gilmore-lie-groups_ F02137	211	1.000	$H^{\prime}=$	$H''=$	hold also for pseudo-Riemannian symmetric spaces. That is, rank can be defined,
gilmore-lie-groups_ F02138	211	1.000	$\mathbf{EXP}(\mathfrak{h}^{\prime})$	$\mathbf{EXP}(\mathfrak{h}'')$	and carries most of the implications listed in Section 12.3.
gilmore-lie-groups_ F02139	211	1.000	T_1, T_2	T_1, T_2	$T_1\mathfrak{g}_{\pm,*}=\pm\mathfrak{g}_{\pm,*}$ (12.23)
gilmore-lie-groups_ F02140	211	0.819	$T_1^2=I$	$T_1^2=I$	$T_2\mathfrak{g}_{*,\pm}=\pm\mathfrak{g}_{*,\pm}$ (12.22)
gilmore-lie-groups_ F02141	211	0.819	$T_2^2=I$	$T_2^2=I$	$T_2\mathfrak{g}_{*,\pm}=\pm\mathfrak{g}_{*,\pm}$ (12.22)
gilmore-lie-groups_ F02142	211	1.000	$T_1\neq T_2$	$T_1\neq T_2$	
gilmore-lie-groups_ F02143	211	1.000	$\mathfrak{g}_{\pm,\pm}$	$\mathfrak{g}_{\pm,\pm}$	
gilmore-lie-groups_ F02144	212	1.000	$\mathfrak{h}^{\prime}, \mathfrak{p}^{\prime}$	$\mathfrak{h}'', \mathfrak{p}''$	The subspaces $\mathfrak{h}, \mathfrak{p}$ obey commutation relations (12.20). The Cartan Killing inner product is indefinite on both $\mathfrak{h}''$ and $\mathfrak{n}''$ as long as $T_1\neq T_2$.
gilmore-lie-groups_ F02145	212	0.996	$\mathfrak{p}^{\prime}$	$\mathfrak{p}''$	For $\mathfrak{su}(2)$ the only two mappings are T_1 = block diag
gilmore-lie-groups_ F02146	212	0.999	$T_1=$	$T_1=$	T_2 = complex conjugation. The eigenspace decomposition is
gilmore-lie-groups_ F02147	212	1.000	$T_2=$	$T_2=$	
gilmore-lie-groups_ F02148	212	1.000	$i\sigma_1$	$i\sigma_1$	T_1 = block matrix decomposition $\begin{pmatrix} -1 & -1 & +1 \\ -1 & +1 & -1 \\ +1 & -1 & -1 \end{pmatrix}$ T_2 = complex conjugation $T_3=T_1T_2$ $\mathfrak{g}_{+,+}=0, \mathfrak{g}_{+,-}=i\sigma_3, \mathfrak{g}_{-,+}=i\sigma_2, \mathfrak{g}_{-,-}=i\sigma_1$. Note tha

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gilmore-lie-groups_ F02149	212	■ 1.000	$i \ \sigma_{\{2\}}$	$i\sigma_2$	$\begin{array}{rcc} T_1 = \text{block matrix decomposition} & -1 & -1 & +1 \\ T_2 = \text{complex conjugation} & -1 & +1 & -1 \\ T_3 = T_1 T_2 & +1 & -1 & -1 \end{array}$ $\mathfrak{g}_{+,+} = 0, \mathfrak{g}_{+,-} = i\sigma_3, \mathfrak{g}_{-,+} = i\sigma_2, \mathfrak{g}_{-,-} = i\sigma_1$. Note tha
gilmore-lie-groups_ F02150	212	■ 1.000	$i \ \sigma_{\{3\}}$	$i\sigma_3$	$\begin{array}{rcc} T_1 = \text{block matrix decomposition} & -1 & -1 & +1 \\ T_2 = \text{complex conjugation} & -1 & +1 & -1 \\ T_3 = T_1 T_2 & +1 & -1 & -1 \end{array}$ $\mathfrak{g}_{+,+} = 0, \mathfrak{g}_{+,-} = i\sigma_3, \mathfrak{g}_{-,+} = i\sigma_2, \mathfrak{g}_{-,-} = i\sigma_1$. Note tha
gilmore-lie-groups_ F02151	212	■ 1.000	$T_{\{3\}}=T_{\{1\}} \ T_{\{2\}}$	$T_3 = T_1 T_2$	$\begin{array}{rcc} T_1 = \text{block matrix decomposition} & -1 & -1 & +1 \\ T_2 = \text{complex conjugation} & -1 & +1 & -1 \\ T_3 = T_1 T_2 & +1 & -1 & -1 \end{array}$ $\mathfrak{g}_{+,+} = 0, \mathfrak{g}_{+,-} = i\sigma_3, \mathfrak{g}_{-,+} = i\sigma_2, \mathfrak{g}_{-,-} = i\sigma_1$. Note tha
gilmore-lie-groups_ F02152	212	■ 1.000	$\mathfrak{g}_{+,+}=0, \ \mathfrak{g}_{+,-}=i \ \sigma_{\{3\}}, \ \mathfrak{g}_{-,-}=i \ \sigma_{\{2\}}, \ \mathfrak{g}_{-,+}=i \ \sigma_{\{1\}}$	$\mathfrak{g}_{+,+} = 0, \mathfrak{g}_{+,-} = i\sigma_3, \mathfrak{g}_{-,+} = i\sigma_2, \mathfrak{g}_{-,-} = i\sigma_1$	for the maximal compact subalgebra $\mathfrak{h}'$ of the noncompact real form $\mathfrak{g}'$.
gilmore-lie-groups_ F02153	212	■ 1.000	$T_{\{i\}}$	T_i	
gilmore-lie-groups_ F02154	212	■ 0.999	$\mathfrak{h}^{\prime}$	$\mathfrak{h}'$	
gilmore-lie-groups_ F02155	212	■ 0.984	$P=G \ / \ K$	$P = G/K$	Riemannian globally symmetric spaces are constructed as quotients of a simple Lie group G by a maximal compact subgroup K . More specifically, they are even-
gilmore-lie-groups_ F02156	212	■ 1.000	$\phi_j\left(B, \ B^{\dagger}\right)$	$\phi_j\left(B, B^{\dagger}\right)$	matrix are zero (singular value decomposition theorem).
gilmore-lie-groups_ F02157	212	■ 0.585	$(p-q)$	$(p-q)$	
gilmore-lie-groups_ F02158	212	■ 0.585	$(q-p)$	$(q-p)$	
gilmore-lie-groups_ F02159	213	■ 1.000	Δ^2	Δ^2	2. The second order Laplace–Beltrami operator Δ^2 is constructed from the second order
gilmore-lie-groups_ F02160	213	■ 1.000	$G \ / \ H=P$	$G/H = P$	$G/H = P$.
gilmore-lie-groups_ F02161	213	■ 1.000	$\Delta^2=g^{ij}\left(\partial_i\partial_j-\Gamma_{ij}^k\partial_k\right)$	$\Delta^2 = g^{ij} \left(\partial_i \partial_j - \Gamma_{ij}^k \partial_k \right)$	tensor on P as $\Delta^2 = g^{ij} (\partial_i \partial_j - \Gamma_{ij}^k \partial_k)$. Show that there is one Laplace–Beltrami or
gilmore-lie-groups_ F02162	213	■ 1.000	$x^2+y^2 \ \leq \ 1$	$x^2 + y^2 \leq 1$	. Show that there is one L
gilmore-lie-groups_ F02163	213	■ 1.000	$S^{\{n\}}$	S^n	Show that the two metric-preserving mappings T_1 and T_2 that satisfy
gilmore-lie-groups_ F02164	213	■ 1.000	$H^{\{n\}}, \ n>2$	$H^n, n > 2$	Show that the two metric-preserving mappings T_1 and T_2 that satisfy
gilmore-lie-groups_ F02165	213	■ 1.000	$T_{\{1\}}^2=T_{\{2\}}^2=I$	$T_1^2 = T_2^2 = I$	generate S_3 , $S_3 = \langle \tau_1, \tau_2 \rangle$ and $S_3 = \langle \tau_1, \tau_2 \rangle$ where $\tau_1 = (1\ 2)$ and $\tau_2 = (1\ 2\ 3)$. Show that these three operators, together with the identity, form a group isomorphic with
gilmore-lie-groups_ F02166	213	■ 1.000	$T_{\{1\}} \ T_{\{2\}}=T_{\{2\}} \ T_{\{1\}}$	$T_1 T_2 = T_2 T_1$	the “four-group” (“vierergruppe”) V_4 . Describe the variety of decompositions of a
gilmore-lie-groups_ F02167	213	■ 1.000	$T_{\{3\}} \ \neq \ I$	$T_3 \neq I$	the “four-group” (“vierergruppe”) V_4 . Describe the variety of decompositions of a
gilmore-lie-groups_ F02168	213	■ 1.000	$\mathfrak{g}=\mathfrak{g}_{+,+}+\mathfrak{g}_{+,-}+\mathfrak{g}_{-,+}+\mathfrak{g}_{-,-}+\mathfrak{g}_{-,-}+\mathfrak{g}_{-,+}+\mathfrak{g}_{+,-}+\mathfrak{g}_{+,+}$	$\mathfrak{g} = \mathfrak{g}_{+,+} + \mathfrak{g}_{+,-} + \mathfrak{g}_{-,+} + \mathfrak{g}_{-,-}$	Discuss dualities.
gilmore-lie-groups_ F02169	213	■ 1.000	$3!/\ 1!=6$	$3!/1! = 6$	Show that the secular equation for the symmetric space $SO(3)/SO(2)$ can be obtained

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gilmore-lie-groups-F02170	213	1.000	$b_{\{3\}}=0$	$b_3 = 0$	There is one independent function
gilmore-lie-groups-F02171	213	1.000	$\phi_{\{j\}}(\frac{p}{h})$	$\phi_j(\mathfrak{p})$	6. The hyperbolic plane H_2^2 is the Riemannian symmetric space $SO(2,1)/SO(2)$ ob-
gilmore-lie-groups-F02172	213	0.992	$\phi_{\{j\}}(\frac{h}{p}, \frac{p}{h})$	$\phi_j(\mathfrak{h}, \mathfrak{p})$	tained by exponentiating a real symmetric matrix in the three-dimensional Lie algebra
gilmore-lie-groups-F02173	213	0.998	$\mathfrak{h}=0$	$\mathfrak{h} = 0$	$\Gamma \cap _{\mathfrak{h}_+} = \Gamma \cap \Gamma \cap \Gamma \cap \Gamma$
gilmore-lie-groups-F02174	213	0.891	$x_{\{0\}}^{\{2\}}-x_{\{1\}}^{\{2\}}-x_{\{2\}}^{\{2\}}=1$	$x_0^2-x_1^2-x_2^2=1$	
gilmore-lie-groups-F02175	213	1.000	$-d\,s^{\{2\}}=d\,x_{\{0\}}^{\{2\}}-d\,x_{\{1\}}^{\{2\}}-$	$-ds^2=dx_0^2-dx_1^2-$	
gilmore-lie-groups-F02176	213	0.973	$d\,x_{\{2\}}^{\{2\}}$	dx_2^2	
gilmore-lie-groups-F02177	214	1.000	$x_{\{1\}},\,x_{\{2\}}$	x_1,x_2	c. Use coordinates x_1,x_2 to parameterize the points in H_2^2 , and show
gilmore-lie-groups-F02178	214	1.000	$(r,\,\theta),\,x_{\{1\}}=r\,\cos\,(\theta),\,x_{\{2\}}=r\,\sin\,(\theta)$	$(r,\theta),x_1=r\cos(\theta),x_2=r\sin(\theta)$	$\Gamma \cap \Gamma \cap \Gamma \cap \Gamma \cap \Gamma$
gilmore-lie-groups-F02179	214	1.000	$S\,O(1,2)$	$SO(1,2)$	1. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$. a. Show that the generators of infinitesimal rotations at a point are $X_{rs} = g_{rt}x^t\partial_s -$
gilmore-lie-groups-F02180	214	1.000	$\left(x_{\{1\}},\,x_{\{2\}}\right)\,\in\,H_{\{2\}}^{\{2\}}$	$(x_1,x_2)\in H_2^2$	1. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$. a. Show that the generators of infinitesimal rotations at a point are $X_{rs} = g_{rt}x^t\partial_s -$
gilmore-lie-groups-F02181	214	1.000	$g_{\{i\,j\}}(x)$	$g_{ij}(x)$	$g_{st}x^t\partial_r,$ $G_{ab;rs} = G_{ab;rs} + G_{ab;rs} + G_{ab;rs} + G_{ab;rs} + G_{ab;rs} + G_{ab;rs} + G_{ab;rs} + G_{ab;rs} + G_{ab;rs} + G_{ab;rs}$
gilmore-lie-groups-F02182	214	1.000	$X_{\{r\,s\}}=g_{\{r\,t\}}\,x^{\{t\}}\,\partial_{\{s\}}-$	$X_{rs} = g_{rt}x^t\partial_s -$	b. Show $[X_{ab},\Delta]=0$, where $\Delta = G^{ab}X_{ab}X_{rs}$ is the Laplace–Beltrami operator on this space. $\mathcal{Q}_{ab} = \text{tr}(\partial_a\partial_b V) - \partial_a\partial_b(V) - \partial_a\partial_b(V)$ and $G_{ab;rs}$ is the image of $\mathcal{Q}_{ab}$.
gilmore-lie-groups-F02183	214	1.000	$g_{\{s\,t\}}\,x^{\{t\}}\,\partial_{\{r\}}$	$g_{st}x^t\partial_r$	c. S
gilmore-lie-groups-F02184	214	1.000	$\left[X_{\{a\,b\}},\,\Delta\right]=0$	$[X_{ab},\Delta] = 0$	c. Show that Δ consists of terms that are both quadratic and linear in the operators ∂_r , and that
gilmore-lie-groups-F02185	214	1.000	$\Delta=G^{\{a\,b\,;\,r\,s\}}\,X_{\{a\,b\}}\,X_{\{r\,s\}}$	$\Delta = G^{ab;rs}X_{ab}X_{rs}$	c. Show that Δ consists of terms that are both quadratic and linear in the operators ∂_r , and that
gilmore-lie-groups-F02186	214	0.546	$G_{\{a\,b\,;\,r\,s\}}=\text{operatorname{tr}}\left\{\frac{d\,e\,f}{\left(X_{\{a\,b\}}\right)\,\frac{d\,e\,f}{\left(X_{\{r\,s\}}\right)}\right\}$	$G_{ab;rs} = \text{tr}\{\mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{ab})\mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{rs})\}$	1. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$.
gilmore-lie-groups-F02187	214	0.546	$G^{\{a\,b\,;\,r\,s\}}$	$G^{ab;rs}$	1. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$.
gilmore-lie-groups-F02188	214	0.546	$G_{\{a\,b\,;\,r\,s\}}$	$G_{ab;rs}$	1. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$.
gilmore-lie-groups-F02189	214	0.989	Δ	Δ	$\Delta = g^{-rs}\partial_r\partial_s - g^{-rs}\mathbf{1}_{rs}^{}\partial_t$
gilmore-lie-groups-F02190	214	1.000	$\partial_{\{r\}}$	∂_r	The f
gilmore-lie-groups-F02191	214	0.765	$\Gamma_{\{r\}\{s\}}^{\{t\}}$	Γ_{rs}^{t}	$\mathbf{1}_{rs} = \frac{1}{2}g^{-rs}(\mathfrak{u}_sg_r\mathfrak{u}_t + \mathfrak{u}_r\mathfrak{g}_s\mathfrak{u}_t - \mathfrak{u}_t\mathfrak{g}_{rs})$
gilmore-lie-groups-F02192	214	1.000	$\left(r,\,\phi_{\{2\}},\,\phi_{\{3\}},\,\ldots,\,\phi_{\{n\}}\right)$	$(r,\phi_2,\phi_3,\dots,\phi_n)$	$dr\wedge d\phi_2\wedge d\phi_3\wedge\cdots\wedge d\phi_n$
gilmore-lie-groups-F02193	214	1.000	$S^{\{n\}}\,\subset\,R^{\{n+1\}}$	$S^n\subset R^{n+1}$	$dr\wedge d\phi_2\wedge d\phi_3\wedge\cdots\wedge d\phi_n$
gilmore-lie-groups-F02194	215	0.995	$S\,O(n+1)$	$SO(n+1)$	c. Compare this with the second order Casimir operator for $SO(n+1)$:
gilmore-lie-groups-F02195	215	1.000	$f_{\{1\}}(\phi)$	$f_1(\phi)$	1. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$.

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gilmore-lie-groups-F02222	216	0.719	$(\partial, \partial) x_{\mu}$	$(\partial, \partial) x_{\mu}$	
gilmore-lie-groups-F02223	216	1.000	$z=x+i y$	$z = x + iy$	$az + b = \begin{bmatrix} a & b \end{bmatrix}$
gilmore-lie-groups-F02224	216	1.000	$P S L(2, \mathbb{R})$	$PSL(2, \mathbb{R})$	
gilmore-lie-groups-F02225	216	1.000	$M, -M \in S L(2 ; \mathbb{R})$	$M, -M \in SL(2; \mathbb{R})$	$z' = \frac{az + b}{cz + d}$
gilmore-lie-groups-F02226	216	1.000	$y^{\prime} > 0$	$y' > 0$	
gilmore-lie-groups-F02227	216	1.000	$y > 0$	$y > 0$	
gilmore-lie-groups-F02228	216	1.000	$y^{\prime} = 0$	$y' = 0$	
gilmore-lie-groups-F02229	216	0.997	$(y=0)$	$(y = 0)$	
gilmore-lie-groups-F02230	217	1.000	$d z^{\prime} = d z / c z+d ^2$	$dz' = dz/ cz + d ^2$	a. Show that $dx = dx/(1-x^2)$. e. Show that the invariant mea
gilmore-lie-groups-F02231	217	1.000	$d \mu = d x d y / y^2$	$d\mu = dxdy/y^2$	f. Show that the distance between two points z_1 and z_2 is
gilmore-lie-groups-F02232	217	1.000	$z_{\{2\}}$	z_2	$z_1 = z_2 = 1$
gilmore-lie-groups-F02233	217	1.000	$w=x+i y$	$w = x + iy$	$ww = x^* + y^* \leq 1$. The unit disk, with a suitable metric, provides a second represen-
gilmore-lie-groups-F02234	217	1.000	$\bar{w} w=w^{\{2\}+y^{\{2\}} \leq 1$	$\bar{w} w = x^2 + y^2 \leq 1$	tion of the hyperbolic plane. The unit disk is mapped onto itself by linear fractional transformations
gilmore-lie-groups-F02235	217	1.000	$M, -M \in S U(1,1)$	$M, -M \in SU(1, 1)$	b. Show that $w = e^{i\psi} \rightarrow w' = e^{i\psi}$. Compute $\psi(\phi)$.
gilmore-lie-groups-F02236	217	1.000	$w=e^{i \phi} \rightarrow w^{\prime}=e^{i \psi}$	$w = e^{i\phi} \rightarrow w' = e^{i\psi}$	$\begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$
gilmore-lie-groups-F02237	217	1.000	$\psi(\phi)$	$\psi(\phi)$	$\begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$
gilmore-lie-groups-F02238	217	1.000	$w_{\{1\}}$	w_1	Show that the mapping from z in the upper half-plane to w in the unit disk given by
gilmore-lie-groups-F02239	217	1.000	$w_{\{2\}}$	w_2	Show that the mapping from z in the upper half-plane to w in the unit disk given by
gilmore-lie-groups-F02240	217	1.000	w	w	a. Compute the inverse of this mapping, and show that it maps the interior of the unit
gilmore-lie-groups-F02241	217	1.000	$z_{\{0\}}$	z_0	
gilmore-lie-groups-F02242	218	1.000	$z_{\{0\}}=i$	$z_0 = i$	b. Choose $z_0 = i$ and $e^{i\phi} = i$ to give the canonical ma
gilmore-lie-groups-F02243	218	1.000	$e^{i \phi}=i$	$e^{i\phi} = i$	b. Choose $z_0 = i$ and $e^{i\phi} = i$ to give the canonical ma
gilmore-lie-groups-F02244	219	1.000	$Y_{\{r\}}$	Y_r	$C_{rs}(\epsilon) = M_r(\epsilon) M_s(\epsilon) C_{ij}(M(\epsilon)^{-1})_k$ (cf. Eq. (4.22)). As long as the change of basis transformation is nonsingular the Lie algebra is unchanged.
gilmore-lie-groups-F02245	219	1.000	$Y_{\{r\}}=$	$Y_r =$	If the transformation becomes singular, the structure constants $C^{\mu}{}_{\nu\lambda}(x)$ may
gilmore-lie-groups-F02246	219	1.000	$M_{\{r\}}^{\{i\}}(\epsilon) X_{\{i\}}$	$M_r^{\ i}(\epsilon) X_i$	If the transformation becomes singular, the structure constants $C_{rs}(\epsilon)$ may still converge to a well-defined limit. It is often the case that the structure con-
gilmore-lie-groups-F02247	219	0.986	$C_{\{r\ s\}}^{\{t\}}(\epsilon) M_{\{r\}}^{\{i\}}(\epsilon) M_{\{s\}}^{\{j\}}(\epsilon) C_{\{i\ j\}}^{\{k\}} \left(M(\epsilon)^{-1}\right)^t \rightarrow C_{rs}^{\ t}(\epsilon)$	$C_{rs}^{\ t}(\epsilon) = M_r^{\ i}(\epsilon) M_s^{\ j}(\epsilon) C_{ij}^{\ k}(M(\epsilon)^{-1})_k^{\ t}$	stant converge to a well-defined limit. It is often the case that the structure con-

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gilmore-lie-groups_ F02248	219	■ 1.000	$C_{rs}^t(\epsilon)$	$C_{rs}^t(\epsilon)$	$C_{rs}^t(0) = \lim_{\epsilon \rightarrow 0} C_{rs}^t(\epsilon)$ (13.1)
gilmore-lie-groups_ F02249	220	■ 0.998	$\mathfrak{g} \rightarrow \mathfrak{g}^{\prime}$	$\mathfrak{g} \rightarrow \mathfrak{g}'$	basis transformation
gilmore-lie-groups_ F02250	220	■ 1.000	$\operatorname{dim}(\mathfrak{h})$	$\dim(\mathfrak{h})$	$\mathfrak{g}$ are then defined for all values of ϵ , including the singular value $\epsilon = 0$.
gilmore-lie-groups_ F02251	220	■ 1.000	$\epsilon \rightarrow 0$	$\epsilon \rightarrow 0$	$[\mathfrak{h}', \mathfrak{h}'] = [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}$
gilmore-lie-groups_ F02252	220	■ 1.000	$\left[\mathfrak{h}, \mathfrak{p}^{\prime}\right] \subseteq \mathfrak{p}^{\prime}$	$[\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}'$	$[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \qquad \rightarrow \qquad [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}$ (13)
gilmore-lie-groups_ F02253	220	■ 0.905	$\mathrm{SO}(3) \rightarrow \operatorname{ISO}(2)$	$\mathrm{SO}(3) \rightarrow \mathrm{ISO}(2)$	utions are
gilmore-lie-groups_ F02254	220	■ 0.998	$L_{\{1\}}\!=\!X_{\{23\}}\!=$	$L_1 = X_{23} =$	$[L_1, L_2] = -L_3$
gilmore-lie-groups_ F02255	220	■ 1.000	$x_{\{2\}}\,\partial_{\{3\}}\!-\!x_{\{3\}}\,\partial_{\{2\}}\!=\!\epsilon_{1jk}x_j\partial_k$	$x_2\partial_3 - x_3\partial_2 = \epsilon_{1jk}x_j\partial_k$	$[L_2, L_3] = -L_1$ (15.0)
gilmore-lie-groups_ F02256	220	■ 1.000	$L_{\{2\}}$	L_2	$[L_2, L_3] = -L_1$ (15.0)
gilmore-lie-groups_ F02257	220	■ 1.000	$L_{\{3\}}$	L_3	$[L_2, L_3] = -L_1$ (15.0)
gilmore-lie-groups_ F02258	221	■ 1.000	$L_{\{1\}}$	L_1	finitesimal generator L_3) the operators L_1 and L_2 go to
gilmore-lie-groups_ F02259	221	■ 1.000	$\mathrm{ISO}(2)=\mathrm{E}(2)$	$\mathrm{ISO}(2) = \mathrm{E}(2)$	the commutation relations of the commutator algebra $\mathfrak{so}(3) \oplus \mathfrak{so}(2) \cong \mathfrak{e}(3)$ are
gilmore-lie-groups_ F02260	221	■ 1.000	$L_{\{3\}}, P_{\{1\}}, P_{\{2\}}$	L_3, P_1, P_2	$E(2)$, or inhomogeneous orthogonal transformations in the plane K^{π} , $\mathrm{ISO}(2)$. This
gilmore-lie-groups_ F02261	221	■ 0.983	$R^2, \mathrm{ISO}(2)$	$R^2, \mathrm{ISO}(2)$	group consists of rotations about the z -axis, generated by L_3 , and displacements of the origin in the x - and y -directions, generated by $P_1 = \partial_x$ and $P_2 = \partial_y$.
gilmore-lie-groups_ F02262	221	■ 1.000	$P_{\{1\}}\!=\!\partial_{\{1\}}$	$P_1 = \partial_1$	$x^2 + y^2 + z^2 = R^2$ in the neighborhood of the north pole $(0, 0, R)$, as shown
gilmore-lie-groups_ F02263	221	■ 1.000	$P_{\{2\}}\!=\!\partial_{\{2\}}$	$P_2 = \partial_2$	$x^2 + y^2 + z^2 = R^2$ in the neighborhood of the north pole $(0, 0, R)$, as shown
gilmore-lie-groups_ F02264	221	■ 0.698	$x^2\!+\!y^2\!+\!z^2\!=\!R^2$	$x^2 + y^2 + z^2 = R^2$	$\mathbb{R}^3 \setminus \{0\} \cong \mathbb{R}^+ \times \mathbb{S}^2$
gilmore-lie-groups_ F02265	221	■ 0.698	$0,0,R$	$0, 0, R$	$\mathbb{R}^3 \setminus \{0\} \cong \mathbb{R}^+ \times \mathbb{S}^2$
gilmore-lie-groups_ F02266	221	■ 1.000	$R \rightarrow \infty$	$R \rightarrow \infty$	$R^+ \times R^+ \cong \mathbb{R}^+$
gilmore-lie-groups_ F02267	221	■ 0.992	$-P_{\{2\}},+P_{\{1\}}$	$-P_2, +P_1$	$d_2 = -R\theta_1$ (15.11)
gilmore-lie-groups_ F02268	221	■ 0.992	$-y$	$-y$	$d_2 = -R\theta_1$ (15.11)
gilmore-lie-groups_ F02269	221	■ 0.992	$+x$	$+x$	$d_2 = -R\theta_1$ (15.11)
gilmore-lie-groups_ F02270	221	■ 1.000	θ_1, θ_2	θ_1, θ_2	As the radius of the spheres becomes very large, the two angles $\theta_1 = 0$ become
gilmore-lie-groups_ F02271	221	■ 1.000	$d_{\{1\}}, d_{\{2\}}$	d_1, d_2	As the radius of the spheres becomes very large, the two angles $\theta_1 = 0$ become
gilmore-lie-groups_ F02272	221	■ 1.000	$R\,\theta_{\{i\}}(i=1,2)$	$R\theta_i(i = 1, 2)$	
gilmore-lie-groups_ F02273	221	■ 1.000	$\theta_2\!=\!d_{\{1\}}\,/\,R$	$\theta_2 = d_1/R$	

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gilmore-lie-groups-F02274	221	■ 1.000	$d_{\{1\}}$	d_1	
gilmore-lie-groups-F02275	221	■ 1.000	$\theta_{\{1\}}=d_{\{2\}} / R$	$\theta_1 = d_2/R$	
gilmore-lie-groups-F02276	221	■ 1.000	$-d_{\{2\}}$	$-d_2$	
gilmore-lie-groups-F02277	222	■ 0.737	$\mathrm{SO}(4) \rightarrow \operatorname{ISO}(3)$	$\mathrm{SO}(4) \rightarrow \mathrm{ISO}(3)$	$V_i = x_i \partial_4 - x_4 \partial_i \qquad 1 \leq i \neq j \neq k \leq 3$
gilmore-lie-groups-F02278	223	■ 0.418	$P_{\{i\}}$	P_i	group, on R^3 .
gilmore-lie-groups-F02279	223	■ 0.418	$i=1,2,3$	$i = 1, 2, 3$	group, on R^3 .
gilmore-lie-groups-F02280	223	■ 1.000	$\mathrm{ISO}(3)$	$ISO(3)$	As in the case $SO(3) \rightarrow ISO(2)$, we can contract the second order Casimir
gilmore-lie-groups-F02281	223	■ 1.000	$\mathrm{SO}(3) \rightarrow \mathrm{ISO}(2)$	$SO(3) \rightarrow ISO(2)$	$\mathcal{C}^2[\mathrm{ISO}(3)] = \lim_{R \rightarrow \infty} (1/R^2)(\mathbf{L} \cdot \mathbf{L} + \mathbf{V} \cdot \mathbf{V})$
gilmore-lie-groups-F02282	223	■ 1.000	$\mathbf{P} \cdot \mathbf{P} = \nabla^2$	$\mathbf{P} \cdot \mathbf{P} = \nabla^2$	
gilmore-lie-groups-F02283	223	■ 1.000	$\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$	$\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$	
gilmore-lie-groups-F02284	224	■ 0.994	$\mathrm{SO}(4,1) \rightarrow \mathrm{ISO}(3,1)$	$\mathrm{SO}(4,1) \rightarrow \mathrm{ISO}(3,1)$	13.3.3 The contraction $\mathrm{SO}(4,1) \rightarrow \mathrm{ISO}(3,1)$
gilmore-lie-groups-F02285	224	■ 0.592	$[\mathrm{SO}(3,1)]$	$[\mathrm{SO}(3,1)]$	The group $ISO(3,1)$ consists of proper Lorentz transformations $[\mathrm{SO}(3,1)]$ that
gilmore-lie-groups-F02286	224	■ 0.997	$\mathrm{SO}(3,2)$	$SO(3,2)$	we choose as infinitesimal generators for the group $SO(4,1)$.
gilmore-lie-groups-F02287	224	■ 0.998	$\epsilon^{\alpha\beta\gamma\mu\nu}$	$\epsilon^{\alpha\beta\gamma\mu\nu}$	$\lim_{R \rightarrow \infty} (1/R)^{\alpha\beta\gamma\mu\nu} = \lim_{R \rightarrow \infty} \epsilon^{\alpha\beta\gamma\mu\nu} (1/R)^{\alpha\beta\gamma\mu\nu}$
gilmore-lie-groups-F02288	224	■ 1.000	$W_{\{\alpha\}}$	W_α	$= \epsilon^{a\rho\gamma\mu\lambda} X_{\beta\gamma}(\partial/\partial x^\mu)$ (13.24)
gilmore-lie-groups-F02289	224	■ 1.000	$W^{\{\alpha\}}$	W^α	$= \epsilon^{a\rho\gamma\mu\lambda} X_{\beta\gamma}(\partial/\partial x^\mu)$ (13.24)
gilmore-lie-groups-F02290	224	■ 0.615	v	v	
gilmore-lie-groups-F02291	225	■ 1.000	$\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma}(\partial/\partial x^\mu)$	$\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma}(\partial/\partial x^\mu)$	The four vector $\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma}(\partial/\partial x^\mu)$ is fairly complicated. Since $W^\alpha W_\alpha$ is invariant,
gilmore-lie-groups-F02292	225	■ 1.000	$W^{\{\alpha\}} W_{\{\alpha\}}$	$W^\alpha W_\alpha$	The four vector $\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma}(\partial/\partial x^\mu)$ is fairly complicated. Since $W^\alpha W_\alpha$ is invariant,
gilmore-lie-groups-F02293	225	■ 1.000	$\mathbf{P} \cdot \mathbf{P} = \sum P_\mu P^\mu = -(mc)^2$	$\mathbf{P} \cdot \mathbf{P} = \sum P_\mu P^\mu = -(mc)^2$	It should be emphasized that if an interpretation is desired, the operator
gilmore-lie-groups-F02294	225	■ 0.862	$\mathfrak{u}(2)$	$\mathfrak{u}(2)$	$\{I_3, -I_1\} = \mathfrak{u}(2)$ (13.28)
gilmore-lie-groups-F02295	225	■ 0.862	$J_{\{3\}}, J_{\{\pm\}}, J_{\{0\}}$	$J_3, J_\pm, J_0$	$\{I_3, -I_1\} = \mathfrak{u}(2)$ (13.28)
gilmore-lie-groups-F02296	225	■ 0.996	$h_{\{3\}}, h_{\{\pm\}}, h_{\{0\}}$	$h_3, h_\pm, h_0$	$\left \begin{smallmatrix} h_3 \\ h_n \end{smallmatrix} \right \left \begin{smallmatrix} 2c^2 \\ 1 \end{smallmatrix} \right \left \begin{smallmatrix} h_3 \\ J_n \end{smallmatrix} \right $
gilmore-lie-groups-F02297	226	■ 1.000	$c \rightarrow 0$	$c \rightarrow 0$	commutation relations (Eq. (13.30)) converge to a well-defined limit satisfied by
gilmore-lie-groups-F02298	226	■ 1.000	$J_{\{0\}} \rightarrow h_{\{0\}}$	$J_0 \rightarrow h_0$	$\lim_{c^2 \rightarrow 0} c^2 = \lim_{c^2 \rightarrow 0} c^2 \left(h_3 - \frac{1}{h_3} \right) = \frac{1}{2} (c I_3 M c I_3) + (c I_3 M c I_3)$
gilmore-lie-groups-F02299	226	■ 1.000	$\left(h_{\{0\}} / 2 c \right)^2$	$(h_0/2c)^2$	in the Lie algebra in the limit $c \rightarrow 0$, (13.30) $\rightarrow 0$ and the remaining operators go to a well-defined limit

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gilmore-lie-groups_ F02326	231	■ 0.992	$\backslash\theta^{\backslash\prime}, \backslash\phi^{\backslash\prime}$	θ', ϕ'	operator on $S^2 = SO(3)/SO(2)$ when restricted to the sphere surface, and this
gilmore-lie-groups_ F02327	231	■ 0.998	$\backslash\mathbf{L} \cdot \backslash\mathbf{L}$	$\mathbf{L} \cdot \mathbf{L}$	Under the contraction $\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$ the representations of the unitar
gilmore-lie-groups_ F02328	231	■ 0.998	$\backslash\nabla^2$	∇^2	Under the contraction $\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$ the representations of the unitar
gilmore-lie-groups_ F02329	231	■ 1.000	$\backslash\mathfrak{u}(2) \rightarrow \backslash\mathfrak{h}_4$	$\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$	
gilmore-lie-groups_ F02330	232	■ 1.000	$N_{\{n\}}$	N_n	c. Carry out steps c–f of the previous problem. The results are obtained by making the following replacements:
gilmore-lie-groups_ F02331	232	■ 0.993	$N_{\{n\}}=1 / \sqrt{2^nn! \sqrt{\pi}}$	$N_n = 1/\sqrt{2^n n! \sqrt{\pi}}$	the following i
gilmore-lie-groups_ F02332	232	■ 0.984	$J_{\{3\}}, J_{\{ \pm \}} \backslash\left(\backslash\left[J_{\{3\}}, J_{\{ \pm \}}\right]= \pm J_{\{ \pm \}}, \backslash\left[J_{\{+ \}}, J_{\{ - \}}\right]=2 J_{\{3\}}\right)$	$J_3, J_{\pm} ([J_3, J_{\pm}] = \pm J_{\pm}, [J_+, J_-] = 2J_3)$	$\lim_{\kappa \rightarrow 0} \epsilon(2J_3) \rightarrow P \quad P' = \partial_\kappa$
gilmore-lie-groups_ F02333	232	■ 0.993	$\backslash\left(P^{\backslash\prime}, T^{\backslash\prime}, V^{\backslash\prime}\right)$	(P', T', V')	4. Contract $SO(n+1)$ with respect to the subgroup $SO(n)$ and show how the invariant
gilmore-lie-groups_ F02334	232	■ 0.991	$\backslash\partial_x, \backslash\partial_t, t \backslash\partial_x$	$\partial_x, \partial_t, t\partial_x$	metric and measure on the sphere $S^n = SO(n+1)/SO(n)$ reduce to the familiar
gilmore-lie-groups_ F02335	232	■ 1.000	$\backslash\mathfrak{a}_{\{1\}}$	$\mathfrak{a}_1$	metric and measure on $R^n = ISO(n)/SO(n)$.
gilmore-lie-groups_ F02336	232	■ 0.985	$\backslash\mathfrak{gal}(1)$	$\mathfrak{gal}(1)$	
gilmore-lie-groups_ F02337	232	■ 1.000	$S^{\{n\}}=S^0(n+1) / S^0(n)$	$S^n = SO(n+1)/SO(n)$	
gilmore-lie-groups_ F02338	232	■ 1.000	$R^{\{n\}}=ISO(n) / S^0(n)$	$R^n = ISO(n)/SO(n)$	
gilmore-lie-groups_ F02339	233	■ 1.000	$\backslash\tau=(\zeta / \zeta) \backslash\tan (\zeta)$	$\tau=(\zeta/ \zeta) \tan(\zeta)$	. Use a faithful matrix representat
gilmore-lie-groups_ F02340	233	■ 1.000	$\backslash\alpha=\lim _{c \rightarrow 0}\{c \rightarrow 0\} \zeta / c$	$\alpha=\lim_{c \rightarrow 0} \zeta/c$	thermal expectation
gilmore-lie-groups_ F02341	233	■ 1.000	$\backslash\langle X\rangle=\backslash\mathrm{tr} X e^{-\backslash\beta \mathcal{H}} / \backslash\mathrm{tr} e^{-\backslash\beta \mathcal{H}}$	$\langle X\rangle=\mathrm{tr} X e^{-\beta \mathcal{H}} / \mathrm{tr} e^{-\beta \mathcal{H}}$	Lie algebra these expectation values can often be computed rather simply.
gilmore-lie-groups_ F02342	233	■ 1.000	$\backslash\left\langle e^{\alpha X}\right\rangle=\backslash\mathrm{tr} e^{\alpha X} e^{-\backslash\beta \mathcal{H}} / \backslash\mathrm{tr} e^{-\backslash\beta \mathcal{H}}$	$\langle e^{\alpha X}\rangle =$	Lie algebra these expectation values can often be computed rather simply.
gilmore-lie-groups_ F02343	233	■ 1.000	$\backslash\mathrm{tr} e^{\alpha X} e^{-\backslash\beta \mathcal{H}} / \backslash\mathrm{tr} e^{-\backslash\beta \mathcal{H}}$	$\mathrm{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \mathrm{tr} e^{-\beta \mathcal{H}}$	a. Assume $\mathcal{H}=\epsilon J_3$ and X is in the Lie algebra $\mathfrak{su}(2)$. Show that in the 2×2 defining
gilmore-lie-groups_ F02344	233	■ 0.998	$\backslash\mathcal{H}=\epsilon J_3$	$\mathcal{H}=\epsilon J_3$	$\Gamma \cosh (A / \gamma) \pm(A / A) \sinh (A / \gamma) \quad(A=-i A) / A \sinh (A / \gamma) \quad 1$
gilmore-lie-groups_ F02345	233	■ 1.000	$2 j+1=2$	$2j+1=2$	d. Show that in the $(2j+1) \times(2j+1)$ dimensional representation,
gilmore-lie-groups_ F02346	234	■ 0.974	$\backslash\mathcal{C}^2=\sum_{i j} X_{i j}^2, \backslash\mathcal{C}^3=\epsilon^{\backslash\text {abcdef }} X_{\{a b\}} X_{\{c d\}} X_{\{e f\}}$	$\mathcal{C}^2=\sum_{ij} X_{ij}^2, \mathcal{C}^3=\epsilon^{\text {abcdef }} X_{ab} X_{cd} X_{ef}$	are analytic continuations of $\mathcal{C}^2=\sum_{ij} X_{ij}^2$, $\mathcal{C}^3=\epsilon^{\text {abcdef }} X_{ab} X_{cd} X_{ef}$, and $\mathcal{C}^4=$
gilmore-lie-groups_ F02347	234	■ 0.974	$\backslash\mathcal{C}^4=$	$\mathcal{C}^4=$	are analytic continuations of $\mathcal{C}^2=\sum_{ij} X_{ij}^2$, $\mathcal{C}^3=\epsilon^{\text {abcdef }} X_{ab} X_{cd} X_{ef}$, and $\mathcal{C}^4=$
gilmore-lie-groups_ F02348	234	■ 0.812	$\backslash\sum_{i j} Y_{i j}^2$	$\sum_{ij} Y_{ij}^2$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij}=\epsilon^{ijkl} X_{cd} X_{ef}$ of the group $SO(6)$ b. Contract $SO(A+2)$ with respect to the subgroups $SO(A) \times$
gilmore-lie-groups_ F02349	234	■ 0.812	$Y_{\{i j\}}=\epsilon^{\{j c d e f\}} X_{\{c d\}} X_{\{e f\}}$	$Y_{ij}=\epsilon^{j c d e f} X_{cd} X_{ef}$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij}=\epsilon^{ijkl} X_{cd} X_{ef}$ of the group $SO(6)$ b. Contract $SO(A+2)$ with respect to the subgroups $SO(A) \times$
gilmore-lie-groups_ F02350	234	■ 0.812	$S^0(6)$	$SO(6)$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij}=\epsilon^{ijkl} X_{cd} X_{ef}$ of the group $SO(6)$ b. Contract $SO(A+2)$ with respect to the subgroups $SO(A) \times$
gilmore-lie-groups_ F02351	234	■ 0.995	$S^0(4) \backslash\otimes S^0(2)$	$SO(4) \otimes SO(2)$	a. Contract $SO(4, 2)$ with respect to the subgroup $SO(4) \otimes SO(2)$. c. Construct the quadratic, cubic, and quartic Casimir operators a

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gilmore-lie-groups-F02352	234	■ 0.933	$A_i=\lim_{\epsilon\rightarrow 0}\epsilon X_{i5}$	$A_i=\lim_{\epsilon\rightarrow 0}\epsilon X_{i5}$	the Casimir operators contract to
gilmore-lie-groups-F02353	234	■ 0.933	$B_i=\lim_{\epsilon\rightarrow 0}\epsilon X_{i6}$	$B_i=\lim_{\epsilon\rightarrow 0}\epsilon X_{i6}$	the Casimir operators contract to
gilmore-lie-groups-F02354	234	■ 1.000	$S\,0(4,2)/[S\,0(4)\,\otimes\,S\,0(2)]$	$SO(4,2)/[SO(4)\otimes SO(2)]$	8. Riemannian symmetric spaces have been classified using the Cartan decomposition.
gilmore-lie-groups-F02355	234	■ 1.000	$I[S\,0(4)\,\otimes\,S\,0(2)]/[S\,0(4)\,\otimes\,S\,0(2)]$	$I[SO(4)\otimes SO(2)]/[SO(4)\otimes SO(2)]$	8. Riemannian symmetric spaces have been classified using the Cartan decomposition.
gilmore-lie-groups-F02356	234	■ 1.000	X_{α}	X_{α}	$g_{\alpha,\beta}$
gilmore-lie-groups-F02357	234	■ 0.987	$Y_{\alpha}=\lim_{\epsilon\rightarrow 0}\epsilon\,X_{\alpha}$	$Y_{\alpha}=\lim_{\epsilon\rightarrow 0}\epsilon\,X_{\alpha}$	$S\,V\,p_{\alpha,\beta}=\lim_{\epsilon\rightarrow 0}\epsilon\,V\,p_{\alpha,\beta}=\epsilon\,V\,p_{\alpha,\beta}$
gilmore-lie-groups-F02358	236	■ 1.000	$ \psi\rangle$	$ \psi\rangle$	among observers forms a group. If observer S' wants to determine what observer
gilmore-lie-groups-F02359	236	■ 1.000	$\langle S\, \,\psi\rangle$	$\langle S\, \,\psi\rangle$	S has seen, he applies the appropriate transformation, $\langle S S'\rangle$, to his mathematical
gilmore-lie-groups-F02360	236	■ 1.000	$\langle S'\, \,\psi\rangle$	$\langle S'\, \,\psi\rangle$	S has seen, he applies the appropriate transformation, $\langle S S'\rangle$, to his mathematical
gilmore-lie-groups-F02361	236	■ 0.998	$\langle S'\, \,S\rangle$	$\langle S'\, \,S\rangle$	
gilmore-lie-groups-F02362	236	■ 1.000	$\langle S\, \,S'\rangle$	$\langle S\, \,S'\rangle$	
gilmore-lie-groups-F02363	237	■ 1.000	$ \psi'\rangle$	$ \psi'\rangle$	$\langle S \psi'\rangle=\langle S$
gilmore-lie-groups-F02364	237	■ 0.997	$S\,0(3))$	$SO(3))$	$\langle S H S\rangle$, and the existence of a $2p_z$ state in a system with spherical symmetry
gilmore-lie-groups-F02365	237	■ 0.997	$\langle S'\, \,H\, \,S'\rangle=\langle S'\, \,S\rangle\langle S H S\rangle\langle S\, \,S'\rangle$	$\langle S'\, \,H\, \,S'\rangle=\langle S'\, \,S\rangle\langle S H S\rangle\langle S\, \,S'\rangle$	$\langle S H S\rangle$, and the existence of a $2p_z$ state in a system with spherical symmetry
gilmore-lie-groups-F02366	237	■ 1.000	$\langle S'\, \,H\, \,S'\rangle=$	$\langle S'\, \,H\, \,S'\rangle=$	implies the existence (by the Principle of Equivalence) of $2p_x$ and $2p_y$ states, as
gilmore-lie-groups-F02367	237	■ 1.000	$\langle S H S\rangle$	$\langle S H S\rangle$	well as arbitrary linear combinations of these three states.
gilmore-lie-groups-F02368	237	■ 1.000	$2\,p_{\{z\}}$	$2p_z$	well as arbitrary linear combinations of these three states.
gilmore-lie-groups-F02369	237	■ 1.000	$2\,p_{\{x\}}$	$2p_x$	
gilmore-lie-groups-F02370	237	■ 1.000	$2\,p_{\{y\}}$	$2p_y$	
gilmore-lie-groups-F02371	237	■ 0.996	$p^{\mu}p_{\mu}=g_{\mu\nu}\,p^{\mu}\,p^{\nu}=(m\,c)^2$	$p^{\mu}p_{\mu}=g_{\mu\nu}p^{\mu}p^{\nu}=(mc)^2$	$E=(\mathbf{p}c)=(mc^2)$ (14.3)
gilmore-lie-groups-F02372	237	■ 1.000	$\mathbf{p}$	$\mathbf{p}$	Interaction of a particle of charge q with the electromagnetic field i
gilmore-lie-groups-F02373	237	■ 1.000	$p_{\mu}\rightarrow\pi_{\mu}=p_{\mu}-(q/c)A_{\mu}$	$p_{\mu}\rightarrow\pi_{\mu}=p_{\mu}-(q/c)A_{\mu}$	field established by a proton, $\Phi=e/r$ and $\mathbf{A}=\mathbf{0}$, so that $E\rightarrow E+e^2/r$. Here
gilmore-lie-groups-F02374	237	■ 0.974	Φ	Φ	r is the proton–electron distance. The Schrödinger prescription for converting a
gilmore-lie-groups-F02375	237	■ 0.997	$\mathbf{B}=\nabla\times\mathbf{A}$	$\mathbf{B}=\nabla\times\mathbf{A}$	dispersion relation to a wave equation is to replace $\mathbf{p}\rightarrow(\hbar/i)\nabla$ and allow the
gilmore-lie-groups-F02376	237	■ 0.997	$\mathbf{E}=-\nabla\Phi-(1/c)(\partial\mathbf{A}/\partial t)$	$\mathbf{E}=-\nabla\Phi-(1/c)(\partial\mathbf{A}/\partial t)$	dispersion relation to a wave equation is to replace $\mathbf{p}\rightarrow(\hbar/i)\nabla$ and allow the
gilmore-lie-groups-F02377	237	■ 0.997	$q=$	$q=$	dispersion relation to a wave equation is to replace $\mathbf{p}\rightarrow(\hbar/i)\nabla$ and allow the

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gilmore-lie-groups_F02378	237	■0.928	-e	$-e$	resulting equation to act on a spacial function $\psi(\mathbf{x})$. This prescription results in the								
gilmore-lie-groups_F02379	237	■1.000	\Phi=e / r	$\Phi = e/r$									
gilmore-lie-groups_F02380	237	■1.000	\mathbf{A}=\mathbf{0}	$\mathbf{A} = \mathbf{0}$									
gilmore-lie-groups_F02381	237	■1.000	E \rightarrow E+e^2 / r	$E \rightarrow E + e^2/r$									
gilmore-lie-groups_F02382	237	■1.000	\mathbf{p} \rightarrow (\hbar / i) \nabla	$\mathbf{p} \rightarrow (\hbar/i)\nabla$									
gilmore-lie-groups_F02383	237	■1.000	\psi(\mathbf{x})	$\psi(\mathbf{x})$									
gilmore-lie-groups_F02384	238	■0.999	\left\langle S^{\prime} \mid H \mid S^{\prime}\right\rangle=\left\langle S \mid H \mid S\right\rangle	$\langle S' H S' \rangle = \langle S H S \rangle$	Schrödinger solved this equation. compared its predictions with the spectral en-								
gilmore-lie-groups_F02385	238	■0.999	\left\langle S^{\prime} \mid S\right\rangle \in S O(3)	$\langle S' S \rangle \in SO(3)$	Schrödinger solved this equation. compared its predictions with the spectral en-								
gilmore-lie-groups_F02386	238	■0.623	m c^2	mc^2	510 000 eV, it makes sense to write $E = mc^2 + W$, where the principal part of the								
gilmore-lie-groups_F02387	238	■0.906	E=m c^2+W	$E = mc^2 + W$	relativistic energy E is the electron rest energy and the nonrelativistic energy W is								
gilmore-lie-groups_F02388	238	■1.000	W	W	a small perturbation of order ($\simeq 0.0025\%$). Under this substitution, and neglecting								
gilmore-lie-groups_F02389	238	■0.884	(\simeq 0.0025 \%)	$(\simeq 0.0025\%)$	the Schrödinger equation:								
gilmore-lie-groups_F02390	238	■1.000	\left(W+e^2 / r\right)^2 / m c^2	$(W + e^2/r)^2/mc^2$									
gilmore-lie-groups_F02391	238	■0.727	[A, B] / i \hbar=\{A, B\}	$[A, B]/i\hbar = \{A, B\}$	14.4 Quantization conditions								
gilmore-lie-groups_F02392	238	■0.860	(r, \theta, \phi)	(r, θ, ϕ)									
gilmore-lie-groups_F02393	239	■1.000	\mathcal{L}^2\left(S^2\right)	$\mathcal{L}^2\left(S^2\right)$	eigenfunctions are the spherical harmonics $Y_m^l(\theta, \phi)$ and its spectrum of eigenvalues								
gilmore-lie-groups_F02394	239	■0.980	\mathcal{L}^2\left(S^2\right) Y_m^l(\theta, \phi)=-l(l+1) Y_m^l(\theta, \phi)	$\mathcal{L}^2\left(S^2\right) Y_m^l(\theta, \phi) = -l(l+1)Y_m^l(\theta, \phi)$	and $-l \leq m \leq +l$. The negative sign and discrete spectrum characteristically indi-								
gilmore-lie-groups_F02395	239	■0.980	(l, m)	(l, m)	and $-l \leq m \leq +l$. The negative sign and discrete spectrum characteristically indi-								
gilmore-lie-groups_F02396	239	■0.980	l=0,1,2, \ldots	$l = 0, 1, 2, \dots$	and $-l \leq m \leq +l$. The negative sign and discrete spectrum characteristically indi-								
gilmore-lie-groups_F02397	239	■0.999	-l \leq m \leq+l	$-l \leq m \leq +l$	cate that S^2 is compact.								
gilmore-lie-groups_F02398	239	■1.000	A, B, C	A, B, C	<table><tr><td>Equation</td><td>A</td><td>B</td><td>C</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2me^2/\hbar^2$</td><td>$2mW/\hbar^2$</td></tr></table>	Equation	A	B	C	Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$
Equation	A	B	C										
Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$										
gilmore-lie-groups_F02399	239	■0.424	-l(l+1)+\left(e^2 / \hbar c\right)^2	$-l(l+1) + (e^2/\hbar c)^2$	There is a standard procedure for solving simple ordinary differential equatic of the type presented in Eq. (14.9). This is the Frobenius method. The steps involv in this method, and the result of each step, are summarized in Table 14.1.								
gilmore-lie-groups_F02400	239	■0.424	2 E e^2 /(\hbar c)^2	$2Ee^2/(\hbar c)^2$	<table><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2me^2/\hbar^2$</td><td>$2mW/\hbar^2$</td></tr></table> There is a standard procedure for solving simple ordinary differential equatic of the type presented in Eq. (14.9). This is the Frobenius method. The steps involv in this method, and the result of each step, are summarized in Table 14.1.	Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$				
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gilmore-lie-groups_F02481	239	■ 0.424	$\left[E^2 - \left(m\,c^2\right)^2\right] / (\hbar c)^2$	$\left[E^2 - \left(m c^2\right)^2\right] / (\hbar c)^2$	Schrödinger $-l(l+1) \frac{2 m e^2 / \hbar^2}{2 m W / \hbar^2}$ There is a standard procedure for solving simple ordinary differential equations of the type presented in Eq. (14.9). This is the Frobenius method. The steps involved in this method, and the result of each step, are summarized in Table 14.1.																				
gilmore-lie-groups_F02482	239	■ 0.424	$2\,m\,e^2 / \hbar^2$	$2 m e^2 / \hbar^2$	Schrödinger $-l(l+1) \frac{2 m e^2 / \hbar^2}{2 m W / \hbar^2}$ There is a standard procedure for solving simple ordinary differential equations of the type presented in Eq. (14.9). This is the Frobenius method. The steps involved in this method, and the result of each step, are summarized in Table 14.1.																				
gilmore-lie-groups_F02483	239	■ 0.424	$2\,\mathrm{m}\,W / \hbar^2$	$2\,\mathrm{m}\,W / \hbar^2$	Schrödinger $-l(l+1) \frac{2 m e^2 / \hbar^2}{2 m W / \hbar^2}$ There is a standard procedure for solving simple ordinary differential equations of the type presented in Eq. (14.9). This is the Frobenius method. The steps involved in this method, and the result of each step, are summarized in Table 14.1.																				
gilmore-lie-groups_F02484	239	■ 1.000	$n =$	$n =$	$\alpha = \frac{e}{\hbar c} = \frac{1}{137.035999704(70)} = 0.007\,297\,352\,531\,3(3\,8) \quad (14.11)$																				
gilmore-lie-groups_F02485	239	■ 1.000	$0, 1, 2, \ldots$	$0, 1, 2, \ldots$																					
gilmore-lie-groups_F02486	239	■ 1.000	$m_{\text{red}}^{-1} = m_e^{-1} + M_p^{-1}$	$m_{\text{red}}^{-1} = m_e^{-1} + M_p^{-1}$																					
gilmore-lie-groups_F02487	240	■ 0.655	$0, \infty$	$0, \infty$	<table><thead><tr><th>Procedure</th><th>Result</th></tr></thead><tbody><tr><td>1 Locate singularities</td><td>$0, \infty$</td></tr><tr><td>2 Determine analytic behavior at singular points</td><td>$r \rightarrow 0: R \simeq r^\gamma, \quad \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty: R \simeq e^{\lambda r}, \quad \lambda^2 + C = 0$</td></tr><tr><td>3 Keep only $\mathcal{L}^2$ solutions</td><td>$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \quad \lambda = -\sqrt{-C}$</td></tr><tr><td>4 Look for solutions with proper asymptotic behavior</td><td>$R = r^\gamma e^{\lambda r} f(r)$</td></tr><tr><td>5 Construct differential equation for $f(r)$</td><td>$\left[(r D^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)\right] f(r) = 0$</td></tr><tr><td>6 Construct recursion relation</td><td>$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$</td></tr><tr><td>7 Look at asymptotic behavior</td><td>$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)</td></tr><tr><td>8 Construct quantization condition</td><td>$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$</td></tr><tr><td>9 Construct explicit solutions</td><td>$E = \frac{m c^2}{\sqrt{1 + (\alpha/N')^2}}, \quad W = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, \quad N = n + l + 1$</td></tr></tbody></table> This is a dimensionless ratio of three physical constants that are fundamental in three “different” areas of physics: e (electromagnetism), $\hbar$ (quantum mechanics),	Procedure	Result	1 Locate singularities	$0, \infty$	2 Determine analytic behavior at singular points	$r \rightarrow 0: R \simeq r^\gamma, \quad \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty: R \simeq e^{\lambda r}, \quad \lambda^2 + C = 0$	3 Keep only $\mathcal{L}^2$ solutions	$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \quad \lambda = -\sqrt{-C}$	4 Look for solutions with proper asymptotic behavior	$R = r^\gamma e^{\lambda r} f(r)$	5 Construct differential equation for $f(r)$	$\left[(r D^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)\right] f(r) = 0$	6 Construct recursion relation	$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$	7 Look at asymptotic behavior	$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)	8 Construct quantization condition	$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$	9 Construct explicit solutions	$E = \frac{m c^2}{\sqrt{1 + (\alpha/N')^2}}, \quad W = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, \quad N = n + l + 1$
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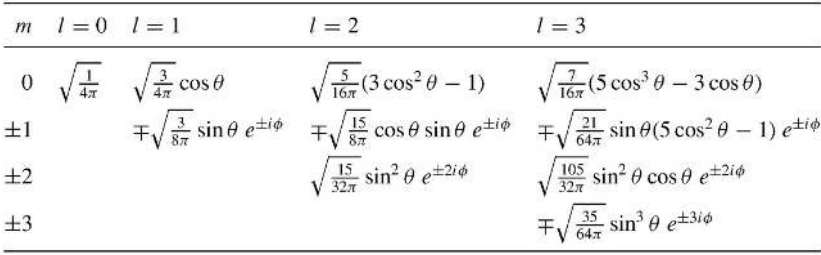
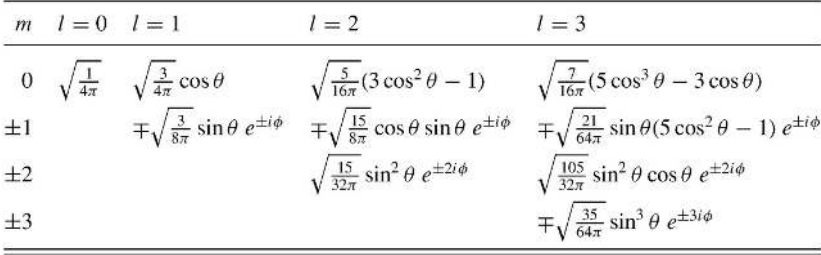
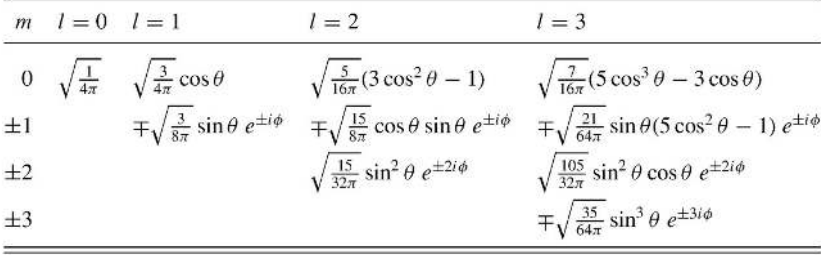
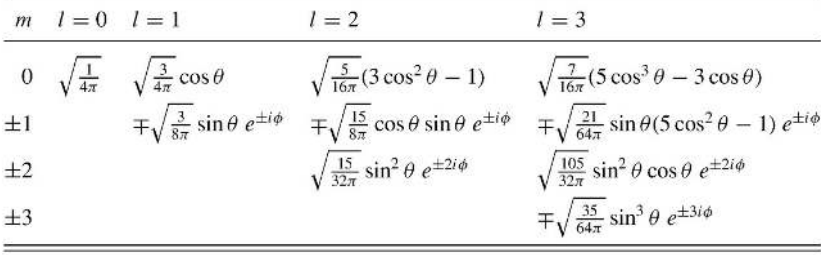
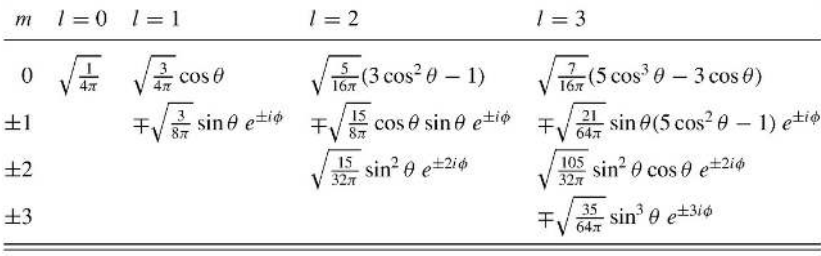
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gilmore-lie-groups_ F02414	240	■ 0.655	$f_{j+1}=-\frac{2 \lambda(j+\gamma)+B}{j(j+1)+2 \gamma(j+1)} f_j$	$f_{j+1}=-\frac{2 \lambda(j+\gamma)+B}{j(j+1)+2 \gamma(j+1)} f_j$	<table><thead><tr><th></th><th>Procedure</th><th>Result</th></tr></thead><tbody><tr><td>1</td><td>Locate singularities</td><td>$0, \infty$</td></tr><tr><td>2</td><td>Determine analytic behavior at singular points</td><td>$r \rightarrow 0: R \simeq r^\gamma, \gamma(\gamma-1)+A=0$ $r \rightarrow \infty: R \simeq e^{\lambda r}, \lambda^2+C=0$</td></tr><tr><td>3</td><td>Keep only $\mathcal{L}^2$ solutions</td><td>$\gamma=\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}, \lambda=-\sqrt{-C}$</td></tr><tr><td>4</td><td>Look for solutions with proper asymptotic behavior</td><td>$R=r^\gamma e^{\lambda r} f(r)$</td></tr><tr><td>5</td><td>Construct differential equation for $f(r)$</td><td>$\left[\left(r D^2+2 \gamma D\right)+(2 \lambda \gamma+B+2 \lambda r D)\right] f(r)=0$</td></tr><tr><td>6</td><td>Construct recursion relation</td><td>$f_{j+1}=-\frac{2 \lambda(j+\gamma)+B}{j(j+1)+2 \gamma(j+1)} f_j$</td></tr><tr><td>7</td><td>Look at asymptotic behavior</td><td>$f \simeq e^{-2 \lambda r}$ if series does not terminate $\simeq e^{+1 \lambda r}$ if series does terminate ($\lambda<0$)</td></tr><tr><td>8</td><td>Construct quantization condition</td><td>$2 \lambda(n+\gamma)+B=0$ or $n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2 \sqrt{-C}}$</td></tr><tr><td>9</td><td>Construct explicit solutions</td><td>$E=\frac{m c^2}{\sqrt{1+(\alpha / N')^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$ $N'=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, N=n+l+1$</td></tr></tbody></table> This is a dimensionless ratio of three physical constants that are fundamental in three “different” areas of physics: e (electromagnetism), $\hbar$ (quantum mechanics),		Procedure	Result	1	Locate singularities	$0, \infty$	2	Determine analytic behavior at singular points	$r \rightarrow 0: R \simeq r^\gamma, \gamma(\gamma-1)+A=0$ $r \rightarrow \infty: R \simeq e^{\lambda r}, \lambda^2+C=0$	3	Keep only $\mathcal{L}^2$ solutions	$\gamma=\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}, \lambda=-\sqrt{-C}$	4	Look for solutions with proper asymptotic behavior	$R=r^\gamma e^{\lambda r} f(r)$	5	Construct differential equation for $f(r)$	$\left[\left(r D^2+2 \gamma D\right)+(2 \lambda \gamma+B+2 \lambda r D)\right] f(r)=0$	6	Construct recursion relation	$f_{j+1}=-\frac{2 \lambda(j+\gamma)+B}{j(j+1)+2 \gamma(j+1)} f_j$	7	Look at asymptotic behavior	$f \simeq e^{-2 \lambda r}$ if series does not terminate $\simeq e^{+1 \lambda r}$ if series does terminate ($\lambda<0$)	8	Construct quantization condition	$2 \lambda(n+\gamma)+B=0$ or $n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2 \sqrt{-C}}$	9	Construct explicit solutions	$E=\frac{m c^2}{\sqrt{1+(\alpha / N')^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$ $N'=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, N=n+l+1$
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gilmore-lie-groups_ F02415	240	—	$\begin{aligned} & \& f \simeq e^{-2 \lambda r} \text { if series does not terminate } \\ & \& \simeq e^{+1 \lambda r} \text { if series does terminate } (\lambda<0) \end{aligned}$	$f \simeq e^{-2 \lambda r}$ if series does not terminate $\simeq e^{+1 \lambda r}$ if series does terminate ($\lambda<0$)	—																														
gilmore-lie-groups_ F02416	240	—	$\begin{aligned} & 2 \lambda(n+\gamma)+B=0 \text { or } \\ & \& n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2 \sqrt{-C}} \end{aligned}$	$2 \lambda(n+\gamma)+B=0$ or $n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2 \sqrt{-C}}$	—																														

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gilmore-lie-groups_ F02417	240	—	$\begin{aligned} & E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2\frac{1}{N^2} \\ & N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}, \quad N = n + l + 1 \end{aligned}$		—
gilmore-lie-groups_ F02418	240	■ 1.000	$\hbar$	$\hbar$	Klein–Gordon equation Schrödinger equation
gilmore-lie-groups_ F02419	241	■ 1.000	$Z=50$	$Z = 50$	
gilmore-lie-groups_ F02420	241	■ 1.000	$\alpha \rightarrow Z \alpha$	$\alpha \rightarrow Z\alpha$	bound state energy $ W_1 = \frac{1}{2}mc^2\alpha$
gilmore-lie-groups_ F02421	241	■ 1.000	$ W_1 = \frac{1}{2}mc^2\alpha^2$	$ W_1 = \frac{1}{2}mc^2\alpha^2$	gies are displayed as a function of the orbital angular momentum l . The relativistic
gilmore-lie-groups_ F02422	241	■ 0.999	$1/N^2$	$1/N^2$	energies of the bound states for the proton–electron system converge to the rest en-
gilmore-lie-groups_ F02423	241	■ 0.999	$N=n+l+1$	$N = n + l + 1$	energies of the bound states for the proton–electron system converge to the rest en-
gilmore-lie-groups_ F02424	241	■ 1.000	$N^{\prime}$	N'	we have instead computed and plotted the bound state spectrum for a single elec-
gilmore-lie-groups_ F02425	241	■ 1.000	$\frac{1}{2}mc^2\alpha^2$	$\frac{1}{2}mc^2\alpha^2$	tron in a potential with positive charge Z . The energies in this case are obtained
gilmore-lie-groups_ F02426	241	■ 1.000	$\frac{1}{2}mc^2(Z\alpha)^2$	$\frac{1}{2}mc^2(Z\alpha)^2$	
gilmore-lie-groups_ F02427	241	■ 1.000	1 s	$1s$	
gilmore-lie-groups_ F02428	242	■ 1.000	$g_i \psi\rangle$	$g_i \psi\rangle$	eigenstate of H with eigenvalue E , then $g_i \psi\rangle$ is also an eigenstate of H with the
gilmore-lie-groups_ F02429	242	■ 1.000	$ \psi\rangle = \psi_{2p_z}(\mathbf{x})$	$ \psi\rangle = \psi_{2p_z}(\mathbf{x})$	the y -axis maps this state to $\psi_{2p_x}(\mathbf{x})$ and a rotation by $\pi/2$ radians about the x -axis
gilmore-lie-groups_ F02430	242	■ 1.000	$\pi/2$	$\pi/2$	the y -axis maps this state to $\psi_{2p_x}(\mathbf{x})$ and a rotation by $\pi/2$ radians about the x -axis
gilmore-lie-groups_ F02431	242	■ 1.000	$\psi_{2p_x}(\mathbf{x})$	$\psi_{2p_x}(\mathbf{x})$	maps this state to $-\psi_{2p_y}(\mathbf{x})$. By invariance (of the hamiltonian) under the rotation
gilmore-lie-groups_ F02432	242	■ 1.000	$-\psi_{2p_y}(\mathbf{x})$	$-\psi_{2p_y}(\mathbf{x})$	group and the principle of equivalence, these new functions describe possible states
gilmore-lie-groups_ F02433	242	■ 1.000	$H = \mathbf{p} \cdot \mathbf{p}/2m - e^2/r$	$H = \mathbf{p} \cdot \mathbf{p}/2m - e^2/r$	terms of the infinitesimal generators of rotations about axis i : $\epsilon_{ijk}x_j\partial_k$. These ge-
gilmore-lie-groups_ F02434	242	■ 1.000	$\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$	$\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$	terms of the infinitesimal generators of rotations about axis i : $\epsilon_{ijk}x_j\partial_k$. These ge-
gilmore-lie-groups_ F02435	242	■ 1.000	$-e^2/r$	$-e^2/r$	ometric operators are proportional to the physical angular momentum operators
gilmore-lie-groups_ F02436	242	■ 1.000	$i: \epsilon_{ijk}x_j\partial_k$	$i: \epsilon_{ijk}x_j\partial_k$	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i)\epsilon_{ijk}x_j\partial_k$. Finite rotations can be expressed as exponentials
gilmore-lie-groups_ F02437	242	■ 1.000	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i)\epsilon_{ijk}x_j\partial_k$	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i)\epsilon_{ijk}x_j\partial_k$	$R(\theta) = e^{\epsilon_{ijk}\theta_i x_j \partial_k} = e^{i\theta \cdot \mathbf{L}/\hbar}$ (14.15)
gilmore-lie-groups_ F02438	242	■ 0.998	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	
gilmore-lie-groups_ F02439	242	■ 0.992	$\mathbf{r} \times \nabla$	$\mathbf{r} \times \nabla$	$[\mathbf{L}_i, \mathbf{L}_j] = i\hbar\epsilon_{ijk}\mathbf{L}_k$ (14.16)
gilmore-lie-groups_ F02440	242	■ 1.000	$\hbar/i$	$\hbar/i$	It is useful to construct linear co

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gilmore-lie-groups_F02441	242	1.000	$\left(L_{+}\right)$	(L_{+})	$[L_{+}, L_{-}] = 2\hbar L_z$ (14.18)																									
gilmore-lie-groups_F02442	242	1.000	$\left(L_{-}\right)$	(L_{-})	$[L_{+}, L_{-}] = 2\hbar L_z$ (14.18)																									
gilmore-lie-groups_F02443	242	1.000	$L_{\pm}=\mathrm{L}_{\mathrm{x}}\pm\mathrm{i}\mathrm{L}_{\mathrm{y}}$	$L_{\pm} = L_x \pm iL_y$	$[L_{+}, L_{-}] = 2\hbar L_z$ (14.18)																									
gilmore-lie-groups_F02444	242	1.000	$L_z=\hbar\frac{1}{2}\left(a_1^{\dagger}a_1-a_2^{\dagger}a_2\right)$, $L_{+}=\hbar a_1^{\dagger}a_2$, $L_{-}=\hbar a_2^{\dagger}a_1$	$L_z = \hbar\frac{1}{2}\left(a_1^{\dagger}a_1 - a_2^{\dagger}a_2\right), L_+ = \hbar a_1^{\dagger}a_2, L_- = \hbar a_2^{\dagger}a_1$																										
gilmore-lie-groups_F02445	242	1.000	$\left n_1\right\rangle\left n_2\right\rangle$	$ n_1n_2\rangle$																										
gilmore-lie-groups_F02446	242	0.838	$\left j\right\rangle_m$	$= \left _m^j\right\rangle$																										
gilmore-lie-groups_F02447	242	0.838	$n_1=0,1,2,\ldots$, $n_2=0,1,2,\ldots$, $n_1+n_2=2j$, $n_1-n_2=2m$	$n_1 = 0, 1, 2, \dots, n_2 = 0, 1, 2, \dots, n_1 + n_2 = 2j, n_1 - n_2 = 2m$																										
gilmore-lie-groups_F02448	243	1.000	$-j\leq m\leq+j$	$-j\leq m\leq+j$	$-j\leq m\leq+j$. These basis vectors describe the finite-dimensional irreducible																									
gilmore-lie-groups_F02449	243	0.994	$(x,y,z)\rightarrow(r,\theta,\phi)$	$((x,y,z)\rightarrow(r,\theta,\phi)$	operators. In spherical coordinates $((x,y,z)\rightarrow(r,\theta,\phi)$ with $x=r\sin\theta\cos\phi)$ these operators are																									
gilmore-lie-groups_F02450	243	0.994	$x=r\sin\theta\cos\phi)$	$x=r\sin\theta\cos\phi)$	operators. In spherical coordinates $((x,y,z)\rightarrow(r,\theta,\phi)$ with $x=r\sin\theta\cos\phi)$ these operators are																									
gilmore-lie-groups_F02451	243	1.000	$Y_{-l}^l(\theta,\phi)$	$Y_{-l}^l(\theta,\phi)$																										
gilmore-lie-groups_F02452	243	1.000	$L_{-}Y_{-l}^l(\theta,\phi)=0$	$L_{-}Y_{-l}^l(\theta,\phi)=0$																										
gilmore-lie-groups_F02453	244	0.998	$l=0$	$l=0$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$
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gilmore-lie-groups_F02455	244	0.998	$l=3$	$l=3$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$
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gilmore-lie-groups_ F02456	244	■ 0.998	$\sqrt{\frac{1}{4 \pi}}$	$\sqrt{\frac{1}{4\pi}}$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$
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gilmore-lie-groups_ F02457	244	■ 0.998	$\sqrt{\frac{3}{4 \pi}} \cos \theta$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$
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gilmore-lie-groups_ F02458	244	■ 0.998	$\sqrt{\frac{5}{16 \pi}} \left(3 \cos ^{2} \theta - 1 \right)$	$\sqrt{\frac{5}{16\pi}} \left(3 \cos^2 \theta - 1 \right)$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$
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gilmore-lie-groups_ F02459	244	■ 0.998	$\sqrt{\frac{7}{16 \pi}} \left(5 \cos ^{3} \theta - 3 \cos \theta \right)$	$\sqrt{\frac{7}{16\pi}} \left(5 \cos^3 \theta - 3 \cos \theta \right)$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$
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gilmore-lie-groups_ F02460	244	■ 0.998	$\mp \sqrt{\frac{3}{8 \pi}} \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$
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gilmore-lie-groups_ F02461	244	■ 0.998	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$	
gilmore-lie-groups_ F02462	244	■ 0.998	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$	
gilmore-lie-groups_ F02463	244	■ 0.998	$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i \phi}$	$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i \phi}$	
gilmore-lie-groups_ F02464	244	■ 0.998	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i \phi}$	
gilmore-lie-groups_ F02465	244	■ 0.998	$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i \phi}$	$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i \phi}$	
gilmore-lie-groups_ F02466	244	■ 1.000	L_{+}	L_{+}	$L_{+} Y_{\downarrow}$
gilmore-lie-groups_ F02467	244	■ 0.815	$(l=0, 1, 2, 3)$	$(l = 0, 1, 2, 3)$	Remark The spectrum of the Casimir invariant for the rotation group SO
gilmore-lie-groups_ F02468	244	■ 0.662	θ, ϕ	θ, ϕ	which these operators act, is compact. By the same token, the spectrum of the square

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gilmore-lie-groups_ F02469	244	■ 1.000	$\hbar^2 l(l+1)$	$\hbar^2 l(l+1)$	and is quantized by integer angular momentum values, measured in units of Planck's constant $\hbar$
gilmore-lie-groups_ F02470	244	■ 1.000	$E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$	$E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$	If we believe the converse, that degeneracy implies symmetry, then we might be led to expect that the hydrogen atom exhibits more symmetry than meets the eye.
gilmore-lie-groups_ F02471	244	■ 1.000	$\sum_{l=0}^{l=N-1} (2l+1) = N^2$	$\sum_{l=0}^{l=N-1} (2l+1) = N^2$	In fact this symmetry, called a dynamical symmetry (Schiff, 1968), exists and is
gilmore-lie-groups_ F02472	244	■ 1.000	$1 / r^2$	$1/r^2$	
gilmore-lie-groups_ F02473	244	■ 1.000	$d \mathbf{p} / d t = -K \mathbf{r} / r^3$	$d\mathbf{p}/dt = -K\mathbf{r}/r^3$	
gilmore-lie-groups_ F02474	245	■ 1.000	$K = G M m, G$	$K = G M m, G$	$K = G M m, G$ is the universal gravitational constant, M and m are the two attracting
gilmore-lie-groups_ F02475	245	■ 1.000	$\mathbf{r} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}}$	$\mathbf{r} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}}$	masses, and $\mathbf{r} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}}$ is the vector from one mass to the other. The time
gilmore-lie-groups_ F02476	245	■ 1.000	$\mathbf{p} \times \mathbf{L}$	$\mathbf{p} \times \mathbf{L}$	derivative of the vector $\mathbf{p} \times \mathbf{L}$ is
gilmore-lie-groups_ F02477	245	■ 0.999	$\mathbf{L}$	$\mathbf{L}$	a constant of motion in any spherically symmetric potential. We also use the force
gilmore-lie-groups_ F02478	245	■ 1.000	$1 / r$	$1/r$	cross product $\mathbf{r} \times \mathbf{L}$ in terms of (generally) nonparallel vectors $\mathbf{r}$ and $\mathbf{L}$. We also use the identity $(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r}) \mathbf{r}/r^3$. The result is that the Laplace–
gilmore-lie-groups_ F02479	245	■ 0.967	$\mathbf{r} \times \mathbf{L}$	$\mathbf{r} \times \mathbf{L}$	Runge–Lenz vector $\mathbf{M}$ is a constant of motion: $d\mathbf{M}/dt = 0$, where
gilmore-lie-groups_ F02480	245	■ 0.967	$\dot{\mathbf{r}}$	$\dot{\mathbf{r}}$	Runge–Lenz vector $\mathbf{M}$ is a constant of motion: $d\mathbf{M}/dt = 0$, where
gilmore-lie-groups_ F02481	245	■ 1.000	$(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r}) \mathbf{r}/r^3$	$(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r}) \mathbf{r}/r^3$	
gilmore-lie-groups_ F02482	245	■ 0.985	$d \mathbf{M} / d t = 0$	$d\mathbf{M}/dt = 0$	$\mathbf{M} = \frac{\mathbf{p} \times \mathbf{L}}{r} - K \frac{\mathbf{r}}{r}$
gilmore-lie-groups_ F02483	245	■ 1.000	$[H, \mathbf{M}] = 0$	$[H, \mathbf{M}] = 0$	$[L_i, M_i] = i \hbar \epsilon_{ijk} M_k$ (14.25)
gilmore-lie-groups_ F02484	245	■ 1.000	L_i, M_j	L_i, M_j	$[L_i, M_i] = i \hbar \epsilon_{ijk} M_k$ (14.25)
gilmore-lie-groups_ F02485	245	■ 0.993	$(E < 0)$	$(E < 0)$	
gilmore-lie-groups_ F02486	245	■ 0.993	$(E > 0)$	$(E > 0)$	
gilmore-lie-groups_ F02487	246	■ 1.000	$\mathbf{M}' = (-m/2H)^{1/2} \mathbf{M}$	$\mathbf{M}' = (-m/2H)^{1/2} \mathbf{M}$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_ F02488	246	■ 1.000	$E > 0$	$E > 0$	of these operators are now
gilmore-lie-groups_ F02489	246	■ 1.000	$- \rightarrow +$	$- \rightarrow +$	of these operators are now
gilmore-lie-groups_ F02490	246	■ 1.000	$SO(4) \rightarrow SO(3, 1)$	$SO(4) \rightarrow SO(3, 1)$	of these operators are now
gilmore-lie-groups_ F02491	246	■ 0.834	$\mathbf{A}$	$\mathbf{A}$	$\mathbf{A} = \frac{1}{r} \mathbf{L} \times \mathbf{M}'$
gilmore-lie-groups_ F02492	246	■ 0.834	$\mathbf{B}$	$\mathbf{B}$	$\mathbf{A} = \frac{1}{r} \mathbf{L} \times \mathbf{M}'$
gilmore-lie-groups_ F02493	246	■ 0.996	$A_3 = \frac{1}{2} (a_1^\dagger a_1 -$	$A_3 = \frac{1}{2} (a_1^\dagger a_1 -$	b_1, b_2 and their creation operators is also introduced.
gilmore-lie-groups_ F02494	246	■ 0.910	$a_2^\dagger a_2$	$a_2^\dagger a_2$	Basis states for a representation of the algebra spanned by the operators $\mathbf{A}$ have the

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gilmore-lie-groups_F02495	246	■ 0.910	$A_{\{+}=a_{\{1}^{\dagger} a_{\{2\}}, A_{\{-}=a_{\{2}^{\dagger} a_{\{1\}}$	$A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1$	Basis states for a representation of the algebra spanned by the operators A have the
gilmore-lie-groups_F02496	246	■ 0.910	$\hbar \rightarrow 1$	$\hbar \rightarrow 1$	Basis states for a representation of the algebra spanned by the operators A have the
gilmore-lie-groups_F02497	246	■ 1.000	$b_{\{1\}}, b_{\{2\}}$	b_1, b_2	states correspond to $p_1 = 2j_a, p_2 = 0; p_1 = 2j_a -$
gilmore-lie-groups_F02498	246	■ 1.000	$\left p_{\{1\}}, p_{\{2\}} \right\rangle$	$ p_1, p_2\rangle$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$, since $\mathbf{A} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{A}$ (cf. Problem 14.15). $i_{\pm} = i_{\pm}$ and the set of states related by the shift operators is $(2i +$
gilmore-lie-groups_F02499	246	■ 1.000	$p_{\{1\}}+p_{\{2\}}=2 j_{\{a\}}$	$p_1 + p_2 = 2j_a$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$, since $\mathbf{A} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{A}$ (cf. Problem 14.15). $i_{\pm} = i_{\pm}$ and the set of states related by the shift operators is $(2i +$
gilmore-lie-groups_F02500	246	■ 1.000	$p_{\{1\}}-p_{\{2\}}=m_{\{a\}}$	$p_1 - p_2 = m_a$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$, since $\mathbf{A} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{A}$ (cf. Problem 14.15). $i_{\pm} = i_{\pm}$ and the set of states related by the shift operators is $(2i +$
gilmore-lie-groups_F02501	246	■ 1.000	$2 j_{\{a\}}+1$	$2j_a + 1$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$, since $\mathbf{A} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{A}$ (cf. Problem 14.15). $i_{\pm} = i_{\pm}$ and the set of states related by the shift operators is $(2i +$
gilmore-lie-groups_F02502	246	■ 0.968	$p_{\{1\}}=2 j_{\{a\}}, p_{\{2\}}=0 ; p_{\{1\}}=2 j_{\{a\}}-1, p_{\{2\}}=1$	$p_1 = 2j_a, p_2 = 0; p_1 = 2j_a - 1, p_2 = 1$	$1)^2$ fold degenerate. where $2i + 1 = N = n + l + 1$.
gilmore-lie-groups_F02503	246	■ 1.000	$\left q_{\{1\}}, q_{\{2\}} \right\rangle$	$ q_1, q_2\rangle$	States with good l and m quantum numbers can be constructed from these states
gilmore-lie-groups_F02504	246	■ 1.000	$q_{\{1\}}+q_{\{2\}}=2 j_{\{b\}}$	$q_1 + q_2 = 2j_b$	States with good l and m quantum numbers can be constructed from these states
gilmore-lie-groups_F02505	246	■ 1.000	$q_{\{1\}}-q_{\{2\}}=m_{\{b\}}$	$q_1 - q_2 = m_b$	States with good l and m quantum numbers can be constructed from these states
gilmore-lie-groups_F02506	246	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$	$\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$	using Clebsch-Gordon coefficients:
gilmore-lie-groups_F02507	246	■ 1.000	$\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$	$\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$	using Clebsch-Gordon coefficients:
gilmore-lie-groups_F02508	246	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$	$\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$	using Clebsch-Gordon coefficients:
gilmore-lie-groups_F02509	246	■ 1.000	$j_{\{a\}}=j_{\{b\}}$	$j_a = j_b$	$\left 1 \lambda \right\rangle = \left 1 0 \right\rangle \left 1 0 \right\rangle \left 1 0 \right\rangle \left 1 \lambda \right\rangle$
gilmore-lie-groups_F02510	246	■ 1.000	$(2 j_{\{a\}}+1)$	$(2j_a + 1)$	$\left 1 \lambda \right\rangle = \left 1 0 \right\rangle \left 1 0 \right\rangle \left 1 0 \right\rangle \left 1 \lambda \right\rangle$
gilmore-lie-groups_F02511	246	■ 1.000	$2 j_{\{a\}}+1=N=n+l+1$	$2j_a + 1 = N = n + l + 1$	$\left \begin{smallmatrix} a \\ m \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j_1 \\ m_a \end{smallmatrix} \right\rangle \left \begin{smallmatrix} j_2 \\ m_b \end{smallmatrix} \right\rangle \left \begin{smallmatrix} j_1 + j_2 \\ m_a + m_b \end{smallmatrix} \right\rangle$
gilmore-lie-groups_F02512	246	■ 0.996	$\operatorname{good} l$	good l	$\left \begin{smallmatrix} j_1 + j_2 \\ m_a + m_b \end{smallmatrix} \right\rangle = \sum_{m_a, m_b} \left \begin{smallmatrix} j_1 \\ m_a \end{smallmatrix} \right\rangle \left \begin{smallmatrix} j_2 \\ m_b \end{smallmatrix} \right\rangle \left \begin{smallmatrix} j_1 + j_2 \\ m_a + m_b \end{smallmatrix} \right\rangle$
gilmore-lie-groups_F02513	247	■ 1.000	$M_{\{+}^{\prime}=$	$M_+^{\prime} =$	spherical harmonics, is determined in a straightforward way. For example, $M_+^{\prime} =$
gilmore-lie-groups_F02514	247	■ 1.000	$A_{\{+}-B_{\{+}=a_{\{1}^{\dagger} a_{\{2\}}-b_{\{1}^{\dagger} b_{\{2\}}$	$A_+ - B_+ = a_1^\dagger a_2 - b_1^\dagger b_2$	$A_+ - B_+ = a_1^\dagger a_2 - b_1^\dagger b_2$, so that
gilmore-lie-groups_F02515	247	■ 0.998	$\mathbf{M}^{\prime}$	$\mathbf{M}^{\prime}$	of a particle in a $1/r$ potential is equivalent to motion of a free particle in the sphere
gilmore-lie-groups_F02516	247	■ 0.608	$\mathbf{W}=\mathbf{L} \times \mathbf{M}$	$\mathbf{W} = \mathbf{L} \times \mathbf{M}$	$\mathbf{W} = \mathbf{L} \times \mathbf{M}$
gilmore-lie-groups_F02517	247	■ 1.000	$\mathbf{r} \cdot \mathbf{L}=0$	$\mathbf{r} \cdot \mathbf{L} = 0$	
gilmore-lie-groups_F02518	247	■ 1.000	$\mathbf{p} \cdot \mathbf{L}=0$	$\mathbf{p} \cdot \mathbf{L} = 0$	
gilmore-lie-groups_F02519	248	■ 0.986	$\mathbf{W}$	$\mathbf{W}$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02520	248	■ 0.986	$p_{\{z\}}=0, p_{\{x\}}=\mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$	$p_z = 0, p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$

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gilmore-lie-groups_F02521	248	■ 0.986	$\mathbf{p}_{\{y\}}=$	$p_y =$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02522	248	■ 1.000	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$. The two nonzero components of the momentum vector are not independent, but obey the constraint
gilmore-lie-groups_F02523	248	■ 1.000	$m K / L$	$m K / L$	center displaced from the origin by $m M / L$.
gilmore-lie-groups_F02524	248	■ 1.000	$m M / L$	$m M / L$	The circle in R^2 is lifted to a circle in
gilmore-lie-groups_F02525	248	■ 1.000	$p_0 = \sqrt{-2 E / m}$	$p_0 = \sqrt{-2 E / m}$	$\mathbf{p} \cdot \mathbf{p} = p_0^2 = -2 p_0$
gilmore-lie-groups_F02526	248	■ 1.000	$\hat{\mathbf{u}} \in S^3 \subset R^4$	$\hat{\mathbf{u}} \in S^3 \subset R^4$	$\mathbf{p} \cdot \mathbf{p} = p_0^2 = -2 p_0$
gilmore-lie-groups_F02527	248	■ 1.000	$\hat{\mathbf{w}}$	$\hat{\mathbf{w}}$	preserves angles (conformal), it is a simple matter to check that $\mathbf{u}$ is a unit vector. The circular trajectory in R^3 (Eq. (14.33)) lifts to a circle in S^3 . Reversibly, circles in
gilmore-lie-groups_F02528	248	■ 0.995	$\hat{\mathbf{u}}$	$\hat{\mathbf{u}}$	Rotations in $SO(4)$ rigidly rotate the sphere S^3 into itself. They rotate circles into
gilmore-lie-groups_F02529	248	■ 1.000	S^3	S^3	circles, which then project down to circular momentum trajectories in the physical
gilmore-lie-groups_F02530	248	■ 0.803	$SO(4) / SO(3)$	$SO(4) / SO(3)$	
gilmore-lie-groups_F02531	248	■ 1.000	p_0	p_0	
gilmore-lie-groups_F02532	249	■ 1.000	$N^2 = (n + l + 1)^2$	$N^2 = (n + l + 1)^2$	irreducible representations of dimension $N^2 = (n + l + 1)^2$.
gilmore-lie-groups_F02533	249	■ 1.000	$-N^3$	$-N^3$	The revealed energies have been plotted as a function of l (vertically) and orbital angular momentum quantum number l (horizontally). In this format, the eigen-
gilmore-lie-groups_F02534	249	■ 1.000	$\mathbf{x}' = M \mathbf{x}$	$\mathbf{x}' = M \mathbf{x}$	condition leads to $M^T G M = G$. It is useful to define a new N -vector $\mathbf{v}$ as a scaled version of the original vector
gilmore-lie-groups_F02535	249	■ 1.000	$(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$	$(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$	$\mathbf{v} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_F02536	249	■ 0.749	$\mathbf{y}$	$\mathbf{y}$	
gilmore-lie-groups_F02537	249	■ 1.000	$\mathbf{y} = \lambda \mathbf{x}$	$\mathbf{y} = \lambda \mathbf{x}$	
gilmore-lie-groups_F02538	249	■ 1.000	$z_1 = \lambda$	$z_1 = \lambda$	
gilmore-lie-groups_F02539	249	■ 1.000	$z_2 =$	$z_2 =$	
gilmore-lie-groups_F02540	249	■ 1.000	$\lambda(\mathbf{x}, \mathbf{x})_N$	$\lambda(\mathbf{x}, \mathbf{x})_N$	The conformal condition defines an inner product in the $N + 2$ dimens
gilmore-lie-groups_F02541	249	■ 1.000	$N + 2$	$N + 2$	
gilmore-lie-groups_F02542	249	■ 0.998	$\mathbf{y}, z_1, z_2$	$\mathbf{y}, z_1, z_2$	
gilmore-lie-groups_F02543	249	■ 0.986	$\mathbf{y}, y_{N+1}, y_{N+2}$	$\mathbf{y}, y_{N+1}, y_{N+2}$	(14.38)
gilmore-lie-groups_F02544	249	■ 0.986	$y_{N+1} = \frac{1}{2}(z_1 + z_2)$	$y_{N+1} = \frac{1}{2}(z_1 + z_2)$	(14.38)
gilmore-lie-groups_F02545	249	■ 0.986	$y_{N+2} = \frac{1}{2}(z_1 - z_2)$	$y_{N+2} = \frac{1}{2}(z_1 - z_2)$	(14.38)

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gilmore-lie-groups_F02546	249	■ 1.000	$R^{\{N\}}$	R^N	
gilmore-lie-groups_F02547	250	■ 0.999	$R^{\{N+2\}}$	R^{N+2}	The connection with the Kepler problem is made as follows. The momenta $\mathbf{p}$ are lifted to the coordinates on a sphere $S^3 \subset \mathbb{R}^4 (E < 0)$ or a two sheeted hyperboloid
gilmore-lie-groups_F02548	250	■ 0.999	$S_{-0}(p+1, q+1)$	$SO(p+1, q+1)$	The connection with the Kepler problem is made as follows. The momenta $\mathbf{p}$ are lifted to the coordinates on a sphere $S^3 \subset \mathbb{R}^4 (E < 0)$ or a two sheeted hyperboloid
gilmore-lie-groups_F02549	250	■ 1.000	$S^{\{3\}} \subset R^{\{4\}} (E < 0)$	$S^3 \subset R^4 (E < 0)$	$\dot{\mathbf{u}} = \frac{1}{2}(p_0^2 - \mathbf{p} \cdot \mathbf{p}) \mathbf{u} + \frac{p_0 \mathbf{p}}{E < 0}$
gilmore-lie-groups_F02550	250	■ 1.000	$H^{\{3\}} \subset R^{\{4\}} (E > 0)$	$H^3 \subset R^4 (E > 0)$	$\dot{\mathbf{u}} = \frac{1}{2}(p_0^2 + \mathbf{p} \cdot \mathbf{p}) \mathbf{u} + \frac{1}{2}(p_0^2 + \mathbf{p} \cdot \mathbf{p}) \mathbf{u}$
gilmore-lie-groups_F02551	250	■ 1.000	$\mathbf{u}^t G \mathbf{u} = 1$	$\mathbf{u}^t G \mathbf{u} = 1$	
gilmore-lie-groups_F02552	251	■ 0.999	$\operatorname{diag}(\pm 1, \pm I_3, -1, +1)$	$\operatorname{diag}(1, \pm I_3, -1, +1)$	lows. The group that leaves invariant the conformal metric $\operatorname{diag}(1, \pm I_3, -1, +1)$ is
gilmore-lie-groups_F02553	251	■ 0.984	$S_{-0}(5, 1)$	$SO(5, 1)$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02554	251	■ 0.984	$E < 0$	$E < 0$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02555	251	■ 0.984	$S_{-0}(2, 4)$	$SO(2, 4)$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02556	251	■ 1.000	$z_{\{1\}} = z_{\{2\}}, y_{\{4\}} = \lambda$	$z_1 = z_2, y_4 = \lambda$	the condition $\mathbf{u}^t G \mathbf{u} = 1$ is satisfied, so that $z_1 = z_2, y_4 = \lambda$ and $y_5 = 0$ (the six
gilmore-lie-groups_F02557	251	■ 1.000	$y_{\{5\}} = 0$	$y_5 = 0$	the condition $\mathbf{u}^t G \mathbf{u} = 1$ is satisfied, so that $z_1 = z_2, y_4 = \lambda$ and $y_5 = 0$ (the six
gilmore-lie-groups_F02558	251	■ 0.934	$\left(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2) \right)$	$\left(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2) \right)$	coordinates are labeled $(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2))$. Transformations that map the algebraic surface to itself must map $v_x = 0$ to $v_x = 0$. It is a
gilmore-lie-groups_F02559	251	■ 0.908	$\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array} \right]$	$\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array} \right]$	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_F02560	251	■ 0.997	$\operatorname{diag}(\pm 1, \pm I_3, -1)$	$\operatorname{diag}(1, \pm I_3, -1)$	It remains to show that this group maps these algebraic surfaces into themselves.
gilmore-lie-groups_F02561	251	■ 0.997	$R^{\{5\}}$	R^5	It remains to show that this group maps these algebraic surfaces into themselves.
gilmore-lie-groups_F02562	251	■ 0.997	$S_{-0}(1, 4)$	$SO(1, 4)$	It remains to show that this group maps these algebraic surfaces into themselves.
gilmore-lie-groups_F02563	251	■ 0.761	$\mathbf{u}'$	$\mathbf{u}'$	$(\mathbf{u}')^t G \mathbf{u}' - 1 = \frac{1}{(C \lambda \mathbf{u} + D)}$
gilmore-lie-groups_F02564	251	■ 1.000	$A^t G A - C^t C = G$	$A^t G A - C^t C = G$	$(\mathbf{u}', \mathbf{u}')_N - 1 = \frac{1}{(C \mathbf{u} + D)(C \mathbf{u} + D)} \quad (14.44)$
gilmore-lie-groups_F02565	251	■ 1.000	u	u	
gilmore-lie-groups_F02566	251	■ 1.000	$(u_{\{0\}} \rightarrow u_{\{4\}})$	$(u_0 \rightarrow u_4)$	
gilmore-lie-groups_F02567	252	■ 0.967	$1 / \sqrt{2 m E }$	$1 / \sqrt{2 m E }$	vector $\mathbf{L}$ and the scaled (by $1 / \sqrt{2 m E }$) Runge–Lenz vector $\mathbf{M}'$ as follows:
gilmore-lie-groups_F02568	252	■ 1.000	$L_{\{i\}}, M_{\{i\}}^{\prime}$	L_i, M'_i	for $E < 0$.
gilmore-lie-groups_F02569	252	■ 1.000	$B_{\{\mu\}}$	B_μ	for $E < 0$.
gilmore-lie-groups_F02570	252	■ 0.993	$a_{\{i\}}, a_{\{j\}}^{\dagger}$	$a_i, a_j^\dagger$	for example

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gilmore-lie-groups-F02571	252	1.000	$b_{\{i\}}, b_{\{j\}}^{\dagger}$	$b_i, b_j^\dagger$	$a_1^\dagger a_2 m_1, m_2; n_1, n_2\rangle = m_1 + 1, m_2 - 1; n_1 n_2\rangle \sqrt{m_1 + 1} \sqrt{m_2}$
gilmore-lie-groups-F02572	252	1.000	$\left m_{\{1\}}, m_{\{2\}}; n_{\{1\}}, n_{\{2\}} \right\rangle$	$ m_1, m_2; n_1, n_2\rangle$	
gilmore-lie-groups-F02573	252	1.000	$j_{\{a\}}=\frac{1}{2}\left(m_{\{1\}}+m_{\{2\}}\right)$	$j_a = \frac{1}{2} (m_1 + m_2)$	
gilmore-lie-groups-F02574	252	1.000	$j_{\{b\}}=\frac{1}{2}\left(n_{\{1\}}+n_{\{2\}}\right)$	$j_b = \frac{1}{2} (n_1 + n_2)$	
gilmore-lie-groups-F02575	253	1.000	$N=2 j_{\{a\}}+1=2 j_{\{b\}}+1=\left(j_{\{a\}}+j_{\{b\}}\right)+1$	$N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$	principal quantum number $N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$ must change
gilmore-lie-groups-F02576	253	1.000	$j_{\{a\}}$	j_a	$j_a = j_b$. Operators that change j_a have the form $a_i^\dagger$ or $a_i^\dagger a_j^\dagger$, but they do not simul-
gilmore-lie-groups-F02577	253	1.000	$j_{\{b\}}$	j_b	taneously change j_b . Only operators that simultaneously add or subtract one exci-
gilmore-lie-groups-F02578	253	0.991	$\left(H_{\{1\}}, H_{\{2\}}, H_{\{3\}}, H_{\{4\}}\right)=\left(a_{\{1\}}^{\dagger} a_{\{1\}}, a_{\{2\}}^{\dagger} a_{\{2\}}, b_{\{1\}}^{\dagger} b_{\{1\}}\right)$	$(H_1, H_2, H_3, H_4) = \left(a_1^\dagger a_1, a_2^\dagger a_2, b_1^\dagger b_1\right.$	$b_2^\dagger b_2)$. The remaining twelve operators have eigenoperator commutation relations with this set:
gilmore-lie-groups-F02579	253	0.829	$b_{\{2\}}^{\dagger} b_{\{2\}}$	$b_2^\dagger b_2$	with this set,
gilmore-lie-groups-F02580	253	1.000	$\pm \frac{1}{2}$	$\pm \frac{1}{2}$	$(a_1^\dagger a_1 + a_2^\dagger a_2) - (b_1^\dagger b_1 + b_2^\dagger b_2)$
gilmore-lie-groups-F02581	253	1.000	$\pi / 4$	$\pi/4$	
gilmore-lie-groups-F02582	253	1.000	$3 \pi / 4$	$3\pi/4$	
gilmore-lie-groups-F02583	253	1.000	$A_{\{3\}}=D_{\{3\}}$	$A_3 = D_3$	$a_1^\dagger a_2$
gilmore-lie-groups-F02584	254	1.000	$a_{\{1\}}^{\dagger} a_{\{1\}}+a_{\{2\}}^{\dagger} a_{\{2\}}+b_{\{1\}}^{\dagger} b_{\{1\}}+b_{\{2\}}^{\dagger} b_{\{2\}}$	$a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$	onal operator $a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$. This is the algebra $\mathfrak{so}(4) + \mathfrak{so}(2)$. The
gilmore-lie-groups-F02585	254	1.000	$\mathfrak{so}(4)+\mathfrak{so}(2)$	$\mathfrak{so}(4) + \mathfrak{so}(2)$	onal operator $a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$. This is the algebra $\mathfrak{so}(4) + \mathfrak{so}(2)$. The
gilmore-lie-groups-F02586	254	0.890	$\mathfrak{so}(4,2)=\mathfrak{su}(2,2)$	$\mathfrak{so}(4, 2) = \mathfrak{su}(2, 2)$	$\mathfrak{so}(4, 2) = \mathfrak{su}(2, 2)$. This is the conformal algebra
gilmore-lie-groups-F02587	254	1.000	$\left(q_{\{1\}}, q_{\{2\}}, q_{\{3\}}, q_{\{4\}}\right)$	(q_1, q_2, q_3, q_4)	$\left[Q_1 \right] \quad \left[q_1 \right] \quad \left[q_1 \quad -q_2 \quad -q_3 \quad q_4 \right] \left[q_1 \right]$
gilmore-lie-groups-F02588	254	0.559	$\left(Q_{\{1\}}, Q_{\{2\}}, Q_{\{3\}}\right)$	(Q_1, Q_2, Q_3)	$\left Q_2 \right \quad \left q_2 \right \quad \left q_2 \quad q_1 \quad -q_4 \quad -q_3 \right $
gilmore-lie-groups-F02589	254	1.000	$Q_{\{4\}}$	Q_4	in R^4 are related by $R = a^2$.
gilmore-lie-groups-F02590	254	1.000	$q_{\{1\}}^2+q_{\{2\}}^2+q_{\{3\}}^2+q_{\{4\}}^2 \neq 0$	$q_1^2 + q_2^2 + q_3^2 + q_4^2 \neq 0$	The other half of the canonical transformation, involving the momenta in the
gilmore-lie-groups-F02591	254	1.000	$R=\sqrt{Q_{\{1\}}^2+Q_{\{2\}}^2+Q_{\{3\}}^2}$	$R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$	real and fictitious spaces, is
gilmore-lie-groups-F02592	254	1.000	$q=\sqrt{q_{\{1\}}^2+q_{\{2\}}^2+q_{\{3\}}^2+q_{\{4\}}^2}$	$q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$	real and fictitious spaces, is
gilmore-lie-groups-F02593	254	1.000	$R=q^2$	$R = q^2$	(P_1, P_2, P_3, P_4)
gilmore-lie-groups-F02594	254	1.000	$P_{\{4\}}=0$	$P_4 = 0$	$(1/4R)(p_1^z + p_2^z + p_3^z + p_4^z)$. With these transformations the hamiltonian

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gilmore-lie-groups_F02595	254	■ 1.000	$P^{\{2\}}=P_{\{1\}}^{\{2\}}+P_{\{2\}}^{\{2\}}+P_{\{3\}}^{\{2\}}=\left(1\ / \ 4\ R\ p^{\{2\}}\right)\cdot\left(\zeta^{\{2\}}\ / \ 4\ R^{\{2\}}\right)\rightarrow$	$P^2 = P_1^2 + P_2^2 + P_3^2 = (1/4Rp^2) - (\zeta^2/4R^2) \rightarrow$	
gilmore-lie-groups_F02596	254	■ 0.980	$(1\ / \ 4\ R)\left(p_{\{1\}}^{\{2\}}+p_{\{2\}}^{\{2\}}+p_{\{3\}}^{\{2\}}+p_{\{4\}}^{\{2\}}\right)$	$(1/4R)\left(p_1^2+p_2^2+p_3^2+p_4^2\right)$	
gilmore-lie-groups_F02597	255	■ 1.000	$\left(q_{\{1\}},\ q_{\{2\}},\ q_{\{3\}},\ q_{\{4\}}\ ;\ p_{\{1\}},\ p_{\{2\}},\ p_{\{3\}},\ p_{\{4\}}\right)$	$(q_1,q_2,q_3,q_4;p_1,p_2,p_3,p_4)$	momenta that
gilmore-lie-groups_F02598	255	■ 1.000	$\zeta=0$	$\zeta = 0$	momenta. Preservin
gilmore-lie-groups_F02599	255	■ 1.000	$M^{\{t\}}\ G_{\{1\}}\ M=G_{\{1\}}$	$M^tG_1M = G_1$	$\left[\begin{array}{cc} 0 & M \\ & \end{array}\right]$
gilmore-lie-groups_F02600	255	■ 1.000	$M^{\{t\}}\ G_{\{2\}}\ M=G_{\{2\}}$	$M^tG_2M = G_2$	
gilmore-lie-groups_F02601	256	■ 0.988	$\operatorname{Sp}(8\ ;\ \mathbb{R})$	$\operatorname{Sp}(8;\mathbb{R})$	group that preserves the symmetric metric is $SO(4,4)$. The group that satisfies both
gilmore-lie-groups_F02602	256	■ 1.000	$S\ O(4,4)$	$SO(4,4)$	metric-preserving conditions is their intersection:
gilmore-lie-groups_F02603	256	■ 0.998	$(q,\ p)$	(q,p)	$\left[\begin{array}{c} s_1 \\ \end{array}\right]=\underline{1}\left[\begin{array}{cc} 1 & 1 \\ \end{array}\right]\left[\begin{array}{c} q_1 \\ \end{array}\right]$
gilmore-lie-groups_F02604	256	■ 0.998	$(s,\ r)$	(s,r)	$\left[\begin{array}{c} s_1 \\ \end{array}\right]=\underline{1}\left[\begin{array}{cc} 1 & 1 \\ \end{array}\right]\left[\begin{array}{c} q_1 \\ \end{array}\right]$
gilmore-lie-groups_F02605	256	■ 1.000	$U(2,2)$	$U(2,2)$	A number of workers have shown that the hamiltonian describing the ir
gilmore-lie-groups_F02606	256	■ 0.825	$\left(V(r)=-e^{\{2\}}\ / \ r\right)$	$(V(r)=-e^2/r)$	
gilmore-lie-groups_F02607	257	■ 1.000	$\mathfrak{so}(4,1)$	$\mathfrak{so}(4,1)$	Five additional operators can be introduced that extend the algebra to $\mathfrak{so}(4,2)$.
gilmore-lie-groups_F02608	257	■ 1.000	$\mathfrak{so}(4,2)$	$\mathfrak{so}(4,2)$	These include one vector operator and two additional operators:
gilmore-lie-groups_F02609	257	■ 1.000	$J_{\{i\}},\ M_{\{i\}},\ A_{\{i\}},\ \Gamma_{\{i\}}(i=1,2,3)$	$J_i,M_i,A_i,\Gamma_i(i=1,2,3)$	The Schrödinger and Klein–Gordon hamiltonians for an electron of charge $-e$
gilmore-lie-groups_F02610	257	■ 0.996	$A_{\{4\}},\ \Gamma_{\{4\}},\ \Gamma_{\{5\}}$	A_4,Γ_4,Γ_5	in the Coulomb field $\Phi(r)=e/r$ of a proton can be expressed in terms of operators
gilmore-lie-groups_F02611	257	■ 1.000	$\Phi(r)=e\ / \ r$	$\Phi(r)=e/r$	
gilmore-lie-groups_F02612	257	■ 1.000	$A_{\{4\}},\ \Gamma_{\{4\}}$	A_4,Γ_4	14.10 Spin angular momentum
gilmore-lie-groups_F02613	257	■ 1.000	$\Gamma_{\{5\}}$	Γ_5	14.10 Spin angular momentum
gilmore-lie-groups_F02614	257	■ 0.860	$(\mathbf{E},\ \mathbf{B})$	$(\mathbf{E},\mathbf{B})$	
gilmore-lie-groups_F02615	257	■ 1.000	$A_{\{\mu\}}=(\phi,\ \mathbf{A})$	$A_\mu=(\phi,\mathbf{A})$	
gilmore-lie-groups_F02616	257	■ 1.000	$q=-e$	$q=-e$	
gilmore-lie-groups_F02617	257	■ 1.000	$H_{\{D\}}\ \psi=E\ \psi$	$H_D\psi = E\psi$	

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gilmore-lie-groups_ F02618	258	■ 0.721	$\frac{p^2}{2m} - \frac{\alpha}{r}$	$\frac{p^2}{2m} - \frac{\alpha}{r}$	H $\frac{p^2}{2m} - \frac{\alpha}{r}$ A₄ $\mathbf{r} \cdot \mathbf{p} - i$ Γ₄ $\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$ Γ₅ $\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$ Θ $r(H_S - W)$ A $\frac{1}{2m}$ B $-W$ C $-\alpha$ $\sqrt{p^2 + m^2} - \frac{\alpha}{r}$ $\mathbf{r} \cdot \mathbf{p} - i$ $\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$ $\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$ $r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$ $A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$ $A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$ 1 $m^2 - E^2$ $-2\alpha E$

In the event a magnetic field **B** is present, the momentum operators **p** should be replaced by $\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$. Under this condition the operators still close under

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F02622	258	■ 0.721	$\frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} - r)$	$\frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} - r)$	H $\frac{p^2}{2m} - \frac{\alpha}{r}$ A₄ $\mathbf{r} \cdot \mathbf{p} - i$ Γ₄ $\frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} - r)$ Γ₅ $\frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} + r)$ Θ $r(H_S - W)$ A $\frac{1}{2m}$ B $-W$ C $-\alpha$ $\sqrt{p^2 + m^2} - \frac{\alpha}{r}$ $\mathbf{r} \cdot \mathbf{p} - i$ $\frac{1}{2}\left(r \mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$ $\frac{1}{2}\left(r \mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$ $r \left\{ \left(H_{KG} + \frac{\alpha}{r} \right)^2 - \left(E + \frac{\alpha}{r} \right)^2 \right\}$ $A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$ $\frac{1}{m^2 - E^2}$ $-2\alpha E$

In the event a magnetic field **B** is present, the momentum operators **p** should be replaced by $\pi = \mathbf{p} - \frac{q}{c} \mathbf{A}$. Under this condition the operators still close under

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F02626	258	■ 0.721	\Theta	Θ	$\begin{array}{ll} \text{H} & \frac{p^2}{2m} - \frac{\alpha}{r} \\ A_4 & \mathbf{r} \cdot \mathbf{p} - i \\ \Gamma_4 & \frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r) \\ \Gamma_5 & \frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r) \\ \Theta & r(H_S - W) \\ A & A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C \\ B & 1/2m \\ C & -W \\ & -\alpha \end{array}$ $\begin{array}{l} \sqrt{p^2 + m^2} - \frac{\alpha}{r} \\ \mathbf{r} \cdot \mathbf{p} - i \\ \frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right) \\ \frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right) \\ r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\} \\ A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C \\ 1 \\ m^2 - E^2 \\ -2\alpha E \end{array}$ In the event a magnetic field $\mathbf{B}$ is present, the momentum operators $\mathbf{p}$ should be replaced by $\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$. Under this condition the operators still close under
gilmore-lie-groups_ F02627	258	■ 0.721	$r\text{left}(H_{\{S\}}-W\text{right})$	$r(H_S - W)$	$\begin{array}{ll} \text{H} & \frac{p^2}{2m} - \frac{\alpha}{r} \\ A_4 & \mathbf{r} \cdot \mathbf{p} - i \\ \Gamma_4 & \frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r) \\ \Gamma_5 & \frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r) \\ \Theta & r(H_S - W) \\ A & A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C \\ B & 1/2m \\ C & -W \\ & -\alpha \end{array}$ $\begin{array}{l} \sqrt{p^2 + m^2} - \frac{\alpha}{r} \\ \mathbf{r} \cdot \mathbf{p} - i \\ \frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right) \\ \frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right) \\ r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\} \\ A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C \\ 1 \\ m^2 - E^2 \\ -2\alpha E \end{array}$ In the event a magnetic field $\mathbf{B}$ is present, the momentum operators $\mathbf{p}$ should be replaced by $\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$. Under this condition the operators still close under
gilmore-lie-groups_ F02628	258	■ 0.721	$r\text{left}\{\{\text{left}(H_{\{K\ G\}}+\frac{\alpha}{r}\}\text{right})^{\{2\}}-\text{left}(E+\frac{\alpha}{r}\}\text{right})^{\{2\}}\text{right}\}$	$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$	$\begin{array}{ll} \text{H} & \frac{p^2}{2m} - \frac{\alpha}{r} \\ A_4 & \mathbf{r} \cdot \mathbf{p} - i \\ \Gamma_4 & \frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r) \\ \Gamma_5 & \frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r) \\ \Theta & r(H_S - W) \\ A & A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C \\ B & 1/2m \\ C & -W \\ & -\alpha \end{array}$ $\begin{array}{l} \sqrt{p^2 + m^2} - \frac{\alpha}{r} \\ \mathbf{r} \cdot \mathbf{p} - i \\ \frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right) \\ \frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right) \\ r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\} \\ A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C \\ 1 \\ m^2 - E^2 \\ -2\alpha E \end{array}$ In the event a magnetic field $\mathbf{B}$ is present, the momentum operators $\mathbf{p}$ should be replaced by $\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$. Under this condition the operators still close under
gilmore-lie-groups_ F02629	258	■ 0.721	$A\text{left}(\Gamma_{\{5\}}+\Gamma_{\{4\}}\text{right})+B\text{left}(\Gamma_{\{5\}}-\Gamma_{\{4\}}\text{right})+C$	$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	$\begin{array}{ll} \text{H} & \frac{p^2}{2m} - \frac{\alpha}{r} \\ A_4 & \mathbf{r} \cdot \mathbf{p} - i \\ \Gamma_4 & \frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r) \\ \Gamma_5 & \frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r) \\ \Theta & r(H_S - W) \\ A & A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C \\ B & 1/2m \\ C & -W \\ & -\alpha \end{array}$ $\begin{array}{l} \sqrt{p^2 + m^2} - \frac{\alpha}{r} \\ \mathbf{r} \cdot \mathbf{p} - i \\ \frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right) \\ \frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right) \\ r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\} \\ A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C \\ 1 \\ m^2 - E^2 \\ -2\alpha E \end{array}$ In the event a magnetic field $\mathbf{B}$ is present, the momentum operators $\mathbf{p}$ should be replaced by $\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$. Under this condition the operators still close under

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F02630	258	■ 0.721	1 / 2 m	$1/2m$	H $\frac{p^2}{2m} - \frac{\alpha}{r}$ A₄ $\mathbf{r} \cdot \mathbf{p} - i$ Γ₄ $\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$ Γ₅ $\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$ Θ $r(H_S - W)$ A $A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$ B $1/2m$ C $-W$ $-\alpha$ $\sqrt{p^2 + m^2} - \frac{\alpha}{r}$ $\mathbf{r} \cdot \mathbf{p} - i$ $\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$ $\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$ $r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$ $A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$ 1 $m^2 - E^2$ $-2\alpha E$

In the event a magnetic field **B** is present, the momentum operators **p** should be replaced by $\pi = \mathbf{p} - \mathcal{A}$. Under this condition the operators still close under

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F02634	258	■ 1.000	$\pi=\mathbf{p}-\frac{q}{c}\mathbf{A}$	$\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$	differential equation of first order:
gilmore-lie-groups_ F02635	258	■ 0.971	γ_i	γ_i	$F = \begin{bmatrix} 0 & -I_2 \\ I_2 & 0 \end{bmatrix}$
gilmore-lie-groups_ F02636	258	■ 0.982	σ_i	σ_i	The latter two are modified to allow a treatment of the Dirac operator along the same lines as the treatment of the Schrödinger and Klein–Gordon operators above.
gilmore-lie-groups_ F02637	259	■ 1.000	$\mathbf{p} \rightarrow \pi=\mathbf{p}-\frac{q}{c}\mathbf{A}$	$\mathbf{p} \rightarrow \pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$	As before, the substitution $\mathbf{p} \rightarrow \pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$ is in order in the event there is a
gilmore-lie-groups_ F02638	259	■ 1.000	J_i, M_i, A_i, Γ_i	J_i, M_i, A_i, Γ_i	three-vectors J_i, M_i, A_i, Γ_i defined in Table 14.5. The Dirac Hamiltonian is ex-
gilmore-lie-groups_ F02639	259	■ 0.990	Γ_4, Γ_5	Γ_4, Γ_5	the relation among these operators in the algebraic representation of the relativistic
gilmore-lie-groups_ F02640	259	■ 1.000	$D=d/dr$	$D = d/dr$	$\Gamma_+ D^{-1} = \Gamma_- \qquad \qquad \Gamma_+ \hat{\alpha} \Gamma_- \hat{\alpha}^\dagger = \Gamma_- \hat{\alpha}^\dagger$
gilmore-lie-groups_ F02641	259	■ 1.000	rD^2+r	$rD^2 + r$	
gilmore-lie-groups_ F02642	259	■ 1.000	rD^2-r	$rD^2 - r$	
gilmore-lie-groups_ F02643	260	■ 1.000	$C^2=\Gamma_5^2-\Gamma_4^2-M_4^2=-a$	$C^2 = \Gamma_5^2 - \Gamma_4^2 - M_4^2 = -a$	relations of this algebra have been described in Problem 11.9. The radial equation Eq. (14.63) is expressed in terms of the three operators as
gilmore-lie-groups_ F02644	260	■ 0.978	$a \rightarrow A$	$a \rightarrow A$	
gilmore-lie-groups_ F02645	260	■ 1.000	$C<0$	$C < 0$	where $u(r) = e^{-\theta} R(r)$. If A is the Casimir invariant of this representation of $\mathfrak{su}(1,1)$, the discrete spectrum of the compact operator Γ_+ is $N = -\frac{1}{2} +$
gilmore-lie-groups_ F02646	260	■ 1.000	$e^{-\theta}+Ce^{\theta}=0$	$e^{-\theta} + Ce^{\theta} = 0$	where $u(r) = e^{-\theta} R(r)$. If A is the Casimir invariant of this representation of $\mathfrak{su}(1,1)$, the discrete spectrum of the compact operator Γ_+ is $N = -\frac{1}{2} +$
gilmore-lie-groups_ F02647	260	■ 1.000	$u(r)=e^{\theta M_4}R(r)$	$u(r) = e^{\theta M_4} R(r)$	
gilmore-lie-groups_ F02648	260	■ 0.798	$N=-\frac{1}{2}+$	$N = -\frac{1}{2} +$	
gilmore-lie-groups_ F02649	260	■ 1.000	$\sqrt{\left(\frac{1}{2}\right)^2-A+1+n}, n=0,1,2, \ldots$	$\sqrt{\left(\frac{1}{2}\right)^2 - A + 1 + n}, n = 0, 1, 2, \dots$	
gilmore-lie-groups_ F02650	261	■ 1.000	$W=-\frac{1}{2}m c^2\alpha^2\left(1/N^2\right)$	$W = -\frac{1}{2}mc^2\alpha^2\left(1/N^2\right)$	Here N is the principal quantum number. The result is that this algebra acts to
gilmore-lie-groups_ F02651	261	■ 1.000	$N=l+1+k, k=0,1,2, \ldots$	$N = l + 1 + k, k = 0, 1, 2, \dots$	Here N is the principal quantum number. The result is that this algebra acts to
gilmore-lie-groups_ F02652	261	■ 1.000	m_l	m_l	the action of these operators. The states connected by the operators of this $\mathfrak{so}(2,1)$
gilmore-lie-groups_ F02653	261	■ 0.449	$ N, \ln\rangle \leftrightarrow N \pm 1, lm\rangle$	$ N, \ln\rangle \leftrightarrow N \pm 1, lm\rangle$	organized in angular momentum towers. These states are organized vertically in Fig. 14.9
gilmore-lie-groups_ F02654	261	■ 1.000	$L_z, L_{\pm}$	$L_z, L_{\pm}$	with the Laplace–Runge–Lenz vector act horizontally on the levels shown in these
gilmore-lie-groups_ F02655	261	■ 1.000	$M_z, M_{\pm}$	$M_z, M_{\pm}$	two figures. The operators $\Gamma_z, \Gamma_{\pm} = \Gamma_4 \pm iM_4$ act vertically on the levels shown
gilmore-lie-groups_ F02656	261	■ 0.997	$\Gamma_z, \Gamma_{\pm}=\Gamma_4 \pm iM_4$	$\Gamma_z, \Gamma_{\pm} = \Gamma_4 \pm iM_4$	hydrogenic states.
gilmore-lie-groups_ F02657	261	■ 1.000	$[\mathbf{L}, \Gamma]=0$	$[\mathbf{L}, \Gamma] = 0$	Remark The shift down operator Γ_- annihilates the ground state in a given an-
gilmore-lie-groups_ F02658	261	■ 1.000	Γ_-	Γ_-	mine the radial wavefunction $\langle r N, l = N - 1\rangle$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F02659	261	0.535	$\Gamma_{-}\langle r \mid N^{l=N-1}\rangle=0$	$\Gamma_{-}\langle r \mid N^{l=N-1}\rangle=0$	
gilmore-lie-groups_F02660	261	1.000	$L_{-}Y_{m=-l}^l(\theta,\phi)=0$	$L_{-}Y_{m=-l}^l(\theta,\phi)=0$	
gilmore-lie-groups_F02661	261	1.000	$\langle r \mid N,l=N-1\rangle$	$\langle r \mid N,l=N-1\rangle$	14.11.2 Scattering sta
gilmore-lie-groups_F02662	261	1.000	$C>0$	$C>0$	where as before $u(r)=e^{\theta M_3}R(r)$. Since the generator Γ_4 is noncompact, it has
gilmore-lie-groups_F02663	261	1.000	$e^{-\theta}-Ce^{\theta}=0$	$e^{-\theta}-Ce^{\theta}=0$	where as before $u(r)=e^{\theta M_3}R(r)$. Since the generator Γ_4 is noncompact, it has
gilmore-lie-groups_F02664	261	1.000	$k\simeq e^{-\theta}$	$k\simeq e^{-\theta}$	
gilmore-lie-groups_F02665	261	1.000	$E=\hbar^2k^2/2m$	$E=\hbar^2k^2/2m$	
gilmore-lie-groups_F02666	262	1.000	$\delta(j)=\arg\left[\Gamma(j+1-i(\alpha/k))\right]$	$\delta(j)=\arg[\Gamma(j+1-i(\alpha/k))]$	where $\delta(j)=\arg[\Gamma(j+1-i(\alpha/k))]$ is part of the scattering phase shift, and the
gilmore-lie-groups_F02667	262	1.000	$j=-\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}$	$j=-\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}$	expression for j is given by $j=-\frac{1}{2}+\sqrt{(\frac{1}{2})^2-A}$
gilmore-lie-groups_F02668	262	1.000	$-1/r$	$-1/r$	extra attraction by adding a term of the form $-1/r^2$ to the potential to represent
gilmore-lie-groups_F02669	262	1.000	$-1/r^2$	$-1/r^2$	penetration of the core electrons. To this end the potential used in the Schrodinger and Klein-Gordon equations is $V(r)=-e^2/r-\mu_l(\hbar^2/2m)/r^2$. This
gilmore-lie-groups_F02670	262	1.000	$V(r)=-e^2/r\rightarrow -e^2/r-\mu_l\left(\hbar^2/2m\right)/r^2$	$V(r)=-e^2/r\rightarrow -e^2/r-\mu_l\left(\hbar^2/2m\right)/r^2$	penetration produces a modification in the radial equation. The modification is encapsulated entirely in the change
gilmore-lie-groups_F02671	262	1.000	$j\rightarrow j^{\prime}=j+\Delta j$	$j\rightarrow j^{\prime}=j+\Delta j$	bound state energy spectrum:
gilmore-lie-groups_F02672	262	1.000	$\Delta j=$	$\Delta j=$	bound state energy spectrum:
gilmore-lie-groups_F02673	262	1.000	$-\mu_l/(2l+1)$	$-\mu_l/(2l+1)$	$mc^2\alpha^2$ $mc^2\alpha^2$
gilmore-lie-groups_F02674	262	1.000	Δj	Δj	The excess phase shift is
gilmore-lie-groups_F02675	263	1.000	$G\supset S\,O(3)$	$G\supset SO(3)$	generators include both the angular momentum operators and the components of
gilmore-lie-groups_F02676	264	0.987	3^{deg}	3^{deg}	incompatible with the first of Galileo’s assumptions. Does this create a problem for
gilmore-lie-groups_F02677	264	1.000	$\mathbf{F}=m\,\mathbf{g}$	$\mathbf{F}=m\mathbf{g}$	experiences an acceleration $\mathbf{g}$ in the “up” direction. If the “rest of the universe” “looks
gilmore-lie-groups_F02678	264	1.000	$\mathbf{A}=\frac{1}{2}\,\mathbf{B}\times\mathbf{r}$	$\mathbf{A}=\frac{1}{2}\mathbf{B}\times\mathbf{r}$	these two fields.
gilmore-lie-groups_F02679	264	0.996	$(r,\theta,\phi)=\left(\theta_3,\theta_2,\theta_1\right)$	$(r,\theta,\phi)=(\theta_3,\theta_2,\theta_1)$	$x=x_1=\theta_3\sin\theta_2\sin\theta_1$
gilmore-lie-groups_F02680	264	1.000	$\mathcal{L}^2\left(S^1\right)=\partial^2/\partial\theta_1^2$	$\mathcal{L}^2\left(S^1\right)=\partial^2/\partial\theta_1^2$	Generalize this result to $\mathcal{L}^2(S^n)$ (rect more generally for $\mathcal{L}^2(\mathbb{S}^n)$)
gilmore-lie-groups_F02681	264	1.000	$\mathcal{L}^2\left(S^3\right)$	$\mathcal{L}^2\left(S^3\right)$	
gilmore-lie-groups_F02682	264	1.000	$\left(\partial/\partial\cos\theta_3\right)^2$	$(\partial/\partial\cos\theta_3)^2$	
gilmore-lie-groups_F02683	264	1.000	$\mathcal{L}^2\left(S^n\right)$	$\mathcal{L}^2\left(S^n\right)$	
gilmore-lie-groups_F02684	265	1.000	$r\rightarrow\infty$	$r\rightarrow\infty$	a. Show that the singular points of Eq. (14.9) occur at $r=0$ and $r\rightarrow\infty$.

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gilmore-lie-groups_F02685	265	1.000	$\gamma=\frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2-A}$	$\gamma = \frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 - A}$	• Show that if $\sqrt{\left(\frac{1}{2}\right)^2 - A}$ is real, the solution with the po
gilmore-lie-groups_F02686	265	1.000	$\lambda= \pm \sqrt{-C}$	$\lambda = \pm \sqrt{-C}$	• Show that if $\sqrt{\left(\frac{1}{2}\right)^2 - A}$ is real, the solution with the po
gilmore-lie-groups_F02687	265	1.000	$\sqrt{\left(\frac{1}{2}\right)^2-A}$	$\sqrt{\left(\frac{1}{2}\right)^2 - A}$	integrable in the neighborhood of $r = 0$. Under what conditions is the solution with the negative sign square integrable? Show that if $C < 0$ the solution $\pm \sqrt{-C}$
gilmore-lie-groups_F02688	265	1.000	$\pm \sqrt{-C}$	$\pm \sqrt{-C}$	• Show that a solution of the form $R(r) = r^\gamma e^{\lambda r} f(r)$ can be found where the function
gilmore-lie-groups_F02689	265	1.000	$R(r)=r^{\gamma} e^{\lambda r} f(r)$	$R(r) = r^\gamma e^{\lambda r} f(r)$	e. Find the equation that the function $f(r)$ satisfies. Show that it is equivalent to the
gilmore-lie-groups_F02690	265	1.000	$f(r)=\sum_{j=0}^{\infty} f_j r^j$	$f(r) = \sum_{j=0}^{\infty} f_j r^j$	recursion relation is
gilmore-lie-groups_F02691	265	1.000	$j \rightarrow \infty$	$j \rightarrow \infty$	$f_{n+3} = \cdots = 0$), show that the quantization conditon $2\lambda(n + \gamma) + B = 0$ must be
gilmore-lie-groups_F02692	265	1.000	$f(r) \rightarrow e^{-2\lambda r}$	$f(r) \rightarrow e^{-2\lambda r}$	$f_{n+3} = \cdots = 0$), show that the quantization conditon $2\lambda(n + \gamma) + B = 0$ must be
gilmore-lie-groups_F02693	265	1.000	$\lambda < 0$	$\lambda < 0$	$f_{n+3} = \cdots = 0$), show that the quantization conditon $2\lambda(n + \gamma) + B = 0$ must be
gilmore-lie-groups_F02694	265	0.963	r^n	r^n	
gilmore-lie-groups_F02695	265	0.963	$f_n \neq 0$	$f_n \neq 0$	
gilmore-lie-groups_F02696	265	0.963	$f_{n+1}=0\left(\Rightarrow f_{n+2}=\right.$	$f_{n+1} = 0 (\Rightarrow f_{n+2} =$	
gilmore-lie-groups_F02697	265	1.000	$f_{n+3}=\cdots=0$	$f_{n+3} = \cdots = 0$	$n + \frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 - A} = \underline{\underline{-\frac{B}{2}}}$
gilmore-lie-groups_F02698	265	1.000	$2 \lambda(n+\gamma)+B=0$	$2\lambda(n + \gamma) + B = 0$	$n + \frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 - A} = \underline{\underline{-\frac{B}{2}}}$
gilmore-lie-groups_F02699	266	1.000	$\frac{1}{r} r^\gamma f(r) e^{\lambda r}$	$\frac{1}{r} r^\gamma f(r) e^{\lambda r}$	interval $(0, \infty)$.
gilmore-lie-groups_F02700	266	0.907	$(0, \infty)$	$(0, \infty)$	6. For a highly
gilmore-lie-groups_F02701	266	1.000	$Z e / r$	Ze/r	the relativistic solution is clearly incorrect? (Hint: set $l = 0$.)
gilmore-lie-groups_F02702	266	1.000	$N^{\prime }=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha ^2}$	$N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}$	$E(n,l) = \frac{mc^2}{\sqrt{1 + \left(\frac{\alpha}{Nl}\right)^2}} \rightarrow mc^2 - mc^2 \frac{1}{2N^2} \alpha^2.$
gilmore-lie-groups_F02703	266	1.000	$e^{\lambda r}$	$e^{\lambda r}$	relativistic $\alpha(n,l) = \sqrt{N^2 \alpha^2 + \alpha^2} \alpha_n$ $N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}$
gilmore-lie-groups_F02704	266	1.000	λ^{-1}	λ^{-1}	relativistic $\alpha(n,l) = \sqrt{N^2 \alpha^2 + \alpha^2} \alpha_n$ $N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}$ nonrelativistic $\alpha(n,l) = N \alpha_n$ $N = n + l + 1$
gilmore-lie-groups_F02705	266	1.000	$a \simeq 1/ \lambda $	$a \simeq 1/ \lambda $	relativistic $\alpha(n,l) = \sqrt{N^2 \alpha^2 + \alpha^2} \alpha_n$ $N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}$ nonrelativistic $\alpha(n,l) = N \alpha_n$ $N = n + l + 1$
gilmore-lie-groups_F02706	266	0.910	(n, l)	(n, l)	relativistic $\alpha(n,l) = \sqrt{N^2 \alpha^2 + \alpha^2} \alpha_n$ $N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}$

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gilmore-lie-groups_ F02716	267	■ 0.953	Ω^{-}	Ω^{-}	<table><thead><tr><th>Particle</th><th>Rest energy (MeV)</th></tr></thead><tbody><tr><td>electron $e^{\pm}$</td><td>0.511</td></tr><tr><td>mu meson $\mu^{\pm}$</td><td>105.7</td></tr><tr><td>tau meson $\tau^{\pm}$</td><td>1784.0</td></tr><tr><td>proton, antiproton $p^{\pm}$</td><td>938.26</td></tr><tr><td>deuteron d^{+}</td><td>1875.6</td></tr><tr><td>tritium t^{+}</td><td>2809.4</td></tr><tr><td>pi meson $\pi^{\pm}$</td><td>139.6</td></tr><tr><td>sigma meson $\Sigma^{\pm}$</td><td>1385.0</td></tr><tr><td>cascade meson Ξ^{-}</td><td>1533.0</td></tr><tr><td>omega Ω^{-}</td><td>1672.0</td></tr></tbody></table>	Particle	Rest energy (MeV)	electron $e^{\pm}$	0.511	mu meson $\mu^{\pm}$	105.7	tau meson $\tau^{\pm}$	1784.0	proton, antiproton $p^{\pm}$	938.26	deuteron d^{+}	1875.6	tritium t^{+}	2809.4	pi meson $\pi^{\pm}$	139.6	sigma meson $\Sigma^{\pm}$	1385.0	cascade meson Ξ^{-}	1533.0	omega Ω^{-}	1672.0
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gilmore-lie-groups_ F02717	267	■ 1.000	$a_{\mathrm{B}}=\hbar^2 / m e^2=0.529 \times 10^{-8} \mathrm{~cm}$	$a_B=\hbar^2 / m e^2=0.529 \times 10^{-8} \mathrm{~cm}$	sizes of the atoms in the (n, l) states differ between the relativistic and nonrelativistic																						
gilmore-lie-groups_ F02718	267	■ 0.906	n, l	n, l	Many charged particles can form hydrogen-like atoms through their electrostatic																						
gilmore-lie-groups_ F02719	267	■ 1.000	$W=-\frac{1}{2} m c^2 \alpha^2 / N^2$	$W=-\frac{1}{2} m c^2 \alpha^2 / N^2$	The motion of a classical nonrelativistic particle in a $1 / r^{\eta}$ radial force field is a conic section: an elliptical orbit for bound states ($E < 0$), hyperbolic for scattering states																						
gilmore-lie-groups_ F02720	267	■ 1.000	$1 / m=1 / m_1+1 / m_2$	$1 / m=1 / m_1+1 / m_2$	The motion of a classical nonrelativistic particle in a $1 / r^{\eta}$ radial force field is a conic section: an elliptical orbit for bound states ($E < 0$), hyperbolic for scattering states																						
gilmore-lie-groups_ F02721	267	■ 1.000	$(E=0)$	$(E=0)$	$J=\frac{1}{2} m \dot{r}^2+\frac{1}{2} m r^2 \dot{\phi}^2$																						
gilmore-lie-groups_ F02722	267	■ 1.000	$1 / r^3$	$1 / r^3$	the trajectory has the form																						
gilmore-lie-groups_ F02723	267	■ 1.000	$\alpha=\sqrt{1-\eta}, \eta=C / K a$	$\alpha=\sqrt{1-\eta}, \eta=C / K a$	potential																						
gilmore-lie-groups_ F02724	267	■ 1.000	$\alpha \simeq 1$	$\alpha \simeq 1$																							
gilmore-lie-groups_ F02725	267	■ 1.000	η	η																							
gilmore-lie-groups_ F02726	268	■ 0.651	$E=\sqrt{\left(m c^2\right)^2+(\mathbf{p} c)^2}-K / r$	$E=\sqrt{\left(m c^2\right)^2+(\mathbf{p} c)^2}-K / r$	a. Expand the relativistic energy $E=\sqrt{\left(m c^2\right)^2+(\mathbf{p} c)^2}-K / r$ to fourth order in $\mathbf{p}$																						
gilmore-lie-groups_ F02727	268	■ 1.000	$E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$	$E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$	and show $E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$.																						
gilmore-lie-groups_ F02728	268	■ 1.000	$-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$	$-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$	b. Replace the quartic term $-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$ by $-\left(W+K / r\right)^2 /\left(2 m c^2\right)$ and ex-																						
gilmore-lie-groups_ F02729	268	■ 1.000	$-\left(W+K / r\right)^2 /\left(2 m c^2\right)$	$-\left(W+K / r\right)^2 /\left(2 m c^2\right)$	b. Replace the quartic term $-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$ by $-\left(W+K / r\right)^2 /\left(2 m c^2\right)$ and ex-																						
gilmore-lie-groups_ F02730	268	■ 1.000	$K^{\prime}$	$K^{\prime}$	c. Argue that the classical motion involves a renormalized coupling $K \rightarrow K^{\prime}$ as w																						

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gilmore-lie-groups_ F02757	269	■ 0.999	$\left\langle l_{m'}^l \left L_+ \right l_m^l \right\rangle = \sqrt{(l+m')(l-m)}$		Use methods similar to those described in Problem 12 to construct the radial wave-
gilmore-lie-groups_ F02758	269	■ 1.000	$\delta_{m',m+1}$		inctions for hydrogenic atoms with extreme orbital angular momentum quantum
gilmore-lie-groups_ F02759	269	■ 1.000	$l=N-1$		$\frac{\partial}{\partial x} \left(\frac{x}{r} \right) = \frac{x \frac{\partial}{\partial x} x + x^2 \frac{\partial}{\partial x} \frac{1}{r}}{r^3} = -\frac{x \wedge x \wedge x}{r^3}$
gilmore-lie-groups_ F02760	269	■ 0.980	$n=0$		$\frac{d\mathbf{r}}{dt} = \frac{\mathbf{r}}{r} \frac{dr}{dt} + \frac{\mathbf{r} \wedge \mathbf{v}}{r^2}$ is the position vector from the sun to a planet, or from the planet to th
gilmore-lie-groups_ F02761	269	■ 0.848	$\mathbf{r}$		
gilmore-lie-groups_ F02762	270	■ 1.000	$\mathbf{M} \cdot \mathbf{L} = 0$		a. $\mathbf{M} \cdot \mathbf{L}$
gilmore-lie-groups_ F02763	270	■ 1.000	$\mathbf{M} \cdot \mathbf{M} = (2\mathbf{L} \cdot \mathbf{L}/m)(\mathbf{p} \cdot \mathbf{p}/2m - K/r) + K^2$		b. $\mathbf{M} \cdot \mathbf{M} = (2\mathbf{L} \cdot \mathbf{L}/m)(\mathbf{p} \cdot \mathbf{p}/2m - K/r) + K$
gilmore-lie-groups_ F02764	270	■ 1.000	$\mathbf{M} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L}/m - Kr$		c. $\mathbf{M} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L}/m -$
gilmore-lie-groups_ F02765	270	■ 0.999	$\mathbf{M} \cdot \mathbf{r} = Mr \cos \theta$		d. $\mathbf{M} \cdot \mathbf{r} = MR \cos \theta$
gilmore-lie-groups_ F02766	270	■ 1.000	$r = (\mathbf{L} \cdot \mathbf{L}/mK/1 + (M/K) \cos \theta)$		e. $r = (\mathbf{L} \cdot \mathbf{L}/mK/1 + (M/K) \cos \theta)$ f. Compare this result to the standa
gilmore-lie-groups_ F02767	270	■ 1.000	L^2/mK		in a $1/r$ potential to conclude that L^2/mK is the exchange rate of the smallest orbit and $\epsilon = M/K$ is the eccentricity of the orbit.
gilmore-lie-groups_ F02768	270	■ 1.000	$\epsilon = M/K$		g. Conclude that the Laplace–Runge–Lenz vector is
gilmore-lie-groups_ F02769	270	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = \left(-1/4\hbar^2\right) (\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$		has a similar expression. Show that the two expressions are equal since
gilmore-lie-groups_ F02770	270	■ 1.000	$\mathbf{B} \cdot \mathbf{B}$		has a similar expression. Show that the two expressions are equal since
gilmore-lie-groups_ F02771	270	■ 0.880	$\mathbf{L} \cdot \mathbf{M} = \mathbf{M} \cdot \mathbf{L} = 0$		17. Show that the
gilmore-lie-groups_ F02772	270	■ 1.000	$p_x = \mathbf{p} \cdot \mathbf{M}/M$		Show that circles in R^3 lift to circles in $S^3 \subset R^4$ under the stereographic projection
gilmore-lie-groups_ F02773	270	■ 1.000	$p_y = \mathbf{p} \cdot \mathbf{W}/W$		Show that circles in R^3 lift to circles in $S^3 \subset R^4$ under the stereographic projection
gilmore-lie-groups_ F02774	270	■ 1.000	$p_x^2 + (p_y - a)^2 = r^2$		Show that circles in R^3 lift to circles in $S^3 \subset R^4$ under the stereographic projection
gilmore-lie-groups_ F02775	270	■ 0.900	$0, a$		or Eq. (14.55). Show that circles in S^3 project back down to circles in R^3 under the
gilmore-lie-groups_ F02776	270	■ 1.000	$x^a y^b z^c$		l in $x_1, x_2, \dots, x_N$ is obtained by replacing $3 \rightarrow N$ in this expression. This is the
gilmore-lie-groups_ F02777	270	■ 0.996	$a+b+c=l$		<i>Bose–Einstein counting statistic.</i>
gilmore-lie-groups_ F02778	270	■ 0.996	$N(l,3)=(l+3-1)/l!(3-1)!$		<i>Bose–Einstein counting statistic.</i>
gilmore-lie-groups_ F02779	270	■ 0.997	$x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{N\}}$		degree l . • $S^l = \{x_1, \dots, x_N \mid x_1^2 + \dots + x_N^2 = 1\}$
gilmore-lie-groups_ F02780	270	■ 0.997	$3 \rightarrow N$		degree l . • $S^l = \{x_1, \dots, x_N \mid x_1^2 + \dots + x_N^2 = 1\}$
gilmore-lie-groups_ F02781	270	■ 1.000	$r^l Y_m^l(\theta, \phi)$		ference between the number of homogeneous polynomials of degree l and $l = 2$ on three variables: $\dim \mathcal{V}^l = N(l+3) = N(l+2+3) = 2l+1$

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gilmore-lie-groups_02782	270	1.000	x, y, z	x, y, z	ference between the number of homogeneous polynomials of degree l and $l = 2$ on three variables: $\dim \mathcal{Y}^l = N(l, 3) - N(l - 2, 3) = 2l + 1$
gilmore-lie-groups_02783	270	1.000	$l-2$	$l - 2$	(Bander and Itzykson, 1966a). Show that the number of spherical harmonics of
gilmore-lie-groups_02784	270	1.000	$\operatorname{dim}\left\{Y_m^l\right\}=N(l, 3)-N(l-2, 3)=2 l+1$	$\dim \left\{Y_m^l\right\}=N(l, 3)-N(l-2, 3)=2 l+1$	degree n is $\dim \left\{\mathcal{Y}_{lm}^n\right\}=N(n, 4)-N(n-2, 4)=(n+1)^2$.
gilmore-lie-groups_02785	270	0.999	$\operatorname{dim}\left\{\mathcal{Y}_{lm}^n\right\}=N(n, 4)-N(n-2, 4)=(n+1)^2$	$\dim \left\{\mathcal{Y}_{lm}^n\right\}=N(n, 4)-N(n-2, 4)=(n+1)^2$	$\psi(\mathbf{x})_{nlm}$ for $n=0, 1, 2, l=0, \ldots, n-1$, and $-l \leq m \leq +l$.
gilmore-lie-groups_02786	270	0.995	$\psi(\mathbf{x})_{nlm}$	$\psi(\mathbf{x})_{nlm}$	
gilmore-lie-groups_02787	270	0.995	$n=0,1,2 ; l=0, \ldots, n-1$	$n=0,1,2 ; l=0, \ldots, n-1$	
gilmore-lie-groups_02788	271	1.000	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$	where a and b are non odd integers. $\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \ldots$. Show homogeneous polynomials satisfy the recursion
gilmore-lie-groups_02789	271	1.000	$N(l, d)=$	$N(l, d)=$	$N(l, d-1)+N(l-1, d)$.
gilmore-lie-groups_02790	271	1.000	$N(l, d-1)+N(l-1, d)$	$N(l, d-1)+N(l-1, d)$	Use this result to derive the formula
gilmore-lie-groups_02791	271	1.000	S^{n-1}	S^{n-1}	$\dim \mathcal{Y}^l\left(S^n\right)=\dim \mathcal{Y}^{l-1}\left(S^n\right)+\dim \mathcal{Y}^l\left(S^{n-1}\right)$
gilmore-lie-groups_02792	271	1.000	$(l+1)^2=l^2+(2 l+1)$	$(l+1)^2=l^2+(2 l+1)$	is $\mathcal{Y}^0\left(S^n\right)=1=\dim \mathcal{Y}^0\left(S^n\right)$. Thus $\mathcal{Y}^l\left(S^n\right)=\frac{(l+n-2) !}{l !}(2 l+n-1) \mathcal{Y}^{l-1}$
gilmore-lie-groups_02793	271	1.000	$\mathcal{Y}^0\left(S^n\right)=1=\operatorname{dim} \mathcal{Y}^0\left(S^n\right)$	$\mathcal{Y}^0\left(S^n\right)=1=\dim \mathcal{Y}^0\left(S^n\right)$	$\dim \mathcal{Y}^0\left(S^n\right)=\frac{(n-1) !}{0 !}(2 n+1) \mathcal{Y}^0$
gilmore-lie-groups_02794	271	1.000	$\operatorname{dim} \mathcal{Y}^l\left(S^n\right)=\frac{(l+n-2) !}{l !(n-1) !}(2 l+n-1)$	$\dim \mathcal{Y}^l\left(S^n\right)=\frac{(l+n-2) !}{l !(n-1) !}(2 l+n-1)$	D-dimensional Coulomb problem In D -dimension the Coulomb problem is Eq. (14.43):
gilmore-lie-groups_02795	271	1.000	S^{D-1}	S^{D-1}	operator with eigenvalue $-\frac{1}{2}(D-1)(D-3)$, and α is a quantity that depends on the Lie algebra of $SO(D)$: it is half the sum over all positive roots of the algebra. For
gilmore-lie-groups_02796	271	1.000	$\mathcal{Y}^l\left(S^{D-1}\right)$	$\mathcal{Y}^l\left(S^{D-1}\right)$	the Lie algebras of the orthogonal roots the coefficient of the sum that is important is
gilmore-lie-groups_02797	271	1.000	$-\left[(l+\alpha)^2-\alpha^2\right]$	$-\left[(l+\alpha)^2-\alpha^2\right]$	$\alpha=D-2$.
gilmore-lie-groups_02798	271	0.709	$SO(D)$	$SO(D)$	a. Show that $\psi(\mathbf{x})=\left(1 / r^{(D-1) / 2}\right) \mathcal{Y}^l(\text { angles })$ is a clever ansatz that reduces the
gilmore-lie-groups_02799	271	1.000	$\alpha=D-2$	$\alpha=D-2$	case and
gilmore-lie-groups_02800	271	1.000	$\psi(\mathbf{x})=\left(1 / r^{(D-1) / 2}\right) \mathcal{Y}^l$	$\psi(\mathbf{x})=\left(1 / r^{(D-1) / 2}\right) \mathcal{Y}^l$	b. Show that the only change in Eq. (14.10) is the replacement
gilmore-lie-groups_02801	272	1.000	$\mathrm{a}-1 / r^2$	$\mathbf{a}-1 / r^2$	$\mathbf{a}-1 / r^2$ perturbation. Show that the bound state energy and scattering phase shifts
gilmore-lie-groups_02802	272	0.999	$l(l+1) \rightarrow l(l+1)-\mu_l$	$l(l+1) \rightarrow l(l+1)-\mu_l$	are given by the substitution $l(l+1) \rightarrow l(l+1)-\mu_l$. Argue that electrons in the s
gilmore-lie-groups_02803	272	0.999	s	s	are given by the substitution $l(l+1) \rightarrow l(l+1)-\mu_l$. Argue that electrons in the s
gilmore-lie-groups_02804	272	0.953	$\mu_0 \gg \mu_1 > \cdots$	$\mu_0 \gg \mu_1 > \cdots$	a -state electrons, ...) so that $\mu_0 \gg \mu_1 > \cdots$
gilmore-lie-groups_02805	272	1.000	$L_{ij}=a_i^{\dagger} a_j-a_j^{\dagger} a_i=-L_{ji}$	$L_{ij}=a_i^{\dagger} a_j-a_j^{\dagger} a_i=-L_{ji}$	b. Show that the Lie algebra of its dynamical symmetry generated by angular momentum operators together with the quadrupole
gilmore-lie-groups_02806	272	1.000	$Q_{ij}=$	$Q_{ij}=$	$a_i a_j+a_j a_i=+2 J_{ij}$. c. Show that one spectrum generating algebra includes the operators $\mathbf{L}$ and $\mathbf{Q}$ as well
gilmore-lie-groups_02807	272	1.000	$a_i^{\dagger} a_j+a_j^{\dagger} a_i=+Q_{ji}$	$a_i^{\dagger} a_j+a_j^{\dagger} a_i=+Q_{ji}$	c. Show that one spectrum generating algebra includes the operators $\mathbf{L}$ and $\mathbf{Q}$ as well as the single boson

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gilmore-lie-groups_02808	272	■ 1.000	$\left[a_{\mathfrak{i}}, a_{\mathfrak{j}}^{\dagger}\right]=1$	$\left[a_i, a_j^{\dagger}\right] = 1$	d. Show that another spectrum generating algebra consists of the operators L and Q as
gilmore-lie-groups_02809	272	■ 1.000	$a_{\mathfrak{i}} a_{\mathfrak{j}}$	$a_i a_j$	an invariant, where parity is even or odd according to whether the number of excitations in the spectrum is even or odd.
gilmore-lie-groups_02810	273	■ 0.999	$\mathbf{E}(\mathbf{x}, t), \mathbf{B}(\mathbf{x}, t)$	$\mathbf{E}(\mathbf{x}, t), \mathbf{B}(\mathbf{x}, t)$	In the momentum representation, obtained by taking a Fourier transform
gilmore-lie-groups_02811	273	■ 0.935	$\mathbf{E}(\mathfrak{k}), \mathbf{B}(\mathfrak{k})$	$\mathbf{E}(k), \mathbf{B}(k)$	each photon momentum state there are only two degrees of freedom,
gilmore-lie-groups_02812	274	■ 1.000	$\mathbf{E}(\mathbf{x}, t)$	$\mathbf{E}(\mathbf{x}, t)$	The quantum description involves arbitrary superpositions of two helicity compo-
gilmore-lie-groups_02813	274	■ 1.000	$\mathbf{B}(\mathbf{x}, t)$	$\mathbf{B}(\mathbf{x}, t)$	The quantum description involves arbitrary superpositions of two helicity compo-
gilmore-lie-groups_02814	274	■ 1.000	$\mathbf{E}(\mathfrak{k})$	$\mathbf{E}(k)$	nents for each momentum vector. The helicity states involve an angular momentum
gilmore-lie-groups_02815	274	■ 1.000	$\mathbf{B}(\mathfrak{k})$	$\mathbf{B}(k)$	nents for each momentum vector. The helicity states involve an angular momentum
gilmore-lie-groups_02816	274	■ 1.000	$\mathfrak{k} \cdot \mathfrak{k}=\mathbf{k} \cdot \mathbf{k}-k_4 k_4=0$	$k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$	nents for each momentum vector. The helicity states involve an angular momentum
gilmore-lie-groups_02817	274	■ 0.798	$k_{\mathfrak{4}}$	k_4	aligned along the direction of motion (helicity +1 and right-handed polarization)
gilmore-lie-groups_02818	274	■ 1.000	$\mathfrak{j}=1$	$j = 1$	terms,
gilmore-lie-groups_02819	274	■ 1.000	$\mathfrak{k} \cdot \mathfrak{k}=0$	$k \cdot k = 0$	
gilmore-lie-groups_02820	275	■ 0.999	$(x, y, z, i \mathfrak{c} \mathfrak{t})$	$(x, y, z, i c t)$	and S' be related by a homogeneous Lorentz transformation
gilmore-lie-groups_02821	275	■ 0.999	$(x, y, z, i \mathfrak{c} \mathfrak{t})^{\prime}$	$(x, y, z, i c t)'$	and S' be related by a homogeneous Lorentz transformation
gilmore-lie-groups_02822	275	■ 0.865	Λ	Λ	
gilmore-lie-groups_02823	275	■ 1.000	$\mathfrak{k} \cdot \mathfrak{a}=\Lambda \mathfrak{k} \cdot$	$k \cdot a = \Lambda k \cdot$	
gilmore-lie-groups_02824	275	■ 1.000	$\Lambda \mathfrak{a}$	Λa	$\left[J_i, J_j\right] = -\epsilon_{i j k} J_k$
gilmore-lie-groups_02825	275	■ 1.000	$\Lambda \in O(3,1)$	$\Lambda \in O(3, 1)$	$\left[J_i, J_j\right] = -\epsilon_{i j k} J_k$
gilmore-lie-groups_02826	275	■ 0.936	$\mathbf{J}, \mathbf{K}$	$\mathbf{J}, \mathbf{K}$	$\left[J_i, K_j\right] = -\epsilon_{i j k} K_k$
gilmore-lie-groups_02827	276	■ 0.988	$\mathfrak{a}=(x, y, z, \mathfrak{c} \mathfrak{t})$	$a = (x, y, z, c t)$	tion can be written as a 5×5 matrix, in terms of the 4-vector $a = (x, y, z, c t)$:
gilmore-lie-groups_02828	276	■ 0.995	$(\partial / \partial \mathfrak{x}, \partial / \partial \mathfrak{y}, \partial / \partial \mathfrak{z}, i \partial / \partial(\mathfrak{c} \mathfrak{t}))$	$(\partial / \partial x, \partial / \partial y, \partial / \partial z, i \partial / \partial(c t))$	
gilmore-lie-groups_02829	276	■ 1.000	$\{\Lambda, \emptyset\}$	$\{\Lambda, 0\}$	role in the derivation of the relativistic covariant field equations.
gilmore-lie-groups_02830	276	■ 1.000	$\{\mathfrak{I}, \mathfrak{a}\}$	$\{I, a\}$	
gilmore-lie-groups_02831	276	■ 1.000	$\Gamma^{\mathfrak{k}}$	Γ^k	Physically, k has a natural interpretation as the four momentum of the photon
gilmore-lie-groups_02832	276	■ 1.000	$ \mathfrak{k}\rangle$	$ k\rangle$	
gilmore-lie-groups_02833	277	■ 1.000	$D_{\mathfrak{2}}=\mathfrak{A}_{\mathfrak{1}}+\mathfrak{A}_{\mathfrak{1}}$	$D_2 = A_1 + A_1$	The Lie algebra $D_2 = A_1 + A_1$ is semisimple: it is the direct sum of two simple
gilmore-lie-groups_02834	277	■ 1.000	$\mathbf{J}^{\mathfrak{(1)}}$	$\mathbf{J}^{(1)}$	can be expressed in a $(2 j+1)(2 j'+1)$ dimensional representation $D^{j j'}$ as follows

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gilmore-lie-groups_ F02869	282	■ 1.000	$Y_{\{1\}}=J_{\{1\}}-K_{\{2\}}, \ Y_{\{2\}}=$	$Y_1 = J_1 - K_2, Y_2 =$	
gilmore-lie-groups_ F02870	282	■ 1.000	$J_{\{2\}}+K_{\{1\}}, \ Y_{\{3\}}=J_{\{3\}}$	$J_2 + K_1, Y_3 = J_3$	
gilmore-lie-groups_ F02871	282	■ 1.000	$\mathbf{b}=0$	$\mathbf{b} = 0$	
gilmore-lie-groups_ F02872	283	■ 1.000	$D_{\{\xi^{\prime}\prime}\xi}\left(H_{k^0}\right)$	$D_{\xi'\xi}\left(H_{k^0}\right)$	and irreducible. The cases (i)-(iv) are discussed here
gilmore-lie-groups_ F02873	283	■ 0.999	$k\cdot k>0$	$k\cdot k>0$	Case (ii). See below
gilmore-lie-groups_ F02874	283	■ 1.000	$j=0,\ \frac{1}{2},1,\ \frac{3}{2},\ \ldots$	$j=0,\frac{1}{2},1,\frac{3}{2},\ldots$	Case (iv) The unitary irreducible representations of $SO(3,1)$ are known but not
gilmore-lie-groups_ F02875	283	■ 1.000	$\kappa=\left(\kappa_1,\ \kappa_2\right)$	$\kappa=(\kappa_1,\kappa_2)$	As before, two cases occur (cf., Cases (i) or (iii) and Case (iv) above):
gilmore-lie-groups_ F02876	283	■ 1.000	$\kappa\in R^2,\ \kappa\cdot\kappa\geq 0$	$\kappa\in R^2,\kappa\cdot\kappa\geq 0$	(i) $\kappa\cdot\kappa>0$ little group = Identity
gilmore-lie-groups_ F02877	283	■ 1.000	$ \kappa\rangle$	$ \kappa\rangle$	(i) $\kappa\cdot\kappa>0$ little group = Identity
gilmore-lie-groups_ F02878	283	■ 0.999	$\left \kappa^{\prime}\right\rangle$	$ \kappa'\rangle$	(ii) $\kappa\cdot\kappa=0$ little group = $ISO(2)$ (15.43)
gilmore-lie-groups_ F02879	283	■ 0.999	$\kappa^{\prime}\cdot\kappa^{\prime}=\kappa\cdot\kappa$	$\kappa'\cdot\kappa'=\kappa\cdot\kappa$	(ii) $\kappa\cdot\kappa=0$ little group = $ISO(2)$ (15.43)
gilmore-lie-groups_ F02880	283	■ 0.999	$\kappa^{\prime}=\left(\kappa_1^{\prime},\kappa_2^{\prime}\right)$	$\kappa'=(\kappa_1',\kappa_2')$	(ii) $\kappa\cdot\kappa=0$ little group = $ISO(2)$ (15.43)
gilmore-lie-groups_ F02881	283	■ 1.000	$\kappa^{\prime}=R(\theta)\kappa$	$\kappa'=R(\theta)\kappa$	
gilmore-lie-groups_ F02882	283	■ 1.000	$\kappa\cdot\kappa$	$\kappa\cdot\kappa$	
gilmore-lie-groups_ F02883	283	■ 0.999	$\kappa\cdot\kappa>0\quad$	$\kappa\cdot\kappa>0$	
gilmore-lie-groups_ F02884	283	■ 0.996	$\kappa\cdot\kappa=0\quad$	$\kappa\cdot\kappa=0$	
gilmore-lie-groups_ F02885	283	■ 0.996	$=ISO(2)$	$=ISO(2)$	
gilmore-lie-groups_ F02886	284	■ 1.000	κ^2	κ^2	The first case presents us with two problems. First, κ^2 is a continuous quantum
gilmore-lie-groups_ F02887	284	■ 1.000	$\kappa^2>0$	$\kappa^2>0$	$\kappa^2>0$ there must be an infinite number of states with this same continuous index,
gilmore-lie-groups_ F02888	284	■ 1.000	$\kappa=0$	$\kappa=0$	for each four-momentum value. Therefore we require $\kappa=0$. This leaves us with the
gilmore-lie-groups_ F02889	284	■ 0.961	$\left(Y_1\rightarrow 0,\ Y_2\rightarrow 0\right)$	$(Y_1\rightarrow 0, Y_2\rightarrow 0)$	following physically allowable representations of the little group ($x_1\rightarrow 0,\ x_2\rightarrow 0$)
gilmore-lie-groups_ F02890	284	■ 1.000	$\left(j,\ j^{\prime}\right)$	(j,j')	To compare these two ways of describing a massless particle we compare
gilmore-lie-groups_ F02891	285	■ 1.000	$\left\{H_{k^0},\ 0\right\}$	$\{H_{k^0},0\}$	A. $\{H_{k^0},0\}$ on $ k^0,\xi\rangle$
gilmore-lie-groups_ F02892	285	■ 1.000	$H_{k^0}=\operatorname{EXP}\left(\Theta J_3+\theta_1 Y_1+\theta_2 Y_2\right)$	$H_{k^0}=\operatorname{EXP}\left(\Theta J_3+\theta_1 Y_1+\theta_2 Y_2\right)$	where $H_{k^0}=\operatorname{EXP}\left(\Theta J_3+\theta_1 Y_1+\theta_2 Y_2\right)$.
gilmore-lie-groups_ F02893	285	■ 0.494	$\left.\left.\left.\left k^0\right\rangle\right _{\mu}^{j,j^{\prime}}\right\rangle\right _{\mu^{\prime}}^{j^{\prime}}$	$\left k^0\right\rangle\left _{\mu}^{j,j^{\prime}}\right\rangle\left _{\mu^{\prime}}^{j^{\prime}}\right\rangle$	on the spin states
gilmore-lie-groups_ F02894	285	■ 0.119	$\left.\left.\left.\left k^0\right\rangle\right _{j_0}^{j_0^0,0}\right\rangle\right _{j_0^0}^{j_0^0,0}$	$\left k^0\right\rangle\left _{j_0^0}^{j_0^0,0}\right\rangle\left _{j_0^0}^{j_0^0,0}\right\rangle$	

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gilmore-lie-groups_F02895	285	■ 0.119	$\xi > 0$	$\xi > 0$	
gilmore-lie-groups_F02896	285	■ 0.119	$j = +\xi$	$j = +\xi$	
gilmore-lie-groups_F02897	285	■ 0.983	$\left k^0 \right\rangle \left \begin{array}{cc} 0 & j' \\ 0 & -j' \end{array} \right\rangle$	$ k^0\rangle \left \begin{array}{cc} 0 & j' \\ 0 & -j' \end{array} \right\rangle$	
gilmore-lie-groups_F02898	285	■ 0.983	$\xi < 0$	$\xi < 0$	
gilmore-lie-groups_F02899	285	■ 0.983	$j^{\prime} = -\xi$	$j' = -\xi$	
gilmore-lie-groups_F02900	286	■ 0.560	$ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle$	$ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle$	basis states $ k; \xi\rangle$ or the direct product states $ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle$.
gilmore-lie-groups_F02901	286	■ 1.000	$\langle k; \xi \psi \rangle$	$\langle k; \xi \psi \rangle$	case and $\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \psi \rangle$ in the second. In both cases the sum extends over all k vectors
gilmore-lie-groups_F02902	286	■ 0.741	$\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \psi \rangle$	$\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \psi \rangle$	for which $\Lambda \vec{k} \cdot \Lambda k = 0, k \neq 0$. In the first case the sum extends over the appropriate
gilmore-lie-groups_F02903	286	■ 1.000	$\Lambda k \cdot \Lambda k = 0, k \neq 0$	$\Lambda k \cdot \Lambda k = 0, k \neq 0$	helicity states ξ ($\xi = \pm 1$ for photons). In the second case the sum extends over the
gilmore-lie-groups_F02904	286	■ 1.000	$\xi = \pm 1$	$\xi = \pm 1$	appropriate values of $\mu, \mu' : -j \leq \mu \leq +j, -j' \leq \mu' \leq +j'$.
gilmore-lie-groups_F02905	286	■ 0.712	$\mu, \mu' : -j \leq \mu \leq +j, -j' \leq \mu' \leq +j'$	$\mu, \mu' : -j \leq \mu \leq +j, -j' \leq \mu' \leq +j'$	we discuss the positive helicity state $\xi = j > 0$ first. The amp
gilmore-lie-groups_F02906	286	■ 1.000	$\xi = j > 0$	$\xi = j > 0$	of the state $ k^0; j\rangle$ in any physical state $ \psi\rangle$ may be arbitrary. This is simply the amplitude of the massless particle of helicity j in the state $ \psi\rangle$. The amplitude
gilmore-lie-groups_F02907	286	■ 1.000	$\langle k^0; j \psi \rangle$	$\langle k^0; j \psi \rangle$	of the state $ k^0; j\rangle$ in any physical state $ \psi\rangle$ may be arbitrary. This is simply the amplitude of the massless particle of helicity j in the state $ \psi\rangle$. The amplitude
gilmore-lie-groups_F02908	286	■ 1.000	$ k^0; j\rangle$	$ k^0; j\rangle$	amplitude of the massless particle of helicity j in the state $ \psi\rangle$. The amplitude $\langle k^0, j 0 \rangle$ in the same physical state $ \psi\rangle$ is the same. The amplitudes of the
gilmore-lie-groups_F02909	286	■ 0.993	$\langle k^0; \begin{array}{cc} j & 0 \\ m & 0 \end{array} \psi \rangle$	$\langle k^0; \begin{array}{cc} j & 0 \\ m & 0 \end{array} \psi \rangle$	allowed in the manifestly covariant representation but not present in the Hilbert
gilmore-lie-groups_F02910	286	■ 0.999	$\langle k^0; \begin{array}{cc} j & 0 \\ m & 0 \end{array} \psi \rangle, m \neq j$	$\langle k^0; \begin{array}{cc} j & 0 \\ m & 0 \end{array} \psi \rangle, m \neq j$	space that carries the unitary irreducible representation. A simple linear way to
gilmore-lie-groups_F02911	286	■ 0.719	$\{ \cdot \}$	$\{ \cdot \}$	are absent in the description of a physical state ($m \neq j$) must vanish.
gilmore-lie-groups_F02912	286	■ 0.719	$(j - j) k_3^0 = 0$	$(j - j) k_3^0 = 0$	are absent in the description of a physical state ($m \neq j$) must vanish.
gilmore-lie-groups_F02913	286	■ 0.999	$(m - j) k_3^0$	$(m - j) k_3^0$	For the negative helicity states $\xi = -j$ we have by a completely similar argument
gilmore-lie-groups_F02914	286	■ 1.000	$\langle k^0; \begin{array}{cc} j & 0 \\ m & 0 \end{array} \psi \rangle$	$\langle k^0; \begin{array}{cc} j & 0 \\ m & 0 \end{array} \psi \rangle$	$\langle k^0, j 0 \rangle = \langle k^0, j 0 \rangle$
gilmore-lie-groups_F02915	286	■ 1.000	$(m - j) k_3^0 \neq 0$	$(m - j) k_3^0 \neq 0$	$\langle k^0, j 0 \rangle = \langle k^0, j 0 \rangle$
gilmore-lie-groups_F02916	286	■ 0.961	$m \neq j$	$m \neq j$	$\{ J_3^0 k_3^0 + J_4^0 k_{2j+1}^0 \} \langle k^0; \begin{array}{cc} j & 0 \\ m & 0 \end{array} \psi \rangle = 0$
gilmore-lie-groups_F02917	286	■ 1.000	$\xi = -j$	$\xi = -j$	
gilmore-lie-groups_F02918	286	■ 0.556	$ k^0\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle$	$ k^0\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle$	

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gilmore-lie-groups_F02974	294	■ 1.000	$\Phi(x)$	$\Phi(x)$	transformation S that brings the curved metric to flat form: $S^t g_{\text{grav}} S = g_{\text{flat}}$. Interpret
gilmore-lie-groups_F02975	294	■ 1.000	$S^t g_{\text{grav}} S = g_{\text{flat}}$	$S^t g_{\text{grav}} S = g_{\text{flat}}$	S in terms of a locally free-falling coordinate transformation.
gilmore-lie-groups_F02976	294	■ 1.000	V	V	charge q at the origin of a sphere of radius a , $\rho(x) = q\delta(x)$, The $\mathbf{E}$ field is spherically
gilmore-lie-groups_F02977	294	■ 0.867	$\mathbf{E}$	$\mathbf{E}$	symmetric, and Gauss' Law reduces to
gilmore-lie-groups_F02978	294	■ 0.523	$a, p(x)=q \delta(x)$	$a, p(x) = q\delta(x)$	$4\pi a^2 \mathbf{E}(a) = 4\pi q$
gilmore-lie-groups_F02979	294	■ 1.000	$ \mathbf{E}(a) =q/ \mathbf{a} $	$ \mathbf{E}(a) = q/ \mathbf{a} $	a. An observation point subtends an angle θ when measured from the center of the
gilmore-lie-groups_F02980	294	■ 0.808	$\oint \mathbf{E} \cdot d\mathbf{S} = \int 2\pi \rho dA$	$\left(\oint \mathbf{E} \cdot d\mathbf{S} = \int 2\pi \rho dA\right)$	sphere S^2 (c.f., Fig. 15.1). Show that its distance a from the north pole is $a = R\theta$
gilmore-lie-groups_F02981	294	■ 1.000	$a=R \sin \theta$	$a = R\theta$	
gilmore-lie-groups_F02982	295	■ 1.000	$2\pi R \sin \theta$	$2\pi R \sin \theta$	and the circumference of a circle of latitude through this point is $2\pi R \sin \theta$. Use
gilmore-lie-groups_F02983	295	■ 1.000	q/a	q/a	conclude that the field is stronger than the q/a term it would have in a plane. Show that this effective strengthening is due to the relative compression of the $\mathbf{E}$
gilmore-lie-groups_F02984	295	■ 1.000	$a(a=R \sin \theta)$	$a(a = R\theta)$	if the observer thinks (s)he is in a flat space, conclude (s)he will think the effective charge density at the distance from the observation point is constant if $a = R\theta$
gilmore-lie-groups_F02985	295	■ 1.000	$a=ct$	$a = ct$	is.
gilmore-lie-groups_F02986	295	■ 1.000	$H^n=S^0(n,1)/S^0(n), R^n=$	$H^n = SO(n,1)/SO(n), R^n =$	$\int_0^{2\pi} \int_0^\pi dr^2 \sin \theta d\theta d\phi$
gilmore-lie-groups_F02987	295	■ 0.999	$ISO(n)/SO(n)$	$ISO(n)/SO(n)$	$ds^2 = \frac{dr^2}{1-kr^2} + r^2 \sum_{i=1}^n (\sin \theta_2 \sin \theta_3 \cdots$
gilmore-lie-groups_F02988	295	■ 1.000	$k=(-1,0,+1)$	$k = (-1,0,+1)$	$x_2 = r \sin \theta_2 \cos \theta_3$
gilmore-lie-groups_F02989	295	■ 1.000	H^n, R^n, S^n	H^n, R^n, S^n	$x_2 = r \sin \theta_2 \cos \theta_3$
gilmore-lie-groups_F02990	295	■ 1.000	H^n, S^n	H^n, S^n	$\oint \mathbf{E} \cdot d\mathbf{S} = \int \Omega \rho(x) dV$
gilmore-lie-groups_F02991	295	■ 1.000	Ω	Ω	$ \mathbf{E} a^{n-1} = q$
gilmore-lie-groups_F02992	295	■ 1.000	$S^{n-1} \subset H^n, R^n$	$S^{n-1} \subset H^n, R^n$	$ \mathbf{E} a^{n-1} = q$
gilmore-lie-groups_F02993	295	■ 1.000	$\int e^{-x^2} dx = \sqrt{\pi}$	$\int e^{-x^2} dx = \sqrt{\pi}$	
gilmore-lie-groups_F02994	295	■ 1.000	$\Omega=2\pi^{n/2}/\Gamma(n/2)$	$\Omega = 2\pi^{n/2}/\Gamma(n/2)$	
gilmore-lie-groups_F02995	296	■ 1.000	H^n	H^n	on R, c, t ?
gilmore-lie-groups_F02996	296	■ 1.000	R, c, t	R, c, t	On the space $R \times I \times S^1$
gilmore-lie-groups_F02997	296	■ 0.839	$(l,b)=\left(263^\circ,48^\circ\right)$	$(l,b) = (263^\circ, 48^\circ)$	or, assume the temperature distribution of the microwave cavity is $\nabla^2 T = -\Delta^l(t)Y^l(\theta,\phi)$. How do you use this information
gilmore-lie-groups_F02998	296	■ 1.000	$T(\theta,\phi;t)=$	$T(\theta,\phi;t) =$	modynamic measurements, argue that this special reference frame is statistically
gilmore-lie-groups_F02999	296	■ 1.000	$\sum_{l,m} A_m^l(t) Y_m^l(\theta,\phi)$	$\sum_{l,m} A_m^l(t) Y_m^l(\theta,\phi)$	determined.
gilmore-lie-groups_F03000	296	■ 0.998	$\Delta U \Delta(1/T) \geq k$	$\Delta U \Delta(1/T) \geq k$	

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gilmore-lie-groups_ F03001	297	■ 1.000	$j=\frac{1}{2}, 1, 2$	$j = \frac{1}{2}, 1, 2$	why, or why not, the special frames defined by $j = \frac{1}{2}, 1, 2$ are the same. What
gilmore-lie-groups_ F03002	297	■ 1.000	$v(t)$	$v(t)$	that it moves through the microwave background radiation with a velocity $v(t)$.
gilmore-lie-groups_ F03003	297	■ 1.000	$v(t) \simeq v\left(t_{\{0\}}\right) e^{\{-\left(t-t_{\{0\}}\right) / \tau\}}$	$v(t) \simeq v\left(t_0\right) e^{-\left(t-t_0\right) / \tau}$	Show that its velocity decays to zero according to $v(t) \simeq v(t_0) e^{-\left(t-t_0\right) / \tau}$ because it is moving through a viscous medium. Estimate τ and present your answer in
gilmore-lie-groups_ F03004	297	■ 1.000	τ	τ	the form τ / T_{p} where T_{p} is the present age of the universe ($T_{\text{p}} \sim 13.7$ BY). To
gilmore-lie-groups_ F03005	297	■ 0.990	τ / T_{p}	τ / T_p	carry out this estimate you may assume the massive object is a black body –
gilmore-lie-groups_ F03006	297	■ 0.990	T_{p}	T_p	carry out this estimate you may assume the massive object is a black body –
gilmore-lie-groups_ F03007	297	■ 0.990	$T_{\text{p}} \simeq 13.7 \mathrm{BY}$	$T_p \simeq 13.7 \mathrm{BY}$	carry out this estimate you may assume the massive object is a black body –
gilmore-lie-groups_ F03008	297	■ 1.000	T_{BH}	T_{BH}	Use the standard relations for a neutral nonrotating black hole $R = 2GM/c^2$,
gilmore-lie-groups_ F03009	297	■ 1.000	$R=2 \mathrm{GM} / c^2\}$	$R = 2GM/c^2$	$T_{BH} = \hbar c^3 / 8 \pi k G M$. You can assume that the mass M is sufficiently large that
gilmore-lie-groups_ F03010	297	■ 1.000	$T_{\text{BH}}=\hbar c^3 / 8 \pi \mathrm{kGM}$	$T_{BH} = \hbar c^3 / 8 \pi k G M$	the temperature T_{BH} can be neglected (set to zero). Assume that the absorption
gilmore-lie-groups_ F03011	297	■ 1.000	$\gamma \pi R^2$	$\gamma \pi R^2$	Note that the problem of slowing down in a viscous medium was discussed by Einstein in another of the papers from his “ <i>annus mirabilis</i> ” the precursor of the
gilmore-lie-groups_ F03012	297	■ 1.000	$\gamma=3^3 / 2^2\}$	$\gamma = 3^3 / 2^2$	Note that the problem of slowing down in a viscous medium was discussed by Einstein in another of the papers from his “ <i>annus mirabilis</i> ” the precursor of the
gilmore-lie-groups_ F03013	298	■ 1.000	$d y / d x=g(x)$	$dy/dx = g(x)$	reductions of the differential equation into other solutions. This observation was used – exploited – by Lie to develop an algorithm for determining
gilmore-lie-groups_ F03014	298	■ 1.000	$y: d y / d x=g(x)$	$y: dy/dx = g(x)$	
gilmore-lie-groups_ F03015	298	■ 1.000	$d y / d x=p$	$dy/dx = p$	
gilmore-lie-groups_ F03016	299	■ 1.000	$F(x, y, p)=0$	$F(x, y, p) = 0$	a surface equation $F(x, y, p) = 0$. The special forms of the surface and constraint
gilmore-lie-groups_ F03017	299	■ 1.000	R, S, T	R, S, T	they are transformed to a set of canonical variables x, y, z to algorithmize the canonical variable $R(r, v)$ is the new independent variable (like r). $S(r, v)$ is the
gilmore-lie-groups_ F03018	299	■ 1.000	$R(x, y)$	$R(x, y)$	new dependent variable (like v), and $T(x, v, n)$ is the new constraint between S and
gilmore-lie-groups_ F03019	299	■ 1.000	$S(x, y)$	$S(x, y)$	new dependent variable (like v), and $T(x, v, n)$ is the new constraint between S and
gilmore-lie-groups_ F03020	299	■ 1.000	$T(x, y, p)$	$T(x, y, p)$	R (like dy/dx). In this new coordinate system the surface and constraint equations
gilmore-lie-groups_ F03021	299	■ 1.000	$d y / d x$	dy/dx	assume the desired forms $F(R, -, T) = 0$ and $dS/dR = f(R, -, T)$. The system
gilmore-lie-groups_ F03022	299	■ 0.999	$F(R,-, T)=0$	$F(R, -, T) = 0$	has been reduced to quadratures, and integration follows immediately.
gilmore-lie-groups_ F03023	299	■ 0.999	$d S / d R=f(R,-, T)$	$dS/dR = f(R, -, T)$	has been reduced to quadratures, and integration follows immediately.
gilmore-lie-groups_ F03024	299	■ 1.000	$y-G(x)=0$	$y - G(x) = 0$	represented by the Taylor series displacement operator $e^{-\partial / \partial x}$, for
gilmore-lie-groups_ F03025	299	■ 1.000	$y+c-G(x)=$	$y + c - G(x) =$	represented by the Taylor series displacement operator $e^{-\partial / \partial x}$, for
gilmore-lie-groups_ F03026	299	■ 1.000	$e^{\{c \partial / \partial y\}}$	$e^{c \partial / \partial y}$	In short, the “trivial” additive constant is in fact a one-parameter group

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gilmore-lie-groups_ F03027	300	■ 1.000	$F(x, y, p)=p-g(x)$	$F(x, y, p) = p - g(x)$	$F(x, y, p) = p - g(x)$ for the particular case at hand. There are two relations
gilmore-lie-groups_ F03028	300	■ 1.000	x, y, p	x, y, p	among the three variables x, y, p . They are given by the surface equation and the constraint equation.
gilmore-lie-groups_ F03029	300	■ 1.000	$\partial/\partial x, \partial/\partial y, \partial/\partial p$	$\partial/\partial x, \partial/\partial y, \partial/\partial p$	on the surface $F(x, y, p)$ defining the ordinary differential equation. It is also
gilmore-lie-groups_ F03030	300	■ 1.000	$F(x, y, p)$	$F(x, y, p)$	useful to express the action of the generator of infinitesimal displacements that
gilmore-lie-groups_ F03031	300	■ 0.798	$\left(d^n y / d x^n, n=1\right)$	$(d^n y / d x^n, n = 1)$	in the equation that defines the first order ordinary differential equation, $F(x, y, p) = 0$ is not of the form $p = g(x)$ so that $\partial F(x, y, p) / \partial p \neq 0$ then we
gilmore-lie-groups_ F03032	300	■ 1.000	$p^m=(d y / d x)^m, m=1$	$p^m = (d y / d x)^m, m = 1$	$F(x, y, p) = 0$, is not of the form $p = g(x)$, so that $\partial F(x, y, p) / \partial p \neq 0$, then we can attempt to find the following
gilmore-lie-groups_ F03033	300	■ 1.000	$p-g(x)$	$p - g(x)$	(ii) A new “canonical” coordinate system (R, S, T) . In this coordinate system $R = R(x, y)$ is the independent variable, $S = S(x, y)$ is the dependent variable, and $T = T(x, y, p)$
gilmore-lie-groups_ F03034	300	■ 1.000	$\frac{\partial}{\partial y} F(x, y, p) \neq 0$	$\frac{\partial}{\partial y} F(x, y, p) \neq 0$	(ii) A new “canonical” coordinate system (R, S, T) . In this coordinate system $R = R(x, y)$ is the independent variable, $S = S(x, y)$ is the dependent variable, and $T = T(x, y, p)$
gilmore-lie-groups_ F03035	300	■ 0.999	(R, S, T)	(R, S, T)	
gilmore-lie-groups_ F03036	300	■ 0.999	$R=R(x, y)$	$R = R(x, y)$	
gilmore-lie-groups_ F03037	300	■ 1.000	$S=S(x, y)$	$S = S(x, y)$	
gilmore-lie-groups_ F03038	300	■ 1.000	$T=T(x, y, p)$	$T = T(x, y, p)$	
gilmore-lie-groups_ F03039	301	■ 1.000	$x \rightarrow x$	$x \rightarrow x$	so that $\xi = 0, \eta = 1$, and $\zeta = 0$.
gilmore-lie-groups_ F03040	301	■ 1.000	$y \rightarrow y+\epsilon$	$y \rightarrow y + \epsilon$	so that $\xi = 0, \eta = 1$, and $\zeta = 0$.
gilmore-lie-groups_ F03041	301	■ 1.000	$\xi=0, \eta=1$	$\xi = 0, \eta = 1$	
gilmore-lie-groups_ F03042	301	■ 1.000	$\zeta(x, y, p)$	$\zeta(x, y, p)$	$\xi = 0, \eta = 1, \zeta = 0$
gilmore-lie-groups_ F03043	301	■ 1.000	$\xi(x, y)$	$\xi(x, y)$	$\xi = 0, \eta = 1, \zeta = 0$
gilmore-lie-groups_ F03044	301	■ 1.000	$\eta(x, y)$	$\eta(x, y)$	$\xi = 0, \eta = 1, \zeta = 0$
gilmore-lie-groups_ F03045	301	■ 1.000	$\eta_x=\partial \eta / \partial x$	$\eta_x = \partial \eta / \partial x$	
gilmore-lie-groups_ F03046	302	■ 1.000	$\xi(x, y), \eta(x, y)$	$\xi(x, y), \eta(x, y)$	and $\zeta(x, y, p)$ that define the infinitesimal generator X . These functions are deter-
gilmore-lie-groups_ F03047	302	■ 1.000	$y: p=p(x, y)$	$y: p = p(x, y)$	ables x and y . The generators of the infinitesimal displacements, $\xi(x, y)$ and $\eta(x, y)$,
gilmore-lie-groups_ F03048	302	■ 1.000	$X F(x, y, p(x, y))=0$	$X F(x, y, p(x, y)) = 0$	are represented by Laurent expansions, or Taylor series expansions if convergent
gilmore-lie-groups_ F03049	302	■ 1.000	d_{ξ}, d_{η}	d_{ξ}, d_{η}	This gives a set of simultaneous <i>linear</i> equations in the expansion amplitudes
gilmore-lie-groups_ F03050	302	■ 0.999	$X F=0$	$X F = 0$	ξ_{ij}, η_{ij} . In general, there are more equations than unknowns. Since the equations
gilmore-lie-groups_ F03051	302	■ 0.999	$\sum C_{ij} x^i y^j=0$	$\sum C_{ij} x^i y^j = 0$	ξ_{ij}, η_{ij} . In general, there are more equations than unknowns. Since the equations
gilmore-lie-groups_ F03052	302	■ 1.000	C_{ij}	C_{ij}	are homogeneous, there are no nontrivial solutions if the rank of this system is equal

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gilmore-lie-groups_ F03053	302	■ 0.999	ξ_{ij}, η_{ij}	ξ_{ij}, η_{ij}	scaling factor) is equal to the corank of this system of equations. This is not larger than one for first order equations but may exceed one for second and higher order
gilmore-lie-groups_ F03054	303	■ 1.000	$X(x, y, p) S(x, y)=1$	$X(x, y, p) S(x, y) = 1$	$\frac{1}{X} = \frac{1}{S} \Rightarrow S = \frac{1}{X}$
gilmore-lie-groups_ F03055	303	■ 1.000	$X S=+1$	$X S = +1$	X is some constant.
gilmore-lie-groups_ F03056	303	■ 1.000	$X S=-1$	$X S = -1$	16.2.4.2 Invariant coordinates: independent variable
gilmore-lie-groups_ F03057	303	■ 1.000	$X S=k \neq 0$	$X S = k \neq 0$	16.2.4.2 Invariant coordinates: independent variable
gilmore-lie-groups_ F03058	303	■ 1.000	$X R=0$	$X R = 0$	$\left(\xi(x, y) \frac{\partial}{\partial x} + \eta(x, y) \frac{\partial}{\partial y}\right) R(x, y) = 0$
gilmore-lie-groups_ F03059	303	■ 1.000	$X T=0$	$X T = 0$	$\left(\xi(x, y) \frac{\partial}{\partial x} + \eta(x, y) \frac{\partial}{\partial y}\right) T(x, y) = 0$
gilmore-lie-groups_ F03060	304	■ 1.000	$R: T=T(R)$	$R: T = T(R)$	$S = \int f(R, -, T(R)) dR + c \quad (16.20)$
gilmore-lie-groups_ F03061	304	■ 1.000	$x=x(R, S), y=y(R, S)$	$x = x(R, S), y = y(R, S)$	more general example
gilmore-lie-groups_ F03062	305	■ 1.000	$x p+y-x y^2=0$	$x p + y - x y^2 = 0$	The shape of the surface depends on both coordinates x and y .
gilmore-lie-groups_ F03063	305	■ 0.980	$0.1 \leq x \leq 1.1$	$0.1 \leq x \leq 1.1$	
gilmore-lie-groups_ F03064	305	■ 0.980	$-2 \leq y \leq+2$	$-2 \leq y \leq +2$	
gilmore-lie-groups_ F03065	305	■ 1.000	$y \rightarrow \alpha y$	$y \rightarrow \alpha y$	ant is $x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$. Since there is a one-parameter invariance
gilmore-lie-groups_ F03066	305	■ 1.000	$x \rightarrow \beta x$	$x \rightarrow \beta x$	ant is $x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$. Since there is a one-parameter invariance
gilmore-lie-groups_ F03067	305	■ 1.000	$\alpha\left(x p+y-(\alpha \beta) x y^2\right)=0$	$\alpha\left(x p + y - (\alpha \beta) x y^2\right) = 0$	group for this differential equation, Lie's methods are guaranteed to work. In fact,
gilmore-lie-groups_ F03068	305	■ 1.000	$\alpha \beta=1$	$\alpha \beta = 1$	group for this differential equation, Lie's methods are guaranteed to work. In fact,
gilmore-lie-groups_ F03069	305	■ 1.000	$x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$	$x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$	unecny. The surface $z = v^2 = v/x$ is shown in Fig. 16.1. The value of a clearly depends
gilmore-lie-groups_ F03070	305	■ 1.000	$X(x, y, p)$	$X(x, y, p)$	new coordinate system in which the surface is independent of the new dependent
gilmore-lie-groups_ F03071	305	■ 1.000	$p=y^2-y / x$	$p = y^2 - y/x$	The determining equation Eq. (16.14) is
gilmore-lie-groups_ F03072	306	■ 1.000	$p: p(x, y)=$	$p: p(x, y) =$	we use the surface equation $F(x, y, p) = 0$ to find an expression for p : $p(x, y) =$
gilmore-lie-groups_ F03073	306	■ 1.000	$-y / x+y^2$	$-y/x + y^2$	$-y/x + y^2$. This is substituted into the determining equation $X F(x, y, p) = 0$ to
gilmore-lie-groups_ F03074	306	■ 1.000	$X F(x, y, p)=0$	$X F(x, y, p) = 0$	$-y/x + y^2$. This is substituted into the determining equation $X F(x, y, p) = 0$ to
gilmore-lie-groups_ F03075	306	■ 1.000	$\eta: \xi=\xi_{00}, \eta=\eta_{00}$	$\eta: \xi = \xi_{00}, \eta = \eta_{00}$	are substituted into Eq. (16.23) we obtain three equations for the two unknowns. The
gilmore-lie-groups_ F03076	306	■ 0.999	$y / x, 1$	$y/x, 1$	ξ_{00}, η_{00} as follows:
gilmore-lie-groups_ F03077	306	■ 0.999	$x y$	$x y$	ξ_{00}, η_{00} as follows:
gilmore-lie-groups_ F03078	306	■ 1.000	ξ_{00}, η_{00}	ξ_{00}, η_{00}	

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gilmore-lie-groups_ F03079	307	■ 1.000	\xi_{01}=\xi_{00}=\eta_{00}=\eta_{10}=0	$\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$	four equations we determine $\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$, and from the coefficient
gilmore-lie-groups_ F03080	307	■ 1.000	x y^{2}	xy^2	of xy^2 we learn $-\xi_{10} - \eta_{01} = 0$ so that, up to some overall scaling factor, we can
gilmore-lie-groups_ F03081	307	■ 1.000	-\xi_{10}-\eta_{01}=0	$-\xi_{10} - \eta_{01} = 0$	of xy^2 we learn $-\xi_{10} - \eta_{01} = 0$ so that, up to some overall scaling factor, we can
gilmore-lie-groups_ F03082	307	■ 1.000	\xi(x, y)=x	$\xi(x, y) = x$	take $\xi(x, y) = x$ and $\eta(x, y) = -y$. Since we have found one nontrivial solution for an infinitesimal generator of a one-parameter group of a first-order equation, we
gilmore-lie-groups_ F03083	307	■ 1.000	\eta(x, y)=-y	$\eta(x, y) = -y$	take $\xi(x, y) = x$ and $\eta(x, y) = -y$. Since we have found one nontrivial solution for an infinitesimal generator of a one-parameter group of a first-order equation, we
gilmore-lie-groups_ F03084	307	■ 1.000	\zeta=-2 p	$\zeta = -2p$	$\frac{\partial}{\partial x} \frac{\partial}{\partial y} \frac{\partial}{\partial p}$
gilmore-lie-groups_ F03085	307	■ 1.000	-y d S(y) / d y=1	$-y dS(y)/dy = 1$	neglecting the negative sign: $S(x, y) = \ln(y)$.
gilmore-lie-groups_ F03086	307	■ 1.000	-\ln (y)	$-\ln(y)$	The invariant coordinates are determined using the method of characteristics:
gilmore-lie-groups_ F03087	307	■ 1.000	S(x, y)=\ln (y)	$S(x, y) = \ln(y)$	$\frac{dx}{x} - \frac{dy}{y} - \frac{dp}{p}$
gilmore-lie-groups_ F03088	307	■ 1.000	y d x=-x d y	$ydx = -xdy$	$-dp/2p$ equal to either of the other two differentials. We set it equal to dx/x to
gilmore-lie-groups_ F03089	307	■ 1.000	d(x y)=0	$d(xy) = 0$	avoid having the second invariant coordinate dependent on y . The equation is
gilmore-lie-groups_ F03090	307	■ 1.000	R(x, y)=x y	$R(x, y) = xy$	avoid having the second invariant coordinate dependent on y . The equation is
gilmore-lie-groups_ F03091	307	■ 1.000	-d p / 2 p	$-dp/2p$	The forward and backward transformations between the two coordinate systems are
gilmore-lie-groups_ F03092	307	■ 1.000	d x / x	dx/x	The forward and backward transformations between the two coordinate systems are
gilmore-lie-groups_ F03093	307	■ 1.000	d x / x=-d p / 2 p	$dx/x = -dp/2p$	$\left(\begin{matrix} R \\ S \end{matrix} \right) = \left(\begin{matrix} xy \\ p \end{matrix} \right) \Rightarrow \left(\begin{matrix} x \\ y \end{matrix} \right) = \left(\begin{matrix} Re^{-\frac{S}{2}} \\ \frac{R}{e^{-\frac{S}{2}}} \end{matrix} \right)$ are given
gilmore-lie-groups_ F03094	307	■ 1.000	(1 / x) d \left(x^{2} p\right)=0	$(1/x)d(x^2p) = 0$	$\left(\begin{matrix} R \\ S \end{matrix} \right) = \left(\begin{matrix} xy \\ p \end{matrix} \right) \Rightarrow \left(\begin{matrix} x \\ y \end{matrix} \right) = \left(\begin{matrix} Re^{-\frac{S}{2}} \\ \frac{R}{e^{-\frac{S}{2}}} \end{matrix} \right)$ are given
gilmore-lie-groups_ F03095	307	■ 1.000	T(x, y, p)=x^{2} p	$T(x, y, p) = x^2p$	$\left(\begin{matrix} R \\ S \end{matrix} \right) = \left(\begin{matrix} xy \\ p \end{matrix} \right) \Rightarrow \left(\begin{matrix} x \\ y \end{matrix} \right) = \left(\begin{matrix} Re^{-\frac{S}{2}} \\ \frac{R}{e^{-\frac{S}{2}}} \end{matrix} \right)$ are given
gilmore-lie-groups_ F03096	308	■ 1.000	T / R+1-	$T/R + 1 -$	for $-\frac{3}{2} \leq R \leq +4$ and $-2 \leq S \leq +2$. The function is a simple ruled surface, independent of S .
gilmore-lie-groups_ F03097	308	■ 0.669	R=0	$R = 0$	independent of S .
gilmore-lie-groups_ F03098	308	■ 0.669	R, S	R, S	independent of S .
gilmore-lie-groups_ F03099	308	■ 1.000	-3 \leq R \leq+4	$-3 \leq R \leq +4$	for $-\frac{3}{2} \leq R \leq +4$ and $-2 \leq S \leq +2$. The function is a simple ruled surface, independent of S .
gilmore-lie-groups_ F03100	308	■ 1.000	-2 \leq S \leq+2	$-2 \leq S \leq +2$	for $-\frac{3}{2} \leq R \leq +4$ and $-2 \leq S \leq +2$. The function is a simple ruled surface, independent of S .
gilmore-lie-groups_ F03101	308	■ 1.000	T=T(R, S)	$T = T(R, S)$	or each coordinate is sometimes called a "symplectic". The new constraint equation is
gilmore-lie-groups_ F03102	308	■ 1.000	R: T(R)=R^{2}-R	$R : T(R) = R^2 - R$	$\frac{dT}{dR} = \frac{1}{R} - \frac{1}{R^2} \implies S = \ln(R) + \frac{1}{R} + c \quad (16.34)$
gilmore-lie-groups_ F03103	309	■ 0.999	(R, S)	(R, S)	The inverse transformation, Eq. (16.30), from (R, S) to (x, y) is finally used to
gilmore-lie-groups_ F03104	309	■ 1.000	x d / d x	xd/dx	Remark: The operator $x d/dx$ is the infinitesimal generator for scaling transformations, since $e^{\lambda x d/dx} x = e^{\lambda} x$. As a result, the infinitesimal generator X has the

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gilmore-lie-groups_ F03105	309	■ 1.000	$e^{\lambda x} d/dx \ x = e^{\lambda x} \ x$	$e^{\lambda x} d/dx \ x = e^{\lambda} x$	ditions, since $e^{\lambda x} = e^{\lambda x} \cdot 1$ is a constant, the infinitesimal generator X has the following effect on the coordinates (x, y, p):
gilmore-lie-groups_ F03106	309	■ 0.766	(x, y, p)	(x, y, p)	the canonical coordinates (R, S, T).
gilmore-lie-groups_ F03107	309	■ 1.000	$\ln(y)$	$\ln(y)$	group parameter, $\ln(e^{-\lambda} y) = \ln(y) - \lambda$. The invariant operators come right out of the scaling transformations: xy and $x^2 p$ are unchanged by the scaling transformations.
gilmore-lie-groups_ F03108	309	■ 1.000	$\ln \left(e^{-\lambda} y \right) = \ln(y) - \lambda$	$\ln \left(e^{-\lambda} y \right) = \ln(y) - \lambda$	the scaling transformations, xy and $x^2 p$ are unchanged by the scaling transformation. None of these operators is unique. The operator $\ln(rv^2)$ is linear and $v^3 \ln$
gilmore-lie-groups_ F03109	309	■ 1.000	$x^2 p$	$x^2 p$	then none of these operators is unique. The operator $\ln(xy)$ is linear and $x^2 p$ is invariant. We have just chosen the most convenient (simplest) solutions to the
gilmore-lie-groups_ F03110	309	■ 1.000	$\ln \left(x y^2 \right)$	$\ln \left(x y^2 \right)$	equations defining the new coordinates.
gilmore-lie-groups_ F03111	309	■ 1.000	$x^3 y p$	$x^3 y p$	equations defining the new coordinates.
gilmore-lie-groups_ F03112	309	■ 1.000	$F \left(x y, x^2 p \right) = 0$	$F \left(x y, x^2 p \right) = 0$	the constraint equation $F(x, y, p) = 0$.
gilmore-lie-groups_ F03113	309	■ 1.000	$x^2 p = h(x y)$	$x^2 p = h(x y)$	10.7.2 Higher degree equations
gilmore-lie-groups_ F03114	309	■ 1.000	$dy/dx = x^{-2} h(x y)$	$dy/dx = x^{-2} h(x y)$	10.7.2 Higher degree equations
gilmore-lie-groups_ F03115	309	■ 1.000	$h(z) = -z + z^2$	$h(z) = -z + z^2$	These methods work equally well with first order equations of higher degree.
gilmore-lie-groups_ F03116	309	■ 1.000	$dy/dx + y^2 - 2/x^2 = 0, h(z) = z^2 - 2$	$dy/dx + y^2 - 2/x^2 = 0, h(z) = z^2 - 2$	These methods work equally well with first order equations of higher degree.
gilmore-lie-groups_ F03117	309	■ 1.000	$y'^2 + y^4 - x^{-4} = 0$	$y'^2 + y^4 - x^{-4} = 0$	
gilmore-lie-groups_ F03118	309	■ 1.000	$R^4 + T^2 = 1$	$R^4 + T^2 = 1$	
gilmore-lie-groups_ F03119	310	■ 1.000	$p = \pm \sqrt{x^{-4} - y^4}$	$p = \pm \sqrt{x^{-4} - y^4}$	$p = \pm \sqrt{x^{-4} - y^4}$, corresponding to the two solution branches in the canonical
gilmore-lie-groups_ F03120	310	■ 1.000	$T = \pm \sqrt{1 - R^4}$	$T = \pm \sqrt{1 - R^4}$	coordinate system $T = \pm \sqrt{1 - R^4}$
gilmore-lie-groups_ F03121	310	■ 1.000	$\zeta(x, y, p) = \eta^{(1)}(x, y, p)$	$\zeta(x, y, p) = \eta^{(1)}(x, y, p)$	the canonical coordinates (R, S, T). Since the constraint equation dS/dR depends
gilmore-lie-groups_ F03122	310	■ 1.000	dS/dR	dS/dR	only on the change of variables, it also can be tabulated, and has been. The simplest
gilmore-lie-groups_ F03123	310	■ 1.000	$S \rightarrow S' = S + c, R' = R$	$S \rightarrow S' = S + c, R' = R$	$e^{+c} y$.
gilmore-lie-groups_ F03124	310	■ 1.000	$\ln(y) \rightarrow \ln(y) + c = \ln(e^c y) =$	$\ln(y) \rightarrow \ln(y) + c = \ln(e^c y) =$	$e^{+c} y$.
gilmore-lie-groups_ F03125	310	■ 0.995	$\ln(\bar{y}(c))$	$\ln(\bar{y}(c))$	
gilmore-lie-groups_ F03126	310	■ 0.995	$xy = \bar{x}(c) \bar{y}(c)$	$xy = \bar{x}(c) \bar{y}(c)$	
gilmore-lie-groups_ F03127	310	■ 0.995	$\bar{x}(c) = e^{-c} x$	$\bar{x}(c) = e^{-c} x$	
gilmore-lie-groups_ F03128	310	■ 0.995	$\bar{y}(c) =$	$\bar{y}(c) =$	
gilmore-lie-groups_ F03129	310	■ 0.999	$e^{+c} y$	$e^{+c} y$	
gilmore-lie-groups_ F03130	310	■ 0.998	$y^{(2)}$	$y^{(2)}$	$\partial x \quad \partial y \quad \partial y^{(1)}$

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gilmore-lie-groups_ F03131	311	■ 1.000	\xi, \eta, \zeta	ξ, η, ζ	Table 16.1. <i>Infinitesimal generators ξ, η, ζ, canonical coordinates R, S, T, and</i>																																																																																																																																																																																																																									
					<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $\mathbf{D}^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \qquad (16.10)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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b(R/T)^{(1/b)}]$</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>$y^2 - 2bx$</td><td>y/b</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g' p^2</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>f(x)</td><td>0</td><td>-f' p</td><td>y</td><td>$F (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>g(y)</td><td>g' p</td><td>x</td><td>$G (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - 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y$	T/R^2	-y	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	-p	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	-p^2	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	g(y)	0	-g' p^2	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	f(x)	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	f(x)	0	-f' p	y	$F (F' f = 1)$	pf	$1/T$	0	g(y)	g' p	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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0	x	1	x	y/x	$xp - y$	T/R^2																																																																																																																																																																																																																								
-y	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	-p	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	-p^2	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
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xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
g(y)	0	-g' p^2	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	f(x)	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
f(x)	0	-f' p	y	$F (F' f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
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x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03134	311	0.852	-1 / b	-1/b	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - R)$</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T/R)/(T + R)$</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>$y^2 - 2bx$</td><td>y/b</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g' p^2</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>f(x)</td><td>0</td><td>-f' p</td><td>y</td><td>$F (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>g(y)</td><td>g' p</td><td>x</td><td>$G (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	1/a	-1/b	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	-p	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	-y	-2p	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	2x	y	-p	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	2y	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	-p^2	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	-y	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	-p	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	-p^2	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	g(y)	0	-g' p^2	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	f(x)	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	f(x)	0	-f' p	y	$F (F' f = 1)$	pf	$1/T$	0	g(y)	g' p	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03135	311	0.852	a x+b y	ax + by	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹/ [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(TR ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03136	311	0.852	b x-a y	bx - ay	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹/ [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(TR ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03137	311	0.852	$(b - a \ T) \ / (a + b \ T)$	$(b - aT)/(a + bT)$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F \ (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G \ (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F \ (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G \ (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03138	311	0.852	$-p$	$-p$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F \ (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G \ (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$																																																																																																																																																																																																																								
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$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
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a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
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R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dy} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dy^{(1)}} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dy^{(n+1)}} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_F03140	311	0.852	$x p$	xp	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$																																																																																																																																																																																																																								
0	x	1	x	y/x	$xp - y$	T/R^2																																																																																																																																																																																																																								
$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
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x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dy} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dy^{(1)}} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dy^{(n+1)}} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03144	311	0.852	y/b	y/b	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dy} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dy^{(1)}} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dy^{(n+1)}} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
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R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $r^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_F03148	311	0.852	$p / x^{\{(a / b-1)\}}$	$p/x^{(a/b-1)}$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\,(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\,(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\,(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\,(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$																																																																																																																																																																																																																								
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$																																																																																																																																																																																																																								
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$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$																																																																																																																																																																																																																								
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a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
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$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
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x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$																																																																																																																																																																																																																								
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R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $r^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03156	311	0.852	$y p$	yp	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$																																																																																																																																																																																																																								
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$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$																																																																																																																																																																																																																								
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a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
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$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
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x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$																																																																																																																																																																																																																								
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R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dy} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dy^{(1)}} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dy^{(n+1)}} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03160	311	0.852	p / x	p/x	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - 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2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$																																																																																																																																																																																																																								
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y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln 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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03166	311	0.852	$T \ / \ R^{\wedge\{2\}}$	T/R^2	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln 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p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\ 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y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$																																																																																																																																																																																																																								
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x^{k+1}	$kx^k y$	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$																																																																																																																																																																																																																								

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R)$</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>$y^2 - 2bx$</td><td>y/b</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x²</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y²</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>f(x)</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>g(y)</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dy} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dy^{(1)}} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dy^{(n+1)}} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	1/a	-1/b	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	-p	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	-y	-2p	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	2x	y	-p	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	2y	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	-p ²	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	-y	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	-p	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	-p ²	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x ²	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y ²	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	g(y)	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	f(x)	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	f(x)	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	g(y)	$g'p$	x	$G (G'g = 1)$	p/g	T	x ^{k+1}	kx ^k y	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy ^k	y ^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	-y	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	-p	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	-p ²	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x ²	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y ²	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	g(y)	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	f(x)	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	f(x)	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	g(y)	$g'p$	x	$G (G'g = 1)$	p/g	T	x ^{k+1}	kx ^k y	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy ^k	y ^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	$kx^k 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y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln 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p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\ 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y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g=1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ 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y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a \ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g=1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ F03180	311	0.852	$e^{\{y\}} * p^{\{b\}}$	$e^y * p^b$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln 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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\,(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\,(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\,(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\,(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$																																																																																																																																																																																																																								
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y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>e^y*p^b</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln 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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03186	311	0.852	$e^{\{f(x)\}}$	$e^{f(x)}$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln 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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln 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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$																																																																																																																																																																																																																								
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y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln 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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>e^y*p^b</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\ (F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\ (G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)}=\frac{\partial}{\partial x}+\frac{dy}{dy}\frac{\partial}{\partial y}+\frac{dy^{(1)}}{dy^{(1)}}\frac{\partial}{\partial y^{(1)}}+\dots+\frac{dy^{(n+1)}}{dy^{(n+1)}}\frac{\partial}{\partial y^{(n+1)}}\qquad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$																																																																																																																																																																																																																								
0	x	1	x	y/x	$xp - y$	T/R^2																																																																																																																																																																																																																								
$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
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xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
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x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F(F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G(G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03194	311	0.852	$y^{\wedge\{2\}}$	y^2	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F(F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G(G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03195	311	0.852	$y \, p - x \, p^2$	$yp - xp^2$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \, (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \, (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \, (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \, (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
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R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03198	311	0.852	$1 / \left(T R^2\right)$	$1/(TR^2)$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\,(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\,(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln 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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\,(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\,(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03202	311	0.852	$y+x\,p$	$y+xp$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln 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y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>e^y*p^b</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\,(F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\,(G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)}=\frac{\partial}{\partial x}+\frac{dy}{dy}\frac{\partial}{\partial y}+\frac{dy^{(1)}}{dy^{(1)}}\frac{\partial}{\partial y^{(1)}}+\dots+\frac{dy^{(n+1)}}{dy^{(n+1)}}\frac{\partial}{\partial y^{(n+1)}}\qquad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\,(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\,(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03203	311	0.852	$(\backslash \ln y) / x$	$(\ln y)/x$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$																																																																																																																																																																																																																								
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$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
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a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
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R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $r^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_F03206	311	0.852	$-g^{\prime\prime} p^2$	$-g'p^2$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
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R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03210	311	0.852	$f(x)$	$f(x)$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$																																																																																																																																																																																																																								
x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$																																																																																																																																																																																																																								
y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$																																																																																																																																																																																																																								
0	x	1	x	y/x	$xp - y$	T/R^2																																																																																																																																																																																																																								
$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

gilmore-lie-groups_ F03213

311

0.852

f p-f^{\prime} y

$f p-f' y$

Infinitesimal generators			Canonical coordinates			Constraint
$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR
0	1	0	x	y	p	T
1	0	0	y	x	p	1/T
1/a	-1/b	0	ax+by	bx-ay	p	(b-aT)/(a+bT)
x	0	-p	y	ln x	xp	1/T
0	y	p	x	ln y	p/y	T
x/a	y/b	(1/b-1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT-aR ^(1/b))
x	y	0	y/x	ln y	p	(T/R)/(T-R)
x	-y	-2p	xy	ln y	x ² p	(T/R)/(T+R)
2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T-R)
x	2y	p	y/x ²	ln y	p/x	(T/R)/(T-2R)
y	0	-p ²	y	x/y	x-y/p	-T/R ²
0	x	1	x	y/x	xp-y	T/R ²
-y	x	1+p ²	√x ² +y ²	tan ⁻¹ (y/x)	(y-xp)/(x+yp)	-T/R
1	y/x	(px-y)/x ²	y/x	x	(xp-y)/x ²	1/T
a	x	1	x ² -2ay	x/a	x-ap	1/(2aT)
a	y	p	x-a ln y	x/a	p/y	(1/a)/(1-aT)
x	b	-p	e ^y /x ^b	y/b	e ^y *p ^b	(bR) ⁻¹ / [1-b(R/T) ^(1/b)]
y	b	-p ²	y ² -2bx	y/b	y-b/p	1/(2bT)
0	e ^{f(x)}	f'e ^f	x	y/e ^f	p-yf'	T/e ^{f(R)}
x ²	xy	y-xp	y/x	1/x	xp-y	1/T
xy	y ²	yp-xp ²	y/x	1/y	y/p-x	1/(TR ²)
xy	0	-yp-xp ²	y	(ln x)/y	y/(xp)-ln x	T/R ²
0	xy	y+xp	x	(ln y)/x	xp/y-ln y	T/R ²
g(y)	0	-g'p ²	y	x/g	1/p-xg'/g	T/g(R)
0	f(x)	f'	x	y/f	fp-f'y	T/f ² (R)
f(x)	0	-f'p	y	F (F'f=1)	pf	1/T
0	g(y)	g'p	x	G (G'g=1)	p/g	T
x ^{k+1}	kx ^k y	x ^k (k ² y/x-p)	y/x ^k	1/x ^k	xp-ky	-k/T
kxy ^k	y ^{k+1}	y ^k (p-k ² xp ² /y)	x/y ^k	1/y ^k	y/p-kx	-k/T

where

$$\boldsymbol{n^{(n)}} = \frac{\partial}{\partial} + \frac{dy}{\partial} \frac{\partial}{\partial} + \frac{dy^{(1)}}{\partial} \frac{\partial}{\partial} + \dots + \frac{dy^{(n+1)}}{\partial} \frac{\partial}{\partial} \quad (16.40)$$

gilmore-lie-groups_ F03214

311

0.852

T / f^2(R)

$T/f^2(R)$

Infinitesimal generators			Canonical coordinates			Constraint
$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR
0	1	0	x	y	p	T
1	0	0	y	x	p	1/T
1/a	-1/b	0	ax+by	bx-ay	p	(b-aT)/(a+bT)
x	0	-p	y	ln x	xp	1/T
0	y	p	x	ln y	p/y	T
x/a	y/b	(1/b-1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT-aR ^(1/b))
x	y	0	y/x	ln y	p	(T/R)/(T-R)
x	-y	-2p	xy	ln y	x ² p	(T/R)/(T+R)
2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T-R)
x	2y	p	y/x ²	ln y	p/x	(T/R)/(T-2R)
y	0	-p ²	y	x/y	x-y/p	-T/R ²
0	x	1	x	y/x	xp-y	T/R ²
-y	x	1+p ²	√x ² +y ²	tan ⁻¹ (y/x)	(y-xp)/(x+yp)	-T/R
1	y/x	(px-y)/x ²	y/x	x	(xp-y)/x ²	1/T
a	x	1	x ² -2ay	x/a	x-ap	1/(2aT)
a	y	p	x-a ln y	x/a	p/y	(1/a)/(1-aT)
x	b	-p	e ^y /x ^b	y/b	e ^y *p ^b	(bR) ⁻¹ / [1-b(R/T) ^(1/b)]
y	b	-p ²	y ² -2bx	y/b	y-b/p	1/(2bT)
0	e ^{f(x)}	f'e ^f	x	y/e ^f	p-yf'	T/e ^{f(R)}
x ²	xy	y-xp	y/x	1/x	xp-y	1/T
xy	y ²	yp-xp ²	y/x	1/y	y/p-x	1/(TR ²)
xy	0	-yp-xp ²	y	(ln x)/y	y/(xp)-ln x	T/R ²
0	xy	y+xp	x	(ln y)/x	xp/y-ln y	T/R ²
g(y)	0	-g'p ²	y	x/g	1/p-xg'/g	T/g(R)
0	f(x)	f'	x	y/f	fp-f'y	T/f ² (R)
f(x)	0	-f'p	y	F (F'f=1)	pf	1/T
0	g(y)	g'p	x	G (G'g=1)	p/g	T
x ^{k+1}	kx ^k y	x ^k (k ² y/x-p)	y/x ^k	1/x ^k	xp-ky	-k/T
kxy ^k	y ^{k+1}	y ^k (p-k ² xp ² /y)	x/y ^k	1/y ^k	y/p-kx	-k/T

where

$$\boldsymbol{n^{(n)}} = \frac{\partial}{\partial} + \frac{dy}{\partial} \frac{\partial}{\partial} + \frac{dy^{(1)}}{\partial} \frac{\partial}{\partial} + \dots + \frac{dy^{(n+1)}}{\partial} \frac{\partial}{\partial} \quad (16.40)$$

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gilmore-lie-groups_F03215	311	0.852	$-f^{\prime}\text{p}$	$-f'p$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>e^y*p^b</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\ (F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\ (G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)}=\frac{\partial}{\partial x}+\frac{dy}{dy}\frac{\partial}{\partial y}+\frac{dy^{(1)}}{dy^{(1)}}\frac{\partial}{\partial y^{(1)}}+\dots+\frac{y^{(n+1)}}{y^{(n+1)}}\frac{\partial}{\partial y^{(n+1)}}\qquad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
Infinitesimal generators			Canonical coordinates			Constraint																																																																																																																																																																																																																								
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x	0	$-p$	y	$\ln x$	xp	$1/T$																																																																																																																																																																																																																								
0	y	p	x	$\ln y$	p/y	T																																																																																																																																																																																																																								
x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$																																																																																																																																																																																																																								
x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$																																																																																																																																																																																																																								
x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$																																																																																																																																																																																																																								
$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$																																																																																																																																																																																																																								
x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$																																																																																																																																																																																																																								
y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$																																																																																																																																																																																																																								
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a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
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$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
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gilmore-lie-groups_F03216	311	0.852	$F\text{left}(F^{\prime}\text{f=1}\text{right})$	$F\left(F'f=1\right)$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln 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y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\ (F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\ (G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)}=\frac{\partial}{\partial x}+\frac{dy}{dy}\frac{\partial}{\partial y}+\frac{dy^{(1)}}{dy^{(1)}}\frac{\partial}{\partial y^{(1)}}+\dots+\frac{y^{(n+1)}}{y^{(n+1)}}\frac{\partial}{\partial y^{(n+1)}}\qquad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03217	311	0.852	$\mathfrak{p} \, \mathfrak{f}$	pf	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \, (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \, (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2y/x - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \, (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \, (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_F03218	311	0.852	$\mathfrak{g}^{\prime \mathfrak{p}} \, \mathfrak{p}$	$g'p$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \, (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \, (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\left(F'f=1\right)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\left(G'g=1\right)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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0	x	1	x	y/x	$xp-y$	T/R^2																																																																																																																																																																																																																								
$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$																																																																																																																																																																																																																								
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a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$																																																																																																																																																																																																																								
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y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>e^y*p^b</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\left(F'f=1\right)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\left(G'g=1\right)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $n^{(n)}=\frac{\partial}{\partial x}+\frac{dy}{dy}\frac{\partial}{\partial y}+\frac{dy^{(1)}}{dy^{(1)}}\frac{\partial}{\partial y^{(1)}}+\dots+\frac{y^{(n+1)}}{y^{(n+1)}}\frac{\partial}{\partial y^{(n+1)}}\qquad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\left(F'f=1\right)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\left(G'g=1\right)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03221	311	0.852	x^{k+1}	x^{k+1}	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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ay$	p	$(b - aT) / (a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1 / T$	0	y	p	x	$\ln y$	p / y	T	x / a	y / b	$(1 / b - 1 / a) p$	y^b / x^a	$b \ln y$	$p / x^{(a / b - 1)}$	$(bT / R) / (bT - aR^{(1 / b)})$	x	y	0	y / x	$\ln y$	p	$(T / R) / (T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T / R) / (T + R)$	$2x$	y	$-p$	y^2 / x	$\ln y$	yp	$(T / R) / (2T - R)$	x	$2y$	p	y / x^2	$\ln y$	p / x	$(T / R) / (T - 2R)$	y	0	$-p^2$	y	x / y	$x - y / p$	$-T / R^2$	0	x	1	x	y / x	$xp - y$	T / R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y / x)$	$(y - xp) / (x + yp)$	$-T / R$	1	y / x	$(px - y) / x^2$	y / x	x	$(xp - y) / x^2$	$1 / T$	a	x	1	$x^2 - 2ay$	x / a	$x - ap$	$1 / (2aT)$	a	y	p	$x - a \ln y$	x / a	p / y	$(1 / a) / (1 - aT)$	x	b	$-p$	e^y / x^b	y / b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R / T)^{(1 / b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y / b	$y - b / p$	$1 / (2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y / e^f	$p - yf'$	$T / e^{f(R)}$	x^2	xy	$y - xp$	y / x	$1 / x$	$xp - y$	$1 / T$	xy	y^2	$yp - xp^2$	y / x	$1 / y$	$y / p - x$	$1 / (T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x) / y$	$y / (xp) - \ln x$	T / R^2	0	xy	$y + xp$	x	$(\ln y) / x$	$xp / y - \ln y$	T / R^2	$g(y)$	0	$-g' p^2$	y	x / g	$1 / p - xg' / g$	$T / g(R)$	0	$f(x)$	f'	x	y / f	$fp - f'y$	$T / f^2(R)$	$f(x)$	0	$-f' p$	y	$F\left(F' f = 1\right)$	pf	$1 / T$	0	$g(y)$	$g' p$	x	$G\left(G' g = 1\right)$	p / g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y / x - p)$	y / x^k	$1 / x^k$	$xp - ky$	$-k / T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2 / y)$	x / y^k	$1 / y^k$	$y / p - kx$	$-k / T$
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gilmore-lie-groups_ F03224	311	0.852	$y / x^{\{k\}}$	y / x^k	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1 / T$</td></tr><tr><td>$1 / a$</td><td>$-1 / b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT) / (a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1 / T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p / y</td><td>T</td></tr><tr><td>x / a</td><td>y / b</td><td>$(1 / b - 1 / a) p$</td><td>y^b / x^a</td><td>$b \ln y$</td><td>$p / x^{(a / b - 1)}$</td><td>$(bT / R) / (bT - aR^{(1 / b)})$</td></tr><tr><td>$x$</td><td>$y$</td><td>0</td><td>$y / x$</td><td>$\ln y$</td><td>$p$</td><td>$(T / R) / (T - R)$</td></tr><tr><td>$x$</td><td>$-y$</td><td>$-2p$</td><td>$xy$</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T / R) / (T + R)$</td></tr><tr><td>$2x$</td><td>$y$</td><td>$-p$</td><td>$y^2 / x$</td><td>$\ln y$</td><td>$yp$</td><td>$(T / R) / (2T - R)$</td></tr><tr><td>$x$</td><td>$2y$</td><td>$p$</td><td>$y / x^2$</td><td>$\ln y$</td><td>$p / x$</td><td>$(T / R) / (T - 2R)$</td></tr><tr><td>$y$</td><td>0</td><td>$-p^2$</td><td>$y$</td><td>$x / y$</td><td>$x - y / p$</td><td>$-T / R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y / x</td><td>$xp - y$</td><td>T / R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y / x)$</td><td>$(y - xp) / (x + yp)$</td><td>$-T / R$</td></tr><tr><td>1</td><td>y / x</td><td>$(px - y) / x^2$</td><td>y / x</td><td>x</td><td>$(xp - y) / x^2$</td><td>$1 / T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x / a</td><td>$x - ap$</td><td>$1 / (2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x / a</td><td>p / y</td><td>$(1 / a) / (1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y / x^b</td><td>y / b</td><td>$e^y * p^b$</td><td>$(bR)^{-1} / [1 - b(R / T)^{(1 / b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y / b$</td><td>$y - b / p$</td><td>$1 / (2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y / e^f</td><td>$p - yf'$</td><td>$T / e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y / x</td><td>$1 / x$</td><td>$xp - y$</td><td>$1 / T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y / x</td><td>$1 / y$</td><td>$y / p - x$</td><td>$1 / (T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x) / y$</td><td>$y / (xp) - \ln x$</td><td>T / R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y) / x$</td><td>$xp / y - \ln y$</td><td>T / R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x / g</td><td>$1 / p - 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ay$	p	$(b - aT) / (a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1 / T$	0	y	p	x	$\ln y$	p / y	T	x / a	y / b	$(1 / b - 1 / a) p$	y^b / x^a	$b \ln y$	$p / x^{(a / b - 1)}$	$(bT / R) / (bT - aR^{(1 / b)})$	x	y	0	y / x	$\ln y$	p	$(T / R) / (T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T / R) / (T + R)$	$2x$	y	$-p$	y^2 / x	$\ln y$	yp	$(T / R) / (2T - R)$	x	$2y$	p	y / x^2	$\ln y$	p / x	$(T / R) / (T - 2R)$	y	0	$-p^2$	y	x / y	$x - y / p$	$-T / R^2$	0	x	1	x	y / x	$xp - y$	T / R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y / x)$	$(y - xp) / (x + yp)$	$-T / R$	1	y / x	$(px - y) / x^2$	y / x	x	$(xp - y) / x^2$	$1 / T$	a	x	1	$x^2 - 2ay$	x / a	$x - ap$	$1 / (2aT)$	a	y	p	$x - a \ln y$	x / a	p / y	$(1 / a) / (1 - aT)$	x	b	$-p$	e^y / x^b	y / b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R / T)^{(1 / b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y / b	$y - b / p$	$1 / (2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y / e^f	$p - yf'$	$T / e^{f(R)}$	x^2	xy	$y - xp$	y / x	$1 / x$	$xp - y$	$1 / T$	xy	y^2	$yp - xp^2$	y / x	$1 / y$	$y / p - x$	$1 / (T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x) / y$	$y / (xp) - \ln x$	T / R^2	0	xy	$y + xp$	x	$(\ln y) / x$	$xp / y - \ln y$	T / R^2	$g(y)$	0	$-g' p^2$	y	x / g	$1 / p - xg' / g$	$T / g(R)$	0	$f(x)$	f'	x	y / f	$fp - f'y$	$T / f^2(R)$	$f(x)$	0	$-f' p$	y	$F\left(F' f = 1\right)$	pf	$1 / T$	0	$g(y)$	$g' p$	x	$G\left(G' g = 1\right)$	p / g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y / x - p)$	y / x^k	$1 / x^k$	$xp - ky$	$-k / T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2 / y)$	x / y^k	$1 / y^k$	$y / p - kx$	$-k / T$
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gilmore-lie-groups_ F03225	311	0.852	$1 / x^{\{k\}}$	$1/x^k$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln 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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ F03229	311	0.852	$y^{\{k+1\}}$	y^{k+1}	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln 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y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\ (F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\ (G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></tbody></table> where $\mathbf{r}^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n+1)}}{dx} \frac{\partial}{\partial y^{(n+1)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03232	311	0.852	$1 / y^{\{k\}}$	$1/y^k$	<table><thead><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr></thead><tbody><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - 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y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></tbody></table> where $D^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \dots + \frac{dy^{(n-1)}}{dx} \frac{\partial}{\partial y^{(n-1)}} + \frac{\partial}{\partial y^{(n)}} \quad (16.40)$	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
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kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								
gilmore-lie-groups_ F03233	311	0.852	$y / p - k x$	$y/p - kx$																																																																																																																																																																																																																										
gilmore-lie-groups_ F03234	312	1.000	$F\left(x, y, y^{\prime \prime}, y^{\prime \prime \prime}\right)=0$	$F\left(x, y, y', y''\right)=0$	ing group (16.36). Then the dependent coordinate is $S = \ln y$ and the surface equa-																																																																																																																																																																																																																									
gilmore-lie-groups_ F03235	312	1.000	$S=\ln y$	$S = \ln y$	tion can be expressed in terms of three invariant coordinates as $F(R, -, T, U) = 0$.																																																																																																																																																																																																																									
gilmore-lie-groups_ F03236	312	1.000	$F(R, -, T, U)=0$	$F(R, -, T, U) = 0$	Here as before R depends only on x and y , $T = T(x, y, y')$, and $U = U(x, y, y', y'')$																																																																																																																																																																																																																									
gilmore-lie-groups_ F03237	312	1.000	$y, T=T\left(x, y, y^{\prime \prime}\right)$	$y, T = T(x, y, y')$	is another invariant coordinate. How does one construct such an invariant coordi- nate? It is simple to see that the derivative dT/dR is invariant under the group. Not																																																																																																																																																																																																																									
gilmore-lie-groups_ F03238	312	1.000	$U=U\left(x, y, y^{\prime \prime}\right)$	$U = U(x, y, y', y'')$	is another invariant coordinate. How does one construct such an invariant coordi- nate? It is simple to see that the derivative dT/dR is invariant under the group. Not																																																																																																																																																																																																																									
gilmore-lie-groups_ F03239	312	1.000	$d T / d R$	dT/dR	only is a invariant, but it is of first degree in the second order term y'' , so																																																																																																																																																																																																																									
gilmore-lie-groups_ F03240	312	1.000	$y^{\prime \prime \prime}$	y''	This is a first order equation in T . Certain forms of the fur																																																																																																																																																																																																																									
gilmore-lie-groups_ F03241	313	1.000	$F\left(x, y, \ldots, y^{\{n\}}\right)=0$	$F\left(x, y, \ldots, y^{(n)}\right) = 0$	equation $F(x, y, \ldots, y^{(n)}) = 0$. As usual, we seek an infinitesimal generator																																																																																																																																																																																																																									
gilmore-lie-groups_ F03242	313	1.000	$T\left(x, y, y^{\{1\}}\right)$	$T\left(x, y, y^{(1)}\right)$	the existence of a Lie symmetry can be used to reduce an n th order equation to an																																																																																																																																																																																																																									
gilmore-lie-groups_ F03243	313	1.000	$d T^{\{j\}} / d R^{\{j\}}, j=0$	$dT^{(j)}/dR^{(j)}, j = 0$	$(n - 1)$ st order equation.																																																																																																																																																																																																																									
gilmore-lie-groups_ F03244	313	1.000	$j=1,2, \ldots, n-1$	$j = 1, 2, \ldots, n - 1$																																																																																																																																																																																																																										
gilmore-lie-groups_ F03245	313	1.000	$y^{\{j+1\}}$	$y^{(j+1)}$																																																																																																																																																																																																																										
gilmore-lie-groups_ F03246	314	1.000	$x^{\{i\}} \rightarrow \lambda x^{\{i\}}, u \rightarrow \alpha u$	$x^i \rightarrow \lambda x^i, u \rightarrow \alpha u$	scaling transformations $x^i \rightarrow \lambda x^i, u \rightarrow \alpha u$. Under the scaling transformation																																																																																																																																																																																																																									

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gilmore-lie-groups_ F03247	314	■ 1.000	\delta(x) \rightarrow \delta(\lambda x)=\lambda^{-n} \delta(x)	$\delta(x) \rightarrow \delta(\lambda x) = \lambda^{-n} \delta(x)$	$\delta(x) \rightarrow \delta(\lambda x) = \lambda^{-n} \delta(x)$, so
gilmore-lie-groups_ F03248	314	■ 1.000	\alpha=\lambda^{2-n}	$\alpha = \lambda^{2-n}$	metries for this equation therefore consist of generators of rotations and scale
gilmore-lie-groups_ F03249	314	■ 1.000	R=R(x, u)	$R = R(x, u)$	combination of the generators in Eq. (16.52). A solution is $R = k x ^{2-n}$. As a result,
gilmore-lie-groups_ F03250	314	■ 1.000	R \sim u x ^{n-2}	$R \sim u x ^{n-2}$	$u \sim x ^{2-n} = k x ^{2-n}$. The constant of proportionality can be computed using the
gilmore-lie-groups_ F03251	314	■ 1.000	u \sim x ^{2-n}=k x ^{2-n}	$u \sim x ^{2-n} = k x ^{2-n}$	$u = x ^{2-n} = k x ^{2-n}$. The constant of proportionality can be computed using the divergence theorem. Both sides of Eq. (16.50) are integrated over the interior of a
gilmore-lie-groups_ F03252	314	■ 1.000	V\left(S^{n-1}\right)=2 \pi^{n / 2} / \Gamma\left(\frac{n}{2}\right)	$V\left(S^{n-1}\right)=2 \pi^{n / 2} / \Gamma\left(\frac{n}{2}\right)$	unit sphere in R^n . The volume integral on the right is +1. The volume integral on
gilmore-lie-groups_ F03253	314	■ 1.000	R^{n}(n \neq 2)	$R^n(n \neq 2)$	is
gilmore-lie-groups_ F03254	314	■ 1.000	u(x, t)	$u(x, t)$	$u(x) = \frac{k}{ x ^{n-2}} \quad k = \frac{1}{(n-2) V / (2 \pi)^n}$ (16.54)
gilmore-lie-groups_ F03255	314	■ 1.000	u \rightarrow \alpha u, t \rightarrow \beta t	$u \rightarrow \alpha u, t \rightarrow \beta t$	is treated similarly (Olver, 1995). It is invariant under X, Y, Z are Lie symmetries. Under the scaling transform
gilmore-lie-groups_ F03256	314	■ 1.000	x^{i} \rightarrow \lambda x^{i}	$x^i \rightarrow \lambda x^i$	invariance under the scaling transformations places the following two constraints on
gilmore-lie-groups_ F03257	314	■ 1.000	\alpha \lambda^n=1	$\alpha \lambda^n = 1$	the three scaling variables (since there is only
gilmore-lie-groups_ F03258	314	■ 1.000	\beta / \lambda^2=1	$\beta / \lambda^2 = 1$	
gilmore-lie-groups_ F03259	315	■ 1.000	n+1	$n+1$	From these relations it is possible to construct $n+1$ additional Lie symmetries, so
gilmore-lie-groups_ F03260	315	■ 1.000	x^{i}, t	x^i, t	which we obtain as before
gilmore-lie-groups_ F03261	315	■ 1.000	R=u t^{n / 2} e^{\left\{ x ^2\right\} / 4 t}	$R = u t^{n / 2} e^{ x ^2 / 4 t}$	which we obtain as before
gilmore-lie-groups_ F03262	315	■ 1.000	F(R,-, T)=	$F(R,-, T) =$	$\int f(R,-, T(R)) d R?$
gilmore-lie-groups_ F03263	315	■ 0.980	\int f(R,-, T(R)) d R	$\int f(R,-, T(R)) d R$	
gilmore-lie-groups_ F03264	316	■ 1.000	\xi, \eta	ξ, η	1850 Galois solve algebraic equations 1874 Lie solve differential equations 1969 Risch integrate in closed form ? – solve determining equations for ξ, η ? – solve determining equations for R, S, T ? – solve $F(R,-, T) = 0$ for R . It is clear that additional algorithms are possible and desirable.
gilmore-lie-groups_ F03265	316	■ 1.000	\zeta(x, y, p)=\eta^{(1)}\left(x, y, y^{(1)}\right)	$\zeta(x, y, p) = \eta^{(1)}\left(x, y, y^{(1)}\right)$	differential equation.
gilmore-lie-groups_ F03266	316	■ 1.000	X=\xi \partial / \partial x+\eta \partial / \partial y+\zeta \partial / \partial p	$X = \xi \partial / \partial x + \eta \partial / \partial y + \zeta \partial / \partial p$	$\eta(x, y)$.
gilmore-lie-groups_ F03267	316	■ 1.000	F=0	$F = 0$	equation is independent of one or dependent parameter $\lambda = \lambda(x, y, z, p)$.
gilmore-lie-groups_ F03268	316	■ 1.000	X R=0, X S=1, X T=0	$X R = 0, X S = 1, X T = 0$	
gilmore-lie-groups_ F03269	316	■ 1.000	F \rightarrow F(R,-, T)=0	$F \rightarrow F(R,-, T) = 0$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image															
gilmore-lie-groups_ F03270	317	■ 1.000	S: S=\int f(R,-, T(R))+c	$S : S = \int f(R, -, T(R)) + c$	(viii) Solve the constraint equation for S : $S = \int f(R, -, T(R)) + c$. (ix) Backsubstitute the original coordinates for the new coordinates x .															
gilmore-lie-groups_ F03271	317	■ 1.000	x=x(R, S)	$x = x(R, S)$	(ix) Backsubstitute the original coordinates for the new coordinates, $x = x(R, S)$ and $y = y(R, S)$ to obtain the solution of the original equation.															
gilmore-lie-groups_ F03272	317	■ 1.000	y=y(R, S)	$y = y(R, S)$	$y = y(R, S)$ to obtain the solution of the original equation.															
gilmore-lie-groups_ F03273	317	■ 1.000	n=0,1	$n = 0, 1$	Construct the invariance group for each of the transformations presented in Table															
gilmore-lie-groups_ F03274	317	■ 1.000	V(\mathbf{x})	$V(\mathbf{x})$	dt^-															
gilmore-lie-groups_ F03275	317	■ 1.000	m \rightarrow \alpha m), \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t	$m \rightarrow \alpha m), \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$	$\alpha^1 R^1, \alpha^{-2} m \frac{d^2 \mathbf{x}}{dt^2} = - R^{k-1} \nabla V(\mathbf{x})$															
gilmore-lie-groups_ F03276	317	■ 1.000	k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})	$k : V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$	$\alpha^p R^p \frac{d^2 \mathbf{x}}{dt^2} = - R^{k-1} \nabla V(\mathbf{x})$															
gilmore-lie-groups_ F03277	317	■ 1.000	\alpha^1 \beta^{2-k}	$\alpha^1 \beta^{2-k}$																
gilmore-lie-groups_ F03278	317	■ 1.000	\gamma^{-2}=1	$\gamma^{-2} = 1$																
gilmore-lie-groups_ F03279	317	■ 1.000	\alpha=1	$\alpha = 1$																
gilmore-lie-groups_ F03280	317	■ 1.000	\gamma^2=\beta^{2-k}	$\gamma^2 = \beta^{2-k}$																
gilmore-lie-groups_ F03281	317	■ 1.000	k=-1, k=0, k,=+1, k=+2	$k = -1, k = 0, k, = +1, k = +2$																
gilmore-lie-groups_ F03282	318	■ 1.000	\gamma^2=\beta^3	$\gamma^2 = \beta^3$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$
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gilmore-lie-groups_ F03283	318	■ 1.000	\gamma^2=\beta^2	$\gamma^2 = \beta^2$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$
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gilmore-lie-groups_ F03284	318	■ 1.000	\gamma^2=\beta^1	$\gamma^2 = \beta^1$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$
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gilmore-lie-groups_ F03285	318	■ 1.000	\gamma^2=\beta^0	$\gamma^2 = \beta^0$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$
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gilmore-lie-groups_ F03286	318	■ 0.785	\gamma^2	γ^2	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and															
gilmore-lie-groups_ F03287	318	■ 0.785	\beta^3	β^3	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and															
gilmore-lie-groups_ F03288	318	■ 0.785	R^{\prime}	R'	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and															

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gilmore-lie-groups_F03289	318	1.000	$T^{\prime}$	T'	period of planet P , and the two planets P and P' have geometrically similar orbits,																					
gilmore-lie-groups_F03290	318	1.000	$P^{\prime}$	P'	period of planet P , and the two planets P and P' have geometrically similar orbits,																					
gilmore-lie-groups_F03291	318	1.000	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$	newton's second law in the absence of forces in an inertial frame, the distance traveled (R) is proportional to the time elapsed (γ). The third line is a statement that																					
gilmore-lie-groups_F03292	318	0.860	(β)	(β)	like the square of the time elapsed. The fourth line is a statement of Hooke's law:																					
gilmore-lie-groups_F03293	318	0.860	(γ)	(γ)	like the square of the time elapsed. The fourth line is a statement of Hooke's law:																					
gilmore-lie-groups_F03294	318	1.000	$V = m \ g \ z$	$V = mgz$	in harmonic motion the period (γ) is independent of the size of the orbit.																					
gilmore-lie-groups_F03295	318	1.000	$\gamma = 1$	$\gamma = 1$	d. Fix $\beta = 1$ and show that the period scales like $\sqrt{M}$ for all homogeneous potentials.																					
gilmore-lie-groups_F03296	318	1.000	$\beta = 1$	$\beta = 1$	independent of its mass in lowest order. ~ If the motion is bounded for all times, show																					
gilmore-lie-groups_F03297	318	1.000	$\sqrt{M}$	$\sqrt{M}$	independent of its mass in lowest order. ~ If the motion is bounded for all times, show																					
gilmore-lie-groups_F03298	318	0.994	$\alpha \beta^2 \gamma^{-2} = \beta^k$	$\alpha \beta^2 \gamma^{-2} = \beta^k$	example, the Euler–Lagrange equations are derived from the variation of an action:																					
gilmore-lie-groups_F03299	318	1.000	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d\mathbf{x} = 0$	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d\mathbf{x} = 0$	$\partial^2 u \quad \partial u$																					
gilmore-lie-groups_F03300	318	1.000	$v_{\{i\}}$	v_i																						
gilmore-lie-groups_F03301	318	1.000	$e^{\epsilon v_i} f(x, t)$	$e^{\epsilon v_i} f(x, t)$																						
gilmore-lie-groups_F03302	319	1.000	$e^{\epsilon v_i} f(x, t) =$	$e^{\epsilon v_i} f(x, t) =$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$</td></tr></table> where $\lambda^2 = 1/(1 + 4\epsilon t)$ xo-dimensional wave equation is	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$
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gilmore-lie-groups_F03303	319	1.000	∂_x	∂_x	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$</td></tr></table> where $\lambda^2 = 1/(1 + 4\epsilon t)$ xo-dimensional wave equation is	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$
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gilmore-lie-groups_ F03305	319	■ 1.000	∂_t	∂_t	v_i Infinitesimal $e^{\epsilon v_i} f(x, t) =$ <table><tr><td>v_1</td><td>∂_x</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>v_2</td><td>∂_t</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>v_3</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>v_4</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table> no-dimensional wave equation is	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03306	319	■ 1.000	$f(x, t-\epsilon)$	$f(x, t - \epsilon)$	v_i Infinitesimal $e^{\epsilon v_i} f(x, t) =$ <table><tr><td>v_1</td><td>∂_x</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>v_2</td><td>∂_t</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>v_3</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>v_4</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table> no-dimensional wave equation is	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$																					
v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$																					
gilmore-lie-groups_ F03307	319	■ 1.000	$u \partial_u$	$u \partial_u$	v_i Infinitesimal $e^{\epsilon v_i} f(x, t) =$ <table><tr><td>v_1</td><td>∂_x</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>v_2</td><td>∂_t</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>v_3</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>v_4</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table> no-dimensional wave equation is	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
v_1	∂_x	$f(x - \epsilon, t)$																					
v_2	∂_t	$f(x, t - \epsilon)$																					
v_3	$u \partial_u$	$e^\epsilon f(x, t)$																					
v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$																					
v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$																					
v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$																					
gilmore-lie-groups_ F03308	319	■ 1.000	$e^{\epsilon} f(x, t)$	$e^\epsilon f(x, t)$	v_i Infinitesimal $e^{\epsilon v_i} f(x, t) =$ <table><tr><td>v_1</td><td>∂_x</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>v_2</td><td>∂_t</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>v_3</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>v_4</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table> no-dimensional wave equation is	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
v_1	∂_x	$f(x - \epsilon, t)$																					
v_2	∂_t	$f(x, t - \epsilon)$																					
v_3	$u \partial_u$	$e^\epsilon f(x, t)$																					
v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$																					
v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$																					
v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$																					
gilmore-lie-groups_ F03309	319	■ 1.000	v_4	v_4	v_i Infinitesimal $e^{\epsilon v_i} f(x, t) =$ <table><tr><td>v_1</td><td>∂_x</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>v_2</td><td>∂_t</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>v_3</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>v_4</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table> no-dimensional wave equation is	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
v_1	∂_x	$f(x - \epsilon, t)$																					
v_2	∂_t	$f(x, t - \epsilon)$																					
v_3	$u \partial_u$	$e^\epsilon f(x, t)$																					
v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$																					
v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$																					
v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$																					
gilmore-lie-groups_ F03310	319	■ 1.000	$x \partial_x + 2t \partial_t$	$x \partial_x + 2t \partial_t$	v_i Infinitesimal $e^{\epsilon v_i} f(x, t) =$ <table><tr><td>v_1</td><td>∂_x</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>v_2</td><td>∂_t</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>v_3</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>v_4</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table> no-dimensional wave equation is	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
v_1	∂_x	$f(x - \epsilon, t)$																					
v_2	∂_t	$f(x, t - \epsilon)$																					
v_3	$u \partial_u$	$e^\epsilon f(x, t)$																					
v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$																					
v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$																					
v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$																					

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F03311	319	■ 1.000	$f(\mathrm{e}^{-\epsilon} x, \mathrm{e}^{-2 \epsilon} t)$	$f\left(e^{-\epsilon} x, e^{-2 \epsilon} t\right)$	v_i Infinitesimal $e^{\epsilon v_i} f(x, t) =$ v_1 ∂_x $f(x - \epsilon, t)$ v_2 ∂_t $f(x, t - \epsilon)$ v_3 $u \partial_u$ $e^{\epsilon} f(x, t)$ v_4 $x \partial_x + 2 t \partial_t$ $f\left(e^{-\epsilon} x, e^{-2 \epsilon} t\right)$ v_5 $2 t \partial_x - x u \partial_u$ $e^{-\epsilon x + \epsilon^2 t} f(x - 2 \epsilon t, t)$ v_6 $4 x t \partial_x + 4 t^2 \partial_t - \left(x^2 + 2 t\right) u \partial_u$ $\lambda e^{-\epsilon \lambda^2 x^2} f\left(\lambda^2 x, \lambda^2 t\right)$ where $\lambda^2 = 1 / (1 + 4 \epsilon t)$ no-dimensional wave equation is

Identifier	Page	Conf.	LaTeX source	Rendered	Image																		
gilmore-lie-groups_ F03317	319	■1.000	$\lambda e^{-\epsilon\lambda^2x^2}f\left(\lambda^2x,\lambda^2t\right)$	$\lambda e^{-\epsilon\lambda^2x^2}f\left(\lambda^2x,\lambda^2t\right)$	v_i Infinitesimal $e^{\epsilon v_i}f(x,t)=$ <table><tr><td>v_1</td><td>∂_x</td><td>$f(x-\epsilon,t)$</td></tr><tr><td>v_2</td><td>∂_t</td><td>$f(x,t-\epsilon)$</td></tr><tr><td>v_3</td><td>$u\partial_u$</td><td>$e^\epsilon f(x,t)$</td></tr><tr><td>v_4</td><td>$x\partial_x+2t\partial_t$</td><td>$f(e^{-\epsilon}x,e^{-2\epsilon}t)$</td></tr><tr><td>$v_5$</td><td>$2t\partial_x-xu\partial_u$</td><td>$e^{-\epsilon x+\epsilon^2t}f(x-2\epsilon t,t)$</td></tr><tr><td>$v_6$</td><td>$4xt\partial_x+4t^2\partial_t-(x^2+2t)u\partial_u$</td><td>$\lambda e^{-\epsilon\lambda^2x^2}f(\lambda^2x,\lambda^2t)$</td></tr></table> where $\lambda^2=1/(1+4\epsilon t)$ no-dimensional wave equation is	v_1	∂_x	$f(x-\epsilon,t)$	v_2	∂_t	$f(x,t-\epsilon)$	v_3	$u\partial_u$	$e^\epsilon f(x,t)$	v_4	$x\partial_x+2t\partial_t$	$f(e^{-\epsilon}x,e^{-2\epsilon}t)$	v_5	$2t\partial_x-xu\partial_u$	$e^{-\epsilon x+\epsilon^2t}f(x-2\epsilon t,t)$	v_6	$4xt\partial_x+4t^2\partial_t-(x^2+2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2x^2}f(\lambda^2x,\lambda^2t)$
v_1	∂_x	$f(x-\epsilon,t)$																					
v_2	∂_t	$f(x,t-\epsilon)$																					
v_3	$u\partial_u$	$e^\epsilon f(x,t)$																					
v_4	$x\partial_x+2t\partial_t$	$f(e^{-\epsilon}x,e^{-2\epsilon}t)$																					
v_5	$2t\partial_x-xu\partial_u$	$e^{-\epsilon x+\epsilon^2t}f(x-2\epsilon t,t)$																					
v_6	$4xt\partial_x+4t^2\partial_t-(x^2+2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2x^2}f(\lambda^2x,\lambda^2t)$																					
gilmore-lie-groups_ F03318	319	■1.000	$\lambda^2=1/(1+4\epsilon t)$	$\lambda^2=1/(1+4\epsilon t)$	v_i Infinitesimal $e^{\epsilon v_i}f(x,t)=$ <table><tr><td>v_1</td><td>∂_x</td><td>$f(x-\epsilon,t)$</td></tr><tr><td>v_2</td><td>∂_t</td><td>$f(x,t-\epsilon)$</td></tr><tr><td>v_3</td><td>$u\partial_u$</td><td>$e^\epsilon f(x,t)$</td></tr><tr><td>v_4</td><td>$x\partial_x+2t\partial_t$</td><td>$f(e^{-\epsilon}x,e^{-2\epsilon}t)$</td></tr><tr><td>$v_5$</td><td>$2t\partial_x-xu\partial_u$</td><td>$e^{-\epsilon x+\epsilon^2t}f(x-2\epsilon t,t)$</td></tr><tr><td>$v_6$</td><td>$4xt\partial_x+4t^2\partial_t-(x^2+2t)u\partial_u$</td><td>$\lambda e^{-\epsilon\lambda^2x^2}f(\lambda^2x,\lambda^2t)$</td></tr></table> where $\lambda^2=1/(1+4\epsilon t)$ no-dimensional wave equation is	v_1	∂_x	$f(x-\epsilon,t)$	v_2	∂_t	$f(x,t-\epsilon)$	v_3	$u\partial_u$	$e^\epsilon f(x,t)$	v_4	$x\partial_x+2t\partial_t$	$f(e^{-\epsilon}x,e^{-2\epsilon}t)$	v_5	$2t\partial_x-xu\partial_u$	$e^{-\epsilon x+\epsilon^2t}f(x-2\epsilon t,t)$	v_6	$4xt\partial_x+4t^2\partial_t-(x^2+2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2x^2}f(\lambda^2x,\lambda^2t)$
v_1	∂_x	$f(x-\epsilon,t)$																					
v_2	∂_t	$f(x,t-\epsilon)$																					
v_3	$u\partial_u$	$e^\epsilon f(x,t)$																					
v_4	$x\partial_x+2t\partial_t$	$f(e^{-\epsilon}x,e^{-2\epsilon}t)$																					
v_5	$2t\partial_x-xu\partial_u$	$e^{-\epsilon x+\epsilon^2t}f(x-2\epsilon t,t)$																					
v_6	$4xt\partial_x+4t^2\partial_t-(x^2+2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2x^2}f(\lambda^2x,\lambda^2t)$																					
gilmore-lie-groups_ F03319	319	■1.000	$D_2=u\,\partial_u$	$D_2=u\partial_u$	$SO(2+1,1+1)=SO(3,2).$																		
gilmore-lie-groups_ F03320	319	■1.000	$SO(2+1,1+1)=SO(3,2)$	$SO(2+1,1+1)=SO(3,2)$	generators. Show that 15 satisfy																		
gilmore-lie-groups_ F03321	319	■1.000	$SO(3+1,1+1)=SO(4,2)$	$SO(3+1,1+1)=SO(4,2)$	Symmetries for this equation is $X=\xi^i\partial_{x^i}+\eta\partial_u$, $t\in\mathbb{R}$, $\xi^i\in\mathbb{R}^4$, $\eta\in\mathbb{R}$. Show that (Stewart 1980)																		
gilmore-lie-groups_ F03322	319	■1.000	$u_{xx}-u_t=0$	$u_{xx}-u_t=0$	$\xi^2=a_1+a_2t+a_3t^2+a_4xt$																		
gilmore-lie-groups_ F03323	319	■1.000	$X=\xi^i\frac{\partial}{\partial x^i}+\eta\frac{\partial}{\partial u}+\cdots=\xi^1\frac{\partial}{\partial x}+\xi^2\frac{\partial}{\partial t}+\eta\frac{\partial}{\partial u}+\cdots$	$X=\xi^i\frac{\partial}{\partial x^i}+\eta\frac{\partial}{\partial u}+\cdots=\xi^1\frac{\partial}{\partial x}+\xi^2\frac{\partial}{\partial t}+\eta\frac{\partial}{\partial u}+\cdots$	$\xi^2=2a_2t+a_4t^2+a_5$																		
gilmore-lie-groups_ F03324	320	■1.000	$h(x,t)$	$h(x,t)$	Here $h(x,t)$ is any function that satisfies the homogeneous heat equation. Construct																		
gilmore-lie-groups_ F03325	320	■0.986	$\left(t\mapsto t'=T(t,x,\epsilon)=t+\right.$	$(t\rightarrow t'=T(t,x,\epsilon)=t+$	$r_{t_2}^{\prime}\qquad\qquad r_{t_2}^{\prime}\qquad\qquad A_{t'}$																		
gilmore-lie-groups_ F03326	320	■1.000	$\left.\epsilon\xi(t,x),x_i\mapsto x_i'=X_i(t,x,\epsilon)=x_i+\epsilon\eta_i(t,x)\right)$	$\epsilon\xi(t,x),x_i\rightarrow x_i'=X_i(t,x,\epsilon)=x_i+\epsilon\eta_i(t,x))$	$I=\int_{t'}L(t',x',x')dt'=\int_{t'}L(t',x',x')\frac{dt'}{dt}dt$																		
gilmore-lie-groups_ F03327	320	■1.000	$dt'/dt=\partial T/\partial t+(\partial T/\partial x_i)dx_i/dt$	$dt'/dt=\partial T/\partial t+(\partial T/\partial x_i)dx_i/dt$	$\int_{t_1}^{t_2}\left(\xi\frac{\partial L}{\partial x_i}+\eta_i\frac{\partial L}{\partial x_i}+\eta_i^{(1)}\frac{\partial L}{\partial x_i}+\frac{a\xi}{x_i}L\right)dt=0$																		
gilmore-lie-groups_ F03328	320	■1.000	$\epsilon=0$	$\epsilon=0$	$J_{t_1}^{t_2}=\int_{t_1}^{t_2}\left(\xi\frac{\partial L}{\partial x_i}+\eta_i\frac{\partial L}{\partial x_i}+\eta_i^{(1)}\frac{\partial L}{\partial x_i}+\frac{a\xi}{x_i}L\right)dt$																		
gilmore-lie-groups_ F03329	321	■1.000	$L[u]=$	$L[u]=$	the dynamics of a system is derivable from an action integral of the form $L[u]=\int_{t_1}^{t_2}\mathcal{L}(x,u)dx$, $x\in R^p$, $u\in R^q$, and suppose the infinitesimal generators that leave the																		
gilmore-lie-groups_ F03330	321	■1.000	$\int\mathcal{L}(x,u)dx,x\in R^p,u\in R^q$	$\int\mathcal{L}(x,u)dx,x\in R^p,u\in R^q$	$\int\mathcal{L}(x,u)dx,x\in R^p,u\in R^q$, and suppose the infinitesimal generators that leave the dynamics invariant have the form																		
gilmore-lie-groups_ F03331	321	■1.000	$d\,\rho(g)$	$d\rho(g)$	structed by reduction of tensor products (Wigner–Stone theorem), and $\phi(g),\psi(g)$																		
gilmore-lie-groups_ F03332	321	■0.995	$\operatorname{Vol}(G)=\int d\,\rho(g),\,\Gamma_{\mu\nu}^{\lambda}(g)$	$\operatorname{Vol}(G)=\int d\rho(g),\Gamma_{\mu\nu}^{\lambda}(g)$	are functions defined on the group manifold. The orthogonality and completeness																		
gilmore-lie-groups_ F03333	321	■1.000	$\phi(g),\,\psi(g)$	$\phi(g),\psi(g)$	relations are																		
gilmore-lie-groups_ F03334	322	■1.000	$\psi(g)=\langle g\mid\psi\rangle$	$\psi(g)=\langle g\mid\psi\rangle$	c. Carry out a Fourier decomposition on the functions $\psi(g)=\langle g \psi\rangle$ and $\langle\frac{\lambda}{\mu\nu} \psi\rangle=$																		

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gilmore-lie-groups-F03335	322	1.000	$\left\langle \right\rangle_{\mu\nu}^{\lambda} \mid \psi \right\rangle$	$\langle \lambda_{\mu\nu} \mid \psi \rangle =$	c. Carry out a Fourier decomposition on the functions $\psi(g) = \langle g \mid \psi \rangle$ and $\langle \lambda_{\mu\nu} \mid \psi \rangle =$
gilmore-lie-groups-F03336	322	1.000	$\int d\rho(g) \left\langle \right\rangle_{\mu\nu}^{\lambda} \mid g \rangle \langle g \mid \psi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} \mid g \rangle \langle g \mid \psi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} \mid g \rangle \langle g \mid \psi \rangle$ (and similarly for $\phi(g) = \langle g \mid \phi \rangle$) using the Dirac represen-
gilmore-lie-groups-F03337	322	1.000	$\phi(g) = \langle g \mid \phi \rangle$	$\phi(g) = \langle g \mid \phi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} \mid g \rangle \langle g \mid \psi \rangle$ (and similarly for $\phi(g) = \langle g \mid \phi \rangle$) using the Dirac represen-
gilmore-lie-groups-F03338	322	1.000	$\int \phi^*(g) \psi(g) d\rho(g)$	$\int \phi^*(g) \psi(g) d\rho(g)$	tation. Write down the Parseval equality for the inner product $\int \phi^*(g) \psi(g) d\rho(g)$
gilmore-lie-groups-F03339	323	1.000	He^{+}	He^{+}	H. A. BETHE AND E. E. SALPETER (1951), <i>Quantum Mechanics of One- and Two-Electron</i>
gilmore-lie-groups-F03340	324	0.868	$A_n; D_n, B_n, C_n, J$	$A_n; D_n, B_n, C_n, J$	1853–1854.
gilmore-lie-groups-F03341	325	0.866	$\mathrm{mx} / \sim$	$\mathrm{mx} / \sim$	<i>Physics, Memoirs of the American Mathemat</i>
gilmore-lie-groups-F03342	327	0.995	$A(p\ q),\ 39, 48$	$A(pq), 39, 48$	$A_2, 151, 160$
gilmore-lie-groups-F03343	327	0.978	$A_{\{1\}},\ 161$	$A_1, 161$	$A_3, 46$
gilmore-lie-groups-F03344	327	0.991	$A_{\{2\}}$	A_2	$A_n, 151, 160, 161$
gilmore-lie-groups-F03345	327	0.865	$A_{\{3\}},\ 46, 161, 162$	$A_3, 46, 161, 162$	$A_3, 151, 160, 161$
gilmore-lie-groups-F03346	327	0.946	$A_{\{n\}},\ 46, 49, 161, 166$	$A_n, 46, 49, 161, 166$	$B_3, 162$
gilmore-lie-groups-F03347	327	0.992	$B_{\{1\}},\ 161$	$B_1, 161$	$B_n, 11$
gilmore-lie-groups-F03348	327	0.749	$B_{\{2\}},\ 151, 160, 161, 164$	$B_2, 151, 160, 161, 164$	$C_1, 161$
gilmore-lie-groups-F03349	327	0.761	$B_{\{3\}},\ 162$	$B_3, 162$	$C_2, 15$
gilmore-lie-groups-F03350	327	0.729	$\boldsymbol{B}_{\{n\}},\ 161, 166, 168$	$B_n, 161, 166, 168$	$C_3, 162$
gilmore-lie-groups-F03351	327	1.000	$C_{\{1\}},\ 161$	$C_1, 161$	$C_n, 11$
gilmore-lie-groups-F03352	327	0.523	$C_{\{2\}},\ 151, 160, 161, 164$	$C_2, 151, 160, 161, 164$	$D_2, 151, 160, 161, 164$
gilmore-lie-groups-F03353	327	0.712	$C_{\{3\}},\ 162$	$C_3, 162$	$D_3, 11$
gilmore-lie-groups-F03354	327	0.997	$C_{\{n\}},\ 161, 166, 168$	$C_n, 161, 166, 168$	$E(2), 91, 207$
gilmore-lie-groups-F03355	327	0.840	$D_{\{2\}},\ 151, 160, 162$	$D_2, 151, 160, 162$	$E(3), 42$
gilmore-lie-groups-F03356	327	0.435	$G\ L(1\ ;\ \mathbb{Q}),\ 40, 47$	$GL(1; \mathbb{Q}), 40, 47$	$GL(n; \mathbb{C}), 47$
gilmore-lie-groups-F03357	327	0.979	$G\ L(2\ ;\ \mathbb{C}),\ 47$	$GL(2; \mathbb{C}), 47$	$GL(n; \mathbb{F}), 34$
gilmore-lie-groups-F03358	327	0.788	$G\ L(2\ ;\ \mathbb{R})$	$GL(2; \mathbb{R})$	$GL(n; \mathbb{Q}), 4$
gilmore-lie-groups-F03359	327	0.731	$G\ L(2\ ;\ \mathbb{Z}),\ 45, 49$	$GL(2; \mathbb{Z}), 45, 49$	$GL(n; \mathbb{R}), 47, 11$
gilmore-lie-groups-F03360	327	0.903	$G\ L(3\ ;\ \mathbb{Z}),\ 45$	$GL(3; \mathbb{Z}), 45$	$GL(n; \mathbb{Z}), 44, 44$
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